THE COMPLETE REFERENCE 2026 EDITION

CCNA Mastery

From Absolute Beginner to Enterprise Network Engineer

A comprehensive, beginner-friendly guide to the Cisco Certified Network Associate (200-301) exam. Twenty focused parts take you from "what is a network?" through enterprise architecture, OSPF, BGP, security, automation, and 100+ hands-on labs. Includes 200+ solved subnetting examples, 1,300+ practice questions, full Cisco CLI configurations, and 5 real-world enterprise projects.

TCP/IP & OSI | IPv4 & IPv6 | Subnetting & VLSM | Switching & VLANs | OSPF & BGP | Cisco IOS | Security & ACLs | Wireless | Automation | SDN & Python

AUDIENCE

Beginners · IT Support · Aspiring Network Engineers

ALIGNED WITH

Cisco CCNA 200-301 Exam Blueprint

20 PARTS · 100+ LABS · 1300+ QUESTIONS ENTERPRISE EDITION

Preface: How to Use This Book

This book is designed to take a complete beginner — someone who has never configured a router or understood what an IP address really is — and guide them step by step to the point where they can confidently sit the Cisco CCNA 200-301 exam, design and configure enterprise networks, troubleshoot complex issues, and qualify for entry-level network engineer roles. Every concept is explained in plain English with real-world analogies. Every chapter includes visual diagrams, lab exercises, troubleshooting scenarios, exam tips, common mistakes, a quiz, and practical challenges. Work through the book linearly the first time, then return to specific parts as reference.

The book is organized into 20 parts. Parts 1-5 build the foundations: what networks are, the OSI and TCP/IP models, Ethernet, and IPv4. Parts 6-10 go deeper: subnetting mastery, IPv6, Cisco devices, the Cisco IOS CLI, and switching. Parts 11-15 cover routing, wireless, network services, security, and automation. Parts 16-20 are practical: troubleshooting, 100 hands-on labs, 5 enterprise projects, exam preparation with 1,300+ practice questions, and a career roadmap.

How to study with this book: Read each chapter actively — don't just skim. Take notes. Work through every lab in Packet Tracer (free from Cisco NetAcad). Try the quiz questions before reading the answers. When you reach Part 19 (Exam Prep), do all 1,300+ practice questions and aim for 90%+ correct before booking the real exam. Build the five enterprise projects in Part 18 — they're worth their weight in gold for job interviews.

Prerequisites: A computer running Windows, macOS, or Linux. Internet access to download Packet Tracer and Wireshark (both free). No prior networking knowledge required. Basic computer literacy (opening a command prompt, browsing the web) is helpful.

Book Conventions

Throughout this book you'll see several types of callout boxes:

Exam Tip (red): Tips specifically for the CCNA 200-301 exam — what to memorize, common traps, time-saving techniques.

Common Mistake (red): Frequent errors that cost points or break networks. Learn from others' pain.

Tip (green): Practical advice that goes beyond the exam — real-world best practices from working engineers.

Warning (yellow): Things that can break production networks. Read carefully before applying configs.

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PART 1

Networking Foundations

What networks are, why they exist, and how they evolved

Chapter 1: What Is a Network?

A computer network is a collection of devices — computers, servers, phones, printers, and increasingly everything from light bulbs to cars — connected together so they can share information and resources. When you check your email, stream a video, or print a document from your laptop to a printer down the hall, you are using a network. Even the simple act of reading this sentence involves dozens of networks passing data between your device and the server where this content is stored. Networks have become so essential to modern life that we rarely notice them — until they stop working.

At its most basic, a network requires three things: devices that want to communicate (called **nodes** or **hosts**), a physical or wireless medium that carries the signal between them (cables, fiber, radio waves), and a set of rules — called a **protocol** — that both devices agree to follow so the message actually makes sense. Without an agreed protocol, two devices might as well be speaking different languages. The most famous of these protocols is the Internet Protocol suite (TCP/IP), and the entire modern Internet runs on it.

Networks exist because sharing is useful. Before networks, if you wanted to give a file to a colleague, you had to copy it onto a floppy disk and physically walk it over — a process jokingly called the 'sneakernet'. Networks let devices share files in milliseconds, share expensive peripherals (one office printer instead of one per desk), centralize data (so 50 accountants can all work on the same ledger), and most importantly today, share a single Internet connection across every device in a home or office. Without networks, the cloud, video streaming, online gaming, and basically every modern digital service would be impossible.

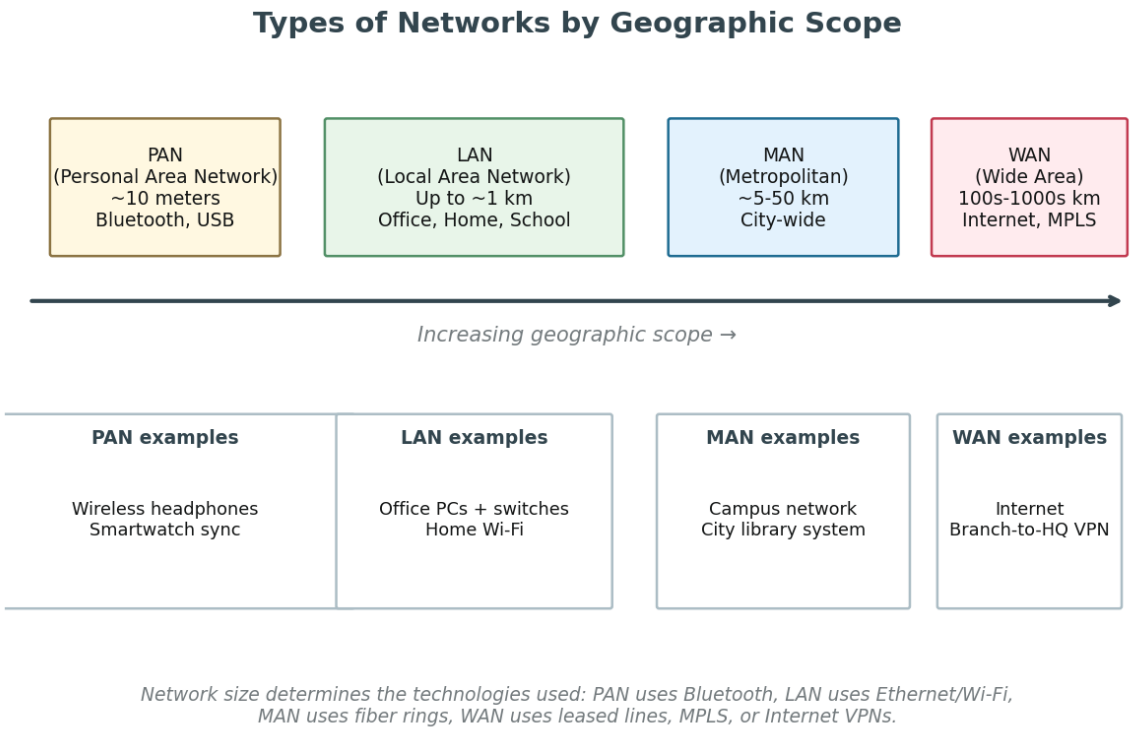


Figure: Networks are categorized by the geographic area they cover — from a personal Bluetooth link to the global Internet itself.

Why Networks Exist: Five Core Reasons

Networks solve real business and human problems. Understanding these underlying motivations makes every other concept in this book easier to grasp, because every protocol, device, and feature exists to serve one of these goals. When you find yourself struggling with a concept, ask yourself: *which of these five needs does this serve?* The answer almost always reveals the purpose.

Reason	Description	Real Example
Resource Sharing	Multiple devices use one expensive resource (printer, storage, server)	Office of 30 people sharing one network-attached printer and file server
Communication	People and systems exchange messages instantly regardless of distance	Email, Slack, VoIP phone calls, video conferencing
Data Centralization	Single source of truth that everyone can access and update	Hospital patient records accessible from any ward
Internet Access Sharing	One WAN connection serves hundreds of internal devices via NAT	Home Wi-Fi router sharing one broadband line across all family devices
Scalability & Redundancy	Add capacity or survive failures by adding more devices	A web farm of 100 servers handles traffic spikes and survives 10 servers failing

A Brief History of Computer Networks

Computer networking is younger than most people realize. The first packet-switched network, ARPANET, sent its first message in October 1969 between UCLA and Stanford. The message was supposed to be 'LOGIN' but the system crashed after two letters, so the very first Internet message was actually 'LO' — a fitting start to a technology that has been imperfect ever since. From that humble beginning, networks grew through several distinct eras, each shaped by the limitations of the technology available at the time.

The 1970s saw the invention of Ethernet by Robert Metcalfe at Xerox PARC (originally at 2.94 Mbps over coaxial cable) and the development of TCP/IP by Vint Cerf and Bob Kahn. The 1980s brought personal computers into offices and homes, creating massive demand for Local Area Networks, and saw the introduction of the OSI reference model in 1984. The 1990s saw the World Wide Web transform the Internet from an academic tool into a household utility, while the 2000s brought broadband, Wi-Fi, and the beginnings of mobile data. The 2010s saw cloud computing, software-defined networking, and 4G mobile make networks ubiquitous and invisible. Today, in the 2020s, we are entering the era of 5G, Wi-Fi 6E/7, automation, and edge computing — networks now carry more traffic from machines to machines than from humans to humans.

> **Exam Tip**
> The CCNA 200-301 blueprint expects you to know the difference between LAN, WAN, MAN, and PAN, and to recognize common network topologies (bus, star, ring, mesh, tree, hybrid). Memorize the geographic scope of each.

Types of Networks by Geographic Scope

Networks are classified primarily by the geographic area they cover. This is not just an academic distinction — the size of the network determines which technologies make sense. A PAN uses Bluetooth because low power and short range are fine for connecting headphones. A WAN uses MPLS, leased lines, or Internet VPNs because long distances require carrier infrastructure. Choosing the wrong technology for the scope leads to wasted money, poor performance, or unreliable service.

Type	Scope	Typical Technology	Example Use
PAN	~10 meters	Bluetooth, USB, NFC	Wireless headphones, smartwatch

Type	Scope	Typical Technology	Example Use
LAN	Up to ~1 km	Ethernet (Cat6), Wi-Fi	Office, home, school lab
MAN	~5-50 km	Fiber rings, metro Ethernet	City library system, campus
WAN	Hundreds to thousands of km	MPLS, leased lines, Internet VPN	Branch office to HQ, the Internet

Network Topologies

A **topology** describes the physical or logical arrangement of devices and the links between them. The topology determines how data flows, how the network handles failures, how easy it is to scale, and how much it costs to build. There are six classic topologies you need to know for the CCNA exam, plus the modern hybrid designs used in real enterprises.

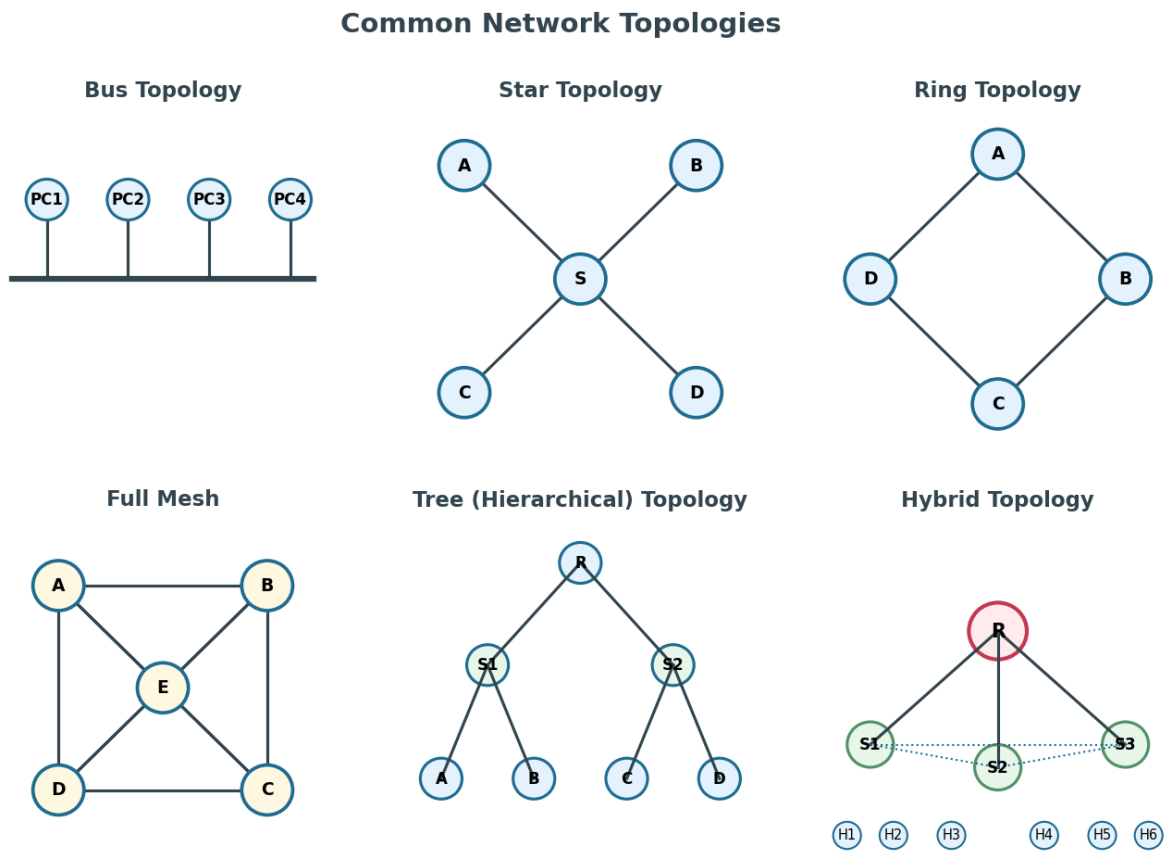


Figure: Six classic network topologies. Modern enterprise networks are typically hybrid designs combining two or more of these patterns.

Topology	Pros	Cons	When to Use
Bus	Cheap, simple	One break kills the network; collisions	Legacy, obsolete
Star	Easy to manage; one failure isolates	Central switch is single point of failure	Modern LANs (default)
Ring	Predictable, no collisions	One break can take down all	Token Ring, FDDI (legacy)
Mesh	Maximum redundancy	Expensive, complex	Backbone, military
Tree	Hierarchical, scales	Root failure = disaster	Enterprise 3-tier design

Topology	Pros	Cons	When to Use
Hybrid	Combines best of multiple	Complex to design and troubleshoot	Real-world enterprises

Data Transmission Basics

Data is transmitted across a network as a stream of bits — ones and zeros. How those bits are physically represented depends on the medium: on copper cable, they are voltage changes; on fiber, pulses of light; on wireless, modulations of radio waves. No matter the medium, three terms describe the performance of any data link: **bandwidth**, **throughput**, and **latency**. Understanding the difference is critical for troubleshooting and network design.

Bandwidth is the theoretical maximum capacity of a link — what it could carry under perfect conditions. A Gigabit Ethernet link has a bandwidth of 1,000 Mbps. Think of bandwidth as the width of a highway: a 10-lane highway can theoretically carry more cars per hour than a 2-lane one. **Throughput** is the actual data transferred per second in real conditions. It is always less than bandwidth because of protocol overhead (headers, ACKs), collisions, retransmissions, and congestion. A 1 Gbps link rarely delivers more than 940 Mbps of usable TCP throughput. **Latency** is the time delay for a single bit to travel from source to destination — measured in milliseconds. Fiber across the Atlantic has ~70 ms latency; satellite has ~600 ms; Wi-Fi adds 1-5 ms; copper LAN is sub-millisecond. Lower latency matters more for voice and gaming than for file downloads.

Metric	Definition	Unit	Analogy
Bandwidth	Theoretical max capacity	Mbps, Gbps	Highway width (lanes)
Throughput	Actual data transferred per second	Mbps	Cars per hour actually passing
Latency	Time for one bit to traverse link	milliseconds	Travel time per car
Jitter	Variation in latency	milliseconds	Inconsistent travel times
Packet Loss	% of packets that never arrive	%	Cars that don't reach destination

Network Components

Every network, from a home Wi-Fi setup to a global enterprise, is built from a small set of fundamental device types. Each type operates at a specific layer of the OSI model and solves a specific problem. Mastering these devices — what they do, how they work, and when to use each — is foundational to everything else in this book.

Device	OSI Layer	Function	Example
Hub	L1	Multi-port repeater; broadcasts to all ports	Obsolete, avoid
Repeater	L1	Regenerates signal to extend distance	Fiber extenders
Bridge	L2	Connects two LAN segments; learns MACs	Legacy, replaced by switches
Switch	L2 (or L3)	Forwards frames based on MAC table	Cisco Catalyst 2960
Router	L3	Forwards packets based on IP routing table	Cisco ISR 4000
Firewall	L3-L7	Filters traffic by ruleset	Cisco Firepower
Access Point	L2	Bridges wireless clients to wired LAN	Cisco Aironet
Wireless Controller	L2-L3	Manages multiple APs centrally	Cisco WLC 3504
Load Balancer	L4-L7	Distributes traffic across servers	F5 BIG-IP, Cisco ACE
Modem	L1-L2	Converts signals between media (e.g., cable to Ethernet)	Cable/DSL modem

Understanding Packets

A **packet** is the fundamental unit of data at Layer 3 (the Network layer). When you send a file across the Internet, it does not travel as one continuous stream — instead, it is broken into thousands of small packets, each carrying a portion of the data plus a header containing addressing and control information. Each packet can take a different path through the network, and they are reassembled at the destination. This approach, called **packet switching**, is vastly more efficient than the older alternative (circuit switching, where a dedicated path is held open for the duration of a call).

Think of packets like mailing the pages of a book in separate envelopes. Each envelope has the destination address (the IP header), a return address (source IP), a sequence number (so the receiver can reassemble them in order), and a portion of the actual book content (the payload). If one envelope is lost, only that one page needs to be re-sent, not the whole book. If a road is closed, the postal service can route different envelopes through different cities. This resilience and efficiency is why every modern network uses packet switching.

Bandwidth, Latency, Throughput — In Depth

These three terms are constantly confused, even by experienced engineers. Getting them straight is critical because each requires different troubleshooting approaches. If a user complains 'the network is slow', your first job is to figure out *which* of these is the actual problem. Adding more bandwidth does not help latency-bound applications like voice. Reducing latency does not help bandwidth-bound applications like file downloads.

Bandwidth is purchased from your ISP or built into your hardware. If you have a 100 Mbps WAN link, that is your ceiling. You cannot exceed it without upgrading the link. **Throughput** is what users actually experience. A 100 Mbps link with 30% packet loss and 500 ms latency might deliver only 10 Mbps of usable TCP throughput because TCP backs off when it senses congestion. **Latency** is dominated by physics — the speed of light in fiber is about 5 microseconds per kilometer, so a 5,000 km link has a hard minimum latency of 25 ms each way, no matter what you do. The only way to reduce it is to move the endpoints closer together (using CDNs, edge computing).

> **Common Mistake**
> Confusing bandwidth with throughput. A "slow" 1 Gbps link is usually a latency or loss problem, not a bandwidth problem. Always measure actual throughput (iperf3) and round-trip latency (ping) before assuming you need a bigger pipe.

Chapter 1 Quiz

Q1. What is the primary purpose of a computer network?

A. To make computers run faster
B. To share resources and communicate between devices
C. To increase hard drive storage
D. To provide battery backup
Answer: B — Networks exist to share resources (files, printers, internet) and enable communication.

Q2. Which network type covers the smallest geographic area?

A. LAN
B. WAN
C. MAN
D. PAN
Answer: D — PAN (Personal Area Network) covers ~10 meters, smaller than LAN/WAN/MAN.

Q3. Which topology provides maximum redundancy by connecting every device to every other?

A. Bus

B. Star

C. Ring

D. Full Mesh

Answer: D — In full mesh, every node connects to every other node — maximum redundancy but expensive.

Q4. What is the difference between bandwidth and throughput?

A. Bandwidth = max capacity; Throughput = actual data transferred per second

B. Bandwidth = speed; Throughput = latency

C. They are the same thing

D. Bandwidth is physical, throughput is logical

Answer: A — Bandwidth is theoretical maximum; throughput is what's actually achieved due to overhead, collisions, etc.

Q5. Which unit measures latency?

A. Mbps

B. ms (milliseconds)

C. Bytes

D. Hz

Answer: B — Latency = delay, measured in milliseconds. Lower = better.

Practical Challenge

> **Hands-On Lab**
>
> On your home network, open a command prompt and run:
>
> **ping 8.8.8.8** — note the round-trip time (latency)
>
> **ping -n 10 8.8.8.8** (Windows) or **ping -c 10 8.8.8.8** (Mac/Linux)
>
> Calculate packet loss percentage: lost packets ÷ total sent × 100
>
> **tracert 8.8.8.8** (Windows) or **traceroute 8.8.8.8** (Mac/Linux)
>
> Count the hops — this is how many routers are between you and Google DNS.

PART 2

OSI Model Deep Dive

Seven layers from bits to applications

Chapter 2: The OSI Reference Model

The OSI (Open Systems Interconnection) model is the most important conceptual framework in all of networking. Developed by the International Organization for Standardization in 1984, it defines seven distinct layers that data passes through as it travels from an application on one computer to an application on another. No real-world protocol implements all seven layers exactly as specified — the TCP/IP model that actually runs the Internet collapses them into four — but OSI remains the universal language that network engineers use to discuss and troubleshoot networks.

The genius of the OSI model is that it isolates concerns. Each layer has a specific job and depends only on the layer below it. This means you can swap out the physical cable (Layer 1) without changing your applications (Layer 7), or upgrade your routing protocol (Layer 3) without affecting how a web browser renders HTML. This separation is what makes modern networking possible — without it, every change would break everything.

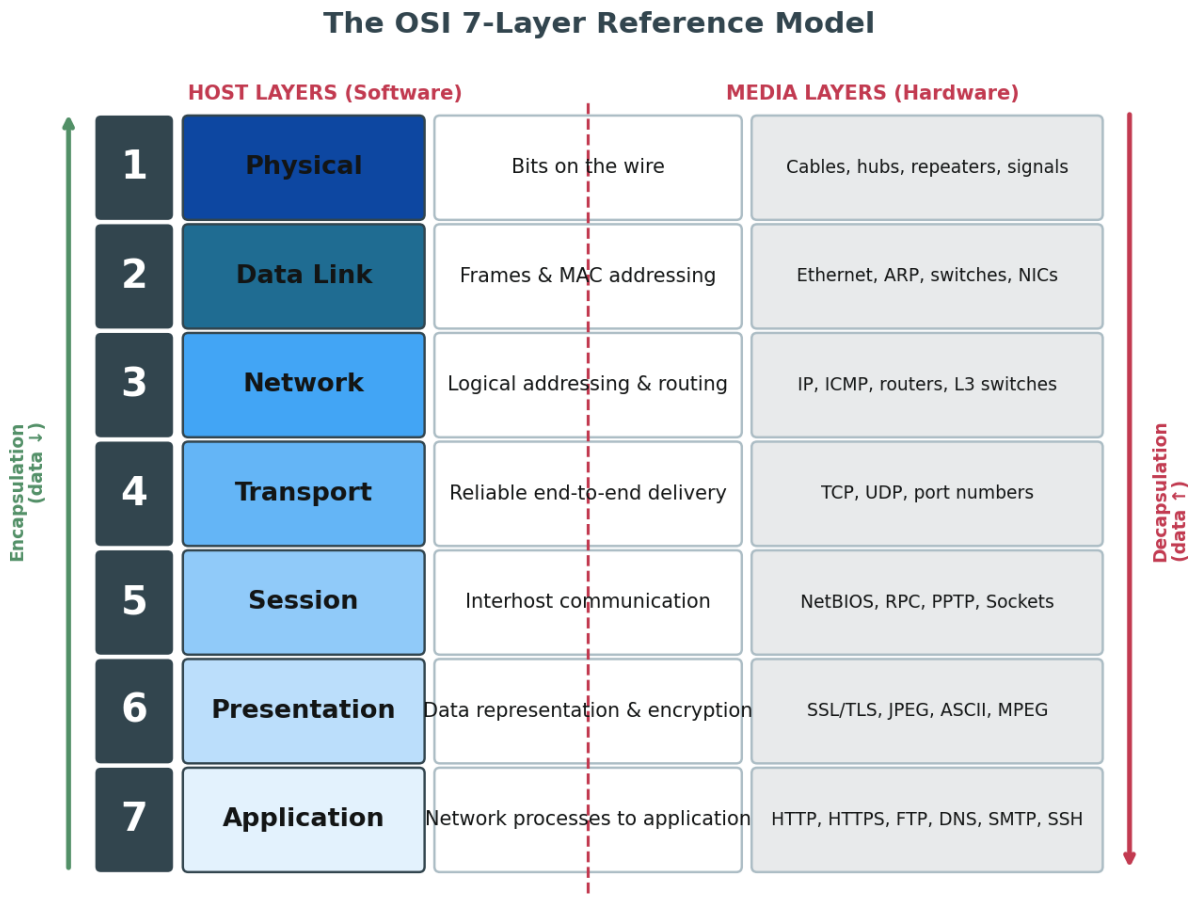


Figure: The seven OSI layers, from Physical (bits on the wire) at the bottom to Application (user-facing protocols) at the top.

Memorizing the Seven Layers

Every CCNA candidate needs to instantly recall the seven layers in order. Two classic mnemonics help — one reading bottom-up, one top-down. Pick the one that sticks for you.

Bottom to top (Layer 1 to 7): **P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way (Physical, Data Link, Network, Transport, Session, Presentation, Application)

Top to bottom (Layer 7 to 1): All **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing

> **Exam Tip**
> For each layer, memorize: (1) the PDU name (bit, frame, packet, segment, data), (2) the devices that operate there, (3) the protocols that live there. CCNA loves questions like "Which OSI layer does ARP operate at?" — answer: Layer 2 (Data Link).

Layer 1: Physical

Transmitting raw bits on the wire

Purpose: Converts bits (1s and 0s) into physical signals — electrical voltages on copper, light pulses on fiber, or radio waves on wireless — and transmits them across the medium. Also defines the connectors, voltage levels, timing, and pin layouts.

Devices: Hubs, repeaters, cables (Cat5e/6/6a, fiber, coax), connectors (RJ-45, LC, SC), NICs (the physical transceiver portion), media converters.

Protocols/Standards: IEEE 802.3 (Ethernet physical variants like 100BASE-TX, 1000BASE-T, 10GBASE-T), IEEE 802.11 (Wi-Fi physical), RS-232, SONET/SDH.

PDU: Bits (binary digits — 0 and 1).

Encapsulation: Layer 1 receives the bits from Layer 2 and converts them to physical signals on the medium.

Common Issues: Broken cables, bad connectors, EMI (electromagnetic interference), attenuation (signal loss over distance), duplex mismatch (technically L2 but symptoms manifest here), wrong cable type (crossover vs straight).

Layer 2: Data Link

Frames, MAC addresses, and local switching

Purpose: Provides reliable node-to-node delivery on the same physical network. Handles framing (packaging data into frames), physical addressing (MAC), error detection (FCS), and media access control (who can transmit when, e.g., CSMA/CD for Ethernet).

Devices: Switches (L2), bridges (legacy), NICs (the MAC layer portion), wireless access points (at L2).

Protocols: Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11), ARP (Address Resolution Protocol — straddles L2/L3), PPP, HDLC, Frame Relay (legacy), STP (Spanning Tree Protocol — L2 control plane).

PDU: Frame (includes destination MAC, source MAC, EtherType, payload, and FCS — a 4-byte CRC for error detection).

Encapsulation: L2 takes the packet from L3, adds source/destination MAC addresses, EtherType, and FCS to form a frame.

Common Issues: Duplex mismatch, MAC flapping, broadcast storms (STP failure), VLAN misconfiguration, CRC errors from bad cabling.

Layer 3: Network

Logical addressing and routing

Purpose: Moves data across multiple networks. Handles logical addressing (IP), routing (selecting the best path), and packet forwarding. This is the layer where the Internet actually internets — connecting millions of separate L2 networks into one global fabric.

Devices: Routers, Layer 3 switches, multilayer firewalls.

Protocols: IPv4, IPv6, ICMP (ping, traceroute), OSPF, EIGRP, RIP, BGP, IPsec (ESP/AH).

PDU: Packet (carries source/destination IP addresses and the payload).

Encapsulation: L3 takes the segment from L4, adds source/destination IP and protocol field to form a packet.

Common Issues: Wrong subnet mask, missing/default route, routing protocol adjacency failure, ACL blocking, asymmetric routing.

Layer 4: Transport

Reliable end-to-end delivery

Purpose: Provides end-to-end communication between applications on different hosts. Handles multiplexing (port numbers identify applications), reliable delivery (TCP), flow control (TCP sliding window), and error recovery (TCP ACKs and retransmissions).

Devices: Firewalls (L4+), load balancers (L4+), stateful inspection devices.

Protocols: TCP (Transmission Control Protocol — reliable, connection-oriented), UDP (User Datagram Protocol — unreliable, connectionless), SCTP (less common).

PDU: Segment (TCP) or Datagram (UDP).

Encapsulation: L4 takes application data, adds source/destination port numbers, sequence numbers, and other TCP/UDP header fields to form a segment.

Common Issues: Firewall blocking port, NAT translation failure, MTU mismatch causing fragmentation, TCP connection refused (port closed), UDP packet loss.

Layer 5: Session

Managing application dialogues

Purpose: Establishes, maintains, and terminates sessions (dialogues) between applications. Handles session checkpointing, resumption after interruption, and dialog control (half-duplex vs full-duplex).

Devices: No traditional network devices operate here — this layer lives inside the operating systems and applications.

Protocols: NetBIOS, RPC, PPTP (control channel), SQL session establishment, SMB session setup. Many modern protocols (HTTP, SSH) effectively incorporate session functions into the application itself.

PDU: Data (no special name).

Encapsulation: Adds session control information to the L6/L7 data.

Common Issues: Session timeouts, idle disconnects, mismatched session parameters.

Layer 6: Presentation

Data translation and encryption

Purpose: Translates, compresses, and encrypts data so the receiving application can understand it. Handles character encoding (ASCII, UTF-8), data compression (gzip), and encryption (TLS, SSL).

Devices: No standalone devices — this layer is implemented in software on endpoints.

Protocols: TLS/SSL (technically spans L4-L7 but commonly placed at L6), JPEG, GIF, MPEG, ASCII, EBCDIC, MIME encoding.

PDU: Data (still no special name).

Encapsulation: Translates/encrypts L7 data before passing to L5.

Common Issues: TLS handshake failures, certificate expiration, encoding mismatches (UTF-8 vs Latin-1), compression bugs.

Layer 7: Application

Network services for end-user applications

Purpose: Provides network services directly to user applications. This is the layer users actually interact with — the layer where web browsing, email, file transfer, and DNS happen.

Devices: Application servers, web servers, email servers, DNS servers, proxies.

Protocols: HTTP, HTTPS, DNS, DHCP, SMTP, POP3, IMAP, FTP, SFTP, SSH, Telnet, SNMP, Syslog, NTP, RADIUS, TACACS+.

PDU: Data (the actual content being exchanged — an HTML page, an email message, a DNS response).

Encapsulation: Generates the data to be sent, then hands it to L6 (or directly to L4 in modern stacks).

Common Issues: Service not running, DNS resolution failure, wrong URL, expired credentials, application bugs.

Encapsulation & Decapsulation: The Packet Journey

When you type a URL in your browser and press Enter, an incredible sequence of events begins. The HTTP request generated by your browser must travel down through all seven OSI layers on your computer, across the network as bits, and back up through all seven layers on the destination server. The downward journey is called **encapsulation** — each layer adds its own header (and sometimes trailer) to the data. The upward journey is **decapsulation** — each layer removes and processes its header.

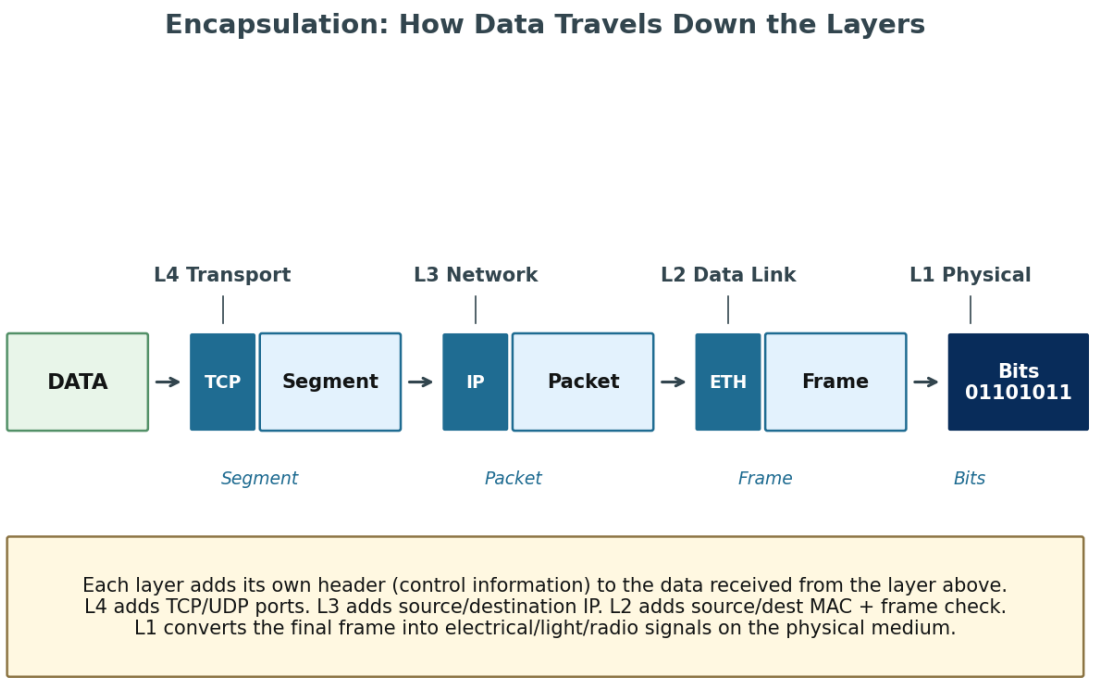


Figure: Encapsulation: each layer wraps the data from above with its own header. The reverse process at the destination is decapsulation.

At Layer 7, your browser creates an HTTP request like GET /index.html HTTP/1.1. This data is handed to Layer 4, which adds TCP header (source port like 51234, destination port 80, sequence number, etc.) — forming a **segment**. The segment goes to Layer 3, which adds an IP header (source IP 192.168.1.10, destination IP 93.184.216.34) — forming a **packet**. The packet goes to Layer 2, which adds Ethernet header (source MAC, destination MAC of the default gateway, EtherType 0x0800 for IPv4) plus a 4-byte FCS — forming a **frame**. Finally Layer 1 converts this frame into electrical signals on the wire, light pulses on fiber, or radio waves on Wi-Fi.

On the receiving end, the process reverses. Layer 1 captures the signals and reconstructs the bit stream. Layer 2 verifies the FCS (drops if corrupted), strips the Ethernet header, and passes the packet up. Layer 3 reads the IP header to confirm the destination matches, then strips it. Layer 4 reads the TCP header, verifies sequence numbers, sends an ACK, and hands the data to Layer 7 — where the web server processes the HTTP request and prepares a response. The whole cycle takes a few milliseconds for nearby servers, longer for cross-continent traffic.

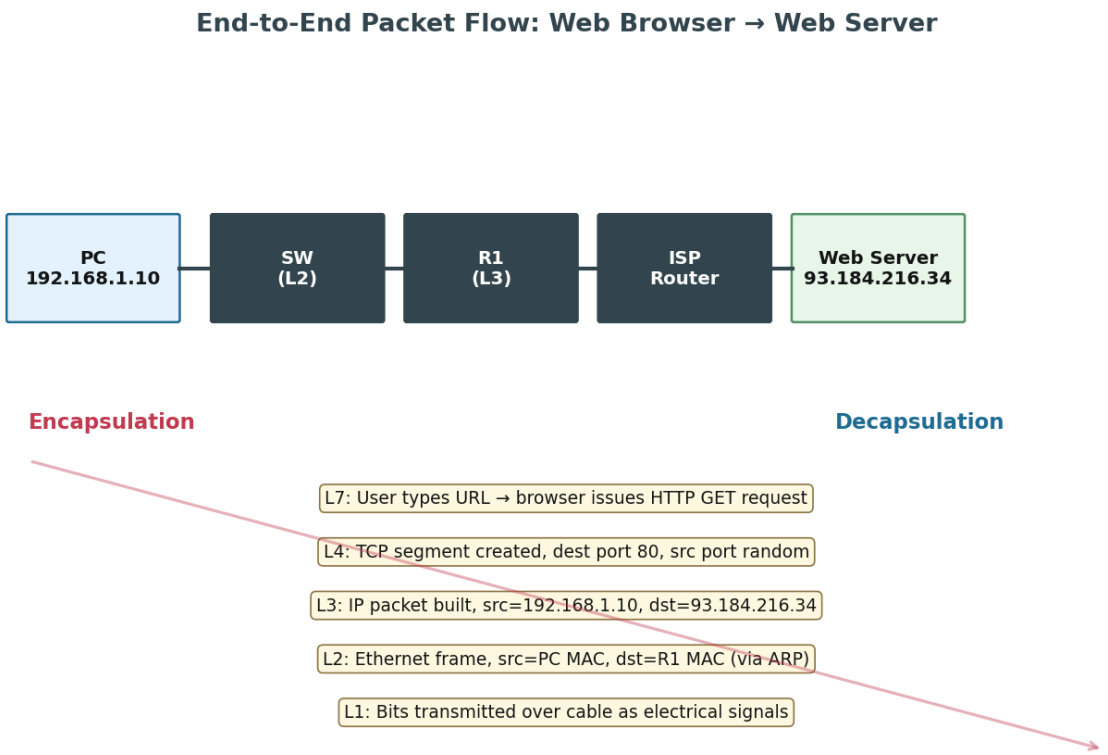


Figure: End-to-end packet flow: data is encapsulated at the source, switched through the LAN, routed across the WAN, and decapsulated at the destination.

> **Troubleshooting Tip**
>
> Always troubleshoot bottom-up: check Layer 1 first (cable plugged in? link lights?), then Layer 2 (VLAN/MAC issues), then Layer 3 (IP/routing), then Layer 4 (firewall/port), finally Layer 7 (application/service). Most issues are L1/L2 — never skip the basics.

Chapter 2 Quiz

Q1. At which OSI layer does a switch primarily operate?

A. Layer 1 - Physical
B. Layer 2 - Data Link
C. Layer 3 - Network
D. Layer 4 - Transport

Answer: B — Switches make forwarding decisions based on MAC addresses, which is a Layer 2 function.

Q2. What is the PDU at Layer 3 (Network)?

A. Bit
B. Frame
C. Packet
D. Segment

Answer: C — Layer 3 PDU = Packet. Layer 2 = Frame. Layer 4 = Segment. Layer 1 = Bit.

Q3. Which layer is responsible for encryption?

A. Layer 3
B. Layer 4
C. Layer 6 (Presentation)
D. Layer 7
Answer: C — Encryption is a Presentation layer function, though modern TLS spans L4-L7.

Q4. How many layers does the OSI model have?

A. 4
B. 5
C. 7
D. 9
Answer: C — OSI has 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q5. Which layer handles logical addressing (IP)?

A. Layer 2
B. Layer 3
C. Layer 4
D. Layer 7
Answer: B — Layer 3 (Network) handles IP addressing and routing.

Practical Challenge

> **Hands-On Lab: OSI Layer Walkthrough**
>
> Open Wireshark and capture traffic while loading a web page. Identify each layer:
>
> 1. Find an HTTP/HTTPS frame — observe the layers in the packet details pane
> 2. Expand Layer 2: source/dest MAC, EtherType
> 3. Expand Layer 3: source/dest IP, TTL, protocol
> 4. Expand Layer 4: source/dest port, TCP flags, sequence numbers
> 5. Expand Layer 7: HTTP method, URL, headers
>
> For each layer, write down one piece of information you observed.

PART 3

TCP/IP Model

The practical model that runs the real Internet

Chapter 3: The TCP/IP Model

While the OSI model is theoretically elegant, the model that actually runs the Internet is the **TCP/IP model**, named after its two most famous protocols. Developed in the 1970s by Vint Cerf and Bob Kahn (with heavy influence from the ARPANET research community), TCP/IP predates OSI and was designed to solve a real, pressing problem: how to connect many different networks — each with their own physical media and protocols — into one giant 'inter-network'. The answer was brilliant in its simplicity: define a single universal addressing scheme (IP) and a universal transport layer (TCP/UDP), and let the underlying networks be whatever they want.

The classic TCP/IP model has four layers. From top to bottom they are: **Application** (combining OSI layers 5, 6, 7), **Transport** (OSI layer 4), **Internet** (OSI layer 3), and **Network Access** (combining OSI layers 1 and 2). A modern five-layer version splits Network Access into Physical and Data Link for clarity. The four-layer version is what the CCNA exam expects you to know.

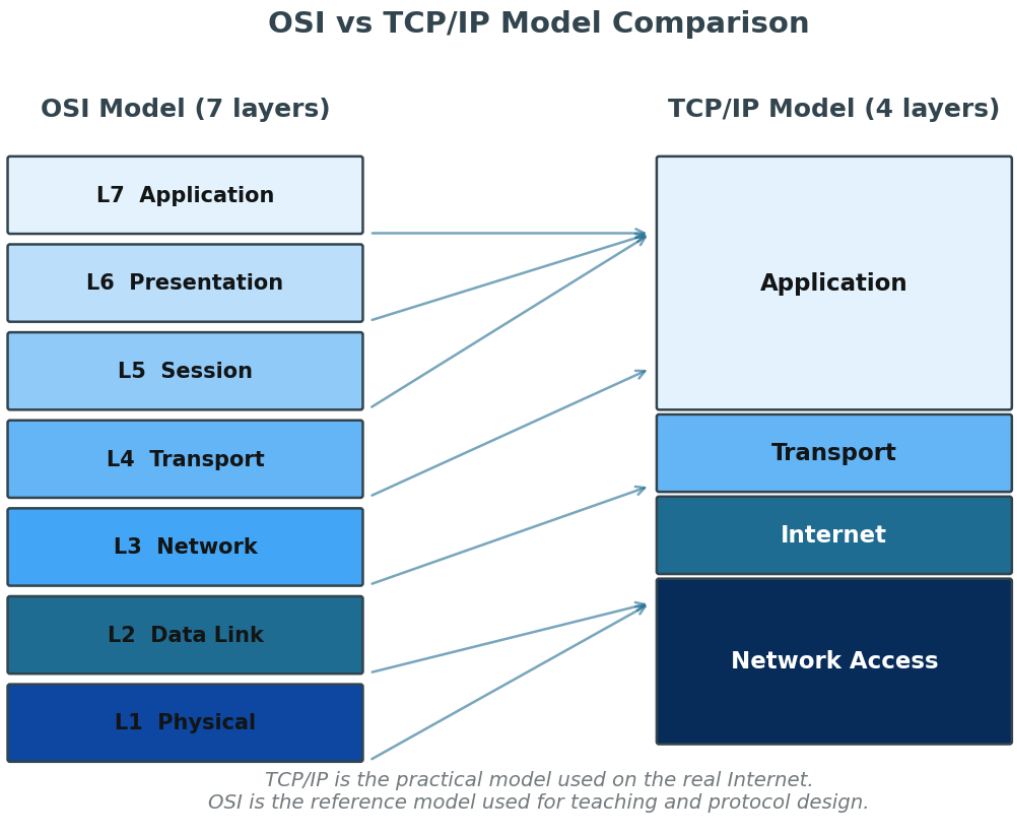


Figure: Comparison of OSI (7 layers) and TCP/IP (4 layers). TCP/IP combines the upper three OSI layers into one Application layer and the bottom two into one Network Access layer.

TCP vs UDP: The Transport Showdown

Layer 4 of TCP/IP offers two very different transport protocols. **TCP** (Transmission Control Protocol) is reliable, connection-oriented, and ordered — it guarantees that data arrives intact, in the right order, or it tells the application it failed. **UDP** (User Datagram Protocol) is unreliable, connectionless, and unordered — it fires off packets and hopes for the best. Each is appropriate for different applications, and choosing the wrong one causes real problems.

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable (ACKs, retransmissions)	Best-effort (no guarantees)
Ordering	Ordered (sequence numbers)	No ordering guarantee
Header size	20 bytes minimum	8 bytes (fixed)
Speed	Slower (overhead)	Faster (no overhead)
Flow control	Yes (sliding window)	No
Congestion control	Yes (slow start, AIMD)	No
Use cases	Web (HTTP/HTTPS), email, file transfer, SSH	DNS, DHCP, SNMP, VoIP, video streaming, gaming

A useful analogy: TCP is like a registered letter with delivery confirmation and tracking — guaranteed to arrive, signed for, in order, but slow and expensive. UDP is like a postcard thrown into a mailbox — fast and cheap, but it might get lost or arrive out of order. For a bank transaction, you want TCP. For a live video stream, you want UDP — by the time a retransmitted packet arrives, the video frame is already obsolete.

Ports and Sockets

An IP address identifies a host, but a host can run many applications simultaneously. How does the operating system know which packet goes to which application? The answer is **port numbers**. A port is a 16-bit number (0-65535) that identifies a specific application or service on a host. Combined with an IP address, it forms a **socket** (IP:port), which uniquely identifies a single conversation endpoint.

IANA (Internet Assigned Numbers Authority) divides ports into three ranges. **Well-known ports** (0-1023) are assigned to common services — HTTP (80), HTTPS (443), SSH (22), DNS (53), SMTP (25). **Registered ports** (1024-49151) are assigned to specific applications by IANA. **Ephemeral/dynamic ports** (49152-65535) are used by clients as temporary source ports for outgoing connections. When you browse to a website, your browser picks a random ephemeral port like 51234 as its source, and connects to port 80 or 443 on the server.

Port	Protocol	Service	Transport
20, 21	FTP	File Transfer	TCP
22	SSH	Secure Shell	TCP
23	Telnet	Remote login (insecure)	TCP
25	SMTP	Email sending	TCP
53	DNS	Domain Name System	UDP/TCP
67, 68	DHCP	IP assignment	UDP
69	TFTP	Trivial FTP	UDP
80	HTTP	Web (unencrypted)	TCP
110	POP3	Email retrieval	TCP
123	NTP	Time sync	UDP
143	IMAP4	Email retrieval (modern)	TCP
161, 162	SNMP	Network monitoring	UDP
389, 636	LDAP/LDAPS	Directory access	TCP
443	HTTPS	Web (encrypted)	TCP
514	Syslog	Logging	UDP
3389	RDP	Remote Desktop	TCP

The TCP Three-Way Handshake

TCP is connection-oriented, which means before any data can be exchanged, the two endpoints must establish a connection. They do this with a three-step process called the **three-way handshake**. Understanding this handshake is critical — it appears in countless Wireshark captures, firewall logs, and exam questions, and it's the foundation of every TCP-based connection you make.

Step 1: The client sends a **SYN** (synchronize) segment with a random sequence number (let's call it x). This says 'I want to talk, and here's my starting sequence number.' Step 2: The server responds with a **SYN-ACK** segment, containing its own random sequence number (y) and an acknowledgment number of x+1 (acknowledging the client's SYN). Step 3: The client sends an **ACK** with acknowledgment number y+1. At this point, both endpoints have agreed on initial sequence numbers and the connection is established. Data can now flow in both directions.

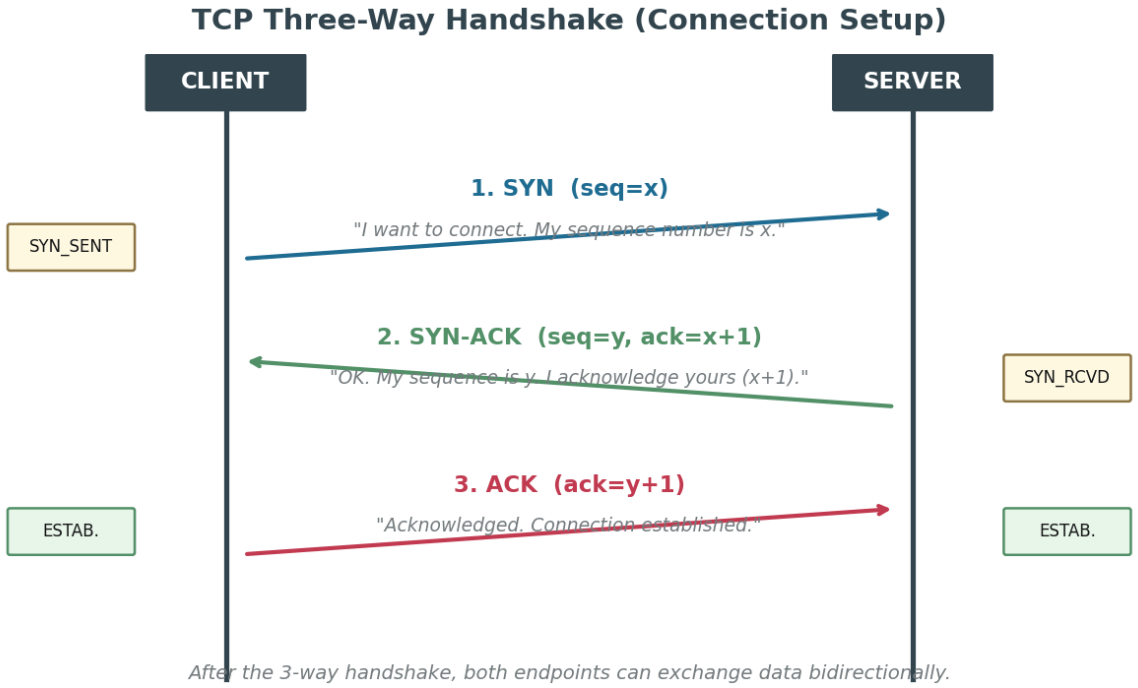


Figure: The TCP three-way handshake: SYN, SYN-ACK, ACK. After this exchange, both endpoints have agreed on initial sequence numbers and the connection is established.

Reliable Communication: How TCP Guarantees Delivery

TCP's reliability comes from a combination of four mechanisms. First, **sequence numbers** — every byte sent is numbered, so the receiver can detect gaps and reorder out-of-order segments. Second, **acknowledgments** — the receiver sends ACKs confirming which bytes it has received. If an ACK doesn't arrive within a timeout, the sender retransmits. Third, **flow control** via the sliding window — the receiver tells the sender how much buffer space it has, so the sender doesn't overwhelm a slow receiver. Fourth, **congestion control** — TCP monitors for packet loss (signaled by timeouts or duplicate ACKs) and reduces its sending rate to avoid congesting the network.

These mechanisms make TCP the choice for any application that needs reliable data transfer: web browsing, email, file transfers, SSH. The cost is overhead — every connection requires the 3-way handshake, every segment carries a 20-byte header (vs 8 for UDP), and retransmissions add latency. For real-time applications like

voice and video where occasional packet loss is preferable to delay, UDP is the right choice.

> **Common Mistake**
> Assuming TCP is always better because it's "reliable". For real-time applications (voice, video, gaming), TCP's retransmissions cause more harm than good — by the time a retransmitted packet arrives, it's too late to be useful. Use UDP for these.

Packet-Level Analysis with Wireshark

The best way to understand TCP/IP is to see it in action. Wireshark is the world's most popular packet analyzer, and it's free. Install it, start a capture on your Wi-Fi or Ethernet interface, then browse to a website. Stop the capture and filter by tcp.port == 443 to see HTTPS traffic. You'll see the three-way handshake (SYN, SYN-ACK, ACK), followed by encrypted application data, followed by the connection teardown (FIN, ACK, FIN, ACK).

```
# Useful Wireshark display filters
tcp.port == 443                    # HTTPS traffic
tcp.flags.syn == 1                 # TCP SYN packets (handshake)
tcp.flags.reset == 1               # TCP RST packets
icmp                               # ping/traceroute
arp                                # ARP requests/replies
dns                                # DNS queries
ip.addr == 8.8.8.8                 # Traffic to/from Google DNS
!(ip.addr == 192.168.1.0/24)       # Exclude local traffic
```

Chapter 3 Quiz

Q1. How many layers does the classic TCP/IP model have?

A. 4
B. 5
C. 7
D. 9
Answer: A — Classic TCP/IP has 4 layers: Application, Transport, Internet, Network Access.

Q2. Which transport protocol is connection-oriented?

A. UDP
B. TCP
C. IP
D. ICMP
Answer: B — TCP is connection-oriented (uses 3-way handshake). UDP is connectionless.

Q3. What is the well-known port for HTTPS?

A. 80
B. 443
C. 22
D. 53
Answer: B — HTTPS uses port 443 (HTTP over TLS).

Q4. How many segments are exchanged in a TCP 3-way handshake?

A. 2
B. 3
C. 4
D. 5
Answer: B — SYN, SYN-ACK, ACK = 3 segments.

Q5. Which protocol is best for live video streaming?

A. TCP
B. UDP
C. ICMP
D. ARP
Answer: B — UDP is preferred for live streaming — lower latency, can tolerate some packet loss.

PART 4

Ethernet Fundamentals

The dominant LAN technology of the modern world

Chapter 4: Ethernet — The King of LANs

If you unplugged every network cable on Earth and looked at the connector, you'd find an RJ-45 plug. Look at the cable itself, and it's Cat5e or Cat6 twisted pair. Look at the protocol the devices are speaking, and it's Ethernet. Ethernet has so thoroughly dominated local area networking that the term 'LAN' is almost synonymous with 'Ethernet LAN'. The technology has survived and evolved since Robert Metcalfe invented it at Xerox PARC in 1973 — from 2.94 Mbps over thick coaxial cable to 400 Gbps over fiber today, with terabit Ethernet on the horizon.

Ethernet's longevity comes from a combination of simplicity, low cost, and continuous evolution. The core ideas have stayed the same for fifty years: devices share a medium, they transmit **frames** addressed by **MAC address**, and collisions are detected and handled. What has changed is the speed, the medium (coax → twisted pair → fiber), and the topology (bus → star with switches). Understanding Ethernet is essential — almost every enterprise network runs on it.

Ethernet Frames

An Ethernet **frame** is the Layer 2 PDU — the unit of data that switches forward. A standard frame consists of seven fields, with the total size ranging from 64 to 1518 bytes (excluding the preamble). Understanding each field helps you troubleshoot and analyze captures.

Field	Size	Purpose
Preamble	7 bytes	Synchronization pattern (10101010...) for receiver clock sync
Start Frame Delimiter (SFD)	1 byte	Marks end of preamble, start of frame (10101011)
Destination MAC	6 bytes	MAC address of intended recipient (or broadcast FF:FF:FF:FF:FF:FF)
Source MAC	6 bytes	MAC address of sender
EtherType / Length	2 bytes	Upper-layer protocol (0x0800=IPv4, 0x86DD=IPv6, 0x0806=ARP) OR frame length
Payload (Data + Pad)	46-1500 bytes	Upper-layer data (the IP packet). Padded if <46 bytes
Frame Check Sequence (FCS)	4 bytes	CRC-32 for error detection — receiver drops frame if mismatch

The minimum frame size of 64 bytes is not arbitrary — it's tied to CSMA/CD collision detection. The frame must be long enough that the sender is still transmitting when the signal reaches the far end of the cable and any collision propagates back. For 100m of copper at 100 Mbps, this works out to 64 bytes. If the payload is less than 46 bytes (so total frame would be less than 64), padding bytes are added. The maximum payload of 1500 bytes is the famous Ethernet **MTU** (Maximum Transmission Unit) — if you've ever heard of MTU problems, this is where they come from.

MAC Addresses

Every Ethernet network interface card (NIC) ever manufactured has a unique **MAC address** burned into it at the factory. This 48-bit address (6 bytes, displayed as 12 hex digits like 00:1A:2B:3C:4D:5E) uniquely identifies the interface worldwide. The first 24 bits (3 bytes) are the **OUI** (Organizationally Unique Identifier) — assigned to the manufacturer (e.g., 00:1A:2B is Cisco, 00:50:56 is VMware). The last 24 bits are a unique device ID chosen by the manufacturer.

MAC addresses have interesting properties you should know for the exam. The **least significant bit** of the first byte is the **I/G bit** (Individual/Group): 0 = unicast (one device), 1 = multicast (group of devices). The **second least significant bit** is the **U/L bit** (Universal/Local): 0 = globally unique (factory-assigned), 1 = locally

administered (manually set, like for VMs or MAC spoofing). The broadcast MAC address is FF:FF:FF:FF:FF:FF — all bits set, received by every device on the segment.

Exam Tip

MAC addresses are 48 bits (6 bytes). First 3 bytes = OUI (manufacturer). Last 3 bytes = unique device ID. Broadcast = FF:FF:FF:FF:FF:FF. Multicast has I/G bit set. Locally-administered MACs have U/L bit set.

Broadcast vs Collision Domains

Two terms that confuse many CCNA candidates — but they're fundamentally different. A **collision domain** is the set of devices that can collide with each other when transmitting simultaneously. On a hub-based network, all devices share one collision domain. On a switched network, each switch port is its own collision domain — full-duplex operation eliminates collisions entirely. A **broadcast domain** is the set of devices that receive a broadcast frame (destination FF:FF:FF:FF:FF:FF). On a flat switched network (no VLANs), all devices are in one broadcast domain. Routers and L3 switches divide broadcast domains.

Aspect	Collision Domain	Broadcast Domain
What it contains	Devices that can collide on transmit	Devices that receive each other's broadcasts
Created by	Hubs (extend), Switches (break)	Switches/hubs (extend), Routers/L3 (break)
Default count	Each switch port = 1	Each VLAN = 1
Why it matters	Collisions slow the network	Broadcasts (ARP, DHCP) consume bandwidth
How to reduce size	Use switches (not hubs)	Use VLANs + routing

ARP — Address Resolution Protocol

Here's a fundamental problem: IP (Layer 3) addresses are used to route packets across networks, but Ethernet frames (Layer 2) are delivered using MAC addresses. When a host wants to send a packet to an IP on its local subnet, how does it know what MAC address to put on the frame? The answer is **ARP** — Address Resolution Protocol. ARP lets a host ask the local network 'Who has IP 192.168.1.20? Tell me your MAC address', and the owner of that IP replies with its MAC.

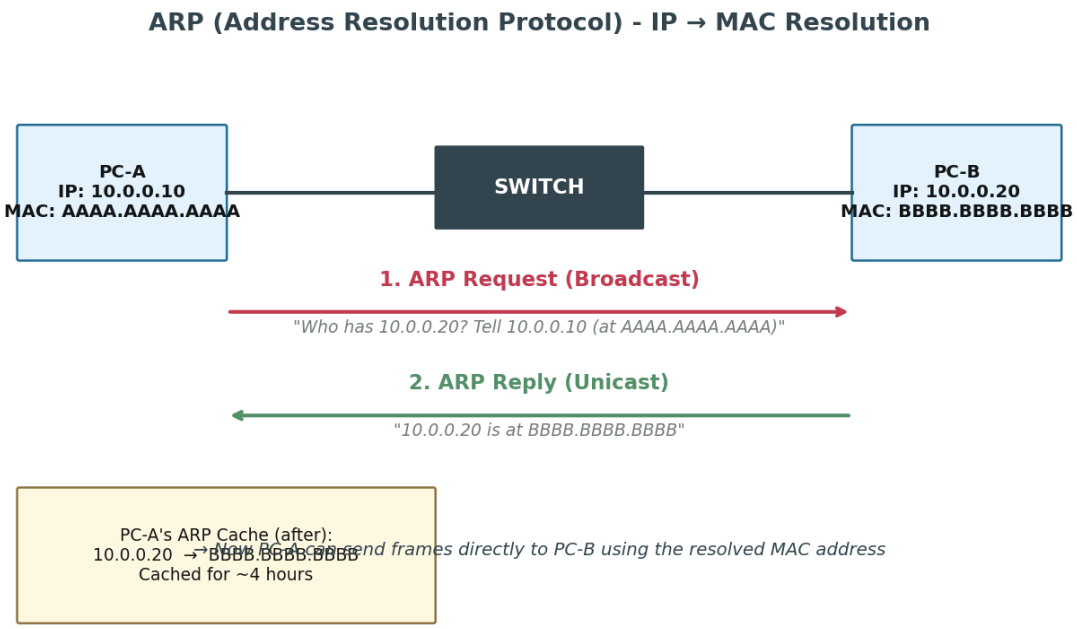


Figure: The ARP process: PC-A broadcasts an ARP request asking who owns 10.0.0.20. PC-B replies with its MAC address. PC-A caches the result for ~4 hours.

The ARP process is two messages: an **ARP Request** (broadcast to FF:FF:FF:FF:FF:FF, asking 'who has IP X?') and an **ARP Reply** (unicast back to the asker, saying 'IP X is at MAC Y'). The asker stores the result in its **ARP cache** for typically 4 hours (Cisco default), so it doesn't need to ask again for subsequent packets to the same destination. View your ARP cache on Windows with arp -a, on Linux/Mac with arp -n or ip neigh, and on Cisco with show ip arp.

CAM Tables — How Switches Learn

Switches don't know which MAC addresses are on which port when they boot up. They learn by examining incoming frames. When a frame arrives on port Fa0/5 with source MAC AA:BB:CC:DD:EE:FF, the switch records 'AA:BB:CC:DD:EE:FF is on Fa0/5' in its **CAM table** (Content-Addressable Memory, also called the MAC address table). Next time it needs to forward a frame to AA:BB:CC:DD:EE:FF, it knows to send it out Fa0/5 only, instead of flooding it everywhere.

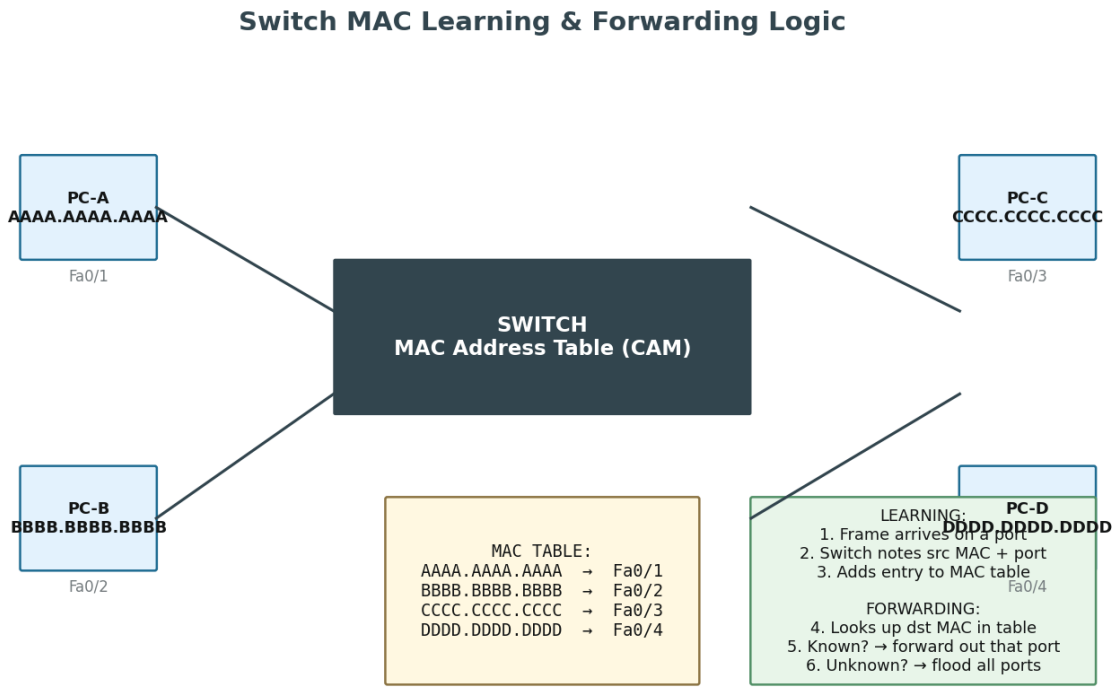


Figure: Switch MAC learning and forwarding logic. The switch examines source MACs to learn, then looks up destination MACs to forward (or floods if unknown).

When a frame arrives with a destination MAC that's not yet in the CAM table (unknown unicast), the switch **floods** it — sends it out every active port in the same VLAN except the port it arrived on. This is called 'unknown unicast flooding'. Broadcast frames (destination FF:FF:FF:FF:FF:FF) are always flooded. CAM entries have a default age of 5 minutes (300 seconds) — if no traffic is seen from a MAC in that time, the entry is removed.

```
# Cisco IOS commands for MAC/ARP tables
show mac address-table                # View CAM (MAC) table
show mac address-table dynamic        # Only dynamically learned entries
show mac address-table vlan 10        # Filter by VLAN
show ip arp                           # View ARP cache (IP→MAC mappings)
clear mac address-table dynamic       # Clear dynamic MAC entries
clear ip arp                          # Clear ARP cache
```

Chapter 4 Quiz

Q1. What is the size of an Ethernet MAC address?

A. 32 bits

B. 48 bits

C. 64 bits

D. 128 bits

Answer: B — MAC addresses are 48 bits (6 bytes), displayed as 12 hex digits.

Q2. Which protocol resolves an IP address to a MAC address?

A. DNS

B. ARP

C. DHCP

D. ICMP

Answer: B — ARP (Address Resolution Protocol) maps IP addresses to MAC addresses on a LAN.

Q3. What is the Layer 2 broadcast MAC address?

A. 255.255.255.255

B. FF:FF:FF:FF:FF:FF

C. 00:00:00:00:00:00

D. 0.0.0.0

Answer: B — FF:FF:FF:FF:FF:FF is the Layer 2 broadcast MAC.

Q4. How does a switch handle a frame with an unknown destination MAC?

A. Drops it

B. Floods it out all ports except the source

C. Sends it to the router

D. Sends an ICMP error

Answer: B — Unknown unicast flooding — switch sends the frame out all ports in the VLAN except the source.

Q5. What is the default CAM table entry age on a Cisco switch?

A. 30 seconds

B. 5 minutes (300 seconds)

C. 1 hour

D. 24 hours

Answer: B — Default MAC age time is 5 minutes (300 seconds) on Cisco switches.

PART 5

IPv4 Mastery

The 32-bit addressing system that runs the Internet

Chapter 5: IPv4 — Structure, Binary, and Subnet Masks

Every device on the Internet needs a unique address so other devices can send data to it. The Internet Protocol version 4 (IPv4), defined in RFC 791 in 1981, uses 32-bit addresses — written as four decimal numbers separated by dots, like 192.168.1.10. Each of the four numbers (called **octets**) represents 8 bits, so it ranges from 0 to 255. There are about 4.3 billion possible IPv4 addresses (2^32 = 4,294,967,296), which sounds like a lot — but the Internet exploded in the 1990s and 2000s, and we ran out. This exhaustion is why IPv6 (with 128-bit addresses) was created, but IPv4 still carries the vast majority of Internet traffic.

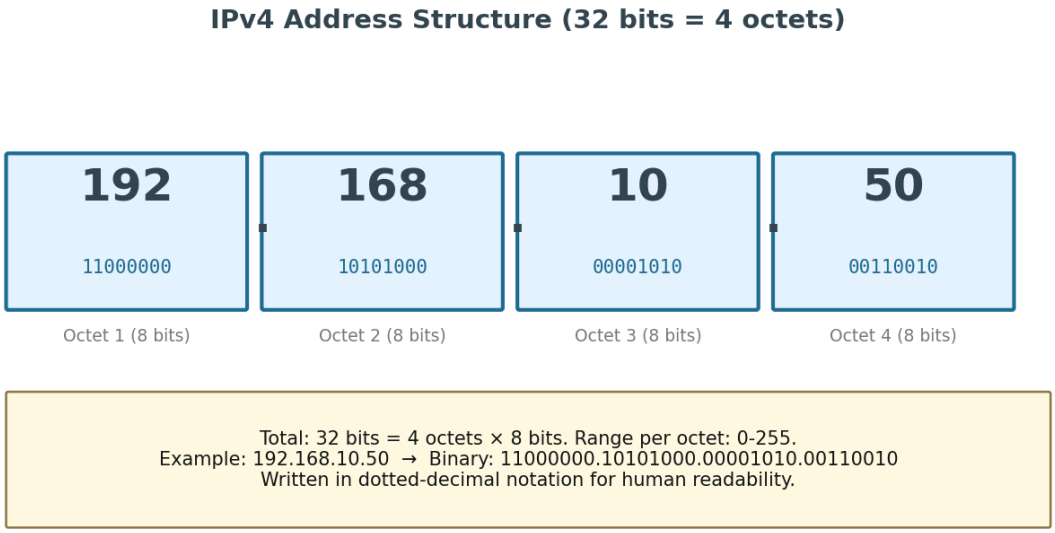


Figure: IPv4 address structure: 32 bits = 4 octets of 8 bits. Each octet ranges 0-255 in decimal.

Binary Basics — Essential for Subnetting

Computers think in binary (1s and 0s), and to truly understand IP addressing and subnetting, you must too. The good news: binary for IP is straightforward because each octet is independent. Each of the 8 bits in an octet represents a power of 2, from 2^7 (128) on the left down to 2^0 (1) on the right. To convert binary to decimal, just add up the values of the bits that are 1.

Bit Position	b7	b6	b5	b4	b3	b2	b1	b0
Power of 2	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal Value	128	64	32	16	8	4	2	1

Example: convert binary 11000000 to decimal. Bits 7 and 6 are 1, all others 0. So: 128 + 64 = **192**. Convert 10101000: 128 + 32 + 8 = **168**. So the IP 192.168.1.10 in binary is 11000000.10101000.00000001.00001010. The reverse — decimal to binary — uses successive subtraction: take 192, the largest power of 2 that fits is 128 (bit 7=1), remainder 64, that fits 64 (bit 6=1), remainder 0 (rest are 0).

Public vs Private IPv4 Addresses

Because IPv4 addresses are scarce, RFC 1918 defined three ranges of **private addresses** that anyone can use inside their own network for free, without registering. These addresses are not routable on the public Internet — if you try to send a packet to 192.168.1.5 from the Internet, your ISP will drop it. To access the Internet, devices with private IPs use **NAT** (Network Address Translation), which translates their private IP to a public IP at the router.

Class	Private Range	CIDR	Default Mask	# of /24s
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8	255.0.0.0	65,536
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12	255.240.0.0	1,024
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16	255.255.0.0	256

Other special IPv4 ranges you should know: **loopback** (127.0.0.0/8, where 127.0.0.1 is 'localhost'), **link-local / APIPA** (169.254.0.0/16, auto-assigned when DHCP fails), **multicast** (224.0.0.0/4), **CGNAT** (100.64.0.0/10, used by carriers for carrier-grade NAT), and **TEST-NET** (192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24 — reserved for documentation).

Network ID, Host ID, and Default Gateway

An IPv4 address has two parts: the **network portion** (which network am I on?) and the **host portion** (which specific device on that network?). The split point is determined by the **subnet mask**. For example, 192.168.1.10/24 has the first 24 bits as network (192.168.1) and the last 8 bits as host (.10).

Every subnet has a **network address** (all host bits = 0, e.g., 192.168.1.0/24) and a **broadcast address** (all host bits = 1, e.g., 192.168.1.255). These two addresses are reserved and cannot be assigned to hosts. The remaining addresses are usable host IPs — for a /24, that's 254 usable hosts (256 total - 2 reserved).

The **default gateway** is the IP address of the router interface on the local subnet — typically the first or last usable host IP (192.168.1.1 or 192.168.1.254 are common choices). When a host wants to send a packet to a destination outside its local subnet, it sends the packet to the default gateway, which forwards it toward the destination. Without a correctly configured default gateway, a host can only talk to other devices on the same subnet.

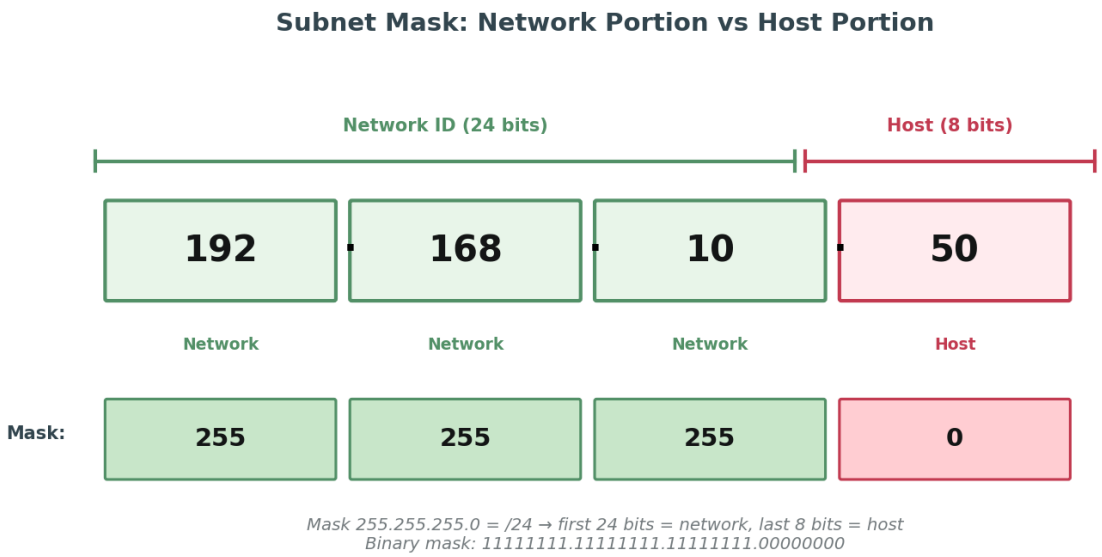


Figure: Subnet mask separates network portion from host portion. /24 = 255.255.255.0 means first 24 bits are network, last 8 are host.

Subnet Masks — Decimal, Binary, and CIDR

The subnet mask can be written three ways: dotted-decimal (255.255.255.0), binary (11111111.11111111.11111111.00000000), or CIDR notation (/24). CIDR (Classless Inter-Domain Routing) is the modern way — it just counts the number of 1 bits in the mask. CIDR is more concise and is what most documentation, configs, and the exam use. You should be able to instantly convert between /prefix and dotted-decimal for all common masks.

CIDR	Decimal	Binary (last octet)	Hosts	Block Size
/24	255.255.255.0	00000000	254	256
/25	255.255.255.128	10000000	126	128
/26	255.255.255.192	11000000	62	64
/27	255.255.255.224	11100000	30	32
/28	255.255.255.240	11110000	14	16
/29	255.255.255.248	11111000	6	8
/30	255.255.255.252	11111100	2	4
/31	255.255.255.254	11111110	2*	2
/32	255.255.255.255	11111111	1	1

** /31 is special (RFC 3021) — used for point-to-point links, both addresses are usable.*

Memorize This Table

The /24-/30 subnet mask table is the single most important thing to memorize for the CCNA exam. Be able to instantly recall: mask in decimal, number of usable hosts, and block size for each prefix. Practice until it's automatic.

Chapter 5 Quiz

Q1. How many bits is an IPv4 address?

A. 32 bits
B. 64 bits
C. 128 bits
D. 16 bits
Answer: A — IPv4 addresses are 32 bits (4 octets of 8 bits).

Q2. Which range is a private IPv4 address space?

A. 11.0.0.0/8
B. 172.16.0.0/12
C. 201.0.0.0/24
D. 8.8.8.0/24
Answer: B — 172.16.0.0/12 (172.16.0.0 - 172.31.255.255) is private per RFC 1918.

Q3. What is the loopback IPv4 address?

A. 127.0.0.1
B. 0.0.0.0
C. 255.255.255.255
D. 192.168.1.1
Answer: A — 127.0.0.0/8 is the loopback range; 127.0.0.1 is the most common loopback address.

Q4. What is the default subnet mask for 192.168.1.0/24?

A. 255.0.0.0
B. 255.255.0.0
C. 255.255.255.0
D. 255.255.255.255
Answer: C — /24 = 255.255.255.0.

Q5. How many usable hosts are in a /24?

A. 256
B. 255
C. 254
D. 253
Answer: C — /24 has 256 addresses; minus network and broadcast = 254 usable.

PART 6

Subnetting Mastery

From binary basics to VLSM and exam-speed shortcuts

Chapter 6: Subnetting — The CCNA Rite of Passage

Subnetting is the skill that separates CCNA passers from failers. It looks intimidating at first, but it's actually just arithmetic — once you internalize a few patterns, you can solve any subnetting problem in seconds without a calculator. This chapter teaches you three methods: the **binary method** (slow but you understand every bit), the **shortcut method** (block sizes, fast for common cases), and the **exam-speed method** (mental math tricks for the test).

Why does subnetting matter? Because in the real world, you almost never have one flat network. You segment for security (separate VLANs for HR, Engineering, Guest), for performance (smaller broadcast domains), for scalability (route between subnets instead of bridging), and for address conservation (don't waste 254 addresses on a 2-router link — use a /30). Mastering subnetting lets you design networks that are efficient, organized, and ready to scale.

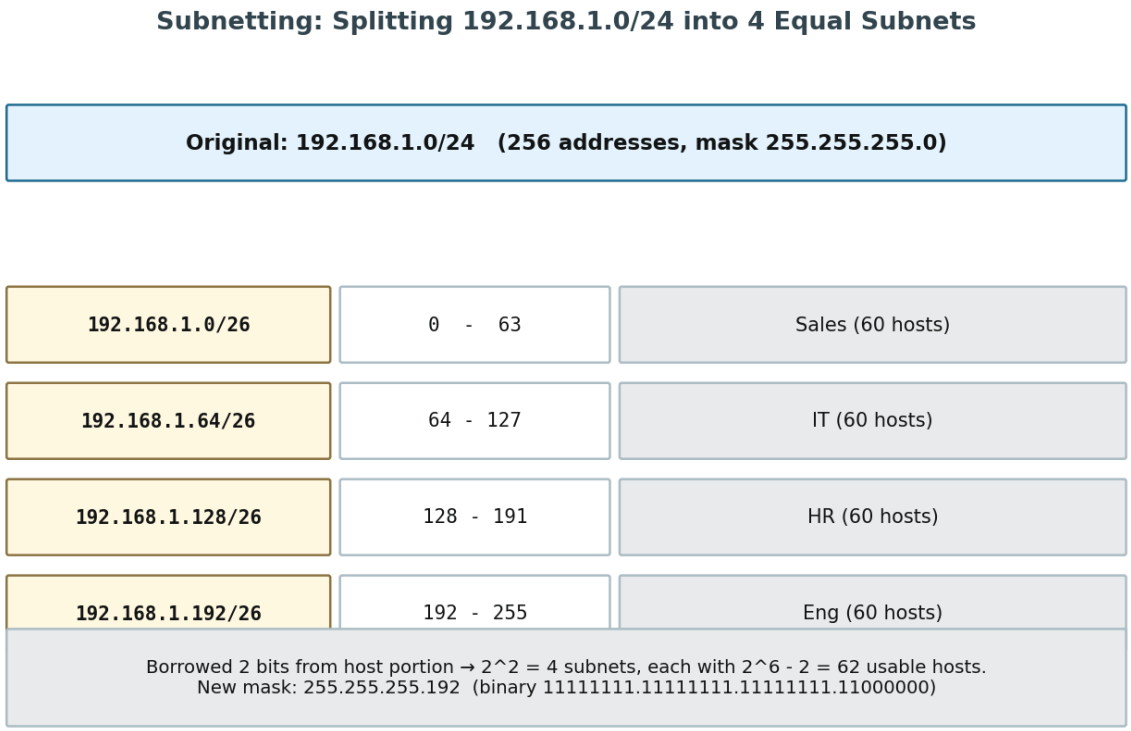


Figure: Subnetting splits one large network into smaller subnets. Here 192.168.1.0/24 is split into 4 /26 subnets, each with 62 usable hosts.

Binary Mathematics — The Foundation

Every subnetting calculation ultimately comes down to bitwise operations. Don't worry — you don't need to do long division in binary. You just need to know: (1) powers of 2 up to 2^12 (4096), (2) how to convert between decimal and binary for octets, (3) how the subnet mask splits the address into network and host bits, and (4) how to compute the number of subnets and hosts from the bit counts.

2^0 = 1	2^1 = 2	2^2 = 4	2^3 = 8	2^4 = 16	2^5 = 32	2^6 = 64	2^7 = 128	2^8 = 256
2^9 = 512	2^10 = 1024	2^11 = 2048	2^12 = 4096	2^13 = 8192	2^14 = 16384	2^15 = 32768	2^16 = 65536	—

Two formulas drive everything in subnetting:

Number of subnets = 2^(borrowed bits), where borrowed bits = new prefix - original prefix. E.g., borrowing 3 bits from /24 gives 2^3 = 8 subnets.

Number of usable hosts per subnet = 2^(host bits) - 2, where host bits = 32 - prefix. The -2 accounts for the network address (all host bits = 0) and broadcast address (all host bits = 1). E.g., /26 has 2^6 - 2 = 62 usable hosts.

CIDR — Classless Inter-Domain Routing

Before 1993, IPv4 used **classful addressing**: Class A (/8), B (/16), and C (/24) were the only mask options. This wasted huge numbers of addresses — a company needing 300 hosts would get a Class B with 65,534 addresses, wasting 65,000+. CIDR (RFC 4632) replaced classful with **classless** addressing: any prefix length from /0 to /32 is valid, and the mask travels with the address as /prefix. This flexibility is what makes the modern Internet possible.

VLSM — Variable Length Subnet Masking

VLSM takes CIDR one step further: within a single organization, different subnets can use different prefix lengths. This is essential for efficient address allocation. Consider a branch office needing: 1 LAN with 60 hosts, 1 LAN with 25 hosts, 1 LAN with 10 hosts, and 2 point-to-point WAN links (2 hosts each).

Without VLSM, you'd use a /26 (62 hosts) for the biggest LAN, but then you'd waste 60 addresses on each WAN link. With VLSM, you give the 60-host LAN a /26, the 25-host LAN a /27, the 10-host LAN a /28, and the WAN links /30s. Total used: less than half a /24, with room to grow. VLSM is essential for any real-world network design.

> **VLSM Procedure**
>
> 1. List all subnets needed, sorted by size (largest first)
> 2. For each subnet, choose the smallest prefix that fits the host count
> 3. Allocate addresses sequentially from the start of the block
> 4. Always start the next subnet at the next aligned boundary
> 5. Document each allocation (subnet, prefix, hosts, purpose)
> 6. Leave remaining space for future growth

The Shortcut Method — Block Sizes

The fastest way to subnet on the exam is the **block size method**. The block size is the number of addresses in each subnet, including network and broadcast. It equals 256 - (the interesting octet of the mask). Once you know the block size, you can list all subnet boundaries instantly.

Prefix	Last Octet of Mask	Block Size	# Hosts	Subnet Boundaries (in last octet)
/24	0	256	254	0 (whole octet)
/25	128	128	126	0, 128
/26	192	64	62	0, 64, 128, 192
/27	224	32	30	0, 32, 64, 96, 128, 160, 192, 224
/28	240	16	14	0, 16, 32, ..., 240 (16 subnets)
/29	248	8	6	0, 8, 16, ..., 248 (32 subnets)
/30	252	4	2	0, 4, 8, ..., 252 (64 subnets)

To find which subnet an IP belongs to: divide the interesting octet by the block size, take the integer part, multiply by block size. Example: what subnet is 192.168.1.135/26 in? Block size 64. 135 ÷ 64 = 2 (integer), 2 × 64 = 128. So subnet is 192.168.1.128, broadcast is 128 + 64 - 1 = 191. Quick mental check: 135 is between 128 and 191, so it's in the 128 subnet.

Exam-Speed Shortcuts

On the CCNA exam, you have about 1 minute per question. Subnetting questions can eat that time fast. These shortcuts let you solve common patterns in under 10 seconds each:

Shortcut 1: The block size is always 2^(host bits) for the interesting octet. Memorize the table above.

Shortcut 2: Usable hosts = block size - 2 (except /31, /32).

Shortcut 3: Number of subnets of size /X in /Y = 2^(X-Y). E.g., /30s in /24 = 2^(30-24) = 64.

Shortcut 4: Wildcard mask = 255.255.255.255 - subnet mask. E.g., /26 = 255.255.255.192, wildcard = 0.0.0.63.

Shortcut 5: To find the broadcast, add (block size - 1) to the network address. E.g., network 192.168.1.128/26, broadcast = 128 + 63 = 191.

200+ Solved Subnetting Examples

The best way to master subnetting is repetition. Below are 200+ worked examples organized by category — basic identification, host range calculations, supernets, design problems, VLSM scenarios, and exam-speed shortcuts. Work through them all.

Total examples: 204

A: Basic /24 — Network Identification

```
Example 1: 192.168.1.10/24
  IP address              : 192.168.1.10
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.1.0
  First usable host       : 192.168.1.1
  Last usable host        : 192.168.1.254
  Broadcast address       : 192.168.1.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 2: 10.0.0.5/24
  IP address              : 10.0.0.5
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.0.0.0
  First usable host       : 10.0.0.1
  Last usable host        : 10.0.0.254
  Broadcast address       : 10.0.0.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 3: 172.16.5.20/24
  IP address              : 172.16.5.20
  Prefix length           : /24
```

```
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.16.5.0
  First usable host       : 172.16.5.1
  Last usable host        : 172.16.5.254
  Broadcast address       : 172.16.5.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 4: 192.168.100.50/24
  IP address              : 192.168.100.50
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.100.0
  First usable host       : 192.168.100.1
  Last usable host        : 192.168.100.254
  Broadcast address       : 192.168.100.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 5: 10.1.1.1/24
  IP address              : 10.1.1.1
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.1.1.0
  First usable host       : 10.1.1.1
  Last usable host        : 10.1.1.254
  Broadcast address       : 10.1.1.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 6: 172.16.10.15/24
  IP address              : 172.16.10.15
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.16.10.0
  First usable host       : 172.16.10.1
  Last usable host        : 172.16.10.254
  Broadcast address       : 172.16.10.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 7: 192.168.0.254/24
  IP address              : 192.168.0.254
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.0.0
  First usable host       : 192.168.0.1
  Last usable host        : 192.168.0.254
  Broadcast address       : 192.168.0.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
```

```
Example 8: 10.10.10.10/24
  IP address              : 10.10.10.10
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.10.10.0
  First usable host       : 10.10.10.1
  Last usable host        : 10.10.10.254
  Broadcast address       : 10.10.10.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 9: 172.16.0.1/24
  IP address              : 172.16.0.1
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.16.0.0
  First usable host       : 172.16.0.1
  Last usable host        : 172.16.0.254
  Broadcast address       : 172.16.0.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 10: 192.168.50.100/24
  IP address              : 192.168.50.100
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.50.0
  First usable host       : 192.168.50.1
  Last usable host        : 192.168.50.254
  Broadcast address       : 192.168.50.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 11: 10.0.0.254/24
  IP address              : 10.0.0.254
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.0.0.0
  First usable host       : 10.0.0.1
  Last usable host        : 10.0.0.254
  Broadcast address       : 10.0.0.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 12: 172.16.254.1/24
  IP address              : 172.16.254.1
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.16.254.0
  First usable host       : 172.16.254.1
  Last usable host        : 172.16.254.254
  Broadcast address       : 172.16.254.255
```

```
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 13: 192.168.1.1/24
  IP address              : 192.168.1.1
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.1.0
  First usable host       : 192.168.1.1
  Last usable host        : 192.168.1.254
  Broadcast address       : 192.168.1.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 14: 10.5.5.5/24
  IP address              : 10.5.5.5
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.5.5.0
  First usable host       : 10.5.5.1
  Last usable host        : 10.5.5.254
  Broadcast address       : 10.5.5.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 15: 172.16.100.100/24
  IP address              : 172.16.100.100
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.16.100.0
  First usable host       : 172.16.100.1
  Last usable host        : 172.16.100.254
  Broadcast address       : 172.16.100.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 16: 192.168.200.200/24
  IP address              : 192.168.200.200
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.200.0
  First usable host       : 192.168.200.1
  Last usable host        : 192.168.200.254
  Broadcast address       : 192.168.200.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 17: 10.100.50.25/24
  IP address              : 10.100.50.25
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.100.50.0
```

```
  First usable host       : 10.100.50.1
  Last usable host        : 10.100.50.254
  Broadcast address       : 10.100.50.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 18: 172.16.30.40/24
  IP address              : 172.16.30.40
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.16.30.0
  First usable host       : 172.16.30.1
  Last usable host        : 172.16.30.254
  Broadcast address       : 172.16.30.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 19: 192.168.1.254/24
  IP address              : 192.168.1.254
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.1.0
  First usable host       : 192.168.1.1
  Last usable host        : 192.168.1.254
  Broadcast address       : 192.168.1.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 20: 10.0.1.1/24
  IP address              : 10.0.1.1
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.0.1.0
  First usable host       : 10.0.1.1
  Last usable host        : 10.0.1.254
  Broadcast address       : 10.0.1.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 21: 172.17.5.5/24
  IP address              : 172.17.5.5
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.17.5.0
  First usable host       : 172.17.5.1
  Last usable host        : 172.17.5.254
  Broadcast address       : 172.17.5.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 22: 192.168.2.50/24
  IP address              : 192.168.2.50
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
```

```
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.2.0
  First usable host       : 192.168.2.1
  Last usable host        : 192.168.2.254
  Broadcast address       : 192.168.2.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 23: 10.50.50.50/24
  IP address              : 10.50.50.50
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.50.50.0
  First usable host       : 10.50.50.1
  Last usable host        : 10.50.50.254
  Broadcast address       : 10.50.50.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 24: 172.16.16.16/24
  IP address              : 172.16.16.16
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.16.16.0
  First usable host       : 172.16.16.1
  Last usable host        : 172.16.16.254
  Broadcast address       : 172.16.16.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 25: 192.168.99.99/24
  IP address              : 192.168.99.99
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.99.0
  First usable host       : 192.168.99.1
  Last usable host        : 192.168.99.254
  Broadcast address       : 192.168.99.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 26: 10.255.255.254/24
  IP address              : 10.255.255.254
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 10.255.255.0
  First usable host       : 10.255.255.1
  Last usable host        : 10.255.255.254
  Broadcast address       : 10.255.255.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
```

```
Example 27: 172.31.31.31/24
  IP address              : 172.31.31.31
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 172.31.31.0
  First usable host       : 172.31.31.1
  Last usable host        : 172.31.31.254
  Broadcast address       : 172.31.31.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
Example 28: 192.168.123.45/24
  IP address              : 192.168.123.45
  Prefix length           : /24
  Subnet mask (decimal)   : 255.255.255.0
  Subnet mask (binary)    : 11111111.11111111.11111111.00000000
  Wildcard mask           : 0.0.0.255
  Network address         : 192.168.123.0
  First usable host       : 192.168.123.1
  Last usable host        : 192.168.123.254
  Broadcast address       : 192.168.123.255
  Usable hosts per subnet : 254
  # of /24 subnets in a /24 : 1
```

B: /25–/30 — Host Range Calculations

```
Example 29: 192.168.1.10/25
  IP address              : 192.168.1.10
  Prefix length           : /25
  Subnet mask (decimal)   : 255.255.255.128
  Subnet mask (binary)    : 11111111.11111111.11111111.10000000
  Wildcard mask           : 0.0.0.127
  Network address         : 192.168.1.0
  First usable host       : 192.168.1.1
  Last usable host        : 192.168.1.126
  Broadcast address       : 192.168.1.127
  Usable hosts per subnet : 126
  # of /25 subnets in a /24 : 2
Example 30: 192.168.1.200/25
  IP address              : 192.168.1.200
  Prefix length           : /25
  Subnet mask (decimal)   : 255.255.255.128
  Subnet mask (binary)    : 11111111.11111111.11111111.10000000
  Wildcard mask           : 0.0.0.127
  Network address         : 192.168.1.128
  First usable host       : 192.168.1.129
  Last usable host        : 192.168.1.254
  Broadcast address       : 192.168.1.255
  Usable hosts per subnet : 126
  # of /25 subnets in a /24 : 2
Example 31: 10.0.0.10/26
  IP address              : 10.0.0.10
  Prefix length           : /26
  Subnet mask (decimal)   : 255.255.255.192
  Subnet mask (binary)    : 11111111.11111111.11111111.11000000
  Wildcard mask           : 0.0.0.63
  Network address         : 10.0.0.0
  First usable host       : 10.0.0.1
```

```
  Last usable host        : 10.0.0.62
  Broadcast address       : 10.0.0.63
  Usable hosts per subnet : 62
  # of /26 subnets in a /24 : 4
Example 32: 10.0.0.200/26
  IP address              : 10.0.0.200
  Prefix length           : /26
  Subnet mask (decimal)   : 255.255.255.192
  Subnet mask (binary)    : 11111111.11111111.11111111.11000000
  Wildcard mask           : 0.0.0.63
  Network address         : 10.0.0.192
  First usable host       : 10.0.0.193
  Last usable host        : 10.0.0.254
  Broadcast address       : 10.0.0.255
  Usable hosts per subnet : 62
  # of /26 subnets in a /24 : 4
Example 33: 172.16.1.50/27
  IP address              : 172.16.1.50
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 172.16.1.32
  First usable host       : 172.16.1.33
  Last usable host        : 172.16.1.62
  Broadcast address       : 172.16.1.63
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 34: 172.16.1.200/27
  IP address              : 172.16.1.200
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 172.16.1.192
  First usable host       : 172.16.1.193
  Last usable host        : 172.16.1.222
  Broadcast address       : 172.16.1.223
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 35: 192.168.5.10/28
  IP address              : 192.168.5.10
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.5.0
  First usable host       : 192.168.5.1
  Last usable host        : 192.168.5.14
  Broadcast address       : 192.168.5.15
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 36: 192.168.5.250/28
  IP address              : 192.168.5.250
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
```

```
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.5.240
  First usable host       : 192.168.5.241
  Last usable host        : 192.168.5.254
  Broadcast address       : 192.168.5.255
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 37: 10.1.1.10/29
  IP address              : 10.1.1.10
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 10.1.1.8
  First usable host       : 10.1.1.9
  Last usable host        : 10.1.1.14
  Broadcast address       : 10.1.1.15
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 38: 10.1.1.15/29
  IP address              : 10.1.1.15
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 10.1.1.8
  First usable host       : 10.1.1.9
  Last usable host        : 10.1.1.14
  Broadcast address       : 10.1.1.15
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 39: 192.168.10.1/30
  IP address              : 192.168.10.1
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 192.168.10.0
  First usable host       : 192.168.10.1
  Last usable host        : 192.168.10.2
  Broadcast address       : 192.168.10.3
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 40: 192.168.10.5/30
  IP address              : 192.168.10.5
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 192.168.10.4
  First usable host       : 192.168.10.5
  Last usable host        : 192.168.10.6
  Broadcast address       : 192.168.10.7
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 41: 10.0.0.1/30
  IP address              : 10.0.0.1
```

```
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.0.0.0
  First usable host       : 10.0.0.1
  Last usable host        : 10.0.0.2
  Broadcast address       : 10.0.0.3
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 42: 10.0.0.2/30
  IP address              : 10.0.0.2
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.0.0.0
  First usable host       : 10.0.0.1
  Last usable host        : 10.0.0.2
  Broadcast address       : 10.0.0.3
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 43: 172.16.0.1/30
  IP address              : 172.16.0.1
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.16.0.0
  First usable host       : 172.16.0.1
  Last usable host        : 172.16.0.2
  Broadcast address       : 172.16.0.3
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 44: 172.16.0.5/30
  IP address              : 172.16.0.5
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.16.0.4
  First usable host       : 172.16.0.5
  Last usable host        : 172.16.0.6
  Broadcast address       : 172.16.0.7
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 45: 192.168.1.130/25
  IP address              : 192.168.1.130
  Prefix length           : /25
  Subnet mask (decimal)   : 255.255.255.128
  Subnet mask (binary)    : 11111111.11111111.11111111.10000000
  Wildcard mask           : 0.0.0.127
  Network address         : 192.168.1.128
  First usable host       : 192.168.1.129
  Last usable host        : 192.168.1.254
  Broadcast address       : 192.168.1.255
  Usable hosts per subnet : 126
```

```
  # of /25 subnets in a /24 : 2
Example 46: 192.168.1.65/26
  IP address              : 192.168.1.65
  Prefix length           : /26
  Subnet mask (decimal)   : 255.255.255.192
  Subnet mask (binary)    : 11111111.11111111.11111111.11000000
  Wildcard mask           : 0.0.0.63
  Network address         : 192.168.1.64
  First usable host       : 192.168.1.65
  Last usable host        : 192.168.1.126
  Broadcast address       : 192.168.1.127
  Usable hosts per subnet : 62
  # of /26 subnets in a /24 : 4
Example 47: 192.168.1.193/26
  IP address              : 192.168.1.193
  Prefix length           : /26
  Subnet mask (decimal)   : 255.255.255.192
  Subnet mask (binary)    : 11111111.11111111.11111111.11000000
  Wildcard mask           : 0.0.0.63
  Network address         : 192.168.1.192
  First usable host       : 192.168.1.193
  Last usable host        : 192.168.1.254
  Broadcast address       : 192.168.1.255
  Usable hosts per subnet : 62
  # of /26 subnets in a /24 : 4
Example 48: 192.168.1.33/27
  IP address              : 192.168.1.33
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 192.168.1.32
  First usable host       : 192.168.1.33
  Last usable host        : 192.168.1.62
  Broadcast address       : 192.168.1.63
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 49: 192.168.1.97/27
  IP address              : 192.168.1.97
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 192.168.1.96
  First usable host       : 192.168.1.97
  Last usable host        : 192.168.1.126
  Broadcast address       : 192.168.1.127
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 50: 192.168.1.161/27
  IP address              : 192.168.1.161
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 192.168.1.160
  First usable host       : 192.168.1.161
```

```
  Last usable host        : 192.168.1.190
  Broadcast address       : 192.168.1.191
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 51: 192.168.1.225/27
  IP address              : 192.168.1.225
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 192.168.1.224
  First usable host       : 192.168.1.225
  Last usable host        : 192.168.1.254
  Broadcast address       : 192.168.1.255
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 52: 192.168.1.17/28
  IP address              : 192.168.1.17
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.16
  First usable host       : 192.168.1.17
  Last usable host        : 192.168.1.30
  Broadcast address       : 192.168.1.31
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 53: 192.168.1.33/28
  IP address              : 192.168.1.33
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.32
  First usable host       : 192.168.1.33
  Last usable host        : 192.168.1.46
  Broadcast address       : 192.168.1.47
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 54: 192.168.1.49/28
  IP address              : 192.168.1.49
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.48
  First usable host       : 192.168.1.49
  Last usable host        : 192.168.1.62
  Broadcast address       : 192.168.1.63
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 55: 192.168.1.65/28
  IP address              : 192.168.1.65
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
```

```
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.64
  First usable host       : 192.168.1.65
  Last usable host        : 192.168.1.78
  Broadcast address       : 192.168.1.79
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 56: 192.168.1.81/28
  IP address              : 192.168.1.81
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.80
  First usable host       : 192.168.1.81
  Last usable host        : 192.168.1.94
  Broadcast address       : 192.168.1.95
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 57: 192.168.1.97/28
  IP address              : 192.168.1.97
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.96
  First usable host       : 192.168.1.97
  Last usable host        : 192.168.1.110
  Broadcast address       : 192.168.1.111
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 58: 192.168.1.113/28
  IP address              : 192.168.1.113
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.112
  First usable host       : 192.168.1.113
  Last usable host        : 192.168.1.126
  Broadcast address       : 192.168.1.127
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 59: 192.168.1.129/28
  IP address              : 192.168.1.129
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.128
  First usable host       : 192.168.1.129
  Last usable host        : 192.168.1.142
  Broadcast address       : 192.168.1.143
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 60: 192.168.1.145/28
  IP address              : 192.168.1.145
```

```
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.144
  First usable host       : 192.168.1.145
  Last usable host        : 192.168.1.158
  Broadcast address       : 192.168.1.159
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 61: 192.168.1.161/28
  IP address              : 192.168.1.161
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.160
  First usable host       : 192.168.1.161
  Last usable host        : 192.168.1.174
  Broadcast address       : 192.168.1.175
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 62: 192.168.1.177/28
  IP address              : 192.168.1.177
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.176
  First usable host       : 192.168.1.177
  Last usable host        : 192.168.1.190
  Broadcast address       : 192.168.1.191
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 63: 192.168.1.193/28
  IP address              : 192.168.1.193
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.192
  First usable host       : 192.168.1.193
  Last usable host        : 192.168.1.206
  Broadcast address       : 192.168.1.207
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 64: 192.168.1.209/28
  IP address              : 192.168.1.209
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.208
  First usable host       : 192.168.1.209
  Last usable host        : 192.168.1.222
  Broadcast address       : 192.168.1.223
  Usable hosts per subnet : 14
```

```
  # of /28 subnets in a /24 : 16
Example 65: 192.168.1.225/28
  IP address              : 192.168.1.225
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.224
  First usable host       : 192.168.1.225
  Last usable host        : 192.168.1.238
  Broadcast address       : 192.168.1.239
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 66: 192.168.1.241/28
  IP address              : 192.168.1.241
  Prefix length           : /28
  Subnet mask (decimal)   : 255.255.255.240
  Subnet mask (binary)    : 11111111.11111111.11111111.11110000
  Wildcard mask           : 0.0.0.15
  Network address         : 192.168.1.240
  First usable host       : 192.168.1.241
  Last usable host        : 192.168.1.254
  Broadcast address       : 192.168.1.255
  Usable hosts per subnet : 14
  # of /28 subnets in a /24 : 16
Example 67: 10.0.0.9/29
  IP address              : 10.0.0.9
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 10.0.0.8
  First usable host       : 10.0.0.9
  Last usable host        : 10.0.0.14
  Broadcast address       : 10.0.0.15
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 68: 10.0.0.17/29
  IP address              : 10.0.0.17
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 10.0.0.16
  First usable host       : 10.0.0.17
  Last usable host        : 10.0.0.22
  Broadcast address       : 10.0.0.23
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 69: 10.0.0.25/29
  IP address              : 10.0.0.25
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 10.0.0.24
  First usable host       : 10.0.0.25
```

```
  Last usable host        : 10.0.0.30
  Broadcast address       : 10.0.0.31
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 70: 10.0.0.33/29
  IP address              : 10.0.0.33
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 10.0.0.32
  First usable host       : 10.0.0.33
  Last usable host        : 10.0.0.38
  Broadcast address       : 10.0.0.39
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 71: 10.0.0.41/29
  IP address              : 10.0.0.41
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 10.0.0.40
  First usable host       : 10.0.0.41
  Last usable host        : 10.0.0.46
  Broadcast address       : 10.0.0.47
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 72: 10.0.0.49/29
  IP address              : 10.0.0.49
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 10.0.0.48
  First usable host       : 10.0.0.49
  Last usable host        : 10.0.0.54
  Broadcast address       : 10.0.0.55
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 73: 172.16.5.1/29
  IP address              : 172.16.5.1
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 172.16.5.0
  First usable host       : 172.16.5.1
  Last usable host        : 172.16.5.6
  Broadcast address       : 172.16.5.7
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 74: 172.16.5.9/29
  IP address              : 172.16.5.9
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
```

```
  Wildcard mask           : 0.0.0.7
  Network address         : 172.16.5.8
  First usable host       : 172.16.5.9
  Last usable host        : 172.16.5.14
  Broadcast address       : 172.16.5.15
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 75: 172.16.5.17/29
  IP address              : 172.16.5.17
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 172.16.5.16
  First usable host       : 172.16.5.17
  Last usable host        : 172.16.5.22
  Broadcast address       : 172.16.5.23
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 76: 172.16.5.25/29
  IP address              : 172.16.5.25
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 172.16.5.24
  First usable host       : 172.16.5.25
  Last usable host        : 172.16.5.30
  Broadcast address       : 172.16.5.31
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 77: 192.168.100.1/30
  IP address              : 192.168.100.1
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 192.168.100.0
  First usable host       : 192.168.100.1
  Last usable host        : 192.168.100.2
  Broadcast address       : 192.168.100.3
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 78: 192.168.100.5/30
  IP address              : 192.168.100.5
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 192.168.100.4
  First usable host       : 192.168.100.5
  Last usable host        : 192.168.100.6
  Broadcast address       : 192.168.100.7
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 79: 192.168.100.9/30
  IP address              : 192.168.100.9
```

```
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 192.168.100.8
  First usable host       : 192.168.100.9
  Last usable host        : 192.168.100.10
  Broadcast address       : 192.168.100.11
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 80: 192.168.100.13/30
  IP address              : 192.168.100.13
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 192.168.100.12
  First usable host       : 192.168.100.13
  Last usable host        : 192.168.100.14
  Broadcast address       : 192.168.100.15
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 81: 10.10.10.1/30
  IP address              : 10.10.10.1
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.10.10.0
  First usable host       : 10.10.10.1
  Last usable host        : 10.10.10.2
  Broadcast address       : 10.10.10.3
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 82: 10.10.10.5/30
  IP address              : 10.10.10.5
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.10.10.4
  First usable host       : 10.10.10.5
  Last usable host        : 10.10.10.6
  Broadcast address       : 10.10.10.7
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 83: 10.10.10.9/30
  IP address              : 10.10.10.9
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.10.10.8
  First usable host       : 10.10.10.9
  Last usable host        : 10.10.10.10
  Broadcast address       : 10.10.10.11
  Usable hosts per subnet : 2
```

```
  # of /30 subnets in a /24 : 64
Example 84: 10.10.10.13/30
  IP address              : 10.10.10.13
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.10.10.12
  First usable host       : 10.10.10.13
  Last usable host        : 10.10.10.14
  Broadcast address       : 10.10.10.15
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 85: 172.16.0.9/30
  IP address              : 172.16.0.9
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.16.0.8
  First usable host       : 172.16.0.9
  Last usable host        : 172.16.0.10
  Broadcast address       : 172.16.0.11
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 86: 172.16.0.13/30
  IP address              : 172.16.0.13
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.16.0.12
  First usable host       : 172.16.0.13
  Last usable host        : 172.16.0.14
  Broadcast address       : 172.16.0.15
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 87: 172.16.0.17/30
  IP address              : 172.16.0.17
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.16.0.16
  First usable host       : 172.16.0.17
  Last usable host        : 172.16.0.18
  Broadcast address       : 172.16.0.19
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 88: 172.16.0.21/30
  IP address              : 172.16.0.21
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.16.0.20
  First usable host       : 172.16.0.21
```

```
  Last usable host        : 172.16.0.22
  Broadcast address       : 172.16.0.23
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 89: 192.168.50.130/25
  IP address              : 192.168.50.130
  Prefix length           : /25
  Subnet mask (decimal)   : 255.255.255.128
  Subnet mask (binary)    : 11111111.11111111.11111111.10000000
  Wildcard mask           : 0.0.0.127
  Network address         : 192.168.50.128
  First usable host       : 192.168.50.129
  Last usable host        : 192.168.50.254
  Broadcast address       : 192.168.50.255
  Usable hosts per subnet : 126
  # of /25 subnets in a /24 : 2
Example 90: 192.168.50.5/25
  IP address              : 192.168.50.5
  Prefix length           : /25
  Subnet mask (decimal)   : 255.255.255.128
  Subnet mask (binary)    : 11111111.11111111.11111111.10000000
  Wildcard mask           : 0.0.0.127
  Network address         : 192.168.50.0
  First usable host       : 192.168.50.1
  Last usable host        : 192.168.50.126
  Broadcast address       : 192.168.50.127
  Usable hosts per subnet : 126
  # of /25 subnets in a /24 : 2
Example 91: 10.0.0.130/25
  IP address              : 10.0.0.130
  Prefix length           : /25
  Subnet mask (decimal)   : 255.255.255.128
  Subnet mask (binary)    : 11111111.11111111.11111111.10000000
  Wildcard mask           : 0.0.0.127
  Network address         : 10.0.0.128
  First usable host       : 10.0.0.129
  Last usable host        : 10.0.0.254
  Broadcast address       : 10.0.0.255
  Usable hosts per subnet : 126
  # of /25 subnets in a /24 : 2
Example 92: 10.0.0.129/25
  IP address              : 10.0.0.129
  Prefix length           : /25
  Subnet mask (decimal)   : 255.255.255.128
  Subnet mask (binary)    : 11111111.11111111.11111111.10000000
  Wildcard mask           : 0.0.0.127
  Network address         : 10.0.0.128
  First usable host       : 10.0.0.129
  Last usable host        : 10.0.0.254
  Broadcast address       : 10.0.0.255
  Usable hosts per subnet : 126
  # of /25 subnets in a /24 : 2
Example 93: 172.16.10.33/27
  IP address              : 172.16.10.33
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
```

```
  Wildcard mask           : 0.0.0.31
  Network address         : 172.16.10.32
  First usable host       : 172.16.10.33
  Last usable host        : 172.16.10.62
  Broadcast address       : 172.16.10.63
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 94: 172.16.10.65/27
  IP address              : 172.16.10.65
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 172.16.10.64
  First usable host       : 172.16.10.65
  Last usable host        : 172.16.10.94
  Broadcast address       : 172.16.10.95
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 95: 172.16.10.97/27
  IP address              : 172.16.10.97
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 172.16.10.96
  First usable host       : 172.16.10.97
  Last usable host        : 172.16.10.126
  Broadcast address       : 172.16.10.127
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 96: 172.16.10.129/27
  IP address              : 172.16.10.129
  Prefix length           : /27
  Subnet mask (decimal)   : 255.255.255.224
  Subnet mask (binary)    : 11111111.11111111.11111111.11100000
  Wildcard mask           : 0.0.0.31
  Network address         : 172.16.10.128
  First usable host       : 172.16.10.129
  Last usable host        : 172.16.10.158
  Broadcast address       : 172.16.10.159
  Usable hosts per subnet : 30
  # of /27 subnets in a /24 : 8
Example 97: 192.168.20.1/29
  IP address              : 192.168.20.1
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 192.168.20.0
  First usable host       : 192.168.20.1
  Last usable host        : 192.168.20.6
  Broadcast address       : 192.168.20.7
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 98: 192.168.20.9/29
  IP address              : 192.168.20.9
```

```
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 192.168.20.8
  First usable host       : 192.168.20.9
  Last usable host        : 192.168.20.14
  Broadcast address       : 192.168.20.15
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 99: 192.168.20.17/29
  IP address              : 192.168.20.17
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 192.168.20.16
  First usable host       : 192.168.20.17
  Last usable host        : 192.168.20.22
  Broadcast address       : 192.168.20.23
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 100: 192.168.20.25/29
  IP address              : 192.168.20.25
  Prefix length           : /29
  Subnet mask (decimal)   : 255.255.255.248
  Subnet mask (binary)    : 11111111.11111111.11111111.11111000
  Wildcard mask           : 0.0.0.7
  Network address         : 192.168.20.24
  First usable host       : 192.168.20.25
  Last usable host        : 192.168.20.30
  Broadcast address       : 192.168.20.31
  Usable hosts per subnet : 6
  # of /29 subnets in a /24 : 32
Example 101: 10.1.2.3/30
  IP address              : 10.1.2.3
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.1.2.0
  First usable host       : 10.1.2.1
  Last usable host        : 10.1.2.2
  Broadcast address       : 10.1.2.3
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 102: 10.1.2.7/30
  IP address              : 10.1.2.7
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.1.2.4
  First usable host       : 10.1.2.5
  Last usable host        : 10.1.2.6
  Broadcast address       : 10.1.2.7
  Usable hosts per subnet : 2
```

```
  # of /30 subnets in a /24 : 64
Example 103: 10.1.2.11/30
  IP address              : 10.1.2.11
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.1.2.8
  First usable host       : 10.1.2.9
  Last usable host        : 10.1.2.10
  Broadcast address       : 10.1.2.11
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 104: 10.1.2.15/30
  IP address              : 10.1.2.15
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 10.1.2.12
  First usable host       : 10.1.2.13
  Last usable host        : 10.1.2.14
  Broadcast address       : 10.1.2.15
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 105: 172.30.0.1/30
  IP address              : 172.30.0.1
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.30.0.0
  First usable host       : 172.30.0.1
  Last usable host        : 172.30.0.2
  Broadcast address       : 172.30.0.3
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 106: 172.30.0.5/30
  IP address              : 172.30.0.5
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.30.0.4
  First usable host       : 172.30.0.5
  Last usable host        : 172.30.0.6
  Broadcast address       : 172.30.0.7
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 107: 172.30.0.9/30
  IP address              : 172.30.0.9
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.30.0.8
  First usable host       : 172.30.0.9
```

```
  Last usable host        : 172.30.0.10
  Broadcast address       : 172.30.0.11
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
Example 108: 172.30.0.13/30
  IP address              : 172.30.0.13
  Prefix length           : /30
  Subnet mask (decimal)   : 255.255.255.252
  Subnet mask (binary)    : 11111111.11111111.11111111.11111100
  Wildcard mask           : 0.0.0.3
  Network address         : 172.30.0.12
  First usable host       : 172.30.0.13
  Last usable host        : 172.30.0.14
  Broadcast address       : 172.30.0.15
  Usable hosts per subnet : 2
  # of /30 subnets in a /24 : 64
```

C: Supernet /20–/23 — Large Blocks

```
Example 109: 172.16.0.1/23
  IP address              : 172.16.0.1
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 172.16.0.0
  First usable host       : 172.16.0.1
  Last usable host        : 172.16.1.254
  Broadcast address       : 172.16.1.255
  Usable hosts per subnet : 510
Example 110: 172.16.2.1/23
  IP address              : 172.16.2.1
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 172.16.2.0
  First usable host       : 172.16.2.1
  Last usable host        : 172.16.3.254
  Broadcast address       : 172.16.3.255
  Usable hosts per subnet : 510
Example 111: 172.16.4.1/23
  IP address              : 172.16.4.1
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 172.16.4.0
  First usable host       : 172.16.4.1
  Last usable host        : 172.16.5.254
  Broadcast address       : 172.16.5.255
  Usable hosts per subnet : 510
Example 112: 172.16.6.1/23
  IP address              : 172.16.6.1
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
```

```
  Wildcard mask           : 0.0.1.255
  Network address         : 172.16.6.0
  First usable host       : 172.16.6.1
  Last usable host        : 172.16.7.254
  Broadcast address       : 172.16.7.255
  Usable hosts per subnet : 510
Example 113: 10.0.0.1/22
  IP address              : 10.0.0.1
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 10.0.0.0
  First usable host       : 10.0.0.1
  Last usable host        : 10.0.3.254
  Broadcast address       : 10.0.3.255
  Usable hosts per subnet : 1,022
Example 114: 10.0.4.1/22
  IP address              : 10.0.4.1
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 10.0.4.0
  First usable host       : 10.0.4.1
  Last usable host        : 10.0.7.254
  Broadcast address       : 10.0.7.255
  Usable hosts per subnet : 1,022
Example 115: 10.0.8.1/22
  IP address              : 10.0.8.1
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 10.0.8.0
  First usable host       : 10.0.8.1
  Last usable host        : 10.0.11.254
  Broadcast address       : 10.0.11.255
  Usable hosts per subnet : 1,022
Example 116: 10.0.12.1/22
  IP address              : 10.0.12.1
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 10.0.12.0
  First usable host       : 10.0.12.1
  Last usable host        : 10.0.15.254
  Broadcast address       : 10.0.15.255
  Usable hosts per subnet : 1,022
Example 117: 192.168.0.1/21
  IP address              : 192.168.0.1
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 192.168.0.0
```

```
  First usable host       : 192.168.0.1
  Last usable host        : 192.168.7.254
  Broadcast address       : 192.168.7.255
  Usable hosts per subnet : 2,046
Example 118: 192.168.8.1/21
  IP address              : 192.168.8.1
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 192.168.8.0
  First usable host       : 192.168.8.1
  Last usable host        : 192.168.15.254
  Broadcast address       : 192.168.15.255
  Usable hosts per subnet : 2,046
Example 119: 172.16.0.1/22
  IP address              : 172.16.0.1
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 172.16.0.0
  First usable host       : 172.16.0.1
  Last usable host        : 172.16.3.254
  Broadcast address       : 172.16.3.255
  Usable hosts per subnet : 1,022
Example 120: 172.16.4.1/22
  IP address              : 172.16.4.1
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 172.16.4.0
  First usable host       : 172.16.4.1
  Last usable host        : 172.16.7.254
  Broadcast address       : 172.16.7.255
  Usable hosts per subnet : 1,022
Example 121: 10.1.0.1/23
  IP address              : 10.1.0.1
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 10.1.0.0
  First usable host       : 10.1.0.1
  Last usable host        : 10.1.1.254
  Broadcast address       : 10.1.1.255
  Usable hosts per subnet : 510
Example 122: 10.1.2.1/23
  IP address              : 10.1.2.1
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 10.1.2.0
  First usable host       : 10.1.2.1
  Last usable host        : 10.1.3.254
```

```
  Broadcast address       : 10.1.3.255
  Usable hosts per subnet : 510

Example 123: 10.1.4.1/23
  IP address              : 10.1.4.1
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 10.1.4.0
  First usable host       : 10.1.4.1
  Last usable host        : 10.1.5.254
  Broadcast address       : 10.1.5.255
  Usable hosts per subnet : 510

Example 124: 10.1.6.1/23
  IP address              : 10.1.6.1
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 10.1.6.0
  First usable host       : 10.1.6.1
  Last usable host        : 10.1.7.254
  Broadcast address       : 10.1.7.255
  Usable hosts per subnet : 510

Example 125: 172.20.0.1/20
  IP address              : 172.20.0.1
  Prefix length           : /20
  Subnet mask (decimal)   : 255.255.240.0
  Subnet mask (binary)    : 11111111.11111111.11110000.00000000
  Wildcard mask           : 0.0.15.255
  Network address         : 172.20.0.0
  First usable host       : 172.20.0.1
  Last usable host        : 172.20.15.254
  Broadcast address       : 172.20.15.255
  Usable hosts per subnet : 4,094

Example 126: 172.20.16.1/20
  IP address              : 172.20.16.1
  Prefix length           : /20
  Subnet mask (decimal)   : 255.255.240.0
  Subnet mask (binary)    : 11111111.11111111.11110000.00000000
  Wildcard mask           : 0.0.15.255
  Network address         : 172.20.16.0
  First usable host       : 172.20.16.1
  Last usable host        : 172.20.31.254
  Broadcast address       : 172.20.31.255
  Usable hosts per subnet : 4,094

Example 127: 172.20.32.1/20
  IP address              : 172.20.32.1
  Prefix length           : /20
  Subnet mask (decimal)   : 255.255.240.0
  Subnet mask (binary)    : 11111111.11111111.11110000.00000000
  Wildcard mask           : 0.0.15.255
  Network address         : 172.20.32.0
  First usable host       : 172.20.32.1
  Last usable host        : 172.20.47.254
  Broadcast address       : 172.20.47.255
  Usable hosts per subnet : 4,094
```

```
Example 128: 172.20.48.1/20
  IP address              : 172.20.48.1
  Prefix length           : /20
  Subnet mask (decimal)   : 255.255.240.0
  Subnet mask (binary)    : 11111111.11111111.11110000.00000000
  Wildcard mask           : 0.0.15.255
  Network address         : 172.20.48.0
  First usable host       : 172.20.48.1
  Last usable host        : 172.20.63.254
  Broadcast address       : 172.20.63.255
  Usable hosts per subnet : 4,094

Example 129: 10.10.0.1/19
  IP address              : 10.10.0.1
  Prefix length           : /19
  Subnet mask (decimal)   : 255.255.224.0
  Subnet mask (binary)    : 11111111.11111111.11100000.00000000
  Wildcard mask           : 0.0.31.255
  Network address         : 10.10.0.0
  First usable host       : 10.10.0.1
  Last usable host        : 10.10.31.254
  Broadcast address       : 10.10.31.255
  Usable hosts per subnet : 8,190

Example 130: 10.10.32.1/19
  IP address              : 10.10.32.1
  Prefix length           : /19
  Subnet mask (decimal)   : 255.255.224.0
  Subnet mask (binary)    : 11111111.11111111.11100000.00000000
  Wildcard mask           : 0.0.31.255
  Network address         : 10.10.32.0
  First usable host       : 10.10.32.1
  Last usable host        : 10.10.63.254
  Broadcast address       : 10.10.63.255
  Usable hosts per subnet : 8,190

Example 131: 10.10.64.1/19
  IP address              : 10.10.64.1
  Prefix length           : /19
  Subnet mask (decimal)   : 255.255.224.0
  Subnet mask (binary)    : 11111111.11111111.11100000.00000000
  Wildcard mask           : 0.0.31.255
  Network address         : 10.10.64.0
  First usable host       : 10.10.64.1
  Last usable host        : 10.10.95.254
  Broadcast address       : 10.10.95.255
  Usable hosts per subnet : 8,190

Example 132: 10.10.96.1/19
  IP address              : 10.10.96.1
  Prefix length           : /19
  Subnet mask (decimal)   : 255.255.224.0
  Subnet mask (binary)    : 11111111.11111111.11100000.00000000
  Wildcard mask           : 0.0.31.255
  Network address         : 10.10.96.0
  First usable host       : 10.10.96.1
  Last usable host        : 10.10.127.254
  Broadcast address       : 10.10.127.255
  Usable hosts per subnet : 8,190

Example 133: 192.168.0.0/22
  IP address              : 192.168.0.0
  Prefix length           : /22
```

```
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 192.168.0.0
  First usable host       : 192.168.0.1
  Last usable host        : 192.168.3.254
  Broadcast address       : 192.168.3.255
  Usable hosts per subnet : 1,022
Example 134: 192.168.4.0/22
  IP address              : 192.168.4.0
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 192.168.4.0
  First usable host       : 192.168.4.1
  Last usable host        : 192.168.7.254
  Broadcast address       : 192.168.7.255
  Usable hosts per subnet : 1,022
Example 135: 192.168.8.0/22
  IP address              : 192.168.8.0
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 192.168.8.0
  First usable host       : 192.168.8.1
  Last usable host        : 192.168.11.254
  Broadcast address       : 192.168.11.255
  Usable hosts per subnet : 1,022
Example 136: 192.168.12.0/22
  IP address              : 192.168.12.0
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 192.168.12.0
  First usable host       : 192.168.12.1
  Last usable host        : 192.168.15.254
  Broadcast address       : 192.168.15.255
  Usable hosts per subnet : 1,022
Example 137: 172.16.0.0/21
  IP address              : 172.16.0.0
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 172.16.0.0
  First usable host       : 172.16.0.1
  Last usable host        : 172.16.7.254
  Broadcast address       : 172.16.7.255
  Usable hosts per subnet : 2,046
Example 138: 172.16.8.0/21
  IP address              : 172.16.8.0
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
```

```
  Wildcard mask           : 0.0.7.255
  Network address         : 172.16.8.0
  First usable host       : 172.16.8.1
  Last usable host        : 172.16.15.254
  Broadcast address       : 172.16.15.255
  Usable hosts per subnet : 2,046
Example 139: 172.16.16.0/21
  IP address              : 172.16.16.0
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 172.16.16.0
  First usable host       : 172.16.16.1
  Last usable host        : 172.16.23.254
  Broadcast address       : 172.16.23.255
  Usable hosts per subnet : 2,046
Example 140: 172.16.24.0/21
  IP address              : 172.16.24.0
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 172.16.24.0
  First usable host       : 172.16.24.1
  Last usable host        : 172.16.31.254
  Broadcast address       : 172.16.31.255
  Usable hosts per subnet : 2,046
Example 141: 10.0.0.0/23
  IP address              : 10.0.0.0
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 10.0.0.0
  First usable host       : 10.0.0.1
  Last usable host        : 10.0.1.254
  Broadcast address       : 10.0.1.255
  Usable hosts per subnet : 510
Example 142: 10.0.2.0/23
  IP address              : 10.0.2.0
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 10.0.2.0
  First usable host       : 10.0.2.1
  Last usable host        : 10.0.3.254
  Broadcast address       : 10.0.3.255
  Usable hosts per subnet : 510
Example 143: 10.0.4.0/23
  IP address              : 10.0.4.0
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 10.0.4.0
```

```
  First usable host       : 10.0.4.1
  Last usable host        : 10.0.5.254
  Broadcast address       : 10.0.5.255
  Usable hosts per subnet : 510
Example 144: 10.0.6.0/23
  IP address              : 10.0.6.0
  Prefix length           : /23
  Subnet mask (decimal)   : 255.255.254.0
  Subnet mask (binary)    : 11111111.11111111.11111110.00000000
  Wildcard mask           : 0.0.1.255
  Network address         : 10.0.6.0
  First usable host       : 10.0.6.1
  Last usable host        : 10.0.7.254
  Broadcast address       : 10.0.7.255
  Usable hosts per subnet : 510
Example 145: 172.18.0.0/22
  IP address              : 172.18.0.0
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 172.18.0.0
  First usable host       : 172.18.0.1
  Last usable host        : 172.18.3.254
  Broadcast address       : 172.18.3.255
  Usable hosts per subnet : 1,022
Example 146: 172.18.4.0/22
  IP address              : 172.18.4.0
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 172.18.4.0
  First usable host       : 172.18.4.1
  Last usable host        : 172.18.7.254
  Broadcast address       : 172.18.7.255
  Usable hosts per subnet : 1,022
Example 147: 172.18.8.0/22
  IP address              : 172.18.8.0
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 172.18.8.0
  First usable host       : 172.18.8.1
  Last usable host        : 172.18.11.254
  Broadcast address       : 172.18.11.255
  Usable hosts per subnet : 1,022
Example 148: 172.18.12.0/22
  IP address              : 172.18.12.0
  Prefix length           : /22
  Subnet mask (decimal)   : 255.255.252.0
  Subnet mask (binary)    : 11111111.11111111.11111100.00000000
  Wildcard mask           : 0.0.3.255
  Network address         : 172.18.12.0
  First usable host       : 172.18.12.1
  Last usable host        : 172.18.15.254
```

```
  Broadcast address       : 172.18.15.255
  Usable hosts per subnet : 1,022

Example 149: 192.168.128.0/20
  IP address              : 192.168.128.0
  Prefix length           : /20
  Subnet mask (decimal)   : 255.255.240.0
  Subnet mask (binary)    : 11111111.11111111.11110000.00000000
  Wildcard mask           : 0.0.15.255
  Network address         : 192.168.128.0
  First usable host       : 192.168.128.1
  Last usable host        : 192.168.143.254
  Broadcast address       : 192.168.143.255
  Usable hosts per subnet : 4,094

Example 150: 192.168.144.0/20
  IP address              : 192.168.144.0
  Prefix length           : /20
  Subnet mask (decimal)   : 255.255.240.0
  Subnet mask (binary)    : 11111111.11111111.11110000.00000000
  Wildcard mask           : 0.0.15.255
  Network address         : 192.168.144.0
  First usable host       : 192.168.144.1
  Last usable host        : 192.168.159.254
  Broadcast address       : 192.168.159.255
  Usable hosts per subnet : 4,094

Example 151: 192.168.160.0/20
  IP address              : 192.168.160.0
  Prefix length           : /20
  Subnet mask (decimal)   : 255.255.240.0
  Subnet mask (binary)    : 11111111.11111111.11110000.00000000
  Wildcard mask           : 0.0.15.255
  Network address         : 192.168.160.0
  First usable host       : 192.168.160.1
  Last usable host        : 192.168.175.254
  Broadcast address       : 192.168.175.255
  Usable hosts per subnet : 4,094

Example 152: 192.168.176.0/20
  IP address              : 192.168.176.0
  Prefix length           : /20
  Subnet mask (decimal)   : 255.255.240.0
  Subnet mask (binary)    : 11111111.11111111.11110000.00000000
  Wildcard mask           : 0.0.15.255
  Network address         : 192.168.176.0
  First usable host       : 192.168.176.1
  Last usable host        : 192.168.191.254
  Broadcast address       : 192.168.191.255
  Usable hosts per subnet : 4,094

Example 153: 10.50.0.0/21
  IP address              : 10.50.0.0
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 10.50.0.0
  First usable host       : 10.50.0.1
  Last usable host        : 10.50.7.254
  Broadcast address       : 10.50.7.255
  Usable hosts per subnet : 2,046
```

```
Example 154: 10.50.8.0/21
  IP address              : 10.50.8.0
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 10.50.8.0
  First usable host       : 10.50.8.1
  Last usable host        : 10.50.15.254
  Broadcast address       : 10.50.15.255
  Usable hosts per subnet : 2,046
Example 155: 10.50.16.0/21
  IP address              : 10.50.16.0
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 10.50.16.0
  First usable host       : 10.50.16.1
  Last usable host        : 10.50.23.254
  Broadcast address       : 10.50.23.255
  Usable hosts per subnet : 2,046
Example 156: 10.50.24.0/21
  IP address              : 10.50.24.0
  Prefix length           : /21
  Subnet mask (decimal)   : 255.255.248.0
  Subnet mask (binary)    : 11111111.11111111.11111000.00000000
  Wildcard mask           : 0.0.7.255
  Network address         : 10.50.24.0
  First usable host       : 10.50.24.1
  Last usable host        : 10.50.31.254
  Broadcast address       : 10.50.31.255
  Usable hosts per subnet : 2,046
```

D: Subnet Design Problems

```
Example 157 (Design Problem):
  Q: Subnet 192.168.1.0/24 into 4 equal subnets for 4 departments.

  Solution:
  Step 1: Need 4 subnets → 2 bits borrowed (2^2 = 4).
  Step 2: New prefix = /24 + 2 = /26.
  Step 3: New mask = 255.255.255.192 (binary 11000000 in last octet).
  Step 4: Each subnet has 2^6 − 2 = 62 usable hosts.
  Step 5: Subnet list:
    • 192.168.1.0/26   (hosts .1 –.62,   broadcast .63)   → Sales
    • 192.168.1.64/26  (hosts .65 –.126, broadcast .127)  → IT
    • 192.168.1.128/26 (hosts .129–.190, broadcast .191)  → HR
    • 192.168.1.192/26 (hosts .193–.254, broadcast .255)  → Eng
Example 158 (Design Problem):
  Q: Subnet 172.16.0.0/16 to create 100 subnets. What prefix should you use?

  Solution:
  Step 1: 2^6 = 64 (too few). 2^7 = 128 (enough). Borrow 7 bits.
  Step 2: New prefix = /16 + 7 = /23.
  Step 3: New mask = 255.255.254.0.
  Step 4: Each subnet has 2^9 − 2 = 510 usable hosts.
  Step 5: First 4 subnets:
    • 172.16.0.0/23   (hosts 172.16.0.1 – 172.16.1.254)
```

- 172.16.2.0/23 (hosts 172.16.2.1 – 172.16.3.254)
- 172.16.4.0/23 (hosts 172.16.4.1 – 172.16.5.254)
- 172.16.6.0/23 (hosts 172.16.6.1 – 172.16.7.254)

Example 159 (Design Problem):
Q: You need subnets that support 50 hosts each. Given 192.168.10.0/24, how many such subnets can you make?

Solution:
Step 1: 50 hosts → 2^6 – 2 = 62 (enough). Need 6 host bits.
Step 2: Borrow 32 – 24 – 6 = 2 bits for subnet.
Step 3: New prefix = /26.
Step 4: Number of /26 subnets in /24 = 2^2 = 4.
Step 5: Each /26 has 62 usable hosts ≥ 50. Confirmed.

Example 160 (Design Problem):
Q: Subnet 10.0.0.0/8 to create 1000 subnets. What is the new prefix?

Solution:
Step 1: 2^10 = 1024 (enough for 1000). Borrow 10 bits.
Step 2: New prefix = /8 + 10 = /18.
Step 3: New mask = 255.255.192.0.
Step 4: Each subnet has 2^14 – 2 = 16,382 usable hosts.

Example 161 (Design Problem):
Q: Design VLSM subnets for: Network 192.168.1.0/24 needs: LAN-A (60 hosts), LAN-B (25 hosts), LAN-C (10 hosts), 3 point-to-point links.

Solution:
Sort largest first:
- LAN-A (60 hosts) → needs 6 host bits → /26 → 192.168.1.0/26 (hosts .1–.62)
- LAN-B (25 hosts) → needs 5 host bits → /27 → 192.168.1.64/27 (hosts .65–.94)
- LAN-C (10 hosts) → needs 4 host bits → /28 → 192.168.1.96/28 (hosts .97–.110)
- Link 1 (2 hosts) → needs 2 host bits → /30 → 192.168.1.112/30 (hosts .113–.114)
- Link 2 (2 hosts) → needs 2 host bits → /30 → 192.168.1.116/30 (hosts .117–.118)
- Link 3 (2 hosts) → needs 2 host bits → /30 → 192.168.1.120/30 (hosts .121–.122)

Remaining free space: 192.168.1.124 – 192.168.1.254 (132 addresses).

Example 162 (Design Problem):
Q: A network has 192.168.5.0/24. Create subnets to support 14 hosts each. How many subnets?

Solution:
14 hosts → need 5 host bits (2^5 – 2 = 30 ≥ 14).
Borrow 32 – 24 – 5 = 3 bits → /27.
Number of /27 subnets in /24 = 2^3 = 8.
Subnets: .0/27, .32/27, .64/27, .96/27, .128/27, .160/27, .192/27, .224/27.
Each has 30 usable hosts.

Example 163 (Design Problem):
Q: What is the broadcast address of 192.168.10.140/26?

Solution:
/26 → mask 255.255.255.192. Block size = 256 – 192 = 64.
Subnet boundaries: 0, 64, 128, 192.
140 falls in the 128 subnet (128 ≤ 140 < 192).
Broadcast = next boundary – 1 = 192 – 1 = 191.
Answer: 192.168.10.191

```
Example 164 (Design Problem):
  Q: What is the network address of 172.16.45.213/20?

  Solution:
  /20 → mask 255.255.240.0. Block size = 256 − 240 = 16 in 3rd octet.
  Find largest multiple of 16 ≤ 45 → 32 (32, 48 — 32 ≤ 45 < 48).
  Network = 172.16.32.0.
  Broadcast = 172.16.47.255.
```

F: VLSM Scenarios

```
Example 165 (VLSM Scenario):
  Q: VLSM design for a branch office. Network 192.168.50.0/24. Need: 1 LAN (40
hosts), 1 LAN (20 hosts), 1 LAN (10 hosts), 2 WAN links (2 hosts each).

  Solution:
  Sort largest first:
    • LAN1 (40 hosts) → /26 (62 hosts) → 192.168.50.0/26   (hosts .1 – .62,   bcast
.63)
    • LAN2 (20 hosts) → /27 (30 hosts) → 192.168.50.64/27  (hosts .65 – .94,  bcast
.95)
    • LAN3 (10 hosts) → /28 (14 hosts) → 192.168.50.96/28  (hosts .97 – .110, bcast
.111)
    • WAN1 (2 hosts)  → /30 (2 hosts)  → 192.168.50.112/30 (hosts .113 – .114, bcast
.115)
    • WAN2 (2 hosts)  → /30 (2 hosts)  → 192.168.50.116/30 (hosts .117 – .118, bcast
.119)
  Remaining free: 192.168.50.120 – 192.168.50.254 (135 addresses for future use).
Example 166 (VLSM Scenario):
  Q: VLSM design for 4 LANs and 3 WAN links. Network 172.16.0.0/16. LAN sizes: 100,
50, 25, 10 hosts. WAN links: 3 × 2 hosts.

  Solution:
    • LAN1 (100 hosts) → /25 → 172.16.0.0/25    (hosts .1 – .126,   bcast .127)
    • LAN2 (50 hosts)  → /26 → 172.16.0.128/26  (hosts .129 – .190, bcast .191)
    • LAN3 (25 hosts)  → /27 → 172.16.0.192/27  (hosts .193 – .222, bcast .223)
    • LAN4 (10 hosts)  → /28 → 172.16.0.224/28  (hosts .225 – .238, bcast .239)
    • WAN1 (2 hosts)   → /30 → 172.16.0.240/30  (hosts .241 – .242, bcast .243)
    • WAN2 (2 hosts)   → /30 → 172.16.0.244/30  (hosts .245 – .246, bcast .247)
    • WAN3 (2 hosts)   → /30 → 172.16.0.248/30  (hosts .249 – .250, bcast .251)
  Remaining free: 172.16.0.252 – 172.16.255.255 (lots of room for growth).
Example 167 (VLSM Scenario):
  Q: You are given 10.10.0.0/16. Subnet it for 4 sites with these needs: Site A=500
hosts, B=200 hosts, C=100 hosts, D=50 hosts. Then assign /30s for 4 WAN links.

  Solution:
    • Site A (500 hosts) → /23 (510 hosts) → 10.10.0.0/23   (10.10.0.1 –
10.10.1.254)
    • Site B (200 hosts) → /24 (254 hosts) → 10.10.2.0/24   (10.10.2.1 –
10.10.2.254)
    • Site C (100 hosts) → /25 (126 hosts) → 10.10.3.0/25   (10.10.3.1 –
10.10.3.126)
    • Site D (50 hosts)  → /26 (62 hosts)  → 10.10.3.128/26 (10.10.3.129 –
10.10.3.190)
    • WAN1 (2 hosts) → /30 → 10.10.3.192/30
    • WAN2 (2 hosts) → /30 → 10.10.3.196/30
    • WAN3 (2 hosts) → /30 → 10.10.3.200/30
    • WAN4 (2 hosts) → /30 → 10.10.3.204/30
  Remaining free: 10.10.3.208 – 10.10.255.255.
```

Example 168 (VLSM Scenario):
  Q: VLSM with 192.168.10.0/24: 3 LANs (30, 12, 6 hosts) + 2 WAN links (2 hosts each). Show all assignments.

  Solution:
    • LAN1 (30 hosts) → /27 (30 hosts) → 192.168.10.0/27 (hosts .1 – .30, bcast .31)
    • LAN2 (12 hosts) → /28 (14 hosts) → 192.168.10.32/28 (hosts .33 – .46, bcast .47)
    • LAN3 (6 hosts) → /29 (6 hosts) → 192.168.10.48/29 (hosts .49 – .54, bcast .55)
    • WAN1 (2 hosts) → /30 (2 hosts) → 192.168.10.56/30 (hosts .57 – .58, bcast .59)
    • WAN2 (2 hosts) → /30 (2 hosts) → 192.168.10.60/30 (hosts .61 – .62, bcast .63)
  Remaining free: 192.168.10.64 – 192.168.10.254 (191 addresses).

E: Exam-Speed Shortcuts

Example 169 (Exam-Speed Shortcut):
  Q: Quickly find the subnet of 192.168.1.135/26.

  SHORTCUT: /26 → 4 subnets of 64 each. Block size = 64.
Find 135 in 3rd octet. 64×2 = 128 ≤ 135 < 192 = 64×3.
→ Subnet = 192.168.1.128, broadcast = 192.168.1.191.

Example 170 (Exam-Speed Shortcut):
  Q: Quickly find the broadcast of 10.1.1.50/27.

  SHORTCUT: /27 → block size 32. Subnet boundaries: 0, 32, 64.
50 falls in 32–63 range. Broadcast = 63.
→ 10.1.1.63

Example 171 (Exam-Speed Shortcut):
  Q: How many /30 subnets fit in a /24?

  SHORTCUT: 32 – 30 = 2 bits for subnet. 2^8 (host bits in /24) / 2^2 = 64 subnets.

Example 172 (Exam-Speed Shortcut):
  Q: How many host bits do you need for 100 hosts?

  SHORTCUT: 2^n – 2 ≥ 100 → n = 7 (2^7 – 2 = 126). So /25 prefix (32-7=25).

Example 173 (Exam-Speed Shortcut):
  Q: Quickly find subnet of 172.16.75.0/22.

  SHORTCUT: /22 → block size 4 in 3rd octet. Boundaries: 0,4,8,...,72,76.
75 falls in 72–75 range (since 75 < 76). Subnet = 172.16.72.0, broadcast = 172.16.75.255.

Example 174 (Exam-Speed Shortcut):
  Q: What is the wildcard mask for /28?

  SHORTCUT: wildcard = inverse of subnet mask. /28 = 255.255.255.240, so wildcard = 0.0.0.15.

Example 175 (Exam-Speed Shortcut):
  Q: Quickly check if 192.168.1.100/26 is in the same subnet as 192.168.1.200/26.

  SHORTCUT: /26 → block size 64.
100 is in 64–127 range. 200 is in 192–255 range. Different subnets → different broadcast domains.

```
Example 176 (Exam-Speed Shortcut):
  Q: What is the network address of 10.50.75.100/19?

  SHORTCUT: /19 → block size 32 in 3rd octet.
75 ÷ 32 = 2 (integer). 32 × 2 = 64 ≤ 75 < 96.
Network = 10.50.64.0, broadcast = 10.50.95.255.

Example 177 (Exam-Speed Shortcut):
  Q: Block size for /29?

  SHORTCUT: 256 − 248 = 8. Block size 8, 6 usable hosts per subnet.

Example 178 (Exam-Speed Shortcut):
  Q: Block size for /28?

  SHORTCUT: 256 − 240 = 16. Block size 16, 14 usable hosts per subnet.

Example 179 (Exam-Speed Shortcut):
  Q: Block size for /27?

  SHORTCUT: 256 − 224 = 32. Block size 32, 30 usable hosts per subnet.

Example 180 (Exam-Speed Shortcut):
  Q: Block size for /26?

  SHORTCUT: 256 − 192 = 64. Block size 64, 62 usable hosts per subnet.

Example 181 (Exam-Speed Shortcut):
  Q: Block size for /25?

  SHORTCUT: 256 − 128 = 128. Block size 128, 126 usable hosts per subnet.

Example 182 (Exam-Speed Shortcut):
  Q: Quickly find the network address of 192.168.55.200/26.

  SHORTCUT: /26 → block size 64. Boundaries 0, 64, 128, 192.
200 falls in 192–255 range → Network = 192.168.55.192, broadcast = 192.168.55.255.

Example 183 (Exam-Speed Shortcut):
  Q: Quickly find the broadcast of 10.10.10.130/27.

  SHORTCUT: /27 → block size 32. Boundaries 0, 32, 64, 96, 128, 160.
130 falls in 128–159 range → Broadcast = 10.10.10.159.

Example 184 (Exam-Speed Shortcut):
  Q: Quickly find subnet of 172.16.100.50/22.

  SHORTCUT: /22 → block size 4 in 3rd octet.
100 ÷ 4 = 25. 4 × 25 = 100 ≤ 100 < 104 → Network = 172.16.100.0, broadcast =
172.16.103.255.

Example 185 (Exam-Speed Shortcut):
  Q: How many usable hosts in /21?

  SHORTCUT: 32 − 21 = 11 host bits → 2^11 − 2 = 2046 hosts.

Example 186 (Exam-Speed Shortcut):
  Q: How many usable hosts in /23?

  SHORTCUT: 32 − 23 = 9 host bits → 2^9 − 2 = 510 hosts.

Example 187 (Exam-Speed Shortcut):
  Q: How many usable hosts in /20?

  SHORTCUT: 32 − 20 = 12 host bits → 2^12 − 2 = 4094 hosts.
```

```
Example 188 (Exam-Speed Shortcut):
  Q: How many usable hosts in /19?

  SHORTCUT: 32 - 19 = 13 host bits → 2^13 - 2 = 8190 hosts.
Example 189 (Exam-Speed Shortcut):
  Q: Wildcard mask for /27?

  SHORTCUT: /27 = 255.255.255.224 → wildcard = 0.0.0.31.
Example 190 (Exam-Speed Shortcut):
  Q: Wildcard mask for /23?

  SHORTCUT: /23 = 255.255.254.0 → wildcard = 0.0.1.255.
Example 191 (Exam-Speed Shortcut):
  Q: Wildcard mask for /22?

  SHORTCUT: /22 = 255.255.252.0 → wildcard = 0.0.3.255.
Example 192 (Exam-Speed Shortcut):
  Q: Wildcard mask for /21?

  SHORTCUT: /21 = 255.255.248.0 → wildcard = 0.0.7.255.
Example 193 (Exam-Speed Shortcut):
  Q: What prefix gives 254 hosts per subnet?

  SHORTCUT: 2^8 - 2 = 254 → 8 host bits → /24 prefix.
Example 194 (Exam-Speed Shortcut):
  Q: What prefix gives 62 hosts per subnet?

  SHORTCUT: 2^6 - 2 = 62 → 6 host bits → /26 prefix.
Example 195 (Exam-Speed Shortcut):
  Q: What prefix gives 30 hosts per subnet?

  SHORTCUT: 2^5 - 2 = 30 → 5 host bits → /27 prefix.
Example 196 (Exam-Speed Shortcut):
  Q: What prefix gives 14 hosts per subnet?

  SHORTCUT: 2^4 - 2 = 14 → 4 host bits → /28 prefix.
Example 197 (Exam-Speed Shortcut):
  Q: What prefix gives 6 hosts per subnet?

  SHORTCUT: 2^3 - 2 = 6 → 3 host bits → /29 prefix.
Example 198 (Exam-Speed Shortcut):
  Q: Quickly determine: are 192.168.1.5/28 and 192.168.1.20/28 in the same subnet?

  SHORTCUT: /28 → block size 16. Boundaries: 0, 16, 32, 48, ...
5 falls in 0-15 range. 20 falls in 16-31 range. Different subnets.
Example 199 (Exam-Speed Shortcut):
  Q: How many /29 subnets fit in a /24?

  SHORTCUT: 32 - 29 = 3 bits for subnet → 2^3 = 8 subnets per /24 wait — let me redo.
Actually /29 has 8 addresses, /24 has 256 addresses → 256/8 = 32 subnets.
Example 200 (Exam-Speed Shortcut):
  Q: How many /27 subnets fit in a /24?

  SHORTCUT: /27 has 32 addresses, /24 has 256 → 256/32 = 8 subnets.
```

```
Example 201 (Exam-Speed Shortcut):
  Q: How many /28 subnets fit in a /24?

  SHORTCUT: /28 has 16 addresses, /24 has 256 → 256/16 = 16 subnets.
Example 202 (Exam-Speed Shortcut):
  Q: Quick check: 10.0.0.1/30 and 10.0.0.5/30 are in same subnet?

  SHORTCUT: /30 → block size 4. Boundaries: 0, 4, 8, 12...
1 is in 0-3 range. 5 is in 4-7 range. Different subnets.
Example 203 (Exam-Speed Shortcut):
  Q: Mask of /31?

  SHORTCUT: /31 = 255.255.255.254. Used for point-to-point links (RFC 3021) — 2
addresses, both usable.
Example 204 (Exam-Speed Shortcut):
  Q: Mask of /32?

  SHORTCUT: /32 = 255.255.255.255. Host-only route (single host).
```

Practice Worksheet

Work through these problems without looking at the answers. Then check yourself with the answer key below.

```
CCNA Subnetting Practice Worksheet

============================================================

1. Given 192.168.10.55/24, find:
     • Subnet mask (decimal)
     • Network address
     • First usable host
     • Last usable host
     • Broadcast address
     • Number of usable hosts

2. Given 10.0.0.100/24, find:
     • Subnet mask (decimal)
     • Network address
     • First usable host
     • Last usable host
     • Broadcast address
     • Number of usable hosts

3. Given 172.16.50.75/24, find:
     • Subnet mask (decimal)
     • Network address
     • First usable host
     • Last usable host
     • Broadcast address
     • Number of usable hosts

4. Given 192.168.1.130/25, find:
     • Subnet mask (decimal)
     • Network address
     • First usable host
     • Last usable host
     • Broadcast address
     • Number of usable hosts

5. Given 10.0.0.200/25, find:
     • Subnet mask (decimal)
     • Network address
     • First usable host
     • Last usable host
     • Broadcast address
     • Number of usable hosts

6. Given 172.16.1.100/26, find:
     • Subnet mask (decimal)
     • Network address
     • First usable host
     • Last usable host
     • Broadcast address
     • Number of usable hosts

7. Given 192.168.1.130/26, find:
     • Subnet mask (decimal)
     • Network address
```

- First usable host
- Last usable host
- Broadcast address
- Number of usable hosts

8. Given 10.0.0.130/26, find:
   - Subnet mask (decimal)
   - Network address
   - First usable host
   - Last usable host
   - Broadcast address
   - Number of usable hosts

9. Given 172.16.0.65/27, find:
   - Subnet mask (decimal)
   - Network address
   - First usable host
   - Last usable host
   - Broadcast address
   - Number of usable hosts

10. Given 192.168.1.100/27, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

11. Given 10.0.0.50/27, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

12. Given 172.16.0.130/28, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

13. Given 192.168.1.50/28, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

14. Given 10.0.0.130/28, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host

- Broadcast address
- Number of usable hosts

15. Given 172.16.0.10/29, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

16. Given 192.168.1.20/29, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

17. Given 10.0.0.6/30, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

18. Given 172.16.0.5/30, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

19. Given 172.16.0.1/23, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

20. Given 10.0.0.1/22, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

21. Given 192.168.0.1/21, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

22. Given 172.20.10.1/20, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

23. Given 10.50.10.1/19, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

24. Given 192.168.4.1/22, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

25. Given 172.16.10.1/21, find:
    - Subnet mask (decimal)
    - Network address
    - First usable host
    - Last usable host
    - Broadcast address
    - Number of usable hosts

Answer Key

```
CCNA Subnetting Worksheet — Answer Key

============================================================

1. 192.168.10.55/24
     Mask: 255.255.255.0   Network: 192.168.10.0
     Hosts: 192.168.10.1 – 192.168.10.254   Bcast: 192.168.10.255
     Usable hosts: 254

2. 10.0.0.100/24
     Mask: 255.255.255.0   Network: 10.0.0.0
     Hosts: 10.0.0.1 – 10.0.0.254   Bcast: 10.0.0.255
     Usable hosts: 254

3. 172.16.50.75/24
     Mask: 255.255.255.0   Network: 172.16.50.0
     Hosts: 172.16.50.1 – 172.16.50.254   Bcast: 172.16.50.255
     Usable hosts: 254

4. 192.168.1.130/25
     Mask: 255.255.255.128   Network: 192.168.1.128
     Hosts: 192.168.1.129 – 192.168.1.254   Bcast: 192.168.1.255
     Usable hosts: 126

5. 10.0.0.200/25
     Mask: 255.255.255.128   Network: 10.0.0.128
     Hosts: 10.0.0.129 – 10.0.0.254   Bcast: 10.0.0.255
     Usable hosts: 126

6. 172.16.1.100/26
     Mask: 255.255.255.192   Network: 172.16.1.64
     Hosts: 172.16.1.65 – 172.16.1.126   Bcast: 172.16.1.127
     Usable hosts: 62

7. 192.168.1.130/26
     Mask: 255.255.255.192   Network: 192.168.1.128
     Hosts: 192.168.1.129 – 192.168.1.190   Bcast: 192.168.1.191
     Usable hosts: 62

8. 10.0.0.130/26
     Mask: 255.255.255.192   Network: 10.0.0.128
     Hosts: 10.0.0.129 – 10.0.0.190   Bcast: 10.0.0.191
     Usable hosts: 62

9. 172.16.0.65/27
     Mask: 255.255.255.224   Network: 172.16.0.64
     Hosts: 172.16.0.65 – 172.16.0.94   Bcast: 172.16.0.95
     Usable hosts: 30

10. 192.168.1.100/27
     Mask: 255.255.255.224   Network: 192.168.1.96
     Hosts: 192.168.1.97 – 192.168.1.126   Bcast: 192.168.1.127
     Usable hosts: 30

11. 10.0.0.50/27
     Mask: 255.255.255.224   Network: 10.0.0.32
     Hosts: 10.0.0.33 – 10.0.0.62   Bcast: 10.0.0.63
```

```
     Usable hosts: 30

12. 172.16.0.130/28
     Mask: 255.255.255.240   Network: 172.16.0.128
     Hosts: 172.16.0.129 – 172.16.0.142   Bcast: 172.16.0.143
     Usable hosts: 14

13. 192.168.1.50/28
     Mask: 255.255.255.240   Network: 192.168.1.48
     Hosts: 192.168.1.49 – 192.168.1.62   Bcast: 192.168.1.63
     Usable hosts: 14

14. 10.0.0.130/28
     Mask: 255.255.255.240   Network: 10.0.0.128
     Hosts: 10.0.0.129 – 10.0.0.142   Bcast: 10.0.0.143
     Usable hosts: 14

15. 172.16.0.10/29
     Mask: 255.255.255.248   Network: 172.16.0.8
     Hosts: 172.16.0.9 – 172.16.0.14   Bcast: 172.16.0.15
     Usable hosts: 6

16. 192.168.1.20/29
     Mask: 255.255.255.248   Network: 192.168.1.16
     Hosts: 192.168.1.17 – 192.168.1.22   Bcast: 192.168.1.23
     Usable hosts: 6

17. 10.0.0.6/30
     Mask: 255.255.255.252   Network: 10.0.0.4
     Hosts: 10.0.0.5 – 10.0.0.6   Bcast: 10.0.0.7
     Usable hosts: 2

18. 172.16.0.5/30
     Mask: 255.255.255.252   Network: 172.16.0.4
     Hosts: 172.16.0.5 – 172.16.0.6   Bcast: 172.16.0.7
     Usable hosts: 2

19. 172.16.0.1/23
     Mask: 255.255.254.0   Network: 172.16.0.0
     Hosts: 172.16.0.1 – 172.16.1.254   Bcast: 172.16.1.255
     Usable hosts: 510

20. 10.0.0.1/22
     Mask: 255.255.252.0   Network: 10.0.0.0
     Hosts: 10.0.0.1 – 10.0.3.254   Bcast: 10.0.3.255
     Usable hosts: 1,022

21. 192.168.0.1/21
     Mask: 255.255.248.0   Network: 192.168.0.0
     Hosts: 192.168.0.1 – 192.168.7.254   Bcast: 192.168.7.255
     Usable hosts: 2,046

22. 172.20.10.1/20
     Mask: 255.255.240.0   Network: 172.20.0.0
     Hosts: 172.20.0.1 – 172.20.15.254   Bcast: 172.20.15.255
     Usable hosts: 4,094

23. 10.50.10.1/19
```

```
     Mask: 255.255.224.0   Network: 10.50.0.0
     Hosts: 10.50.0.1 – 10.50.31.254   Bcast: 10.50.31.255
     Usable hosts: 8,190

24. 192.168.4.1/22
     Mask: 255.255.252.0   Network: 192.168.4.0
     Hosts: 192.168.4.1 – 192.168.7.254   Bcast: 192.168.7.255
     Usable hosts: 1,022

25. 172.16.10.1/21
     Mask: 255.255.248.0   Network: 172.16.8.0
     Hosts: 172.16.8.1 – 172.16.15.254   Bcast: 172.16.15.255
     Usable hosts: 2,046
```

Mock Subnetting Test (20 Questions, 15 Minutes)

```
CCNA Mock Subnetting Test (20 questions, 15 minutes)

============================================================

1. What is the broadcast address of 192.168.1.95/27?
    Answer: 192.168.1.127

2. How many usable hosts in a /28?
    Answer: 14

3. What is the network address of 172.16.45.213/20?
    Answer: 172.16.32.0

4. What mask is /22 in dotted decimal?
    Answer: 255.255.252.0

5. How many /30 subnets fit in a /24?
    Answer: 64

6. Wildcard mask for /26?
    Answer: 0.0.0.63

7. What is the first usable host of 192.168.10.64/27?
    Answer: 192.168.10.65

8. What is the last usable host of 10.0.0.96/28?
    Answer: 10.0.0.110

9. You need 500 hosts per subnet. What is the smallest prefix?
    Answer: /23

10. 192.168.1.10/26 belongs to which subnet?
    Answer: 192.168.1.0/26

11. What is the block size for /27?
    Answer: 32

12. What is the broadcast of 10.10.10.5/30?
    Answer: 10.10.10.7

13. How many host bits in a /20?
    Answer: 12

14. Is 192.168.1.50/26 in the same subnet as 192.168.1.100/26?
    Answer: No (50 is in .0 subnet, 100 is in .64 subnet)

15. What mask gives 126 usable hosts?
    Answer: 255.255.255.128 (/25)

16. Network of 172.20.155.10/19?
    Answer: 172.20.128.0

17. Last usable host of 192.168.50.130/25?
    Answer: 192.168.50.254

18. Number of usable hosts in /23?
    Answer: 510
```

```
19. What is the prefix for a mask of 255.255.255.240?
    Answer: /28

20. Wildcard mask for /23?
    Answer: 0.0.1.255
```

Chapter 6 Quiz

Q1. What is the broadcast address of 192.168.1.95/27?

A. 192.168.1.95
B. 192.168.1.96
C. 192.168.1.127
D. 192.168.1.128
Answer: C — /27 = block size 32. 95 falls in 64-95 range. Broadcast = 95 (since 96 is next boundary, bcast = 95). Actually recheck: 95 IS the broadcast of 64-95 subnet (96 would be next net).

Q2. How many usable hosts are in a /28?

A. 14
B. 16
C. 30
D. 62
Answer: A — /28 = 4 host bits → 2^4 - 2 = 14 usable hosts.

Q3. What is the network address of 172.16.45.213/20?

A. 172.16.32.0
B. 172.16.45.0
C. 172.16.48.0
D. 172.16.0.0
Answer: A — /20 = block size 16 in 3rd octet. 45÷16=2 (int), 2×16=32. Network = 172.16.32.0.

Q4. What mask is /22 in dotted decimal?

A. 255.255.252.0
B. 255.255.248.0
C. 255.255.254.0
D. 255.255.240.0
Answer: A — /22 = 255.255.252.0 (last octet of mask: 11111100 = 252).

Q5. How many /30 subnets fit in a /24?

A. 16
B. 32
C. 64
D. 128
Answer: C — /30 in /24: 2^(30-24) = 2^6 = 64 subnets.

PART 7

IPv6 — The Future of IP

128-bit addressing for the next billion devices

Chapter 7: IPv6 — Why, What, and How

IPv4 has 4.3 billion addresses. In 2026, the world has 8 billion humans and 30+ billion connected devices. The math doesn't work. IPv6 solves this with 128-bit addresses — that's 2^128 = 340 undecillion (3.4×10^38) addresses. To put that in perspective: every person on Earth could be assigned 5×10^28 IPv6 addresses and we'd still have plenty left over. IPv6 isn't just 'bigger IPv4' — it's a redesign that simplifies the header, eliminates broadcasts, mandates IPsec, and includes built-in auto-configuration.

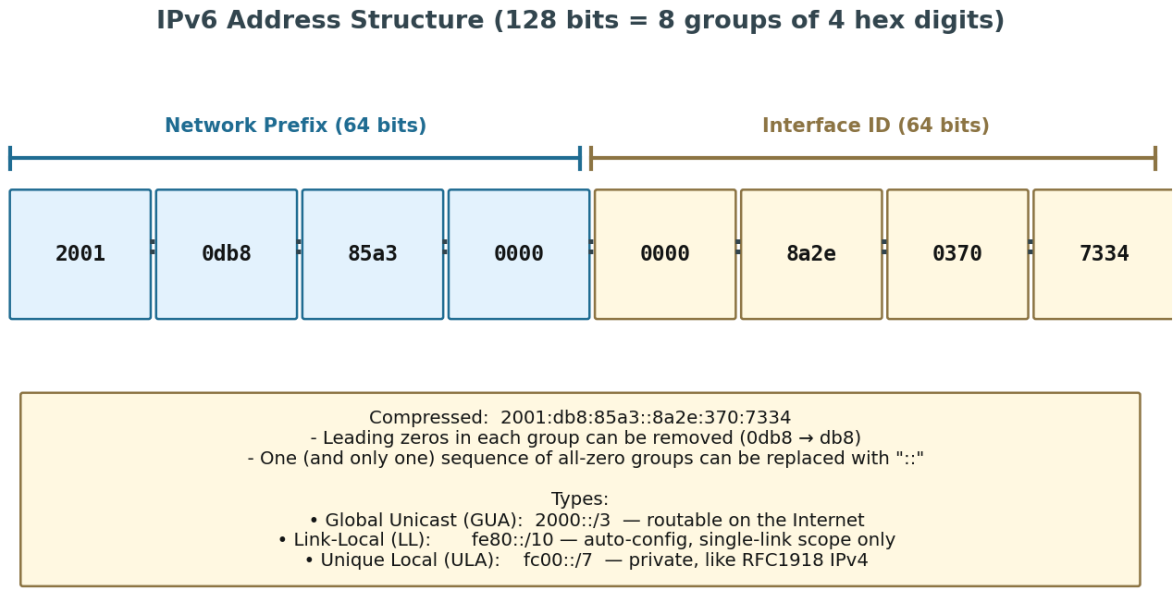


Figure: IPv6 address structure: 128 bits = 8 groups of 4 hex digits. Network prefix (64 bits) + Interface ID (64 bits).

IPv6 Address Structure

An IPv6 address is 128 bits, written as eight groups of four hex digits separated by colons: 2001:0db8:85a3:0000:0000:8a2e:0370:7334. Two simplification rules make this more readable. First, leading zeros in each group can be dropped: 0db8 becomes db8, 0370 becomes 370. Second, one (and only one) consecutive run of all-zero groups can be replaced with double colons ::. So our example becomes 2001:db8:85a3::8a2e:370:7334 — much cleaner.

The address is logically split into two 64-bit halves: the **network prefix** (first 64 bits, assigned by ISP or locally) and the **interface ID** (last 64 bits, identifying the host on the subnet). This /64 boundary is mandatory in IPv6 — subnets are always /64, unlike IPv4 where any prefix is allowed.

IPv6 Address Types

Type	Prefix	Scope	Description
Global Unicast (GUA)	2000::/3	Internet-routable	Equivalent to public IPv4. Assigned by ISP.
Unique Local (ULA)	fc00::/7	Private (site-local)	Equivalent to RFC 1918 IPv4. Not routable on Internet.
Link-Local	fe80::/10	Single link only	Auto-configured on every interface. Used for neighbor discovery.
Multicast	ff00::/8	Group of nodes	Replaces IPv4 broadcast. Various scopes (link, site, org, global).
Loopback	::1/128	Local host	Equivalent to IPv4 127.0.0.1.
Unspecified	::/128	N/A	Equivalent to IPv4 0.0.0.0. Used as source before address assignment.
Anycast	Same as unicast	Nearest of group	One address shared by multiple nodes. Routed to topologically nearest.

SLAAC — Stateless Address AutoConfiguration

IPv6 introduces **SLAAC**, a way for hosts to automatically configure their own addresses without DHCP. When a host boots and connects to an IPv6 network, it sends an ICMPv6 **Router Solicitation** (RS) to the all-routers multicast address (FF02::2). A router replies with a **Router Advertisement** (RA) containing the network prefix (e.g., 2001:db8:1::/64) and some options. The host combines this prefix with its own 64-bit interface ID (derived from MAC or randomly generated) to form its global unicast address.

SLAAC has major advantages over DHCP: zero infrastructure required (just a router sending RAs), no DHCP server to maintain, and hosts can have multiple addresses for privacy. The downside: no centralized tracking of who has which address. For environments needing accountability, **stateful DHCPv6** is still an option.

EUI-64 — Deriving Interface ID from MAC

EUI-64 (Extended Unique Identifier) is a method for deriving a 64-bit interface ID from a 48-bit MAC address. The process is simple: take the 48-bit MAC, split it in half (24 bits + 24 bits), insert FFFE in the middle (making 64 bits total), and flip the U/L bit (bit 1 of the first byte).

Example: MAC 00:1A:2B:3C:4D:5E. Split: 001A2B and 3C4D5E. Insert FFFE: 001A2BFFFE3C4D5E. Flip U/L bit (bit 1 of 00 → 02): 021A:2BFF:FE3C:4D5E. Combined with prefix 2001:db8:1::/64, the full IPv6 address is 2001:db8:1::21A:2BFF:FE3C:4D5E. Modern operating systems use random interface IDs instead of EUI-64 for privacy (so the same MAC doesn't generate the same address everywhere), but EUI-64 is still tested on the CCNA.

ICMPv6 — Replacement for ARP

IPv6 has no ARP. Instead, it uses ICMPv6 **Neighbor Discovery Protocol** (NDP, RFC 4861). When a host needs to find the MAC for an IPv6 neighbor, it sends a **Neighbor Solicitation** (NS) message to the solicited-node multicast address (FF02::1:FFxx:xxxx, derived from the target IPv6). The owner of that IPv6 responds with a **Neighbor Advertisement** (NA) containing its MAC. The host caches this in its neighbor cache (the IPv6 equivalent of the ARP cache).

```
# Cisco IOS IPv6 commands
ipv6 unicast-routing                                # Enable IPv6 routing (global)
interface GigabitEthernet0/0
  ipv6 enable                                       # Enable IPv6 (auto link-local)
```

```
  ipv6 address 2001:db8:1::1/64                    # Assign global IPv6
  ipv6 address fe80::1 link-local                  # Set specific link-local
  ipv6 nd prefix 2001:db8:1::/64                   # Advertise prefix via RA
  ipv6 nd other-config-flag                        # Tell hosts to use DHCPv6 for DNS

show ipv6 interface brief                          # All interfaces, IPv6 summary
show ipv6 route                                    # IPv6 routing table
show ipv6 neighbors                                # IPv6 neighbor cache (ARP
equivalent)
show ipv6 router                                   # Router Advertisement info
ping ipv6 2001:db8:1::10                           # Test IPv6 reachability
traceroute ipv6 2001:db8:1::10                     # IPv6 traceroute
```

Chapter 7 Quiz

Q1. How many bits is an IPv6 address?

A. 32 bits

B. 64 bits

C. 128 bits

D. 256 bits

Answer: C — IPv6 addresses are 128 bits, written as 8 groups of 4 hex digits.

Q2. What is the IPv6 link-local address prefix?

A. 2000::/3

B. FE80::/10

C. FF00::/8

D. FC00::/7

Answer: B — FE80::/10 is link-local. Auto-configured on every IPv6 interface.

Q3. What does SLAAC stand for?

A. StateLess Address AutoConfiguration

B. Stateful Local Address Authentication Code

C. Single Link Auto Configuration

D. Standard Local Area Connection

Answer: A — SLAAC = Stateless Address AutoConfiguration. Uses ICMPv6 Router Advertisements.

Q4. Does IPv6 use ARP?

A. Yes, same as IPv4

B. No — uses ICMPv6 Neighbor Discovery instead

C. Yes, but only for /64 subnets

D. Only on Wi-Fi

Answer: B — IPv6 uses ICMPv6 Neighbor Discovery Protocol (NDP) for MAC resolution.

Q5. What is the IPv6 loopback address?

A. ::1

B. ::0

C. FE80::1

D. FF00::1

Answer: A — ::1/128 is the IPv6 loopback address (equivalent to 127.0.0.1 in IPv4).

PART 8

Cisco Devices

Switches, routers, firewalls, APs, and WLCs

Chapter 8: Cisco Network Devices

Cisco dominates enterprise networking with about 50% market share. Whether you work at a bank, hospital, university, or tech company, you will encounter Cisco gear. This chapter covers the major device types, their roles, and how they fit into a typical enterprise architecture. Even if your future employer uses Juniper, Arista, or HPE, the concepts transfer directly — Cisco just happens to be the common language of networking.

Cisco Switches

A Cisco switch is a Layer 2 device that forwards Ethernet frames based on MAC addresses. Modern Cisco switches (the Catalyst line) often support Layer 3 features too, making them 'multilayer switches'. The switch's job is to learn which MAC addresses live on which ports (by examining the source MAC of incoming frames), and then forward each frame only out the port where the destination MAC lives. This is far more efficient than a hub, which floods every frame out every port.

Model Series	Target	Key Features	Typical Use
Catalyst 1000	Small business	L2, basic L3, web-managed	Small office, retail
Catalyst 2960-X	Entry enterprise	L2, PoE, stackable	Access layer wiring closet
Catalyst 9200	Modern access	L2/L3, PoE+, StackWise-160	Modern access layer
Catalyst 9300	Enterprise access	L2/L3, UPOE, StackWise-480	Premium access, IoE
Catalyst 9400/9500	Distribution/Core	L3, modular, MACsec	Distribution, core, DC
Catalyst 9800	Wireless controller	WLC + switching	Wireless + access
Nexus 9000	Data center	ACI-capable, VXLAN, 25/100G	Data center spine-leaf

Switch MAC Learning and Forwarding Logic

When a frame arrives on a switch port, the switch does three things. First, it examines the **source MAC** and adds an entry to its MAC table (CAM table) recording that this MAC is reachable via this port. This is 'learning'. Second, it examines the **destination MAC** and looks it up in the CAM table. If found, it forwards the frame out only that port — this is 'known unicast forwarding'. If not found (unknown unicast), it floods the frame out all ports in the same VLAN except the port it arrived on. Third, it computes the FCS (CRC-32) on the frame and drops it if the FCS doesn't match (corrupted frame).

Cisco Routers

A Cisco router is a Layer 3 device that forwards IP packets between different networks. Routers maintain a **routing table** — a list of known destination networks and the best next-hop to reach each. When a packet arrives, the router looks up the destination IP in its routing table (using longest-prefix-match to find the most specific route), then forwards the packet out the appropriate interface toward the next-hop.

Model Series	Target	Key Features
ISR 1000	Small branch	ADSL/cable, basic VPN, Wi-Fi option
ISR 4000	Branch/Enterprise edge	Modular, multi-Gbps, IPS, WAAS
Catalyst 8000	Modern branch/SMB	SD-WAN, cloud-ready, 1-10G
ASR 1000	Enterprise edge/WAN aggregation	Carrier-grade, 10-100G
Catalyst 8200/8300/8500	SD-WAN edge	IOS-XE, SD-WAN native
CSR 1000v	Cloud router	Virtual router for AWS/Azure/GCP

Packet Forwarding — Step by Step

When a router receives a packet, the forwarding process is:

Step 1: Frame arrives on an interface. Router checks the FCS — if corrupted, drop the frame.

Step 2: Router checks the destination MAC — if it's not the router's MAC (or broadcast), drop the frame.

Step 3: Router strips the Layer 2 frame, leaving the IP packet.

Step 4: Router decrements TTL by 1. If TTL reaches 0, drop the packet and send ICMP Time Exceeded back to source.

Step 5: Router looks up destination IP in routing table (longest-prefix-match). If no match and no default route, drop and send ICMP Destination Unreachable.

Step 6: Router re-encapsulates the packet in a new Layer 2 frame for the outgoing interface (new source MAC = router's, new destination MAC = next-hop's, found via ARP).

Step 7: Router transmits the new frame out the outgoing interface.

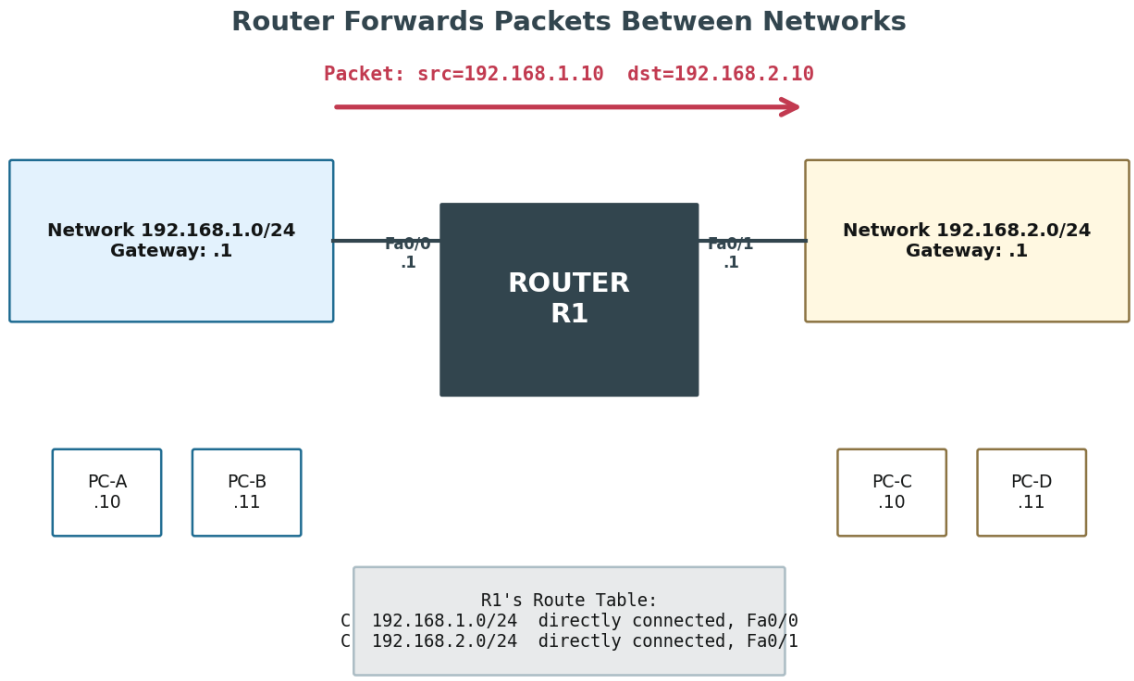


Figure: Router forwards packets between networks. Each router makes an independent decision based on its own routing table.

Firewalls

A firewall filters traffic between security zones (inside, outside, DMZ) based on a ruleset. Modern firewalls (next-gen firewalls, NGFW) are much more than packet filters — they do stateful inspection, application identification, intrusion prevention (IPS), URL filtering, malware analysis, and user identity awareness. Cisco's flagship firewall line is **Cisco Secure Firewall** (formerly Firepower), running either the ASA software (legacy) or Firepower Threat Defense (FTD, modern).

Access Points and Wireless LAN Controllers

Cisco wireless has two main architectures. **Autonomous APs** are standalone — each AP is independently configured and managed. Suitable for small deployments. **Lightweight APs (LAPs)** are managed centrally by a **Wireless LAN Controller (WLC)** using the CAPWAP protocol. The WLC handles configuration, RF management, security, and roaming — the APs are essentially 'dumb' radios. This is the dominant enterprise architecture today.

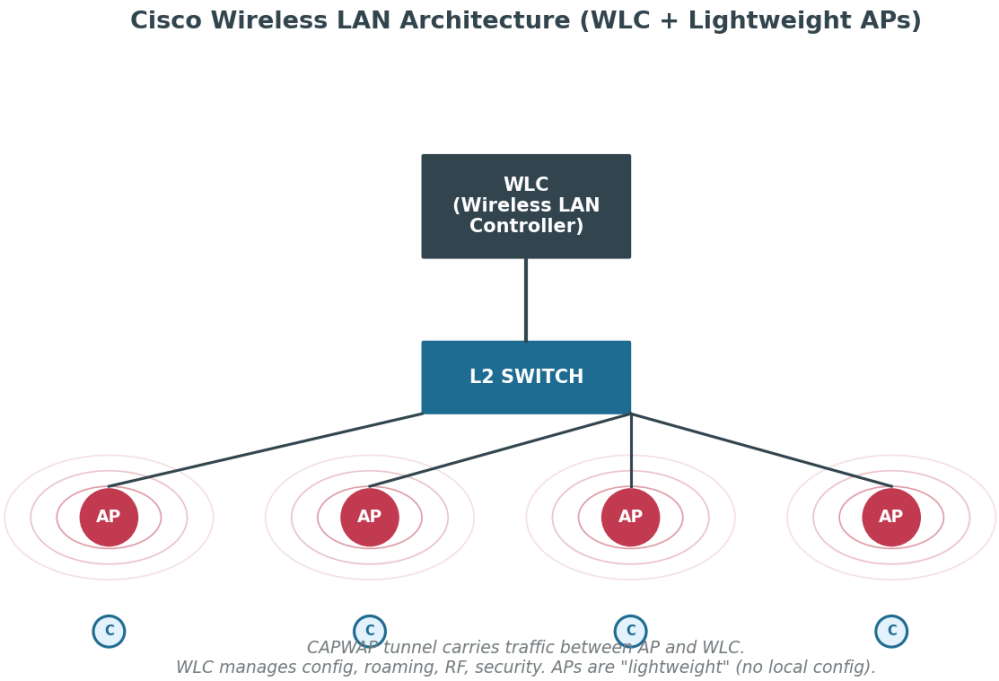


Figure: Cisco wireless architecture: WLC manages multiple lightweight APs via CAPWAP. APs handle RF, WLC handles policy.

Chapter 8 Quiz

Q1. At which OSI layer does a router operate?

A. Layer 2

B. Layer 3

C. Layer 4

D. Layer 7

Answer: B — Routers make forwarding decisions based on IP addresses, which is Layer 3.

Q2. How does a switch learn MAC addresses?

A. By examining destination MAC of incoming frames

B. By examining source MAC of incoming frames

C. By querying the router

D. Manually configured only

Answer: B — Switch learns by examining the SOURCE MAC of each incoming frame and associating it with the ingress port.

Q3. What does the router do when TTL reaches 0?

A. Forwards the packet anyway

B. Drops the packet and sends ICMP Time Exceeded

C. Sends the packet back to source

D. Increases TTL and forwards

Answer: B — TTL=0 → packet dropped, ICMP Time Exceeded sent back to source.

Q4. Which protocol does Cisco use for AP-WLC communication?

A. LWAPP only

B. CAPWAP

C. Telnet

D. HTTP

Answer: B — CAPWAP (Control and Provisioning of Wireless Access Points) is the modern standard.

Q5. Which Cisco product line is designed for data centers with spine-leaf topology?

A. Catalyst 2960

B. Catalyst 9300

C. Nexus 9000

D. ISR 4000

Answer: C — Cisco Nexus 9000 series supports ACI, VXLAN, and spine-leaf data center designs.

PART 9

Cisco IOS

The command-line interface that runs Cisco devices

Chapter 9: Mastering Cisco IOS

Cisco IOS (Internetwork Operating System) is the operating system that runs on most Cisco routers and switches. It's a command-line interface (CLI) — no mouse, no windows, just typed commands. While modern Cisco devices also support web GUIs, REST APIs, and SDN controllers, the CLI remains the most powerful, flexible, and exam-relevant way to configure Cisco devices. Every CCNA candidate must become fluent in IOS.

IOS Command Modes

IOS has a hierarchy of command modes. Each mode gives access to a different set of commands. The prompt changes to indicate which mode you're in.

Mode	Prompt	Purpose	How to Enter
User EXEC	Router>	View-only commands (ping, show)	Default on login
Privileged EXEC	Router#	All show commands, save, reload, debug	Type "enable"
Global Configuration	Router(config)#	Configure device-wide settings	Type "configure terminal"
Interface Configuration	Router(config-if)#	Configure a specific interface	Type "interface Gi0/0"
Line Configuration	Router(config-line)#	Configure console/VTY/AUX	Type "line vty 0 4"
Subinterface	Router(config-subif)#	Configure a subinterface (router-on-stick)	Type "interface Gi0/0.10"
Router Config	Router(config-router)#	Configure routing protocol	Type "router ospf 1"
VLAN Config	Switch(config-vlan)#	Configure a VLAN	Type "vlan 10"

To move back one level, type exit. To jump all the way back to Privileged EXEC from any config mode, press **Ctrl+Z** (or type end). The keyboard shortcut **Ctrl+Shift+6** (often written as ^^) aborts the current command — useful for stopping a ping, traceroute, or DNS lookup that's hanging.

Common CLI Shortcuts

Shortcut	Action
Tab	Auto-complete a partial command
?	List available commands (context-sensitive help)
Ctrl+A	Move cursor to beginning of line
Ctrl+E	Move cursor to end of line
Ctrl+U	Delete to beginning of line
Ctrl+K	Delete to end of line
Ctrl+W	Delete previous word
Ctrl+Z	Exit config mode to Privileged EXEC
Ctrl+Shift+6	Abort current command (ping, traceroute, DNS lookup)
Up arrow	Previous command in history
Down arrow	Next command in history
show history	Show recent commands

Basic Device Configuration

Every new Cisco device needs the same initial configuration: hostname, banner, passwords, IP on management interface, default gateway, and SSH access. Here's a complete first-time setup script:

```
! === Basic Device Setup ===
enable
configure terminal

! Hostname
hostname SW-Floor1

! Domain name (required for SSH keys)
ip domain-name example.com

! Encrypted passwords
enable secret MyEnableSecret123!
username admin privilege 15 secret AdminPass123!

! Console password
line console 0
  password ConsolePass123!
  login local
  logging synchronous
exit

! VTY (SSH) access
line vty 0 15
  transport input ssh
  login local
  logging synchronous
exit

! Generate RSA keys (enables SSH)
crypto key generate rsa modulus 2048

! Enable SSH v2 only
ip ssh version 2

! Management IP (on a L2 switch)
interface vlan 1
  ip address 192.168.1.100 255.255.255.0
  no shutdown
exit
ip default-gateway 192.168.1.1

! Banner (legal warning)
banner motd ^C
========================================
AUTHORIZED ACCESS ONLY. All activity is
monitored and logged. Unauthorized access
will be prosecuted to the full extent
of the law.
========================================
^C

! Save config
end
write memory
```

Interface Configuration Examples

Interface configuration is one of the most common IOS tasks. Here are real examples for switches and routers:

```
! === Switch Access Port (PC to switch) ===
interface FastEthernet0/1
  description Sales-PC-01
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
  spanning-tree bpduguard enable
  no shutdown

! === Switch Trunk Port (switch-to-switch) ===
interface GigabitEthernet0/1
  description Trunk-to-SW-Floor2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30
  switchport trunk native vlan 99
  no shutdown

! === Router Interface (LAN gateway) ===
interface GigabitEthernet0/0
  description LAN-Gateway
  ip address 192.168.1.1 255.255.255.0
  no shutdown

! === Router Subinterface (router-on-a-stick) ===
interface GigabitEthernet0/0.10
  encapsulation dot1q 10
  ip address 192.168.10.1 255.255.255.0
! Repeat for each VLAN

! === L3 Switch Routed Port ===
interface GigabitEthernet1/0/1
  no switchport
  ip address 10.0.0.1 255.255.255.252
  no shutdown

! === Loopback (router ID, management) ===
interface Loopback0
  ip address 1.1.1.1 255.255.255.255
```

Verification Commands — The Show Family

The show commands are your primary tool for inspecting device state. Here are the most-used ones, organized by purpose:

```
! === Basic Verification ===
show version                  # IOS version, uptime, hardware, config register
show running-config           # Current config in RAM
show startup-config           # Saved config in NVRAM
show clock                    # Current time

! === Interface Verification ===
show ip interface brief       # All interfaces, IP, status (concise)
show interfaces               # Detailed stats + errors
show interfaces description   # Port descriptions + status
show interfaces status        # Status, speed, duplex (per port)
show interfaces counters      # In/out bytes, packets, errors

! === Layer 2 Verification ===
```

```
show mac address-table        # CAM table (MAC → port)
show vlan brief               # VLANs, names, ports
show interfaces trunk         # Trunk ports + allowed VLANs
show spanning-tree            # STP state per VLAN
show cdp neighbors            # Directly connected Cisco devices
show lldp neighbors           # LLDP neighbors (vendor-neutral)

! === Layer 3 Verification ===
show ip route                 # IP routing table
show ip arp                   # ARP cache (IP → MAC)
show ip protocols             # Running routing protocols
show ip ospf neighbor         # OSPF adjacencies
show ip bgp summary           # BGP peers

! === Security ===
show access-lists             # ACLs + match counters
show port-security            # Port security state
show port-security interface  # Per-port port security
```

Device Management — Saving, Booting, Backing Up

Cisco devices store configuration in two places: **RAM** (running-config, lost on reboot) and **NVRAM** (startup-config, persistent). To make changes permanent, you must save running-config to startup-config. IOS images are stored in **Flash** memory (persistent, like a hard drive).

```
! === Save Configuration ===
copy running-config startup-config   # The classic way
write memory                          # Shortcut (same as above)
write                                 # Even shorter shortcut

! === Backup to TFTP server ===
copy running-config tftp://192.168.1.10/sw-config.txt
copy startup-config tftp://192.168.1.10/sw-startup.txt
copy flash:c2960-universalk9.bin tftp://192.168.1.10/

! === Restore from TFTP ===
copy tftp://192.168.1.10/sw-config.txt running-config
copy tftp://192.168.1.10/sw-startup.txt startup-config

! === Boot configuration ===
boot system flash:c2960-universalk9.bin  # Specify which IOS to boot

! === Configuration register ===
config-register 0x2102   # Normal boot (load IOS, load startup-config)
config-register 0x2142   # Password recovery (skip startup-config)
config-register 0x2100   # ROMmon mode (no IOS boot)

! === Reload ===
reload                   # Reboot immediately
reload in 15             # Reboot in 15 minutes
reload at 22:00          # Reboot at 10pm

! === Erase and start fresh ===
erase startup-config     # Delete saved config
delete flash:vlan.dat    # Delete VLAN database (older switches)
reload                   # Reboot with empty config
```

Common Mistake

Forgetting to save! After making changes, always run "write memory" or "copy run start". If you reload without saving, all your changes are lost. Set up automatic config archive to prevent this: "archive" + "write-memory" + "time-period 60".

Chapter 9 Quiz

Q1. Which command moves you from User EXEC to Privileged EXEC?

A. enable
B. configure terminal
C. privilege
D. admin
Answer: A — Typing 'enable' moves you from Router> (User EXEC) to Router# (Privileged EXEC).

Q2. Where is the running-config stored?

A. NVRAM
B. RAM
C. Flash
D. TFTP server
Answer: B — Running-config lives in RAM — lost on reboot. Startup-config lives in NVRAM.

Q3. Which command saves the running configuration?

A. copy run start
B. save config
C. write nvram
D. commit
Answer: A — copy running-config startup-config (or 'write memory') saves the config.

Q4. What does Ctrl+Shift+6 do?

A. Auto-complete a command
B. Abort the current command
C. Save the configuration
D. Move to beginning of line
Answer: B — Ctrl+Shift+6 aborts the current command (ping, traceroute, DNS lookup).

Q5. Which configuration register value is used for password recovery?

A. 0x2102
B. 0x2142
C. 0x2100
D. 0x102
Answer: B — 0x2142 = bypass startup-config (used in password recovery).

PART 10

Switching

VLANs, trunking, STP, EtherChannel — the heart of LANs

Chapter 10: Advanced Switching

Switches are the workhorses of every LAN. But a modern enterprise switch does far more than forward frames based on MAC addresses. It segments the network into virtual LANs (VLANs), carries multiple VLANs across trunks, prevents Layer 2 loops with Spanning Tree Protocol, bundles links for redundancy with EtherChannel, and routes between VLANs with Layer 3 switching. This chapter covers all of these essential features.

VLANs — Virtual LANs

A **VLAN** is a logical grouping of devices in the same broadcast domain, regardless of their physical location. Without VLANs, every device on a switch is in one big broadcast domain — every ARP, DHCP discover, and broadcast frame reaches every device. With VLANs, you can put Sales on VLAN 10, IT on VLAN 20, HR on VLAN 30, and Guest on VLAN 99 — each isolated from the others. Devices in different VLANs can't communicate without a router (or L3 switch) between them.

VLANs provide three major benefits. **Security**: segmentation isolates sensitive traffic (HR payroll) from less trusted networks (guest Wi-Fi). **Performance**: smaller broadcast domains mean less broadcast traffic per device. **Manageability**: VLANs map to organizational structure (Sales VLAN, IT VLAN), making it easy to apply policies per department.

```
! === Create VLANs ===
vlan 10
  name SALES
vlan 20
  name IT
vlan 30
  name HR
vlan 99
  name MGMT
exit

! === Assign access ports to VLANs ===
interface range Fa0/1 - 10
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
  spanning-tree bpduguard enable

interface range Fa0/11 - 20
  switchport mode access
  switchport access vlan 20
  spanning-tree portfast
  spanning-tree bpduguard enable

! === Voice VLAN (for Cisco IP phones) ===
interface Fa0/5
  switchport mode access
  switchport access vlan 10
  switchport voice vlan 110
  spanning-tree portfast

! === Verify ===
show vlan brief
show vlan id 10
show interfaces switchport
```

Trunking and 802.1Q

When you need to carry multiple VLANs between two switches (or switch to router, or switch to virtualization host), you use a **trunk**. A trunk adds a 4-byte **802.1Q tag** to each frame, identifying which VLAN it belongs to. The receiving switch reads the tag, strips it, and forwards the frame to the appropriate VLAN. The **native VLAN** is special — frames in this VLAN are sent untagged (no 802.1Q header). Default native VLAN is 1, but best practice is to change it to an unused VLAN for security.

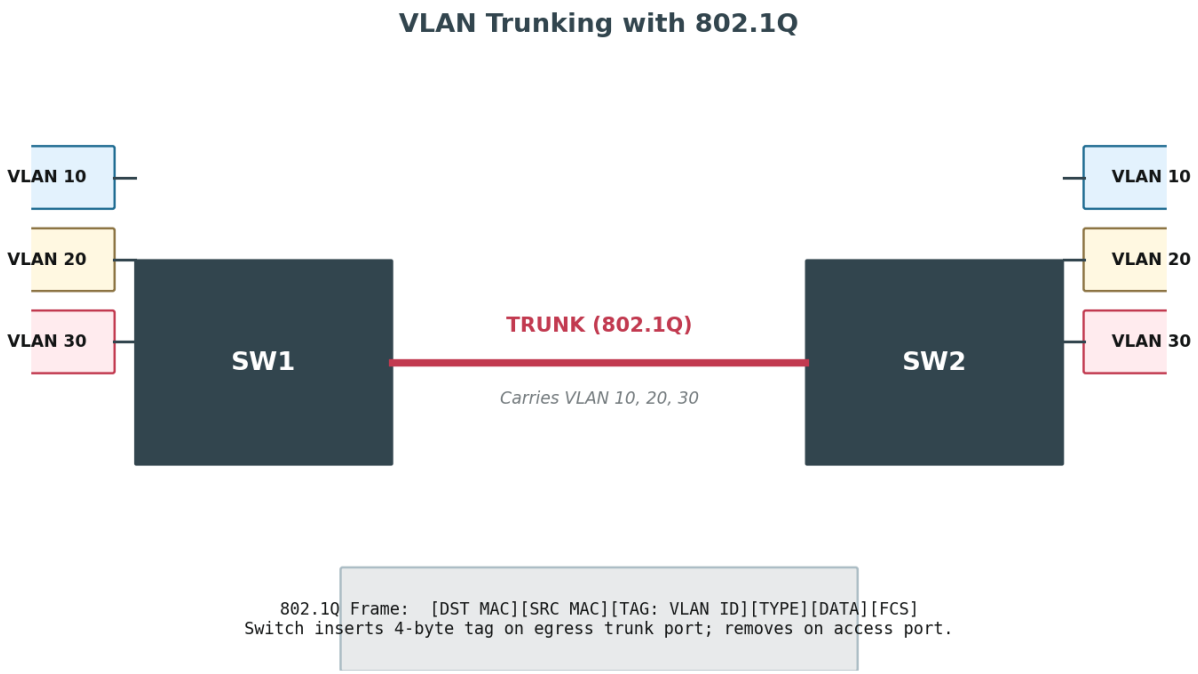


Figure: VLAN trunking with 802.1Q. A single physical link carries multiple VLANs, each tagged. The native VLAN travels untagged.

```
! === Configure trunk port ===
interface GigabitEthernet0/1
  description Trunk-to-Core
  switchport trunk encapsulation dot1q   # Some switches need this
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,99
  switchport trunk native vlan 99
  switchport nonegotiate                 # Disable DTP (security best practice)

! === Verify trunk ===
show interfaces trunk
show interfaces Gi0/1 switchport
show interfaces Gi0/1 trunk
```

Inter-VLAN Routing

VLANs isolate broadcast domains, but devices in different VLANs can't communicate without a Layer 3 device. There are two ways to do inter-VLAN routing: **router-on-a-stick** (an external router with subinterfaces) and **L3 switch with SVIs** (Switch Virtual Interfaces). The L3 switch approach is far more common in modern networks because it's faster (hardware routing) and doesn't require an external router.

```
! === Router-on-a-stick ===
! On the router:
```

```
interface GigabitEthernet0/0
  no shutdown
interface GigabitEthernet0/0.10
  encapsulation dot1q 10
  ip address 192.168.10.1 255.255.255.0
interface GigabitEthernet0/0.20
  encapsulation dot1q 20
  ip address 192.168.20.1 255.255.255.0

! === L3 Switch with SVIs (preferred) ===
! On the switch:
ip routing                              # Enable L3 routing
interface vlan 10
  ip address 192.168.10.1 255.255.255.0
  no shutdown
interface vlan 20
  ip address 192.168.20.1 255.255.255.0
  no shutdown

! === Verify ===
show ip route
show ip interface brief
```

Spanning Tree Protocol (STP)

Redundancy is good — until it causes a loop. In a switched network with redundant links, a broadcast frame would loop forever (broadcast storm), bringing down the network in seconds. **STP** (IEEE 802.1D) prevents this by electing one root bridge and then calculating the best path to root from every other switch. Redundant links are placed in **blocking** state — they don't forward traffic, but they're ready to take over if the primary link fails.

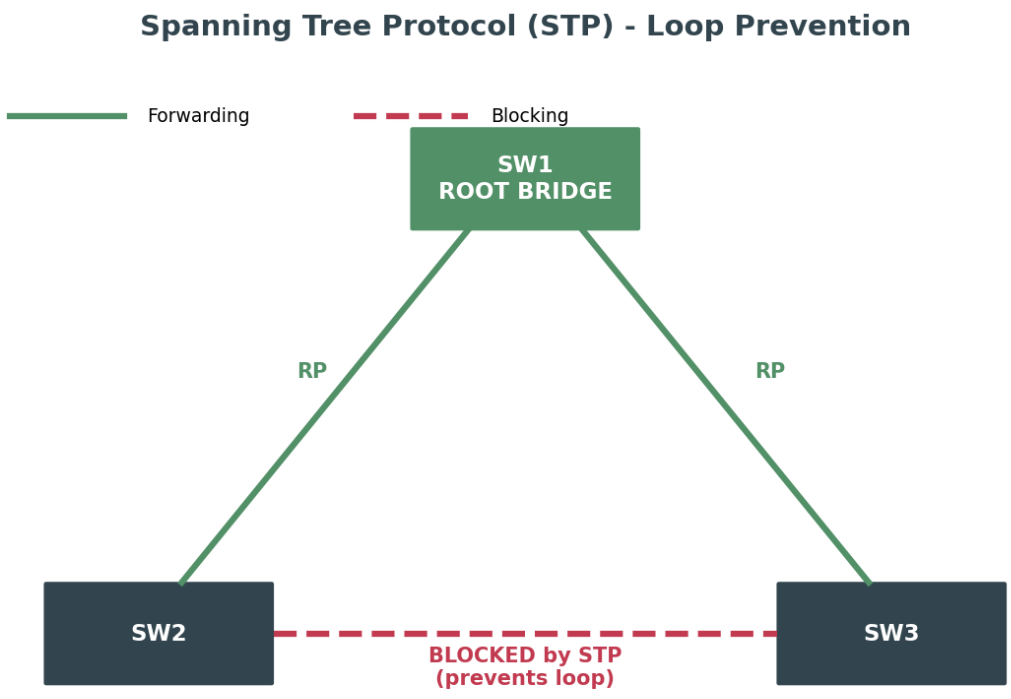


Figure: STP prevents loops by blocking redundant links. The root bridge is elected, and each switch finds its best path to root.

STP elects a **root bridge** (the switch with the lowest bridge ID = priority + MAC). All other switches calculate the lowest-cost path to the root. The port on that path becomes the **root port** (forwarding). On each network segment, one switch is elected **designated bridge** — its port on that segment is the **designated port** (forwarding). All other ports are **alternate ports** (blocking). The original 802.1D converges slowly — up to 50 seconds. **RSTP** (Rapid STP, 802.1w) converges in under a second by using proposal/agreement handshake.

State	802.1D Time	Receives BPDUs	Learns MACs	Forwards Data
Blocking	20s max age	Yes	No	No
Listening	15s forward delay	Yes	No	No
Learning	15s forward delay	Yes	Yes	No
Forwarding	Stable	Yes	Yes	Yes

```
! === STP Configuration ===
! Set root bridge (primary)
spanning-tree vlan 1 root primary
spanning-tree vlan 1 priority 24576      # Or set explicitly

! Set backup root (secondary)
spanning-tree vlan 1 root secondary

! Enable Rapid PVST+ (recommended)
spanning-tree mode rapid-pvst

! PortFast on access ports (skip listening/learning)
interface range Fa0/1 - 24
```

```
  spanning-tree portfast
  spanning-tree bpduguard enable          # Err-disable if BPDU received

! BPDU Guard globally (all PortFast ports)
spanning-tree portfast bpduguard default

! BPDU Filter (don't send/receive BPDUs - use cautiously!)
! spanning-tree portfast bpdufilter default

! Root Guard (prevent external switches from becoming root)
interface Gi0/1
  spanning-tree guard root

! Loop Guard (detect unidirectional links)
interface Gi0/1
  spanning-tree guard loop

! Verify
show spanning-tree
show spanning-tree summary
show spanning-tree vlan 1
show spanning-tree interface Gi0/1 detail
```

EtherChannel — Link Aggregation

When you need more bandwidth or redundancy between two switches, you could add another cable — but STP would block it. **EtherChannel** (also called LAG or port channel) solves this by bundling multiple physical links into one logical link. STP sees the bundle as one interface, so it doesn't block. If one link fails, traffic redistributes across the remaining ones — typically within milliseconds. EtherChannel supports 2-8 active links, with optional 8 standby (LACP only).

```
! === LACP EtherChannel (recommended) ===
interface range Gi0/1 - 4
  channel-group 1 mode active            # active = initiate LACP
  channel-protocol lacp

! Configure the port-channel interface (the logical bundle)
interface port-channel 1
  description LAG-to-Core-Switch
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30

! === PAgP EtherChannel (Cisco proprietary) ===
interface range Gi0/1 - 4
  channel-group 1 mode desirable          # desirable = initiate PAgP

! === Static (no negotiation) ===
interface range Gi0/1 - 4
  channel-group 1 mode on                 # No negotiation, both ends must be 'on'

! Load balancing method (default: src-dst-ip)
port-channel load-balance src-dst-ip

! Verify
show etherchannel summary
show etherchannel 1 port-channel
show lacp neighbor
```

Packet Tracer Lab: VLANs, Trunking, and Inter-VLAN Routing

Lab Topology

Devices: 2 switches (SW1, SW2), 1 router (R1), 4 PCs

Connections:

- SW1 Gi0/1 ↔ SW2 Gi0/1 (TRUNK, carries VLANs 10, 20)
- SW1 Gi0/2 ↔ R1 Gi0/0 (TRUNK to router-on-a-stick)
- PC1 (VLAN 10) on SW1 Fa0/1, PC2 (VLAN 20) on SW1 Fa0/2
- PC3 (VLAN 10) on SW2 Fa0/1, PC4 (VLAN 20) on SW2 Fa0/2

IP Plan:

- VLAN 10: 192.168.10.0/24, gateway .1
- VLAN 20: 192.168.20.0/24, gateway .1

Test: PC1 (VLAN 10) should ping PC4 (VLAN 20) after inter-VLAN routing is configured.

```
# === SW1 Configuration ===
vlan 10; name SALES
vlan 20; name IT
exit

interface range Fa0/1
  switchport mode access; switchport access vlan 10; spanning-tree portfast
interface Fa0/2
  switchport mode access; switchport access vlan 20; spanning-tree portfast

interface Gi0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  switchport nonegotiate

interface Gi0/2
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  switchport nonegotiate

# === SW2 Configuration (similar) ===
vlan 10; name SALES
vlan 20; name IT
interface Fa0/1; switchport mode access; switchport access vlan 10; spanning-tree
portfast
interface Fa0/2; switchport mode access; switchport access vlan 20; spanning-tree
portfast
interface Gi0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  switchport nonegotiate

# === R1 Configuration (router-on-a-stick) ===
interface Gi0/0
  no shutdown
interface Gi0/0.10
  encapsulation dot1q 10
  ip address 192.168.10.1 255.255.255.0
interface Gi0/0.20
  encapsulation dot1q 20
  ip address 192.168.20.1 255.255.255.0
```

```
# === PC Configuration ===
# PC1: 192.168.10.10/24, gateway 192.168.10.1
# PC2: 192.168.20.10/24, gateway 192.168.20.1
# PC3: 192.168.10.11/24, gateway 192.168.10.1
# PC4: 192.168.20.11/24, gateway 192.168.20.1

# === Verification ===
# From PC1: ping 192.168.10.11 (same VLAN, should work)
# From PC1: ping 192.168.20.11 (different VLAN, requires routing)
# On SW1: show vlan brief, show interfaces trunk
# On R1: show ip route, show ip interface brief
```

Chapter 10 Quiz

Q1. What does VLAN stand for?

A. Virtual LAN — logical grouping of devices in same broadcast domain

B. Variable Length Address Number

C. Vector Logical Area Network

D. VLAN Logical Area Network

Answer: A — VLAN = Virtual LAN. Logically groups devices into broadcast domains regardless of physical location.

Q2. What is the default native VLAN on Cisco switches?

A. VLAN 0

B. VLAN 1

C. VLAN 1002

D. VLAN 4094

Answer: B — Default native VLAN = 1. Best practice: change to unused VLAN for security.

Q3. What protocol prevents Layer 2 loops?

A. ARP

B. STP

C. DHCP

D. OSPF

Answer: B — STP (Spanning Tree Protocol) prevents loops by blocking redundant paths.

Q4. What is the default STP bridge priority?

A. 0

B. 1

C. 32768

D. 65535

Answer: C — Default priority = 32768 (with VLAN ID added for PVST+).

Q5. Which protocol is the open standard for EtherChannel?

A. PAgP

B. LACP

C. DTP

D. CDP

Answer: B — LACP (Link Aggregation Control Protocol) is the IEEE 802.3ad standard. PAgP is Cisco proprietary.

PART 11

Routing

Static, dynamic, and the protocols that move packets

Chapter 11: IP Routing — Static and Dynamic

Routing is the process of forwarding packets from one network to another. Every router maintains a **routing table** — a list of destination networks and the best next-hop to reach each. When a packet arrives, the router looks up the destination IP, finds the matching route (using longest-prefix-match), and forwards the packet to the next-hop. Routes can be configured manually (**static routing**) or learned automatically from other routers via **dynamic routing protocols** (OSPF, EIGRP, RIP, BGP).

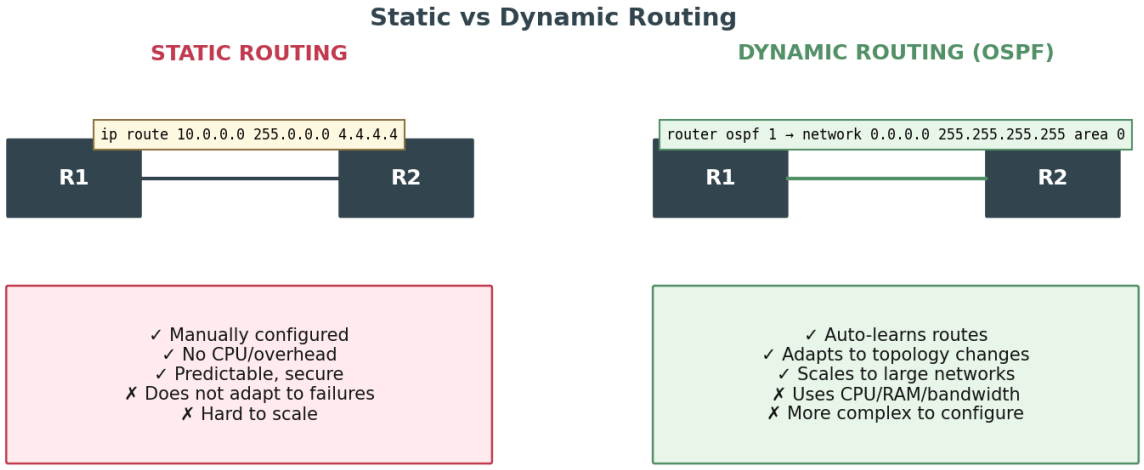


Figure: Static routing: manually configured, no overhead, no adaptation. Dynamic routing: auto-learns, adapts to failures, uses CPU/bandwidth.

Static Routing

Static routes are manually configured by an administrator. They're simple, predictable, use no CPU or bandwidth, and are secure (no protocol to attack). The downside: they don't adapt to topology changes. If a link fails, the static route keeps pointing at it until you manually fix it. Static routes are ideal for small networks, stub networks, default routes, and as backups (floating static).

```
! === Basic static route ===
ip route 192.168.2.0 255.255.255.0 10.0.0.2

! === Default route (IPv4) ===
ip route 0.0.0.0 0.0.0.0 203.0.113.1

! === Default route (IPv6) ===
ipv6 route ::/0 2001:db8::1

! === Floating static (backup, only used when primary fails) ===
! 250 = AD higher than OSPF (110), so this is backup
ip route 192.168.5.0 255.255.255.0 10.0.1.2 250

! === Specify exit interface instead of next-hop ===
! (Works for point-to-point; problematic on multi-access - causes excessive ARP)
ip route 192.168.5.0 255.255.255.0 Serial0/0/0

! === Verify ===
```

```
show ip route
show ip route static
show ip route 192.168.2.0
```

Administrative Distance

When a router learns the same route from multiple sources (e.g., static and OSPF), it needs a tiebreaker. **Administrative Distance** (AD) is Cisco's trustworthiness ranking — lower = more trusted. The route with the lowest AD wins and is installed in the routing table.

Source	Default AD	Notes
Directly connected	0	Most trusted (interface is up)
Static route	1	Manually configured
EIGRP summary route	5	EIGRP summarized routes
eBGP	20	External BGP
EIGRP (internal)	90	Cisco proprietary (now RFC 7868)
IGRP	100	Legacy (removed in IOS 12.2)
OSPF	110	Open standard link-state
IS-IS	115	Common in service provider backbones
RIP	120	Old distance-vector
EIGRP (external)	170	Redistributed from another protocol
iBGP	200	Internal BGP (less trusted than IGP)
Unreachable	255	Route not trusted at all

Exam Tip

Memorize the AD table. Common exam question: "Router learns 192.168.1.0/24 from static (AD 1), OSPF (AD 110), and EIGRP (AD 90). Which wins?" Answer: static (lowest AD). To override, use floating static with higher AD.

Longest Prefix Match

When multiple routes match the destination IP, the router chooses the one with the **longest prefix** (most specific). For example, if the routing table contains both 192.168.0.0/16 and 192.168.1.0/24, and a packet arrives for 192.168.1.50, the /24 route wins because it's more specific (24 > 16). This rule is fundamental to how the Internet works — it allows default routes (0.0.0.0/0) to coexist with specific routes.

Dynamic Routing Protocols Overview

Dynamic routing protocols let routers automatically learn about networks from each other. When a link fails, the protocol re-converges and finds an alternate path. There are three categories: **interior gateway protocols** (IGPs) used inside an organization (OSPF, EIGRP, RIP, IS-IS), **exterior gateway protocols** (EGPs) used between organizations (BGP — the protocol of the Internet), and **routing information protocols** (less common).

Protocol	Type	Metric	AD	Convergence	Best For
RIPv2	Distance-vector	Hop count	120	Slow (30s updates)	Legacy, small networks
OSPF	Link-state	Cost (bandwidth)	110	Fast (<1s)	Open standard, large enterprise
EIGRP	Advanced DV	Composite (BW+delay)	90	Very fast (<1s)	Cisco shops, fast convergence
IS-IS	Link-state	Cost	115	Fast	Service provider backbones

Protocol	Type	Metric	AD	Convergence	Best For
BGP	Path-vector	Path attributes	20 (eBGP)/200 (iBGP)	Slow (minutes)	Internet routing between ASes

OSPF — Open Shortest Path First

OSPF is the most popular open-standard IGP. It's a link-state protocol — each router floods information about its own links (Link State Advertisements, LSAs) to all routers in the area. Every router builds a complete map of the topology (the Link State Database, LSDB) and runs Dijkstra's Shortest Path First (SPF) algorithm to compute the best path to every destination. OSPF scales to very large networks by using a hierarchical area design with area 0 (backbone) at the center.

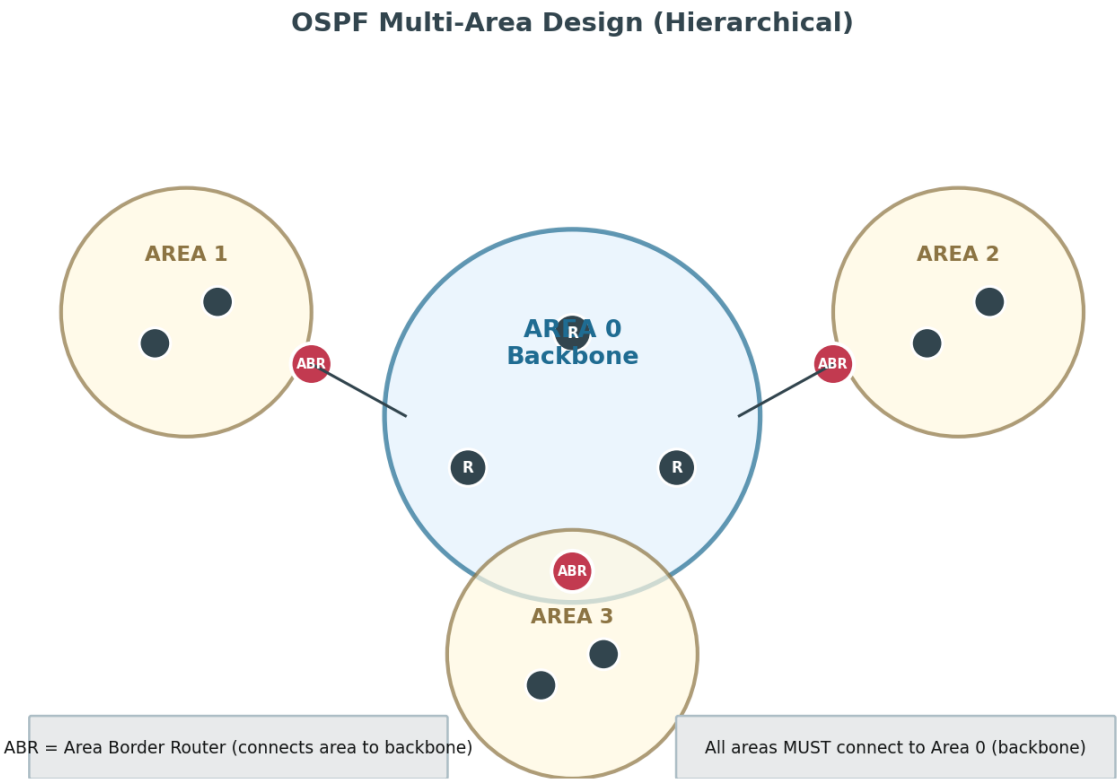


Figure: OSPF multi-area design. All areas must connect to area 0 (backbone). ABRs (Area Border Routers) connect areas.

```
! === OSPF Configuration (classic syntax) ===
router ospf 1
  router-id 1.1.1.1                        # Manually set RID
  network 192.168.1.0 0.0.0.255 area 0     # Advertise this subnet in area 0
  network 192.168.2.0 0.0.0.255 area 0
  passive-interface default                # Don't send hellos on any interface
  no passive-interface Gi0/0               # Except this one (WAN link)
  auto-cost reference-bandwidth 10000      # Update ref-bw for 10G+ links (Mbps)
  default-information originate            # Inject default route into OSPF

! === OSPF Configuration (modern, interface-based) ===
interface Gi0/0
  ip ospf 1 area 0
  ip ospf cost 10                          # Manually set cost
  ip ospf priority 0                       # Never be DR/BDR
```

```
  ip ospf hello-interval 5              # 5s hello (default 10)
  ip ospf dead-interval 20              # 20s dead (default 40, 4x hello)

! === Verify ===
show ip ospf neighbor                   # See neighbors + states
show ip ospf interface brief            # OSPF config per interface
show ip ospf database                   # Link State Database
show ip route ospf                      # OSPF routes in RIB
show ip ospf border-routers             # ABRs and ASBRs
```

EIGRP — Enhanced Interior Gateway Routing Protocol

EIGRP is Cisco's advanced distance-vector protocol (now RFC 7868, so other vendors can implement it). It uses the DUAL (Diffusing Update Algorithm) to guarantee loop-free paths and achieve very fast convergence. EIGRP's metric is a composite of bandwidth and delay by default, but can also include load, reliability, and MTU. EIGRP keeps a topology table of all learned routes and their metrics, plus a **feasible successor** (backup route) for fast failover.

```
! === EIGRP Configuration (classic) ===
router eigrp 100                        # AS number (must match on all routers)
  network 192.168.1.0 0.0.0.255
  network 192.168.2.0 0.0.0.255
  no auto-summary                       # Required for VLSM support
  passive-interface default
  no passive-interface Gi0/0

! === EIGRP Configuration (named mode, modern) ===
router eigrp ENTERPRISE
  address-family ipv4 unicast autonomous-system 100
    af-interface default
      passive-interface
    exit-af-interface
    af-interface Gi0/0
      no passive-interface
    exit-af-interface
    network 192.168.1.0 0.0.0.255
    network 192.168.2.0 0.0.0.255

! === Verify ===
show ip eigrp neighbors
show ip eigrp topology                  # All routes incl. feasible successors
show ip route eigrp
show ip protocols
```

RIP — Routing Information Protocol

RIP is the oldest routing protocol still in use. It's a distance-vector protocol that uses hop count as its metric, with a maximum of 15 hops (16 = unreachable). RIPv2 added VLSM support, multicast updates, and MD5 authentication. RIPng supports IPv6. RIP is rarely used in production anymore but still appears on the CCNA exam because it teaches fundamental distance-vector concepts like split horizon, route poisoning, and hold-down timers.

```
! === RIPv2 Configuration ===
router rip
  version 2
  no auto-summary
```

```
  network 192.168.1.0
  network 192.168.2.0
  passive-interface default
  no passive-interface Gi0/0

! === Verify ===
show ip route rip
show ip protocols
show ip rip database
```

BGP — Border Gateway Protocol (Introduction)

BGP is the routing protocol that holds the Internet together. It's a path-vector protocol used between autonomous systems (ASes) — the organizational units that make up the Internet. Each AS has a unique number (ASN). BGP doesn't use a simple metric like OSPF; instead, it considers many attributes (AS-path length, local preference, MED, origin, weight) when selecting the best path. BGP uses TCP port 179 for reliable transport. For the CCNA, you need to understand the basics — full BGP mastery is a CCNP topic.

```
! === Basic BGP Configuration ===
router bgp 65001                          # Local AS number
  bgp router-id 1.1.1.1
  neighbor 203.0.113.2 remote-as 65002    # eBGP peer
  neighbor 192.168.1.2 remote-as 65001    # iBGP peer
  network 192.168.10.0 mask 255.255.255.0  # Advertise this prefix
  network 192.168.20.0 mask 255.255.255.0

! === Verify ===
show ip bgp summary                       # Peer state + prefix counts
show ip bgp                               # Full BGP table
show ip bgp neighbors                     # Detailed peer info
show ip route bgp                         # BGP routes in RIB
```

First Hop Redundancy Protocols (FHRP)

When hosts use a default gateway, that gateway becomes a single point of failure. **FHRPs** solve this by providing a virtual IP that floats between two or more routers. The host uses the virtual IP as its gateway; if the active router fails, the standby takes over within seconds. Three FHRP protocols exist: **HSRP** (Cisco proprietary), **VRRP** (open standard, RFC 5798), and **GLBP** (Cisco, with load balancing).

```
! === HSRP Configuration ===
interface GigabitEthernet0/0
  ip address 192.168.1.2 255.255.255.0
  standby 1 ip 192.168.1.1                # Virtual IP (gateway for hosts)
  standby 1 priority 110                  # Higher = preferred active (default 100)
  standby 1 preempt                       # Take over when higher priority
  standby 1 track Gi0/1 decrement 20      # If upstream link fails, drop priority

! On the second router:
interface GigabitEthernet0/0
  ip address 192.168.1.3 255.255.255.0
  standby 1 ip 192.168.1.1
  standby 1 priority 100                  # Default, becomes standby

! === Verify ===
show standby
show standby brief
```

Chapter 11 Quiz

Q1. What is the default AD of a static route?

A. 0
B. 1
C. 90
D. 120
Answer: B — Static route AD = 1. Directly connected = 0 (most trusted).

Q2. What is the default AD of OSPF?

A. 90
B. 100
C. 110
D. 120
Answer: C — OSPF AD = 110.

Q3. What is the longest prefix match rule?

A. When multiple routes match, the one with the longest prefix (most specific) wins
B. The route with the highest metric wins
C. The route with the lowest AD wins
D. The most recently learned route wins
Answer: A — Longest prefix match: a /32 beats a /24 beats a /0. Most specific always wins.

Q4. Which routing protocol uses cost based on bandwidth as its metric?

A. RIP
B. OSPF
C. EIGRP
D. BGP
Answer: B — OSPF uses cost = reference-bandwidth / interface-bandwidth.

Q5. What is the default HSRP priority?

A. 50
B. 100
C. 150
D. 200
Answer: B — Default HSRP priority = 100. Higher = preferred active router.

PART 12

Wireless Networking

Wi-Fi standards, architecture, security, and troubleshooting

Chapter 12: Wireless Networking

Wi-Fi has become so essential that we forget how recently it was a luxury. The first consumer Wi-Fi (802.11b, 11 Mbps) shipped in 1999. Today, Wi-Fi 6E delivers multi-gigabit speeds in the 6 GHz band, and Wi-Fi 7 is on the horizon. Modern enterprise wireless is more than just APs — it's a complex system of controllers, RF management, roaming protocols, security, and guest access. This chapter covers the standards, architecture, and troubleshooting you need for the CCNA and the real world.

Wi-Fi Standards

Wi-Fi standards are defined by IEEE 802.11 working groups. The Wi-Fi Alliance (a trade group) assigns friendly names like 'Wi-Fi 6' to make the standards more marketable. Each generation brings faster speeds, better efficiency, and (usually) new frequency bands.

Standard	Wi-Fi Name	Year	Bands (GHz)	Max Speed	Key Feature
802.11b	Wi-Fi 1	1999	2.4	11 Mbps	First mass-market Wi-Fi
802.11a	Wi-Fi 2	1999	5	54 Mbps	OFDM, less interference
802.11g	Wi-Fi 3	2003	2.4	54 Mbps	OFDM in 2.4 GHz
802.11n	Wi-Fi 4	2009	2.4, 5	600 Mbps	MIMO, channel bonding
802.11ac	Wi-Fi 5	2014	5	3.5 Gbps	MU-MIMO (downlink), wide channels
802.11ax	Wi-Fi 6/6E	2019/2021	2.4, 5, 6	9.6 Gbps	OFDMA, BSS coloring, TWT
802.11be	Wi-Fi 7	2024	2.4, 5, 6	46 Gbps	MLO, 320 MHz channels, 4K-QAM

Wireless Architecture: Autonomous vs Lightweight

There are two main wireless architectures. **Autonomous APs** are standalone — each AP runs its own configuration, handles its own RF, authenticates clients independently. Suitable for small deployments (1-5 APs). **Lightweight APs (LAPs)** are managed by a central **Wireless LAN Controller (WLC)** using CAPWAP. The WLC handles configuration, RF management, security, roaming, and guest access — the AP is essentially a 'dumb radio'. This is the dominant enterprise architecture because it scales to thousands of APs and centralizes management.

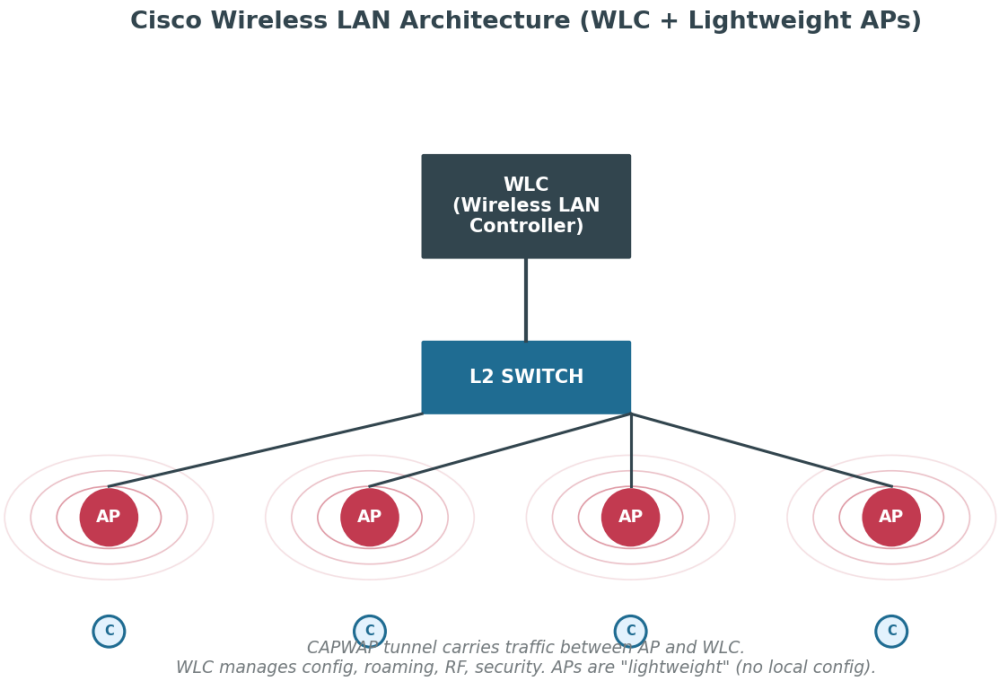


Figure: Cisco WLC + lightweight AP architecture. CAPWAP carries control and data between AP and WLC.

CAPWAP Protocol

CAPWAP (Control and Provisioning of Wireless Access Points, RFC 5415/5416) is the protocol between lightweight APs and the WLC. It uses two channels: **control** (UDP 5246, encrypted with DTLS) for management traffic, and **data** (UDP 5247, optionally DTLS-encrypted) for client traffic tunneled from AP to WLC. The data channel carries 802.11 frames encapsulated inside CAPWAP, so the WLC can apply policy (VLAN mapping, ACLs, QoS) before sending them to the wired network.

Roaming and Fast Transition

Roaming is when a wireless client moves from one AP to another without losing connection. The challenge: the new AP needs to know about the client's session (keys, QoS, VLAN) to continue the connection seamlessly. Without fast roaming, the handoff can take 100+ milliseconds — fine for data, but disruptive for voice. **802.11r** (Fast BSS Transition) reduces this to under 50ms by pre-distributing keys. **802.11k** helps clients find better APs via neighbor reports. **802.11v** lets the WLC suggest better APs to clients (BSS Transition Management).

Wireless Security

Wi-Fi security has evolved through several generations. **WEP** (1997) is broken — crackable in minutes. **WPA** (2003) was a stopgap with TKIP. **WPA2** (2004) introduced AES-CCMP and is still widely used. **WPA3** (2018) replaces the PSK with SAE (Simultaneous Authentication of Equals), preventing offline dictionary attacks. WPA3 also mandates PMF (Protected Management Frames, 802.11w) to prevent deauth attacks. For enterprises, WPA2/WPA3 Enterprise mode uses 802.1X + RADIUS for per-user authentication.

Standard	Year	Authentication	Encryption	Status
WEP	1997	Static WEP key	RC4 (broken)	Deprecated
WPA	2003	PSK or 802.1X	TKIP (RC4-based)	Deprecated
WPA2	2004	PSK or 802.1X	AES-CCMP	Acceptable (use with PMF)

Standard	Year	Authentication	Encryption	Status
WPA3	2018	SAE or 802.1X	AES-GCMP-256	Recommended
OWE	2018	None (DH key exchange)	AES-GCMP	Open networks (RFC 8110)

Wireless Troubleshooting

Wireless problems fall into four categories: **coverage** (weak signal, dead zones), **capacity** (too many clients on one AP), **interference** (other Wi-Fi, microwaves, Bluetooth), and **authentication** (wrong password, RADIUS failure, cert issues). Here's a systematic troubleshooting approach:

Step 1: Verify client sees the SSID. If not, the AP may be down, out of range, or the SSID is hidden. Use a Wi-Fi analyzer app to confirm.

Step 2: Check signal strength. RSSI should be -65 dBm or better for voice, -70 for data. If weaker, add APs, adjust power, or remove obstacles.

Step 3: Check authentication. Wrong password? Cert issue? RADIUS server down? Check WLC logs for the specific client.

Step 4: Check for interference. Use Cisco CleanAir or a spectrum analyzer. Common culprits: microwaves (2.4 GHz), Bluetooth, USB 3.0 cables (interfere with 2.4 GHz Wi-Fi).

Step 5: Check capacity. If only some clients have issues, the AP may be overloaded. Add APs, enable band steering to 5 GHz, or implement load balancing.

Chapter 12 Quiz

Q1. Which Wi-Fi standard is also known as Wi-Fi 6?

A. 802.11ac
B. 802.11ax
C. 802.11n
D. 802.11be
Answer: B — Wi-Fi 6 = 802.11ax. Wi-Fi 7 = 802.11be.

Q2. Which protocol does Cisco use for AP-WLC communication?

A. CAPWAP
B. LWAPP only
C. Telnet
D. HTTP
Answer: A — CAPWAP (Control and Provisioning of Wireless Access Points) uses UDP 5246 (control) and 5247 (data).

Q3. Which wireless security standard introduced SAE?

A. WPA2
B. WPA3
C. WEP
D. WPA
Answer: B — WPA3 replaces PSK with SAE, preventing offline dictionary attacks.

Q4. Which roaming standard reduces handoff time for voice clients?

A. 802.11r (Fast BSS Transition)

B. 802.11k

C. 802.11v

D. 802.11w

Answer: A — 802.11r = Fast BSS Transition. Reduces roaming time from 100s of ms to <50ms.

Q5. What does RSSI measure?

A. Received Signal Strength Indicator

B. Radio Signal Strength Index

C. Relative Signal Strength Indicator

D. Receiver Signal Source Index

Answer: A — RSSI measures signal strength in dBm. Closer to 0 = stronger (-30 dBm excellent, -80 dBm weak).

PART 13

Network Services

DHCP, DNS, NTP, syslog, SNMP, and friends

Chapter 13: Network Services

Network services are the protocols that make a network usable. Without DHCP, every device would need manual IP configuration. Without DNS, you'd be typing IP addresses instead of domain names. Without NTP, log timestamps would be useless and certificates would fail. Without syslog and SNMP, you'd be flying blind. This chapter covers each major service, how it works, and how to configure it on Cisco devices.

DHCP — Dynamic Host Configuration Protocol

DHCP automates IP configuration. Without it, every device on a network would need its IP, mask, gateway, and DNS manually configured — completely unscalable. DHCP uses a 4-message process called **DORA**: Discover, Offer, Request, Acknowledge.

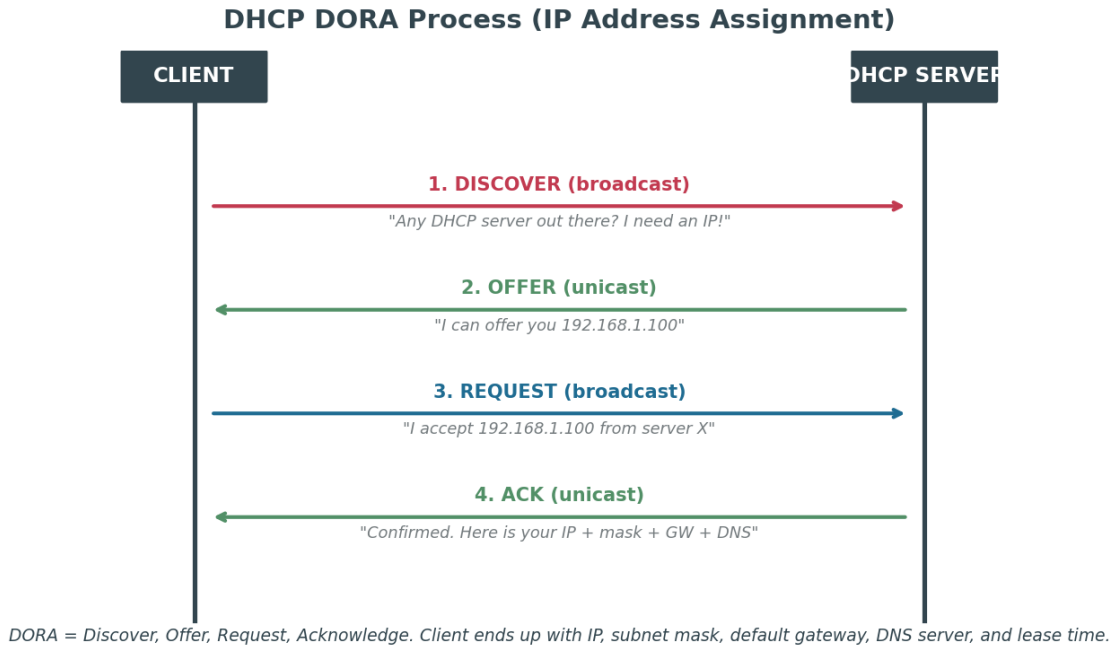


Figure: DHCP DORA process: client broadcasts DISCOVER, server sends OFFER, client broadcasts REQUEST (accepting one offer), server sends ACK with full config.

```
! === Cisco router as DHCP server ===
ip dhcp excluded-address 192.168.1.1 192.168.1.10   # Reserve gateway + servers
ip dhcp excluded-address 192.168.1.250 192.168.1.254

ip dhcp pool VLAN10-Pool
  network 192.168.1.0 255.255.255.0
  default-router 192.168.1.1
  dns-server 8.8.8.8 1.1.1.1
  domain-name example.com
  lease 1           # 1 day lease (default)
exit

! === DHCP Relay (when DHCP server is on different subnet) ===
interface GigabitEthernet0/0
  ip address 192.168.1.1 255.255.255.0
  ip helper-address 10.0.0.5      # Forward DHCP to server at 10.0.0.5

! === Verify ===
```

```
show ip dhcp binding              # Active leases (IP, MAC, expiry)
show ip dhcp pool                 # Pool stats (used, available)
show ip dhcp conflict             # IPs found to be in use
show ip dhcp server statistics    # DHCP message counters
```

DNS — Domain Name System

DNS translates human-readable names (example.com) into IP addresses (93.184.216.34). It's a hierarchical, distributed database — no single server knows all names. Instead, root servers (.) know about TLD servers (.com, .org), which know about authoritative servers (example.com), which know the actual records.

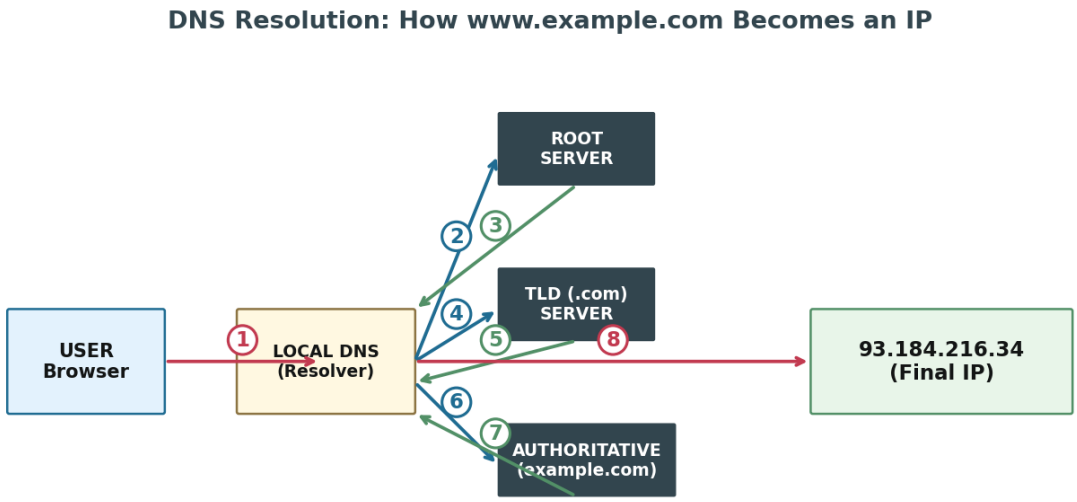


Figure: DNS recursive resolution: client asks local resolver, which queries root → TLD → authoritative, then returns final IP to client.

Common DNS record types you should know: **A** (hostname → IPv4), **AAAA** (hostname → IPv6), **CNAME** (alias to another name), **MX** (mail server, with priority), **NS** (authoritative nameserver), **PTR** (reverse lookup, IP → name), **TXT** (arbitrary text, used for SPF, DKIM, domain verification), **SOA** (Start of Authority, zone metadata), **SRV** (service record, used by AD, SIP).

```
! === Configure DNS on Cisco router ===
ip domain-lookup                          # Enable DNS resolution (default)
ip name-server 8.8.8.8 1.1.1.1            # Configure DNS servers
ip domain-name example.com                # Default domain (for short names)

! === Test DNS ===
ping www.example.com                      # Should resolve then ping

! === View DNS cache ===
show hosts                                # Cached DNS entries
```

NTP — Network Time Protocol

NTP synchronizes clocks across devices. Accurate time is critical for: syslog correlation (you need to know *when* events happened), certificate validation (TLS certs are time-bound), Kerberos authentication (5-minute skew tolerance), and protocol operations (OSPF, BGP use timestamps). NTP uses UDP port 123 and can achieve millisecond accuracy across the Internet, sub-millisecond on LANs.

```
! === NTP Configuration ===
ntp server 192.168.1.5                    # Primary NTP server
ntp server 192.168.1.6 prefer             # Preferred source
ntp source GigabitEthernet0/0             # Use this interface as source
ntp update-calendar                       # Update hardware clock too

! === NTP Authentication (optional, for security) ===
ntp authentication-key 1 md5 MyKey123!
ntp authenticate
ntp trusted-key 1

! === Verify ===
show ntp status                           # Sync state, stratum, offset
show ntp associations                     # Peer list with stratum, role
show clock detail                         # Current time + source
```

Syslog — Centralized Logging

Syslog (RFC 5424) sends log messages from network devices to a central logging server. Uses UDP 514 by default (or TCP for reliable delivery). Syslog messages have a severity level 0-7 (0 = emergency, 7 = debugging) and a facility (which subsystem generated the message).

Severity	Name	Description	Example
0	Emergency	System unusable	Kernel panic
1	Alert	Immediate action needed	Disk full
2	Critical	Critical condition	Redundant power supply failed
3	Error	Error condition	Interface down
4	Warning	Warning condition	Link flapping
5	Notification	Normal but significant	OSPF neighbor up
6	Informational	Informational	User logged in
7	Debugging	Debug-level messages	Debug output

```
! === Syslog Configuration ===
service timestamps log datetime msec localtime show-timezone
logging 192.168.1.10                      # Syslog server IP
logging trap informational                # Send severity 0-6 (not 7=debug)
logging source-interface GigabitEthernet0/0
logging buffered 4096 informational       # Also buffer locally for show log

! === Verify ===
show logging                              # View buffered logs
show logging history                      # SNMP trap logs
```

SNMP — Simple Network Management Protocol

SNMP lets a Network Management System (NMS) poll devices for stats and receive unsolicited traps (alerts). Uses UDP 161 (queries) and 162 (traps). SNMPv1 and v2c use community strings (essentially passwords) sent in cleartext — very insecure. **SNMPv3** adds user-based authentication and encryption — always use v3 in production.

```
! === SNMPv2c Configuration (simple but insecure) ===
snmp-server community MyPublicRO RO        # Read-only community
snmp-server community MyPrivateRW RW       # Read-write community
snmp-server location "HQ Data Center"      # sysLocation
snmp-server contact "Network Team"         # sysContact
snmp-server host 192.168.1.10 version 2c MyPublicRO  # Trap destination

! === SNMPv3 Configuration (secure, recommended) ===
snmp-server group ADMINS v3 priv read WRITE notify WRITE
snmp-server user admin1 ADMINS v3 auth sha AuthPass123! priv aes 128 PrivPass123!
snmp-server host 192.168.1.10 version 3 priv admin1

! === Verify ===
show snmp                                  # SNMP agent info
show snmp host                             # Trap destinations
```

Other Essential Services

Service	Port	Transport	Purpose
HTTP	80	TCP	Web (unencrypted)
HTTPS	443	TCP	Web (TLS encrypted)
SSH	22	TCP	Secure remote CLI
Telnet	23	TCP	Insecure remote CLI (avoid)
FTP	20, 21	TCP	File transfer (cleartext)
TFTP	69	UDP	Trivial file transfer (no auth, used for IOS)
SMTP	25, 587	TCP	Email sending
POP3	110, 995	TCP	Email retrieval (legacy)
IMAP4	143, 993	TCP	Email retrieval (modern)
RDP	3389	TCP	Remote Desktop (Windows)
LDAP	389, 636	TCP	Directory access (AD)
RADIUS	1812, 1813	UDP	AAA authentication
TACACS+	49	TCP	AAA authentication (Cisco)

Chapter 13 Quiz

Q1. Which DHCP message does the client broadcast to find available servers?

A. Discover

B. Offer

C. Request

D. Acknowledge

Answer: A — DISCOVER = client broadcasts to find DHCP servers.

Q2. Which port does DNS use?

A. 53
B. 67
C. 80
D. 443
Answer: A — DNS uses port 53 (UDP for queries, TCP for zone transfers/large responses).

Q3. What is the NTP default port?

A. 53
B. 123
C. 514
D. 161
Answer: B — NTP uses UDP port 123.

Q4. Which SNMP version introduced encryption?

A. v1
B. v2c
C. v3
D. v4
Answer: C — SNMPv3 added authentication (MD5/SHA) and encryption (DES/3DES/AES).

Q5. Which syslog severity level is 'warning'?

A. 2
B. 3
C. 4
D. 5
Answer: C — Level 4 = Warning. (0=Emergency ... 7=Debugging)

PART 14

Security Fundamentals

CIA, AAA, ACLs, VPNs, firewalls, and zero trust

Chapter 14: Network Security Fundamentals

Network security is no longer optional — every device, every protocol, every design decision must consider security. The CCNA 200-301 blueprint dedicates 15% of the exam to security, reflecting how central it has become. This chapter covers the core concepts: the CIA triad, AAA, ACLs, port security, VPNs, firewalls, IDS/IPS, and the modern zero trust model.

The CIA Triad

Every security control serves one of three goals, known as the **CIA Triad**: **Confidentiality** (only authorized users can read data — encryption provides this), **Integrity** (data has not been altered — hashing provides this), and **Availability** (data and services are accessible when needed — redundancy and DDoS protection provide this). Every security mechanism maps to one or more of these three.

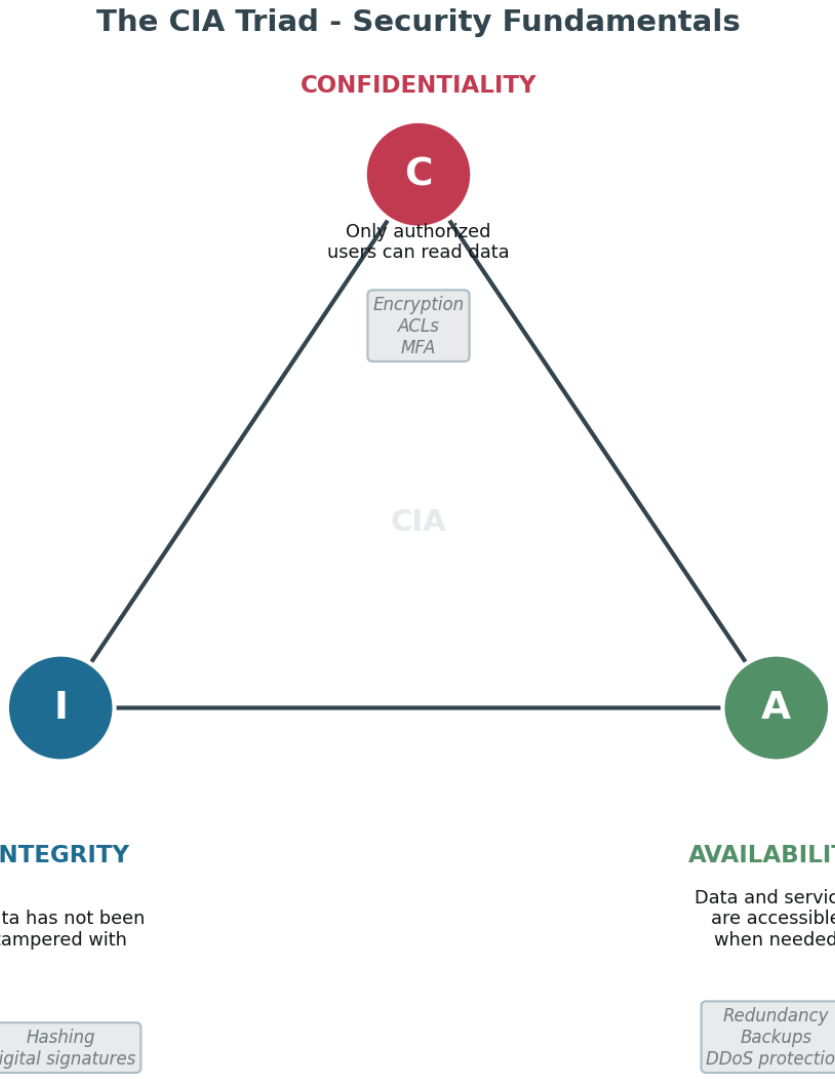


Figure: The CIA Triad — three pillars of security: Confidentiality, Integrity, Availability. Every control serves at least one.

AAA — Authentication, Authorization, Accounting

AAA is the framework for controlling access to network devices and services. **Authentication** proves who you are (username + password, certificate, biometric). **Authorization** determines what you can do (which commands, which VLANs, which routes). **Accounting** records what you did (for audit, billing, forensics). Two protocols implement AAA: **RADIUS** (open standard, UDP 1812/1813, encrypts only password) and **TACACS+** (Cisco proprietary, TCP 49, encrypts entire body, supports per-command authorization).

Aspect	RADIUS	TACACS+
Standard	Open (RFC 2865)	Cisco proprietary
Transport	UDP 1812/1813	TCP 49
Encryption	Password only	Entire packet body
Authorization	Coarse (whole session)	Fine (per-command)
Best for	Network access (VPN, Wi-Fi, 802.1X)	Device administration

ACLs — Access Control Lists

An **ACL** is an ordered list of permit/deny rules used to filter traffic. ACLs can also be used for route filtering, NAT, QoS marking, and SNMP access control. There are two main types: **Standard** (numbers 1-99, 1300-1999) matches only source IP, and **Extended** (numbers 100-199, 2000-2699) matches source/dest IP, protocol, ports, and more.

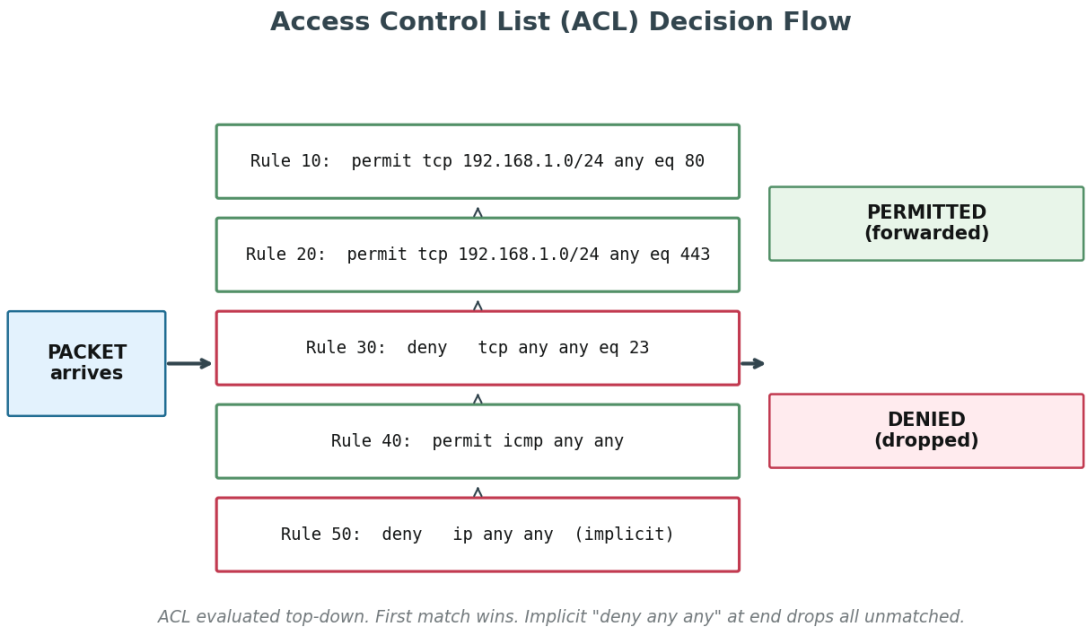


Figure: ACL decision flow: rules evaluated top-down, first match wins, implicit deny at end.

```
! === Standard ACL (source IP only) ===
access-list 10 permit 192.168.1.0 0.0.0.255
access-list 10 deny any log

! === Extended ACL (full matching) ===
access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80
access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 443
```

```
access-list 100 permit icmp any any
access-list 100 deny ip any any log

! === Named ACL (preferred for readability) ===
ip access-list extended BLOCK_TELNET
  remark Block Telnet from sales subnet
  deny tcp 192.168.10.0 0.0.0.255 any eq 23
  permit ip any any

! === Apply ACL to interface ===
interface GigabitEthernet0/0
  ip access-group 100 in

! === Verify ===
show access-lists
show ip access-lists
show ip interface Gi0/0 | include access list
```

Port Security

Port security limits which MAC addresses can send traffic through a switch port. This prevents MAC flooding attacks (where an attacker sends thousands of fake MACs to overflow the CAM table) and unauthorized device connection. Three violation modes: **protect** (silently drops), **restrict** (drops + logs + SNMP trap), **shutdown** (err-disables port — default).

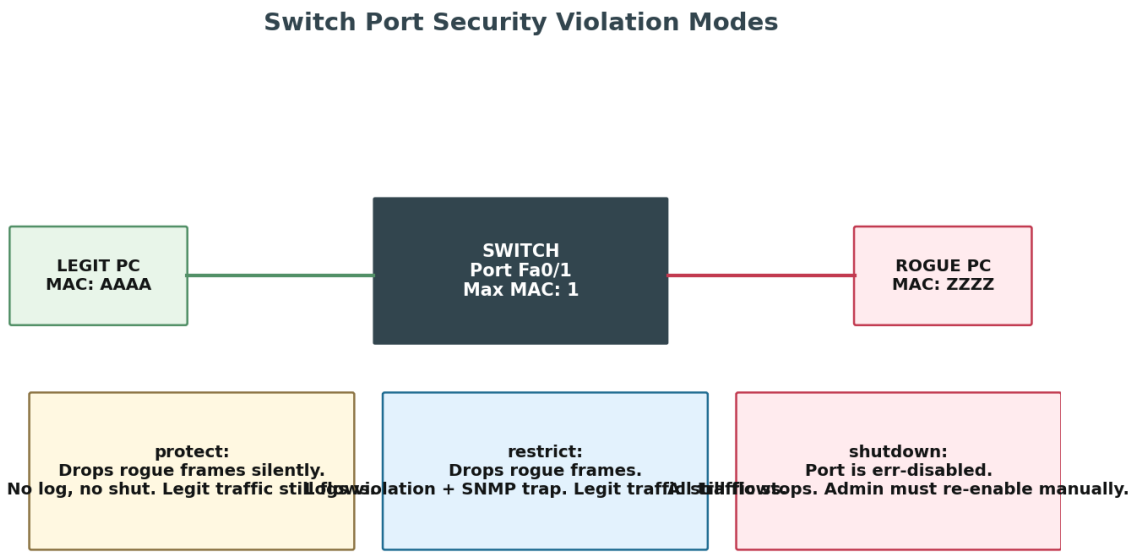


Figure: Port security violation modes: protect (silent drop), restrict (drop+log), shutdown (err-disable port).

```
! === Port Security Configuration ===
interface FastEthernet0/1
  switchport mode access
  switchport port-security                    # Enable port security
  switchport port-security maximum 2          # Max 2 MACs (default 1)
  switchport port-security mac-address sticky # Learn MACs dynamically + save
  switchport port-security violation restrict # restrict = drop + log

! === Auto-recovery from err-disable ===
errdisable recovery cause psecure-violation
```

```
errdisable recovery interval 300              # Recover after 5 minutes

! === Verify ===
show port-security
show port-security interface Fa0/1
show port-security address
```

VPN — Virtual Private Networks

A **VPN** creates an encrypted tunnel over an untrusted network (typically the Internet), allowing remote sites or users to securely access corporate resources. **IPsec** is the standard protocol suite for site-to-site VPNs. **SSL/TLS VPN** (Cisco AnyConnect) is common for remote-access VPNs because it works through browsers and firewalls (uses HTTPS port 443).

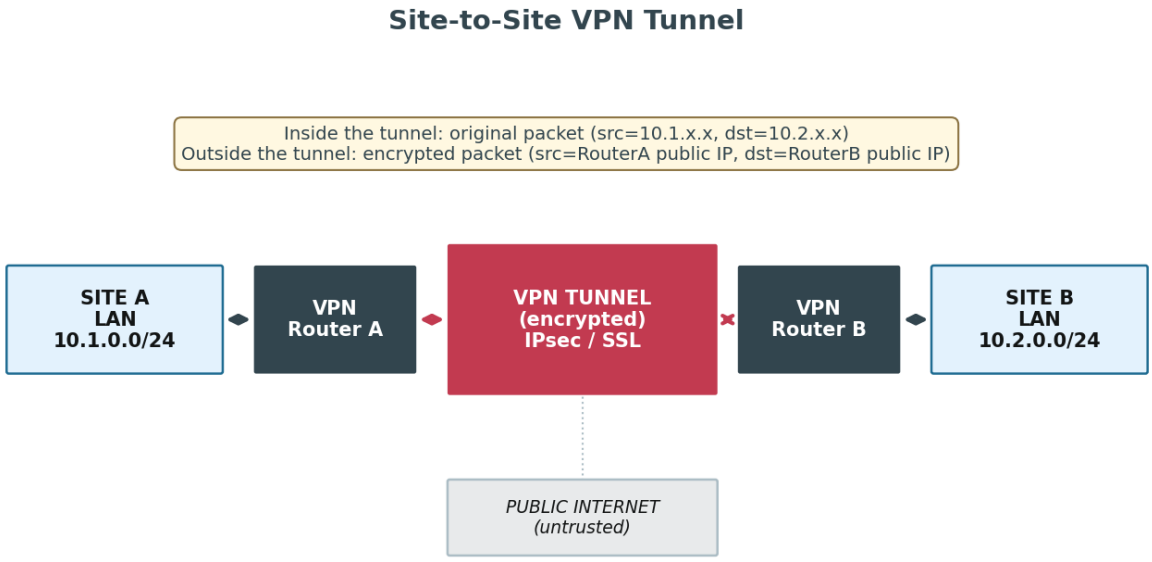


Figure: Site-to-site VPN: traffic between two LANs is encrypted and tunneled over the public Internet.

Firewalls, IDS, and IPS

A **firewall** filters traffic between security zones based on a ruleset. Modern Next-Gen Firewalls (NGFW) add application awareness, intrusion prevention, URL filtering, and user identity. **IDS** (Intrusion Detection System) is passive — analyzes traffic and alerts on threats but doesn't block. **IPS** (Intrusion Prevention System) is active — sits inline and blocks malicious traffic in real-time. Most NGFWs include IPS functionality.

Zero Trust

Zero Trust is the modern security model that assumes no implicit trust — every access request is verified, regardless of where it comes from. The classic 'castle and moat' model assumed the internal network was safe — once you got past the firewall, you were trusted. Zero Trust rejects this: users, devices, and applications must authenticate and authorize for every transaction. Core principles: verify explicitly (use identity + device + context), use least privilege access, and assume breach (segment, monitor, encrypt).

Network Hardening Best Practices

Area	Best Practice	Cisco Command
Passwords	Use enable secret, not enable password	enable secret MySecret!
Encryption	Encrypt all passwords in config	service password-encryption
SSH	Disable Telnet, use SSH v2 only	transport input ssh; ip ssh version 2
Console	Require login on console	login local under line con 0
Banners	Legal warning before login	banner login ^C AUTHORIZED ONLY ^C
Unused ports	Shut them down	shutdown under interface
CDP/LLDP	Disable on edge ports	no cdp enable on edge ports
HTTP	Disable HTTP server, use HTTPS	no ip http server; ip http secure-server
Logging	Send to syslog, set trap level	logging 10.0.0.5; logging trap informational
NTP	Sync time for accurate logs	ntp server 10.0.0.5
ACLs	Restrict management (VTY) access	ip access-group MGMT in under line vty

Chapter 14 Quiz

Q1. What does the 'C' in CIA triad stand for?

A. Confidentiality
B. Capability
C. Connection
D. Compliance
Answer: A — C = Confidentiality. Only authorized users can read data. Encryption provides this.

Q2. Which AAA protocol encrypts the entire packet body?

A. RADIUS
B. TACACS+
C. Both
D. Neither
Answer: B — TACACS+ encrypts the entire body. RADIUS only encrypts the password.

Q3. Which ACL type matches source IP only?

A. Standard
B. Extended
C. Reflexive
D. Dynamic
Answer: A — Standard ACL (1-99, 1300-1999) matches only source IP.

Q4. What is the default port security violation mode?

A. protect
B. restrict
C. shutdown
D. disable
Answer: C — Default = shutdown. Port is err-disabled, requires manual recovery.

Q5. Which security model assumes no implicit trust?

A. Castle and moat

B. Zero Trust

C. Perimeter security

D. Defense in depth

Answer: B — Zero Trust: 'never trust, always verify'. Every access is authenticated/authorized.

PART 15

Automation & Programmability

SDN, APIs, Python, Ansible, and the future of networking

Chapter 15: Network Automation and Programmability

The traditional way of configuring networks — typing CLI commands into each device one at a time — doesn't scale. Modern networks have thousands of devices, change daily, and require consistency that humans can't reliably deliver. **Network automation** uses software to configure, monitor, and operate networks at scale. The CCNA 200-301 reflects this shift with 10% of the exam dedicated to automation and programmability.

Software-Defined Networking (SDN)

SDN separates the control plane (the brain — routing decisions, MAC learning, STP) from the data plane (the muscle — actual packet forwarding). In traditional networking, each device has both planes tightly integrated. In SDN, a central **controller** holds the control plane, and devices become simple forwarders. This centralization enables programmability, automation, and consistent policy.

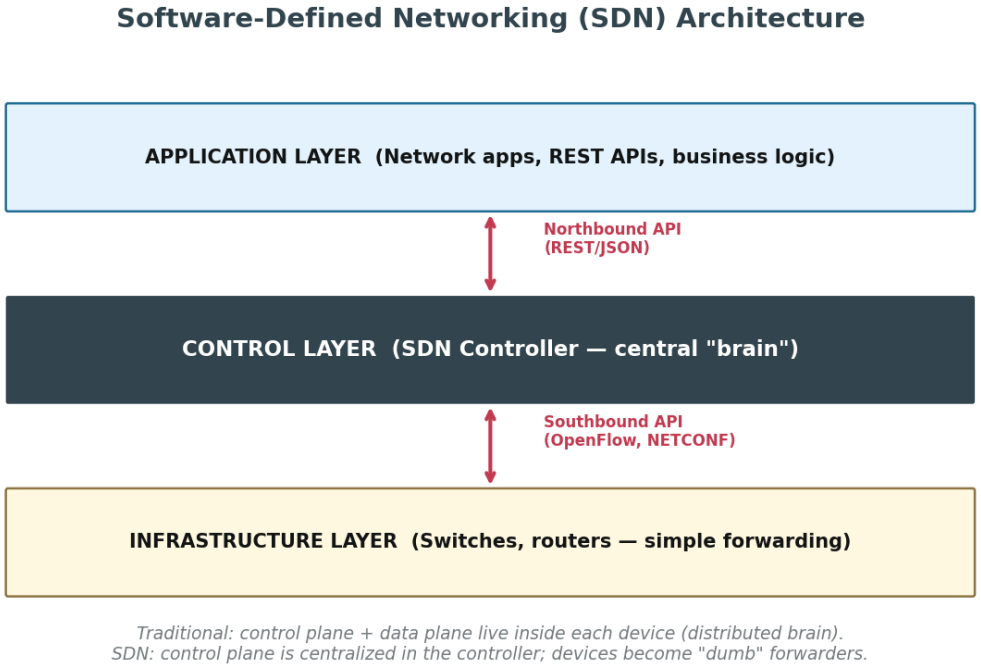


Figure: SDN architecture: Application Layer ↔ Control Layer (controller) ↔ Infrastructure Layer (devices). Northbound and southbound APIs.

SDN has three layers: **Application Layer** (network apps, automation scripts, business logic), **Control Layer** (the SDN controller — the 'brain'), and **Infrastructure Layer** (switches, routers — simple forwarders). Applications talk to the controller via **northbound APIs** (typically REST). The controller talks to devices via **southbound APIs** (OpenFlow, NETCONF, RESTCONF).

Cisco DNA Center and SD-Access

Cisco DNA Center (Digital Network Architecture) is Cisco's SDN controller for campus networks. It provides automation, assurance (monitoring/analytics), and fabric (SD-Access). **SD-Access** is Cisco's campus SDN solution built on DNA Center, using LISP for control plane, VXLAN for data plane, and ISE for security policy. The promise: define intent ('Sales VLAN can reach Internet but not HR'), and the controller pushes the right configs to every device automatically.

APIs — REST, NETCONF, RESTCONF

An **API** (Application Programming Interface) lets software talk to other software. Network APIs let automation tools talk to network devices. Three protocols dominate modern networking:

REST (Representational State Transfer) — HTTP-based, uses GET/POST/PUT/DELETE verbs on URLs. Most modern cloud APIs are REST. Data formats: JSON (most common) or XML.

NETCONF (Network Configuration Protocol, RFC 6241) — XML over SSH (TCP 830). IETF standard. Uses YANG data models. Supports transactions (commit/rollback).

RESTCONF (RFC 8040) — REST-like API over HTTPS (443) that exposes NETCONF data stores. Uses YANG models. Supports JSON or XML. Modern alternative to NETCONF for HTTP-friendly automation.

Data Formats: JSON, XML, YAML

Automation tools need to exchange structured data. Three formats dominate: **JSON** (JavaScript Object Notation — lightweight, uses braces and brackets, most common in REST APIs), **XML** (eXtensible Markup Language — uses tags, more verbose, used in NETCONF), and **YAML** (YAML Ain't Markup Language — uses indentation, most human-readable, used in Ansible and Kubernetes).

Format	Syntax	Use Case	Pros	Cons
JSON	{} and []	REST APIs	Lightweight, universal	No comments, strict syntax
XML	...	NETCONF, SOAP	Schema validation, attributes	Verbose, harder to read
YAML	Indentation	Ansible, K8s	Very readable, comments	Whitespace-sensitive, slow to parse

```
// === JSON Example ===
{
  "interface": "GigabitEthernet0/0",
  "description": "Uplink to Core",
  "ip_address": {
    "address": "192.168.1.1",
    "netmask": "255.255.255.0"
  },
  "enabled": true
}

<!-- === XML Example === -->
<interface>
  <name>GigabitEthernet0/0</name>
  <description>Uplink to Core</description>
  <ip-address>
    <address>192.168.1.1</address>
    <netmask>255.255.255.0</netmask>
  </ip-address>
  <enabled>true</enabled>
</interface>

# === YAML Example ===
interface: GigabitEthernet0/0
description: Uplink to Core
ip_address:
  address: 192.168.1.1
  netmask: 255.255.255.0
enabled: true
```

Python for Network Automation

Python is the dominant language for network automation. Simple syntax, huge ecosystem, and libraries specifically for networking. The most popular libraries:

Netmiko — wraps Paramiko (SSH) for network devices. Send CLI commands to Cisco, Juniper, Arista, etc. Easy to learn.

NAPALM — Cross-vendor abstraction. Same API for get_facts(), get_interfaces(), etc., across vendors.

Nornir — Multi-threaded framework for parallel SSH. Faster than serial Netmiko for large fleets.

Requests — HTTP library for REST API calls. Used with RESTCONF, Cisco DNA Center, Meraki APIs.

ncclient — NETCONF client library for Python.

```
#!/usr/bin/env python3
# === Netmiko Example: SSH to Cisco device and run commands ===
from netmiko import ConnectHandler

device = {
    'device_type': 'cisco_ios',
    'host': '192.168.1.1',
    'username': 'admin',
    'password': 'AdminPass123!',
}

with ConnectHandler(**device) as net_connect:
    output = net_connect.send_command('show ip interface brief')
    print(output)

    # Send config changes
    config_commands = [
        'interface Gi0/0',
        'description Configured by Python',
        'no shutdown',
    ]
    net_connect.send_config_set(config_commands)

# === Requests Example: RESTCONF to Cisco device ===
import requests
import json
from requests.auth import HTTPBasicAuth

url = 'https://192.168.1.1/restconf/data/ietf-interfaces:interfaces'
headers = {
    'Accept': 'application/yang-data+json',
    'Content-Type': 'application/yang-data+json',
}
auth = HTTPBasicAuth('admin', 'AdminPass123!')

response = requests.get(url, headers=headers, auth=auth, verify=False)
print(json.dumps(response.json(), indent=2))
```

Ansible — Agentless Automation

Ansible (Red Hat) is the most popular configuration management tool for networking. It's **agentless** — uses SSH (or NETCONF, RESTCONF) to talk to devices, no daemon needed on the device. Playbooks are written in **YAML** — declarative, idempotent, human-readable. Ansible has built-in modules for Cisco IOS, IOS-XE, NX-OS, Juniper, Arista, and more.

```
# === Ansible playbook: configure interfaces on Cisco switches ===
---
- name: Configure access ports
  hosts: switches
  connection: network_cli
  gather_facts: no

  vars:
    ansible_network_os: cisco.ios.ios
    ansible_user: admin
    ansible_password: AdminPass123!

  tasks:
    - name: Configure VLAN 10 on access ports
      cisco.ios.ios_config:
        lines:
          - switchport mode access
          - switchport access vlan 10
          - spanning-tree portfast
          - spanning-tree bpduguard enable
        parents: interface {{ item }}
      loop:
        - FastEthernet0/1
        - FastEthernet0/2
        - FastEthernet0/3
        - FastEthernet0/4

    - name: Save configuration
      cisco.ios.ios_command:
        commands: write memory

# === Inventory file (inventory.ini) ===
[switches]
192.168.1.10
192.168.1.11
192.168.1.12

# === Run ===
# ansible-playbook -i inventory.ini configure-switches.yml
```

Version Control with Git

Git is the de-facto version control system for code and configuration. Every network engineer should know the basics: git init (start a repo), git clone (copy a repo), git add + git commit (save changes), git push (upload to remote), git pull (download from remote), git branch + git merge (parallel development). Host your repos on GitHub, GitLab, or Bitbucket. Storing network configs in Git gives you history, rollback, code review, and collaboration.

Chapter 15 Quiz

Q1. What does SDN separate?

A. Control plane from data plane

B. Hardware from software

C. LAN from WAN

D. Wireless from wired

Answer: A — SDN separates the control plane (decisions) from the data plane (forwarding). The controller holds the control plane.

Q2. Which port does NETCONF use?

A. 22

B. 443

C. 830

D. 8080

Answer: C — NETCONF uses TCP port 830 over SSH.

Q3. Which data format uses indentation (whitespace)?

A. JSON

B. XML

C. YAML

D. All

Answer: C — YAML uses indentation to define structure. No braces or tags. Whitespace-sensitive.

Q4. Which Python library is used for SSH network automation?

A. Netmiko

B. Requests

C. BeautifulSoup

D. NumPy

Answer: A — Netmiko wraps Paramiko for SSH to network devices.

Q5. Is Ansible agentless?

A. Yes — uses SSH/WinRM, no agent on target

B. No — requires agent

C. Both modes

D. Only on Windows

Answer: A — Ansible is agentless — uses SSH (Linux) or WinRM (Windows). No daemon needed on target.

PART 16

Troubleshooting Mastery

Systematic methods, tools, and 100+ scenarios

Chapter 16: The Art of Network Troubleshooting

Troubleshooting is what separates junior engineers from senior ones. Configuration you can look up; troubleshooting requires understanding the system deeply enough to diagnose problems quickly under pressure. The good news: there's a systematic method that works for almost any network problem. Master it, and you'll be the engineer everyone calls when things break.

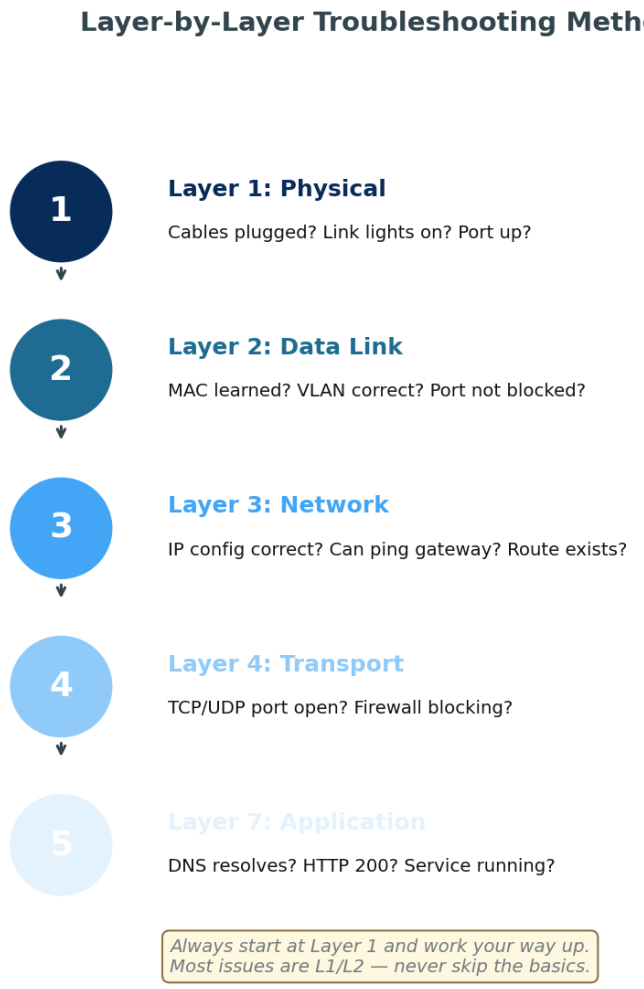


Figure: Layer-by-layer troubleshooting method: start at Layer 1 (physical) and work up. Most issues are L1/L2 — never skip the basics.

The Layer-by-Layer Method

Always troubleshoot **bottom-up**: start at Layer 1 and work your way up. This is faster than top-down because most issues are physical or data link — a loose cable, a wrong VLAN, a duplex mismatch. If you start at the application layer ('the website is down') and skip the basics, you'll waste hours chasing software bugs when the real issue is an unplugged cable.

Layer	What to Check	Commands / Tools
L1 Physical	Cable plugged in? Link lights on? Correct cable type? EMI?	show interfaces, look at LEDs, swap cable
L2 Data Link	Same VLAN? Port up/up? Duplex match? Port security? STP blocking?	show interfaces status, show vlan, show port-security, show spanning-tree

Layer	What to Check	Commands / Tools
L3 Network	Correct IP/mask? Default gateway reachable? Route exists? ACL blocking?	ping gateway, show ip route, show ip interface brief, show access-lists
L4 Transport	TCP/UDP port open? Firewall blocking? NAT issue?	telnet , show ip nat translations, firewall logs
L7 Application	Service running? DNS resolves? Cert valid? App config correct?	nslookup, curl, application logs, ps aux

Essential Troubleshooting Tools

Tool	Purpose	Example Command
ping	Test L3 reachability and latency	ping 8.8.8.8 (Windows: -n count, Linux: -c count)
traceroute / tracert	Discover path packets take	traceroute 8.8.8.8 (Linux/Mac) / tracert 8.8.8.8 (Windows)
arp -a / ip neigh	View ARP cache (IP→MAC)	arp -a (Windows) / ip neigh (Linux)
nslookup / dig	DNS queries	nslookup example.com / dig example.com
netstat -an	View open ports and connections	netstat -an | findstr :80 (Windows)
telnet / nc	Test TCP port reachability	telnet 192.168.1.1 80 / nc -zv 192.168.1.1 80
Wireshark	Capture and analyze packets	GUI: capture on interface, filter with display filters
tcpdump	Command-line packet capture (Linux)	tcpdump -i eth0 -nn port 80
ipconfig /all	View Windows IP config	ipconfig /all (Windows)
show running-config	Cisco device config	show run | section interface
show ip route	Cisco routing table	show ip route 192.168.1.0
show interfaces	Cisco interface stats + errors	show interfaces Gi0/0

100+ Troubleshooting Scenarios

Below are 100+ real-world troubleshooting scenarios. Each includes the symptom, likely cause, and how to verify/fix. Work through these to build your troubleshooting muscle memory.

Scenario 1: Symptom: PC cannot reach any network resource.

Causes: Likely causes: (1) Cable unplugged. (2) Port shut down. (3) Wrong VLAN. (4) IP/mask wrong. (5) Port security violation.

Fix: Fix: Check link lights. Run 'show interfaces status'. Verify 'show vlan brief'. Check 'show port-security'. Run 'ipconfig' on PC.

Scenario 2: Symptom: PC can reach local subnet but not Internet.

Causes: Likely causes: (1) Wrong default gateway. (2) Default gateway down. (3) NAT misconfigured. (4) DNS not working.

Fix: Fix: 'ping '. 'tracert 8.8.8.8'. Verify 'show ip route' on gateway. Check NAT: 'show ip nat translations'. Try 'nslookup google.com'.

Scenario 3: Symptom: PC can ping 8.8.8.8 but cannot browse websites.

Causes: Likely causes: (1) DNS not working. (2) Proxy misconfigured. (3) Firewall blocking port 80/443.

Fix: Fix: 'nslookup google.com' — if it fails, DNS is the issue. Try 'ping google.com' (uses DNS). Check proxy settings in browser. Try 'telnet google.com 443'.

Scenario 4: Symptom: Two PCs in same VLAN cannot ping each other.

Causes: Likely causes: (1) Different subnets. (2) Port security blocking one. (3) STP blocking port. (4) ACL blocking ICMP. (5) One PC firewall blocking.

Fix: Fix: Verify both PCs have IP in same subnet. Check 'show mac address-table' for both MACs. 'show port-security interface'. 'show spanning-tree vlan X'.

Scenario 5: Symptom: PCs in different VLANs cannot communicate.

Causes: Likely causes: (1) Inter-VLAN routing not configured. (2) SVI down. (3) ip routing not enabled. (4) ACL blocking.

Fix: Fix: Verify 'ip routing' is on L3 switch. Check 'show ip interface brief' — SVIs should be up/up. Check 'show ip route' — VLAN subnets should be 'C' (connected). Verify ACLs.

Scenario 6: Symptom: Trunk link between switches not passing VLAN traffic.

Causes: Likely causes: (1) Trunk not configured. (2) Allowed VLAN list excludes needed VLANs. (3) Native VLAN mismatch. (4) DTP negotiation failed.

Fix: Fix: 'show interfaces trunk'. Verify allowed VLANs: 'switchport trunk allowed vlan'. Check native VLAN matches on both ends. Try hard-coding 'switchport mode trunk' + 'switchport nonegotiate'.

Scenario 7: Symptom: OSPF neighborship not forming.

Causes: Likely causes: (1) Hello/dead interval mismatch. (2) Area ID mismatch. (3) Authentication mismatch. (4) Subnet mask mismatch (broadcast networks). (5) Passive interface. (6) MTU mismatch.

Fix: Fix: 'show ip ospf neighbor'. 'show ip ospf interface'. Compare hello/dead. Verify area IDs match. Check authentication. Verify masks. 'show ip ospf' for passive. Try 'ip ospf mtu-ignore' as test.

Scenario 8: Symptom: OSPF routes not appearing in routing table.

Causes: Likely causes: (1) Neighbors not FULL. (2) ACL filtering. (3) Better AD route from another source. (4) Stub area blocking external. (5) Route summarization hiding specifics.

Fix: Fix: 'show ip ospf neighbor' (must be FULL). 'show ip ospf rib' (local RIB). 'show ip route'. 'show access-lists'. Check if area is stub.

Scenario 9: Symptom: Default route not being advertised into OSPF.

Causes: Likely causes: (1) 'default-information originate' missing. (2) Router doesn't have a default route itself (need 'always' keyword).

Fix: Fix: Add 'default-information originate' under 'router ospf'. If router has no default route, use 'default-information originate always'.

Scenario 10: Symptom: DHCP clients not getting IPs.

Causes: Likely causes: (1) DHCP server down. (2) DHCP relay (helper) missing. (3) DHCP pool exhausted. (4) ACL blocking UDP 67/68. (5) IP conflict.

Fix: Fix: Verify DHCP server running. Check 'ip helper-address' on router interface facing clients. 'show ip dhcp pool' (check used/available). 'show ip dhcp binding'. 'show ip dhcp conflict'.

Scenario 11: Symptom: Cannot SSH to Cisco device.

Causes: Likely causes: (1) SSH not enabled. (2) RSA keys not generated. (3) VTY not configured for SSH. (4) ACL blocking port 22. (5) Wrong username/password.

Fix: Fix: Verify 'ip ssh version 2' configured. 'show crypto key mypubkey rsa' — if empty, run 'crypto key generate rsa'. 'show run | section line vty' — should say 'transport input ssh'. Check ACLs.

Scenario 12: Symptom: Inter-VLAN routing works but no Internet access for VLANs.

Causes: Likely causes: (1) Default route on L3 switch missing. (2) NAT not configured for new VLANs. (3) ACL blocking.

Fix: Fix: Add 'ip route 0.0.0.0 0.0.0.0 ' on L3 switch. Add new VLAN subnet to NAT: 'ip nat inside source list interface overload'. Update ACL to include new subnet.

Scenario 13: Symptom: STP keeps reconverging (network flapping).

Causes: Likely causes: (1) Flapping link. (2) Duplex mismatch. (3) Bad cable. (4) Excessive BPDUs from another switch. (5) Unidirectional link.

Fix: Fix: 'show spanning-tree summary' (count topology changes). 'show logging' (look for link up/down messages). Check duplex settings. Try replacing cable. Enable UDLD aggressive.

Scenario 14: Symptom: EtherChannel not forming.

Causes: Likely causes: (1) Configuration mismatch between member ports. (2) Different LACP modes. (3) Different speeds/duplex. (4) Different VLANs allowed.

Fix: Fix: 'show etherchannel summary' (look for 'I' = in use vs 's' = suspended). All member ports must match: speed, duplex, VLAN, trunk config. Use 'active' on both ends for LACP.

Scenario 15: Symptom: HSRP active/active (both routers think they're active).

Causes: Likely causes: (1) HSRP hello blocked. (2) ACL blocking UDP 1985. (3) Different HSRP group or virtual IP. (4) Network partition.

Fix: Fix: Verify both routers see each other: 'show standby'. Check ACLs. Verify same group, same virtual IP. Check connectivity between them.

Scenario 16: Symptom: Port security err-disabled a port.

Causes: Likely causes: (1) More MACs than max. (2) Unauthorized device plugged in. (3) Hub or switch plugged into access port.

Fix: Fix: 'show interfaces status' (look for 'err-disabled'). 'show port-security interface '. To recover: 'shutdown' then 'no shutdown'. Or enable auto-recovery: 'errdisable recovery cause psecure-violation'.

Scenario 17: Symptom: CRC errors incrementing on an interface.

Causes: Likely causes: (1) Bad cable. (2) EMI. (3) Duplex mismatch. (4) Failing NIC or port.

Fix: Fix: 'show interfaces ' (look for CRC, runts, giants). Replace cable. Check duplex: 'show interfaces | include Duplex'. Try different port or NIC.

Scenario 18: Symptom: Latency high during peak hours.

Causes: Likely causes: (1) Bandwidth saturated. (2) CPU high on a device. (3) Buffer exhaustion. (4) QoS not prioritizing important traffic.

Fix: Fix: 'show interfaces' (look for high load + drops). 'show processes cpu sorted'. Enable QoS to prioritize voice. Add bandwidth (upgrade link). Consider WAN optimization.

Scenario 19: Symptom: Wireless clients cannot connect to specific SSID.

Causes: Likely causes: (1) SSID hidden and not entered. (2) Wrong password. (3) RADIUS server down (WPA2-Enterprise). (4) MAC not in allowed list. (5) Client incompatible (e.g., 802.11ax only AP).

Fix: Fix: Check WLC for client association attempts. Verify SSID config. Test RADIUS: 'test aaa group radius legacy'. Check WLC client exclusion list.

Scenario 20: Symptom: Wireless client keeps disconnecting.

Causes: Likely causes: (1) Weak signal. (2) Interference. (3) Roaming issue. (4) Client driver bug. (5) WLC client timeout.

Fix: Fix: Check RSSI at client location. Run spectrum analysis (Cisco CleanAir). Enable 802.11r/k/v for fast roaming. Update client drivers. Check WLC client session timeouts.

Scenario 21: Symptom: VPN tunnel not establishing.

Causes: Likely causes: (1) Peer IPs wrong. (2) Pre-shared key mismatch. (3) IKE proposal mismatch. (4) NAT-T issue. (5) ACL blocking UDP 500/4500.

Fix: Fix: 'show crypto isakmp sa'. 'show crypto ipsec sa'. Enable 'debug crypto isakmp' (carefully). Verify PSKs match exactly. Verify IKE proposals. Check for NAT between peers.

Scenario 22: Symptom: BGP neighbor stuck in Idle or Active state.

Causes: Likely causes: (1) No route to peer. (2) Peer not configured. (3) TCP 179 blocked by firewall. (4) Wrong AS number. (5) Source interface mismatch.

Fix: Fix: 'show ip bgp summary'. Ping peer. 'show ip route '. Verify 'neighbor X.X.X.X remote-as Y' on both sides. Check for TCP 179 in firewall logs. Use 'update-source' to specify source interface.

Scenario 23: Symptom: Cisco device time is wrong.

Causes: Likely causes: (1) NTP not configured. (2) NTP server unreachable. (3) Wrong timezone.

Fix: Fix: 'show ntp status'. 'show ntp associations'. 'ntp server '. Set timezone: 'clock timezone PST -8'. Set DST: 'clock summer-time PDT recurring'.

Scenario 24: Symptom: Syslog messages not reaching syslog server.

Causes: Likely causes: (1) Syslog server not running. (2) Wrong IP. (3) Trap level too high. (4) Firewall blocking UDP 514. (5) Source interface wrong.

Fix: Fix: 'show logging'. Verify 'logging ' configured. Check 'logging trap '. Verify syslog server listening. Test with 'logger' command on Linux.

Scenario 25: Symptom: SNMP polling not working.

Causes: Likely causes: (1) Wrong community string. (2) ACL blocking UDP 161. (3) SNMP not enabled. (4) Wrong SNMP version.

Fix: Fix: 'show snmp'. 'show snmp host'. Verify community string matches NMS. Check ACL on router. Test with 'snmpwalk -v2c -c ' from NMS.

Scenario 26: Symptom: ACL is blocking traffic that should be allowed.

Causes: Likely causes: (1) Rules in wrong order. (2) Wildcard mask wrong. (3) Implicit deny hitting. (4) ACL applied in wrong direction.

Fix: Fix: 'show access-lists' (look at hit counters). 'show ip interface '. Reorder rules (most specific first). Verify wildcard masks. Verify direction (in vs out).

Scenario 27: Symptom: ACL applied but traffic still flows that should be blocked.

Causes: Likely causes: (1) ACL not applied to right interface. (2) Wrong direction. (3) ACL permits the traffic unintentionally. (4) Routing bypasses ACL.

Fix: Fix: 'show ip interface | include access list'. Verify ACL is on correct interface and direction. 'show access-lists' for hit counters. Check if traffic routes around the ACL interface.

Scenario 28: Symptom: SSH login slow on Cisco device.

Causes: Likely causes: (1) DNS lookup for client IP failing (reverse DNS). (2) AAA server slow. (3) Crypto operations slow.

Fix: Fix: Disable reverse DNS: 'no ip domain-lookup'. Verify AAA server response time. Use 'transport input ssh' only (not Telnet+SSH). Consider stronger crypto hardware.

Scenario 29: Symptom: Console access works but SSH doesn't.

Causes: Likely causes: (1) SSH not enabled. (2) VTY not configured. (3) ACL on VTY. (4) Wrong credentials.

Fix: Fix: From console: 'show ip ssh'. 'show run | section line vty'. Verify 'login local' and 'transport input ssh'. Check 'username X secret Y' exists. Check ACL.

Scenario 30: Symptom: VTY login blocked after failed attempts.

Causes: Likely causes: (1) Login block-for configured. (2) ACL CoPP blocking. (3) SSH brute-force protection.

Fix: Fix: 'show login failures'. 'show login'. Wait for block period or use console to clear: 'clear login'.

Scenario 31: Symptom: HSRP keeps switching active routers every few seconds.

Causes: Likely causes: (1) Hello traffic being lost. (2) Network flapping. (3) Preempt bouncing. (4) HSRP authentication mismatch causing flapping.

Fix: Fix: 'show standby'. Check for preempt loop. Add 'standby delay minimum 30'. Verify authentication matches. Check network stability.

Scenario 32: Symptom: NAT translations not happening.

Causes: Likely causes: (1) 'ip nat inside'/'ip nat outside' not set. (2) ACL doesn't match traffic. (3) Pool exhausted. (4) Route missing.

Fix: Fix: 'show ip nat translations' (empty = no translations). 'show ip nat statistics'. Verify inside/outside on interfaces. Verify ACL matches source IPs. Check pool usage.

Scenario 33: Symptom: Telnet to device port 80 succeeds but HTTPS fails.

Causes: Likely causes: (1) HTTP server enabled but HTTPS not. (2) RSA keys not generated. (3) ACL blocking 443.

Fix: Fix: 'show ip http server status'. Generate RSA keys: 'crypto key generate rsa modulus 2048'. Enable: 'ip http secure-server'. Check ACL on VTY/HTTP.

Scenario 34: Symptom: OSPF routes from area 1 not appearing in area 0.

Causes: Likely causes: (1) ABR not configured correctly. (2) Area 1 is stub/totally stubby and routes are being summarized/filtered. (3) Type 4 LSA missing.

Fix: Fix: 'show ip ospf border-routers'. Verify ABR has interfaces in both areas. 'show ip ospf database'. Check area types match on all routers in area 1.

Scenario 35: Symptom: EIGRP neighborship flaps.

Causes: Likely causes: (1) Hello hold timer expired. (2) Unidirectional link. (3) K-values mismatch. (4) AS number mismatch. (5) Authentication issue.

Fix: Fix: 'show ip eigrp neighbors' (look for Q count, SRTT). Verify K-values match: 'show ip protocols'. Check AS number. Verify authentication. Check link stability (UDLD).

Scenario 36: Symptom: Wi-Fi speed is much slower than expected.

Causes: Likely causes: (1) Far from AP (weak signal). (2) Interference. (3) 2.4 GHz only. (4) Old client. (5) AP overloaded. (6) Channel width narrow.

Fix: Fix: Check RSSI (should be > -65 dBm). Run spectrum analysis. Enable band steering to 5 GHz. Reduce clients per AP. Increase channel width to 40 or 80 MHz. Check for co-channel interference.

Scenario 37: Symptom: CDP/LLDP neighbors not showing.

Causes: Likely causes: (1) CDP/LLDP disabled. (2) Neighbor doesn't support it. (3) ACL blocking. (4) Layer 1 issue (link down).

Fix: Fix: 'show cdp run'. 'show lldp'. Verify enabled globally and per-interface. 'cdp run' / 'lldp run'. Check neighbor is non-Cisco (use LLDP for non-Cisco).

Scenario 38: Symptom: Configuration changes don't take effect.

Causes: Likely causes: (1) Forgot to 'no shutdown'. (2) Wrong interface. (3) Typo in command. (4) Change in wrong config mode.

Fix: Fix: 'show running-config' to verify change is there. 'show ip interface brief' for status. Check you're in right interface config mode. Sometimes need to bounce interface.

Scenario 39: Symptom: Switch CPU high.

Causes: Likely causes: (1) Process loops (debugging on). (2) Broadcast storm. (3) ARP storm. (4) Excessive ACL evaluation. (5) Hardware forwarding failed.

Fix: Fix: 'show processes cpu sorted'. Turn off debug: 'undebug all'. 'show interfaces' (look for high broadcast/multicast). Check STP. Verify CEF: 'show ip cef'.

Scenario 40: Symptom: Interface status 'up/down' (line up, protocol down).

Causes: Likely causes: (1) Layer 2 issue — encapsulation mismatch, no keepalives, DLCIs mismatched (Frame Relay), no clock rate on serial.

Fix: Fix: Verify encapsulation matches on both ends. Check keepalive setting. For serial: 'clock rate' on DCE end. For Ethernet: usually physical or duplex issue.

Scenario 41: Symptom: Interface status 'down/down' (both down).

Causes: Likely causes: (1) Cable unplugged. (2) Wrong cable type. (3) Other end is shut down. (4) Hardware failure.

Fix: Fix: Check cable. Verify both ends 'no shutdown'. Try different cable. Try different port. Check transceiver for fiber.

Scenario 42: Symptom: Routing protocol process consuming high CPU.

Causes: Likely causes: (1) SPF running too often. (2) LSAs flapping. (3) Network unstable. (4) Full mesh iBGP causing lots of updates.

Fix: Fix: 'show ip ospf' (SPF run count). Throttle SPF. Stabilize network (find flapping link). Use route reflectors for iBGP. Summarize to reduce LSA count.

Scenario 43: Symptom: Traceroute shows some hops as '* * *'.

Causes: Likely causes: (1) Router configured not to respond to TTL-exceeded (ICMP rate limit). (2) Firewall blocking ICMP. (3) Asymmetric routing. (4) Router is the destination and only final hop responds.

Fix: Fix: Usually normal — some routers rate-limit ICMP. Try 'traceroute -T -p 80' (TCP traceroute) for different result. Check ICMP filtering on suspected hops.

Scenario 44: Symptom: PC gets APIPA (169.254.x.x) address.

Causes: Likely causes: (1) DHCP server unreachable. (2) DHCP relay (helper) missing. (3) Pool exhausted. (4) Client NIC issue.

Fix: Fix: Verify DHCP server. Check 'ip helper-address' on gateway. 'show ip dhcp pool' — if 'used' = 'total', pool is exhausted. Try ipconfig /release && ipconfig /renew on PC.

Scenario 45: Symptom: Cannot reach IPv6-only website.

Causes: Likely causes: (1) Your network is IPv4-only. (2) IPv6 not enabled. (3) IPv6 routing not configured. (4) IPv6 ACL blocking.

Fix: Fix: 'ping -6 ipv6.google.com'. Check 'show ipv6 interface brief'. Verify 'ipv6 unicast-routing'. 'show ipv6 route'. Check IPv6 ACLs.

Scenario 46: Symptom: Wi-Fi connected but no Internet.

Causes: Likely causes: (1) Got IP but DNS not working. (2) Captive portal not completed. (3) WLC has client in exclusion list. (4) ACL on WLAN blocking.

Fix: Fix: Check 'ipconfig' — IP, gateway, DNS. Try 'nslookup google.com'. Open browser to any HTTP site — should redirect to captive portal. Check WLC for client state. Check WLAN ACLs.

Scenario 47: Symptom: Slow file transfer over WAN.

Causes: Likely causes: (1) Latency + TCP windowing. (2) Bandwidth saturated. (3) Packet loss. (4) MTU fragmentation. (5) QoS deprioritizing.

Fix: Fix: 'ping' for latency/loss. Try with bigger TCP window (e.g., 'iperf3 -w 1M'). Check WAN interface utilization. Set MTU correctly (avoid fragmentation). Check QoS policies.

Scenario 48: Symptom: After config change, lost SSH access to remote device.

Causes: Likely causes: (1) ACL blocked your IP. (2) VTY config changed. (3) Interface shut down. (4) Routing broken.

Fix: Fix: If you used 'configure replace' before — rollback. Otherwise: console in, 'show running-config | section line vty', check ACLs. Configure 'archive' + 'time-period' for auto-saves next time. Use 'reload in 10' before risky changes — auto-reboot if you don't cancel.

Scenario 49: Symptom: OSPF adjacency stuck in EXSTART/EXCHANGE.

Causes: Likely causes: (1) MTU mismatch. (2) Router ID duplication. (3) Interface flap during initialization.

Fix: Fix: 'show ip ospf neighbor'. Check MTU on both sides. Verify unique RIDs. 'clear ip ospf process'. Try 'ip ospf mtu-ignore' as a workaround.

Scenario 50: Symptom: VLAN deleted but ports still assigned to it.

Causes: Likely causes: (1) Ports were not moved before VLAN deletion. (2) vlan.dat not synced.

Fix: Fix: 'show vlan brief' — VLAN should show 'active' but ports listed are inactive. Reassign ports to active VLAN or recreate VLAN. 'show mac address-table' will show ports as inactive.

Scenario 51: Symptom: BPDU Guard err-disables port.

Causes: Likely causes: (1) Switch/hub plugged into access port. (2) PC running bridge software. (3) Misconfiguration.

Fix: Fix: 'show interfaces status' (err-disabled). 'show spanning-tree interface '. Verify no switch should be on that port. Recover: 'shutdown' + 'no shutdown'. Auto-recovery: 'errdisable recovery cause bpduguard'.

Scenario 52: Symptom: TFTP file transfer fails.

Causes: Likely causes: (1) TFTP server not running. (2) Firewall blocking UDP 69. (3) File doesn't exist. (4) Permission denied. (5) File too large (TFTP has size limit).

Fix: Fix: Verify TFTP server listening (netstat). Check firewalls allow UDP 69. Verify file path. Check file permissions. Use 'copy tftp://10.0.0.5/file.txt flash:' syntax.

Scenario 53: Symptom: Cannot copy IOS image to flash (no space).

Causes: Likely causes: (1) Flash full. (2) Old images taking space.

Fix: Fix: 'show flash:'. Delete old images: 'delete flash:old-image.bin'. Verify space: 'show flash:'. Then 'copy tftp://10.0.0.5/new-image.bin flash:'.

Scenario 54: Symptom: Switch cannot ping its own gateway.

Causes: Likely causes: (1) 'ip default-gateway' not set (L2 switch). (2) SVI down. (3) Routing issue on gateway.

Fix: Fix: For L2 switch: 'ip default-gateway '. 'show ip interface brief' — VLAN interface should be up/up. Check if VLAN has active ports. Verify gateway router has route to switch's subnet.

Chapter 16 Quiz

Q1. Which layer should you troubleshoot first?

A. Layer 1 (Physical)
B. Layer 7 (Application)
C. Layer 3 (Network)
D. Layer 4 (Transport)
Answer: A — Always start at Layer 1 and work up. Most issues are physical or data link.

Q2. What does 'ping' test?

A. Layer 3 reachability
B. DNS resolution
C. Port connectivity
D. Bandwidth
Answer: A — Ping uses ICMP to test Layer 3 reachability and round-trip latency.

Q3. Which command shows OSPF neighbors?

A. show ip ospf neighbor
B. show ospf neighbors
C. show ip ospf
D. show neighbor ospf
Answer: A — show ip ospf neighbor shows all OSPF neighbors and their states.

Q4. PC gets APIPA (169.254.x.x). What's the likely cause?

A. DHCP server unreachable
B. Wrong gateway
C. DNS failure
D. Cable unplugged
Answer: A — 169.254.0.0/16 = APIPA. Client couldn't reach DHCP server.

Q5. Interface shows 'up/down'. What does this mean?

A. Layer 1 up, Layer 2 down (encapsulation, keepalive, clocking issue)
B. Layer 1 down
C. Layer 3 down
D. Hardware failure
Answer: A — Up/down = line protocol down. Layer 1 OK, Layer 2 issue (encapsulation, keepalive, no clock).

PART 17

Packet Tracer Labs

100 hands-on labs from beginner to enterprise

Chapter 17: 100 Hands-On Packet Tracer Labs

Reading about networking is not enough — you must build networks with your own hands. **Cisco Packet Tracer** is a free network simulator that lets you build topologies, configure devices, and test connectivity without buying hardware. Download it from Cisco NetAcad (free account). This chapter contains 100 labs organized by difficulty — start at Lab 1 and work through them in order.

Beginner Labs (1-30): Device Setup & Basic Connectivity

Lab 1: Single Switch, Two PCs

Objective: Connect 2 PCs to 1 switch, configure IPs, ping between them.

Topology: PC0 (192.168.1.10/24) ↔ SW0 ↔ PC1 (192.168.1.11/24)

Tasks: Configure IP on both PCs. Ping PC0 → PC1.

Lab 2: Two Switches, Four PCs

Objective: Connect 2 switches with a trunk, 2 PCs per switch.

Topology: PC0,PC1 ↔ SW0 ↔ SW1 ↔ PC2,PC3

Tasks: All PCs in 192.168.1.0/24. Ping PC0 → PC3.

Lab 3: Add a Router

Objective: Add router between two switches for inter-subnet routing.

Topology: PC0 (192.168.1.x) ↔ SW0 ↔ R1 ↔ SW1 ↔ PC1 (192.168.2.x)

Tasks: Configure router interfaces as gateways. Ping across subnets.

Lab 4: Configure Hostnames and Banners

Objective: Set hostname, banner, and save config on all devices.

Topology: Any 3-device topology

Tasks: hostname SW-Floor1, banner motd, write memory.

Lab 5: Set Console and VTY Passwords

Objective: Configure console and SSH passwords on a switch.

Topology: 1 PC, 1 switch (with console cable)

Tasks: Configure console login, enable SSH, test SSH from PC.

Lab 6: Generate RSA Keys for SSH

Objective: Generate RSA keys and enable SSH v2.

Topology: 1 PC, 1 switch

Tasks: crypto key generate rsa modulus 2048, ip ssh version 2, test SSH.

Lab 7: Create VLANs

Objective: Create 3 VLANs (Sales, IT, HR) and assign ports.

Topology: 1 switch, 3 PCs

Tasks: vlan 10/20/30, switchport access vlan, verify with show vlan brief.

Lab 8: VLAN Trunking

Objective: Configure trunk between 2 switches carrying multiple VLANs.

Topology: 2 switches, 4 PCs

Tasks: switchport mode trunk, switchport trunk allowed vlan.

Lab 9: Native VLAN Configuration

Objective: Configure non-default native VLAN on trunk.

Topology: 2 switches

Tasks: switchport trunk native vlan 99. Verify both ends match.

Lab 10: Inter-VLAN Routing (Router-on-a-Stick)

Objective: Configure router with subinterfaces for inter-VLAN routing.

Topology: 1 router, 1 switch, 3 PCs (different VLANs)

Tasks: encapsulation dot1q, ping between VLANs.

Lab 11: Inter-VLAN Routing (L3 Switch SVI)

Objective: Same goal but using L3 switch with SVIs.

Topology: 1 L3 switch, 2 L2 switches, 4 PCs

Tasks: ip routing, interface vlan X, ip address.

Lab 12: Static Routes

Objective: Configure static routes between 2 routers.

Topology: 2 routers, 2 PCs

Tasks: ip route , verify with show ip route.

Lab 13: Default Route

Objective: Configure default route and verify Internet access.

Topology: 1 router, 1 PC, 'Internet' simulated by another router

Tasks: ip route 0.0.0.0 0.0.0.0, verify with show ip route.

Lab 14: Floating Static Route

Objective: Configure primary + backup routes with different ADs.

Topology: 2 routers with 2 paths between them

Tasks: Primary: ip route X Y Z. Backup: ip route X Y Z 250.

Lab 15: DHCP Server on Router

Objective: Configure router as DHCP server for attached LAN.

Topology: 1 router, 1 switch, 2 PCs

Tasks: ip dhcp pool, network, default-router, dns-server, exclude addresses.

Lab 16: DHCP Relay

Objective: Configure DHCP relay on router to forward requests to remote DHCP server.

Topology: 1 router (relay), 1 DHCP server (different subnet), PCs

Tasks: ip helper-address, verify PC gets IP.

Lab 17: CDP Discovery

Objective: Use CDP to discover directly connected devices.

Topology: Multiple Cisco devices

Tasks: show cdp neighbors, show cdp neighbors detail.

Lab 18: LLDP Discovery

Objective: Enable and verify LLDP.

Topology: Cisco devices + 1 non-Cisco (or just Cisco with LLDP)

Tasks: lldp run, show lldp neighbors.

Lab 19: Port Security — Sticky MAC

Objective: Configure port security with sticky learning.

Topology: 1 switch, 2 PCs

Tasks: switchport port-security, switchport port-security mac-address sticky.

Lab 20: Port Security — Violation Modes

Objective: Test protect, restrict, and shutdown modes.

Topology: 1 switch, 3 PCs

Tasks: Configure each mode, plug in unauthorized MAC, observe behavior.

Lab 21: Standard ACL

Objective: Create standard ACL to permit/deny by source IP.

Topology: 1 router, 2 LANs

Tasks: access-list 10 permit/deny, ip access-group in/out.

Lab 22: Extended ACL

Objective: Create extended ACL for protocol+ports.

Topology: 1 router, web server, PCs

Tasks: access-list 100 permit tcp X Y eq 80, deny tcp X Y eq 23.

Lab 23: Named ACL

Objective: Create named ACL with sequence numbers.

Topology: Same as Lab 22

Tasks: ip access-list extended NAME, sequence numbers.

Lab 24: NAT Overload (PAT)

Objective: Configure NAT overload for Internet access.

Topology: 1 router with inside/outside interfaces, PCs

Tasks: ip nat inside source list 1 interface Gi0/0 overload.

Lab 25: Static NAT

Objective: Configure 1:1 NAT for a server.

Topology: 1 router, 1 server

Tasks: ip nat inside source static local-IP global-IP.

Lab 26: Port Forward (Static PAT)

Objective: Forward external port to internal server.

Topology: 1 router, 1 web server

Tasks: ip nat inside source static tcp local-IP 80 global-IP 80.

Lab 27: NTP Client

Objective: Configure router to sync time from NTP server.

Topology: 1 router, 1 NTP server

Tasks: ntp server X.X.X.X, show ntp status.

Lab 28: Syslog

Objective: Send router logs to syslog server.

Topology: 1 router, 1 syslog server

Tasks: logging X.X.X.X, logging trap informational, show logging.

Lab 29: SNMPv2c

Objective: Configure SNMP for monitoring.

Topology: 1 router, 1 SNMP management station

Tasks: snmp-server community, snmp-server host, snmp-server location.

Lab 30: Backup Config to TFTP

Objective: Save and restore config via TFTP.

Topology: 1 router, 1 TFTP server

Tasks: copy running-config tftp://X/config.txt, copy tftp://X/config.txt running-config.

Intermediate Labs (31-70): Switching, Routing, DHCP

Lab 31: STP Root Bridge Election

Objective: Force specific switch as root via priority.

Topology: 3 switches in triangle

Tasks: spanning-tree vlan 1 priority 24576, verify with show spanning-tree.

Lab 32: STP PortFast and BPDU Guard

Objective: Configure PortFast + BPDU Guard on access ports.

Topology: 1 switch, 2 PCs

Tasks: spanning-tree portfast, spanning-tree bpduguard enable.

Lab 33: Rapid PVST+

Objective: Switch from PVST+ to Rapid PVST+.

Topology: 3 switches

Tasks: spanning-tree mode rapid-pvst, verify convergence time.

Lab 34: EtherChannel LACP

Objective: Bundle 2-4 links with LACP.

Topology: 2 switches, 4 cables

Tasks: channel-group 1 mode active, channel-protocol lacp.

Lab 35: EtherChannel PAgP

Objective: Same with PAgP (Cisco proprietary).

Topology: 2 Cisco switches

Tasks: channel-group 1 mode desirable.

Lab 36: EtherChannel Static (mode on)

Objective: Configure static LAG without negotiation.

Topology: 2 switches

Tasks: channel-group 1 mode on.

Lab 37: Layer 3 EtherChannel

Objective: Configure L3 port-channel for switch-to-router link.

Topology: 1 L3 switch, 1 router

Tasks: no switchport on port-channel, ip address.

Lab 38: SPAN (Port Mirroring)

Objective: Mirror traffic from one port to monitoring port.

Topology: 1 switch, 1 sniffer PC

Tasks: monitor session 1 source interface Fa0/1, monitor session 1 destination interface Fa0/24.

Lab 39: OSPF Single Area

Objective: Configure OSPF area 0 between 3 routers.

Topology: 3 routers in triangle

Tasks: router ospf 1, network X.X.X.X 0.0.0.255 area 0.

Lab 40: OSPF Multi-Area

Objective: Configure OSPF with area 0 backbone + area 1.

Topology: 4 routers in chain

Tasks: ABR has interfaces in both areas.

Lab 41: OSPF Passive Interface

Objective: Configure LAN interfaces as passive.

Topology: OSPF routers

Tasks: passive-interface default, no passive-interface Gi0/0.

Lab 42: OSPF Default Route Injection

Objective: Inject default route into OSPF.

Topology: 1 edge router, 2 internal routers

Tasks: default-information originate on edge.

Lab 43: OSPF Authentication (MD5)

Objective: Enable OSPF MD5 auth on interfaces.

Topology: 2 OSPF routers

Tasks: ip ospf message-digest-key 1 md5 KEY, ip ospf authentication message-digest.

Lab 44: OSPF Cost Tuning

Objective: Adjust OSPF cost to influence path selection.

Topology: 2 paths between 2 routers

Tasks: ip ospf cost X, auto-cost reference-bandwidth 10000.

Lab 45: EIGRP Basic

Objective: Configure EIGRP AS 100 between 3 routers.

Topology: 3 routers

Tasks: router eigrp 100, network X.X.X.X 0.0.0.255, no auto-summary.

Lab 46: EIGRP Authentication

Objective: Configure EIGRP MD5 authentication.

Topology: 2 EIGRP routers

Tasks: key chain, ip authentication mode eigrp 100 md5.

Lab 47: EIGRP Stub

Objective: Configure branch router as EIGRP stub.

Topology: Hub-spoke EIGRP

Tasks: eigrp stub connected summary.

Lab 48: RIPv2

Objective: Configure RIPv2 (for learning purposes).

Topology: 3 routers

Tasks: router rip, version 2, no auto-summary, network.

Lab 49: Route Redistribution (OSPF↔EIGRP)

Objective: Redistribute between OSPF and EIGRP.

Topology: 1 router in both protocols

Tasks: redistribute ospf 1 metric ... under eigrp, redistribute eigrp 100 subnets under ospf.

Lab 50: Floating Static with IP SLA Tracking

Objective: Use IP SLA to track primary, failover to floating static.

Topology: 1 router with 2 paths

Tasks: track 1 ip sla 1, ip route X Y Z track 1.

Lab 51: HSRP Basic

Objective: Configure HSRP between 2 routers.

Topology: 2 routers, 1 switch, PCs

Tasks: standby 1 ip X, standby 1 priority 110, standby 1 preempt.

Lab 52: HSRP with Interface Tracking

Objective: HSRP with upstream link tracking.

Topology: 2 HSRP routers, each with uplink

Tasks: standby 1 track Gi0/0 30.

Lab 53: VRRP

Objective: Configure VRRP (open standard).

Topology: 2 routers

Tasks: vrrp 1 ip X, vrrp 1 priority 110.

Lab 54: DHCP Server with Multiple Pools

Objective: Configure router as DHCP for multiple VLANs.

Topology: 1 router, 3 VLANs

Tasks: ip dhcp pool VLAN10, VLAN20, VLAN30 — different network/default-router per pool.

Lab 55: DHCP Snooping

Objective: Enable DHCP snooping on switches.

Topology: 1 switch, 1 DHCP server, PCs

Tasks: ip dhcp snooping, ip dhcp snooping vlan 10, trust uplink port.

Lab 56: Dynamic ARP Inspection (DAI)

Objective: Enable DAI using DHCP snooping table.

Topology: Same as Lab 55

Tasks: ip arp inspection vlan 10.

Lab 57: IP Source Guard

Objective: Enable IP Source Guard on access ports.

Topology: Same as Lab 55

Tasks: ip verify source on access ports.

Lab 58: 802.1X with RADIUS

Objective: Configure port-based 802.1X authentication.

Topology: 1 switch, 1 RADIUS server, 1 PC with supplicant

Tasks: aaa new-model, aaa authentication dot1x default group radius, dot1x system-auth-control.

Lab 59: Voice VLAN

Objective: Configure voice VLAN for IP phones.

Topology: 1 switch, 1 IP phone, 1 PC

Tasks: switchport voice vlan 110, mls qos trust cos.

Lab 60: Private VLANs (PVLAN)

Objective: Configure isolated/community PVLANs.

Topology: 1 PVLAN-capable switch, multiple PCs

Tasks: vlan 100 private-vlan primary, vlan 101 private-vlan isolated, association.

Lab 61: VTP Configuration

Objective: Configure VTP server/client/transparent.

Topology: 3 switches

Tasks: vtp mode server/client/transparent, vtp domain NAME, vtp password PASS.

Lab 62: VTP Pruning

Objective: Enable VTP pruning to reduce unnecessary traffic.

Topology: VTP domain

Tasks: vtp pruning on server.

Lab 63: IPv6 Static Routing

Objective: Configure IPv6 static routes.

Topology: 2 routers, IPv6 only

Tasks: ipv6 route 2001:db8:X::/64 next-hop.

Lab 64: IPv6 OSPFv3

Objective: Configure OSPFv3 for IPv6.

Topology: 3 routers, IPv6

Tasks: ipv6 router ospf 1, ipv6 ospf 1 area 0 on interfaces.

Lab 65: IPv6 SLAAC

Objective: Verify SLAAC auto-config on hosts.

Topology: 1 IPv6 router, 1 PC

Tasks: Router sends RA, PC auto-configures IPv6.

Lab 66: NAT64

Objective: Configure NAT64 for IPv6-only to reach IPv4 server.

Topology: 1 NAT64 router

Tasks: ipv6 unicast-routing, nat64 prefix, nat64 v6v4 list.

Lab 67: IPv6 DHCP (Stateful)

Objective: Configure stateful DHCPv6.

Topology: 1 router, 1 PC

Tasks: ipv6 dhcp pool, ipv6 nd managed-config-flag.

Lab 68: GRE Tunnel

Objective: Configure GRE tunnel between 2 routers.

Topology: 2 routers with Internet between them

Tasks: interface Tunnel0, tunnel source Gi0/0, tunnel destination, ip address.

Lab 69: Site-to-Site IPsec VPN

Objective: Configure IPsec VPN between 2 sites.

Topology: 2 routers as VPN gateways

Tasks: crypto isakmp policy, crypto ipsec transform-set, crypto map.

Lab 70: Zone-Based Firewall (ZBF)

Objective: Configure Cisco IOS ZBF.

Topology: 1 router with inside/outside zones

Tasks: zone security, zone-pair, policy-map.

Advanced Labs (71-100): Enterprise Design

Lab 71: Multi-Area OSPF with Summarization

Objective: Configure multi-area OSPF with route summarization at ABR.

Topology: 4+ routers

Tasks: area range at ABR.

Lab 72: OSPF Totally Stubby Area

Objective: Configure totally stubby area to reduce LSA flooding.

Topology: ABR + 2 internal routers in stub area

Tasks: area 1 stub no-summary on ABR, area 1 stub on internals.

Lab 73: OSPF NSSA

Objective: Configure NSSA for area that needs external routes.

Topology: ABR + ASBR in same area

Tasks: area 1 nssa on all routers in area, redistribute on ASBR.

Lab 74: OSPF Virtual Link

Objective: Connect disconnected area to backbone via transit area.

Topology: 4 routers in chain

Tasks: area X virtual-link .

Lab 75: BGP eBGP Peering

Objective: Configure eBGP between 2 autonomous systems.

Topology: 2 routers, different ASNs

Tasks: router bgp 65001, neighbor X remote-as 65002, network.

Lab 76: BGP iBGP

Objective: Configure iBGP within an AS.

Topology: 3 routers in same AS

Tasks: router bgp 65001, neighbor Y remote-as 65001.

Lab 77: BGP Route Reflector

Objective: Configure RR to avoid iBGP full mesh.

Topology: 1 RR + 3 RR clients

Tasks: neighbor X route-reflector-client.

Lab 78: BGP AS Path Filtering

Objective: Filter BGP routes by AS path.

Topology: BGP router receiving many routes

Tasks: ip as-path access-list, neighbor X filter-list.

Lab 79: BGP Local Preference

Objective: Influence outbound BGP path selection.

Topology: Router with multiple eBGP paths

Tasks: neighbor X route-map, set local-preference.

Lab 80: BGP MED

Objective: Influence inbound BGP path selection with MED.

Topology: 2 paths into your AS from neighbors

Tasks: set metric in route-map outbound.

Lab 81: BGP Aggregation

Objective: Aggregate BGP prefixes.

Topology: Router with multiple specific prefixes

Tasks: aggregate-address X Y summary-only.

Lab 82: OSPF↔BGP Redistribution

Objective: Redistribute OSPF into BGP and vice versa.

Topology: ASBR router

Tasks: redistribute ospf 1 under router bgp, redistribute bgp under router ospf.

Lab 83: 3-Tier Campus Network

Objective: Build full Access/Distribution/Core topology.

Topology: 9 switches, multiple PCs

Tasks: Configure VLANs, trunks, EtherChannel, STP root placement.

Lab 84: Dual Stack IPv4/IPv6

Objective: Run IPv4 and IPv6 simultaneously.

Topology: 3 routers, PCs

Tasks: Configure both IPv4 and IPv6 on all interfaces, dual routing protocols.

Lab 85: IPv6 Tunnel (6to4)

Objective: Configure 6to4 tunnel over IPv4.

Topology: 2 IPv6 islands, IPv4 between

Tasks: interface Tunnel0, tunnel mode ipv6ip, 6to4 addressing.

Lab 86: QoS — Marking with DSCP

Objective: Mark voice traffic with EF DSCP.

Topology: 1 router, voice traffic

Tasks: class-map voice, match port 16384-32767, policy-map, set dscp ef.

Lab 87: QoS — LLQ for Voice

Objective: Configure LLQ with strict priority for voice.

Topology: 1 router with WAN link

Tasks: priority queue for voice, CBWFQ for other classes.

Lab 88: QoS — Policing

Objective: Police traffic to limit bandwidth.

Topology: 1 router

Tasks: police cir 1000000 bc 1875 conform-action transmit exceed-action drop.

Lab 89: QoS — Shaping

Objective: Shape traffic to smooth bursts.

Topology: 1 router with WAN

Tasks: shape average 1000000.

Lab 90: Multicast Routing (PIM)

Objective: Configure PIM sparse-mode for multicast.

Topology: 3 routers, multicast source + receivers

Tasks: ip multicast-routing, ip pim sparse-mode on interfaces, RP config.

Lab 91: DHCP High Availability

Objective: Configure 2 DHCP servers for redundancy.

Topology: 2 routers as DHCP, 1 switch, PCs

Tasks: Split scope between servers, or use DHCP failover.

Lab 92: BFD for Fast Failure Detection

Objective: Configure BFD with OSPF.

Topology: 2 OSPF routers

Tasks: bfd, ip ospf bfd, sub-second detection.

Lab 93: IP SLA for Tracking

Objective: Use IP SLA to track reachability and failover routes.

Topology: 1 router with primary + backup paths

Tasks: ip sla 1 icmp-echo X frequency 1, track 1 ip sla 1, floating static with track.

Lab 94: NetFlow for Traffic Analysis

Objective: Configure NetFlow to export traffic stats.

Topology: 1 router, 1 NetFlow collector

Tasks: ip flow-export, ip route-cache flow, verify with show ip cache flow.

Lab 95: StackWise Virtual

Objective: Simulate switch stacking (in Packet Tracer, configure dual switches as one).

Topology: 2 stacking-capable switches

Tasks: stack configuration, verify with show switch.

Lab 96: Cisco DNA Center REST API

Objective: Use Postman to query DNA Center API.

Topology: DNA Center simulator

Tasks: GET /dna/intent/api/v1/network-device with auth token.

Lab 97: NETCONF/RESTCONF

Objective: Configure device via NETCONF/RESTCONF.

Topology: IOS-XE device with NETCONF enabled

Tasks: netconf-yang, restconf. Use Python ncclient or curl.

Lab 98: Ansible Network Automation

Objective: Automate config of 5 switches with Ansible.

Topology: Ansible control node, 5 switches

Tasks: Inventory + playbook with cisco.ios.ios_config module.

Lab 99: Python Netmiko Script

Objective: Push config to multiple devices via Netmiko.

Topology: Python host, multiple switches

Tasks: from netmiko import ConnectHandler, loop through devices, send_config_set.

Lab 100: Complete Enterprise Network

Objective: Build full enterprise: 2 sites, OSPF, BGP, VLANs, security, Wi-Fi, automation.

Topology: 10+ devices

Tasks: Multi-day lab integrating all previous labs. Document topology, IP plan, configs.

PART 18

Real-World Enterprise Projects

Five complete network designs from scratch

Chapter 18: Real-World Enterprise Projects

This chapter walks you through five complete enterprise network designs. Each project includes requirements, topology diagram, IP plan, VLAN plan, full configurations, and testing/troubleshooting steps. These projects mirror real-world scenarios you'll encounter as a network engineer. Build them in Packet Tracer or GNS3 — the experience is invaluable.

Project 1: Small Business Network (50 users)

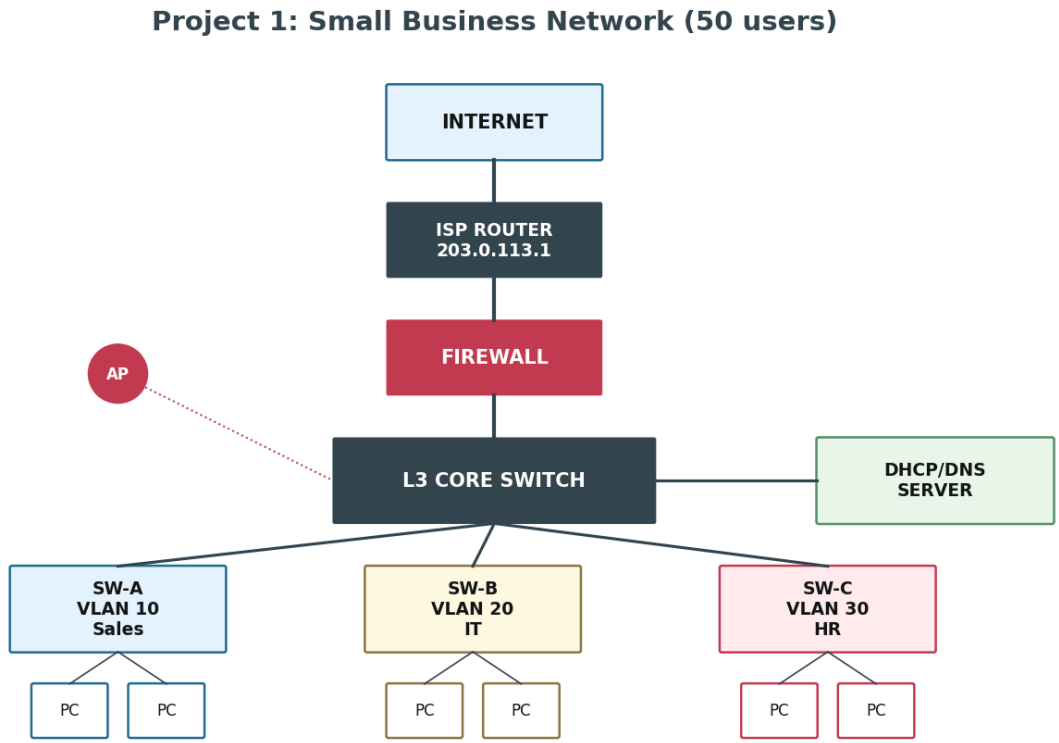


Figure: Project 1: Small business network. Internet → firewall → core switch → access switches with VLANs for Sales, IT, HR.

Requirements:

- 50 users across 3 departments (Sales 25, IT 15, HR 10)
- Internet access via single ISP
- DHCP for automatic IP assignment
- VLAN segmentation per department
- Basic security (firewall, ACLs)
- Wireless for guests (isolated)
- Local file and print server

IP Plan:

VLAN	Subnet	Gateway	DHCP Range	Purpose
10 (Sales)	192.168.10.0/24	192.168.10.1	.50-.99	25 users
20 (IT)	192.168.20.0/24	192.168.20.1	.50-.99	15 users
30 (HR)	192.168.30.0/24	192.168.30.1	.50-.99	10 users
99 (MGMT)	192.168.99.0/24	192.168.99.1	N/A (static)	Switch/router management
100 (GUEST)	192.168.100.0/24	192.168.100.1	.50-.200	Guest Wi-Fi (isolated)
200 (SERVERS)	192.168.200.0/24	192.168.200.1	N/A (static)	File/print server

Configuration (Core L3 Switch):

```
! VLANs and SVIs
vlan 10; name SALES
vlan 20; name IT
vlan 30; name HR
vlan 99; name MGMT
vlan 100; name GUEST
vlan 200; name SERVERS
exit

ip routing

interface vlan 10
  ip address 192.168.10.1 255.255.255.0
  no shutdown
interface vlan 20
  ip address 192.168.20.1 255.255.255.0
  no shutdown
interface vlan 30
  ip address 192.168.30.1 255.255.255.0
  no shutdown
interface vlan 99
  ip address 192.168.99.1 255.255.255.0
  no shutdown
interface vlan 100
  ip address 192.168.100.1 255.255.255.0
  no shutdown
interface vlan 200
  ip address 192.168.200.1 255.255.255.0
  no shutdown

! DHCP pools
ip dhcp excluded-address 192.168.10.1 192.168.10.49
ip dhcp excluded-address 192.168.20.1 192.168.20.49
ip dhcp excluded-address 192.168.30.1 192.168.30.49
ip dhcp excluded-address 192.168.100.1 192.168.100.49

ip dhcp pool SALES
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 192.168.200.10 8.8.8.8
ip dhcp pool IT
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.1
  dns-server 192.168.200.10 8.8.8.8
ip dhcp pool HR
  network 192.168.30.0 255.255.255.0
  default-router 192.168.30.1
  dns-server 192.168.200.10 8.8.8.8
ip dhcp pool GUEST
  network 192.168.100.0 255.255.255.0
  default-router 192.168.100.1
  dns-server 8.8.8.8
  lease 2

! Default route to firewall
ip route 0.0.0.0 0.0.0.0 192.168.99.2

! ACLs - Guest isolation
```

```
ip access-list extended GUEST_ISOLATION
  deny ip 192.168.100.0 0.0.0.255 192.168.10.0 0.0.0.255
  deny ip 192.168.100.0 0.0.0.255 192.168.20.0 0.0.0.255
  deny ip 192.168.100.0 0.0.0.255 192.168.30.0 0.0.0.255
  deny ip 192.168.100.0 0.0.0.255 192.168.200.0 0.0.0.255
  permit ip 192.168.100.0 0.0.0.255 any

interface vlan 100
  ip access-group GUEST_ISOLATION in
```

Testing & Troubleshooting:

1. Verify each PC gets correct IP from DHCP (ipconfig /all).
2. Ping gateway from each VLAN.
3. Ping between VLANs (e.g., Sales PC → file server).
4. Ping Internet (8.8.8.8) — verify NAT on firewall.
5. Connect to guest Wi-Fi, verify Internet works AND cannot ping corporate VLANs.
6. Check 'show ip dhcp binding' for lease assignments.
7. Check 'show ip route' for all directly connected routes.
8. Troubleshoot: if PC can't get IP — verify DHCP pool, 'ip helper-address' if relay needed.

Project 2: School Campus Network

Scenario: Multi-building campus with 5 departments (Admin, Faculty, Student, Library, Guest). 1500 users total. Wireless coverage everywhere. Security: students isolated from admin, internet filtered for students.

Design: 3-tier (Core/Distribution/Access). OSPF area 0 in core, OSPF area 1 per building. WLC + 30 lightweight APs. ISE for 802.1X student/staff authentication. Firewall with URL filtering for student VLAN. VPN for remote staff.

Key configs: Multi-area OSPF, HSRP on distribution switches, EtherChannel uplinks, VLANs per department, WLC + APs, 802.1X with ISE, ACLs to isolate student VLAN from admin, site-to-site VPN for remote access.

Project 3: Hospital Network

Scenario: 500-bed hospital. Critical: patient monitoring (IoT medical devices), electronic health records (EHR), imaging (PACS), guest Wi-Fi. Requirements: high availability (5-nines), HIPAA compliance (audit logs, encryption), strict segmentation.

Design: Fully redundant (dual everything). VLANs: MEDICAL (patient devices), CLINICAL (staff PCs), IMAGING (PACS servers), VOICE (VoIP), MGMT, GUEST. QoS prioritizes medical traffic. MACsec for link encryption. Syslog + NTP for audit. ISE for device profiling (medical devices don't support 802.1X, use MAB).

Key configs: HSRP/VRRP on all gateways, dual ISPs, OSPF with fast convergence (BFD), QoS LLQ for voice and medical, MACsec on uplinks, ISE profiling for medical devices, syslog to central SIEM, NTP stratum 2 source.

Project 4: Corporate Office Infrastructure

Scenario: 5-floor corporate HQ, 1000 employees. Multi-floor wiring, OSPF in core, ACLs for department access, VPN for remote workers, wireless coverage, video conferencing infrastructure.

Design: Each floor = 1 access switch stack + 1 WAP. Distribution switches per wing. Core = 2 L3 switches with HSRP. OSPF area 0 in core, area N per floor. VPN concentrator for remote access. QoS for video conferencing (DSCP AF41).

Key configs: StackWise for access switches, OSPF multi-area, HSRP for gateway redundancy, ACLs per department, IPsec remote-access VPN (AnyConnect), QoS marking on access ports (trust IP phone, re-mark PC), WLC for wireless.

Project 5: ISP-Style Network Simulation

Scenario: Simulate a small ISP with multiple customer routers, OSPF internally, BGP to upstream 'Internet', redundancy throughout.

Design: ISP core = OSPF area 0 with multiple L3 devices. BGP to upstream (eBGP), BGP to customers (eBGP per customer). Route reflectors for iBGP. Customers get /29 or /30 WAN IPs and a /24 public block (simulated). Full redundancy: dual paths, HSRP on customer edge.

Key configs: OSPF for ISP backbone, iBGP with route reflectors, eBGP to upstream and customers, route filtering (prefix-lists), BGP attributes for path selection (local-pref, MED), BFD for fast failure detection, redundant customer edge with HSRP.

Bonus: Lab Diary

Document every project in a lab diary: topology diagram, IP plan, configs, commands used, problems encountered, solutions. This becomes your portfolio for job interviews — show employers what you've built, not just what you know.

PART 19

CCNA Exam Preparation

Blueprint, strategies, and 1000+ practice questions

Chapter 19: CCNA 200-301 Exam Preparation

The CCNA 200-301 is Cisco's current entry-level certification, launched in February 2020. It's a single exam (200-301) covering six domains, lasting 120 minutes, with about 100-110 questions. Pass score is around 825/1000 (Cisco doesn't publish exact). The exam costs $300 USD. This chapter covers the blueprint, exam strategies, time management, common traps, and 1000+ practice questions across all six domains.

CCNA 200-301 Blueprint

Domain	Topic	Weight	Approx. Questions
1	Network Fundamentals	20%	20-22
2	Network Access	20%	20-22
3	IP Connectivity	25%	25-27
4	IP Services	10%	10-11
5	Security Fundamentals	15%	15-16
6	Automation & Programmability	10%	10-11

Exam Strategies

Strategy 1: Master subnetting first. Subnetting appears in many questions across multiple domains. If you can solve subnet problems in seconds (not minutes), you free up time for harder questions. Practice until it's automatic.

Strategy 2: Process of elimination. On multiple-choice questions, eliminate obviously wrong answers first. If you can eliminate 2 of 4 options, your random guess improves from 25% to 50%.

Strategy 3: Read the FULL question. Cisco loves trick questions where one word changes the answer. Read every word, especially 'NOT', 'BEST', 'FIRST', 'LEAST'. Missing these costs easy points.

Strategy 4: Don't get stuck. You have ~70 seconds per question. If a question is taking 3+ minutes, mark it for review, pick your best guess, and move on. Better to miss one hard question than run out of time on 5 easy ones at the end.

Strategy 5: Lab simulations take time. The exam includes drag-and-drop and lab simulation questions. These can eat 5-10 minutes each. Do them carefully but efficiently — they're worth a lot of points.

Time Management

120 minutes / 100 questions = ~70 seconds per question. Plan your time:

- **First pass (90 minutes):** Answer every question, flag the hard ones.
- **Second pass (25 minutes):** Review flagged questions, second-guess carefully.
- **Final (5 minutes):** Review lab/drag-drop answers for silly mistakes.

Common Exam Traps

• **'Best' vs 'first' vs 'most':** Cisco asks 'which is BEST?' — there may be multiple correct answers, pick the most comprehensive.
• **Wildcard vs subnet mask:** ACLs use wildcard (0.0.0.255 for /24). OSPF 'network' command uses wildcard. Don't confuse with subnet mask.
• **'Default' values:** Memorize defaults (STP priority 32768, OSPF hello 10s/dead 40s, HSRP priority 100, etc.).
• **Administrative Distance:** Common trap — 'which route wins?' Need to know AD table cold.
• **Longest prefix match:** Multiple routes match? Longest prefix wins.
• **Implicit deny:** Every ACL ends with 'deny any any'. Don't forget this when predicting ACL behavior.
• **VLAN 1 is default:** Default native VLAN, default management VLAN. Best practice to change but the default is 1.
• **Public vs private IPs:** Memorize RFC 1918 ranges. 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.

Memory Techniques

• **OSI mnemonic:** 'All People Seem To Need Data Processing' (top-down) or 'Please Do Not Throw Sausage Pizza Away' (bottom-up).
• **Subnet mask table:** Write it down on scratch paper as soon as exam starts. /24-/30 with masks, hosts, block sizes.
• **AD table:** Direct=0, Static=1, eBGP=20, EIGRP=90, OSPF=110, RIP=120, iBGP=200.
• **TCP vs UDP:** 'TCP is reliable, UDP is fast.' TCP = web/email/file. UDP = DNS/DHCP/voice/video.
• **Common ports:** SSH=22, Telnet=23, SMTP=25, DNS=53, DHCP=67/68, TFTP=69, HTTP=80, HTTPS=443, RDP=3389.

1000+ Practice Questions

Below are **1317** practice questions across all six CCNA 200-301 domains. Each includes the correct answer and explanation. Work through them in order, marking any you get wrong for re-study.

Domain 1: Network Fundamentals

429 questions

Q1. What is the primary purpose of a computer network?

A. To make computers run faster
B. To share resources and communicate between devices
C. To increase hard drive storage
D. To provide battery backup
Answer: B — Networks exist to share resources (files, printers, internet) and enable communication between devices.

Q2. Which network type covers the smallest geographic area?

A. LAN
B. WAN
C. MAN
D. PAN
Answer: D — PAN (Personal Area Network) covers ~10 meters, smaller than LAN/WAN/MAN.

Q3. What does LAN stand for?

A. Large Access Network
B. Local Area Network
C. Logical Address Node
D. Linked Array Network
Answer: B — LAN = Local Area Network, typically within a single building or campus.

Q4. Which topology connects all devices to a central device like a switch?

A. Bus
B. Ring
C. Star
D. Mesh
Answer: C — In a star topology, all devices connect to a central node (typically a switch).

Q5. Which topology provides the most redundancy by connecting every device to every other?

A. Bus
B. Star
C. Ring
D. Full Mesh
Answer: D — In full mesh, every node connects to every other node — maximum redundancy but expensive.

Q6. What is the unit of data at the Network layer of the OSI model?

A. Bit
B. Frame
C. Packet
D. Segment
Answer: C — Layer 3 (Network) PDU is called a Packet.

Q7. At which OSI layer does a switch primarily operate?

A. Layer 1 - Physical
B. Layer 2 - Data Link
C. Layer 3 - Network
D. Layer 4 - Transport
Answer: B — Switches make forwarding decisions based on MAC addresses, which is a Layer 2 function.

Q8. At which OSI layer does a router operate?

A. Layer 2
B. Layer 3
C. Layer 4
D. Layer 7
Answer: B — Routers make forwarding decisions based on IP addresses, which is a Layer 3 function.

Q9. What is the primary function of the Physical layer (Layer 1)?

A. Routing packets
B. Addressing MAC
C. Transmitting raw bits on the wire
D. Establishing sessions
Answer: C — Layer 1 is responsible for transmitting raw bits over a physical medium.

Q10. How many layers does the OSI model have?

A. 4
B. 5
C. 7
D. 9

Answer: C — The OSI reference model has 7 layers: Application, Presentation, Session, Transport, Network, Data Link, Physical.

Q11. How many layers does the TCP/IP model have?

A. 4
B. 5
C. 6
D. 7

Answer: A — The classic TCP/IP model has 4 layers: Application, Transport, Internet, Network Access.

Q12. Which protocol is connection-oriented?

A. UDP
B. TCP
C. IP
D. ICMP

Answer: B — TCP establishes a connection via 3-way handshake and guarantees delivery. UDP is connectionless.

Q13. What is the well-known port for HTTP?

A. 21
B. 25
C. 80
D. 443

Answer: C — HTTP uses port 80. HTTPS uses port 443.

Q14. What is the well-known port for HTTPS?

A. 80
B. 443
C. 22
D. 53

Answer: B — HTTPS (HTTP over TLS) uses port 443.

Q15. What is the well-known port for SSH?

A. 21
B. 22
C. 23
D. 25

Answer: B — SSH uses port 22. Telnet uses port 23 (insecure, avoid).

Q16. What is the well-known port for DNS?

A. 53
B. 67
C. 68
D. 80

Answer: A — DNS uses port 53 (both TCP for zone transfers and UDP for queries).

Q17. Which protocol uses port 67 and 68?

A. DNS
B. DHCP
C. HTTP
D. SMTP
Answer: B — DHCP server uses port 67, DHCP client uses port 68.

Q18. What is the size of an IPv4 address?

A. 32 bits
B. 64 bits
C. 128 bits
D. 16 bits
Answer: A — IPv4 addresses are 32 bits (4 octets of 8 bits).

Q19. What is the size of an IPv6 address?

A. 32 bits
B. 64 bits
C. 128 bits
D. 256 bits
Answer: C — IPv6 addresses are 128 bits, written as 8 groups of 4 hex digits.

Q20. Which IPv4 address class does 192.168.1.0 belong to?

A. Class A
B. Class B
C. Class C
D. Class D
Answer: C — 192.0.0.0 to 223.255.255.255 is Class C.

Q21. Which range is a private IPv4 address space?

A. 11.0.0.0/8
B. 172.16.0.0/12
C. 201.0.0.0/24
D. 8.8.8.0/24
Answer: B — 172.16.0.0/12 (172.16.0.0 - 172.31.255.255) is private per RFC 1918.

Q22. What is the loopback IPv4 address?

A. 127.0.0.1
B. 0.0.0.0
C. 255.255.255.255
D. 192.168.1.1
Answer: A — 127.0.0.0/8 is the loopback range; 127.0.0.1 is the most common loopback address.

Q23. What is the IPv6 loopback address?

A. ::1
B. ::0
C. FE80::1
D. FF00::1
Answer: A — ::1 is the IPv6 loopback address (equivalent to 127.0.0.1 in IPv4).

Q24. What does the acronym MAC stand for?

A. Machine Access Code
B. Media Access Control
C. Memory Address Cache
D. Main Authentication Channel
Answer: B — MAC = Media Access Control — the unique hardware address burned into a NIC.

Q25. How many bits is a MAC address?

A. 32 bits
B. 48 bits
C. 64 bits
D. 128 bits
Answer: B — MAC addresses are 48 bits (6 bytes), displayed as 12 hex digits like AA:BB:CC:DD:EE:FF.

Q26. Which protocol resolves IP to MAC?

A. DNS
B. ARP
C. DHCP
D. ICMP
Answer: B — ARP (Address Resolution Protocol) maps IP addresses to MAC addresses on a LAN.

Q27. Which protocol resolves MAC to IP?

A. ARP
B. RARP
C. DNS
D. DHCP
Answer: B — RARP (Reverse ARP) resolves MAC to IP. Today DHCP has largely replaced RARP.

Q28. What is the binary value of decimal 192?

A. 10101010
B. 11000000
C. 11111111
D. 10000000
Answer: B — 192 = 128 + 64 = 11000000 in binary.

Q29. What is the decimal value of binary 11111111?

A. 127
B. 192
C. 255
D. 256
Answer: C — 11111111 = 128+64+32+16+8+4+2+1 = 255.

Q30. What is the default subnet mask for a Class B network?

A. 255.0.0.0
B. 255.255.0.0
C. 255.255.255.0
D. 255.255.255.255
Answer: B — Class B default mask is /16 = 255.255.0.0.

Q31. What is the binary representation of /24?

A. 11111111.11111111.00000000.00000000
B. 11111111.11111111.11111111.00000000
C. 11111111.11111111.11111111.11111111
D. 11111111.00000000.00000000.00000000
Answer: B — /24 = first 24 bits are 1s = 255.255.255.0.

Q32. What is the broadcast address for 192.168.1.0/24?

A. 192.168.1.0
B. 192.168.1.1
C. 192.168.1.254
D. 192.168.1.255
Answer: D — Broadcast = all host bits set to 1 = 192.168.1.255.

Q33. How many usable hosts are in a /24?

A. 256
B. 255
C. 254
D. 253
Answer: C — /24 has 256 addresses; minus network and broadcast = 254 usable.

Q34. What is the first usable host IP of 192.168.10.0/24?

A. 192.168.10.0
B. 192.168.10.1
C. 192.168.10.255
D. 192.168.10.254
Answer: B — Network = .0 (reserved), first usable = .1.

Q35. What is the last usable host IP of 10.0.0.0/24?

A. 10.0.0.254
B. 10.0.0.255
C. 10.0.0.0
D. 10.0.0.256
Answer: A — Broadcast = .255 (reserved), last usable = .254.

Q36. What does CIDR stand for?

A. Classless Inter-Domain Routing
B. Classful Internal Domain Routing
C. Centralized IP Domain Resolver
D. Common Internet Distribution Range
Answer: A — CIDR = Classless Inter-Domain Routing. Replaced classful networking; introduced /prefix notation.

Q37. Which subnet mask is equivalent to /26?

A. 255.255.255.128
B. 255.255.255.192
C. 255.255.255.224
D. 255.255.255.240
Answer: B — /26 = 26 ones = 255.255.255.192 (last octet 11000000 = 192).

Q38. How many usable hosts are in a /30?

A. 1
B. 2
C. 4
D. 6
Answer: B — /30 = 4 addresses (2 host bits), minus network/broadcast = 2 usable. Common for point-to-point links.

Q39. What is the wildcard mask for /24?

A. 0.0.0.0
B. 0.0.0.255
C. 0.0.255.255
D. 255.255.255.0
Answer: B — Wildcard = inverse of subnet mask. /24 = 255.255.255.0, wildcard = 0.0.0.255.

Q40. Which command sends 4 ICMP echo requests on Windows?

A. ping -t 4 host
B. ping -n 4 host
C. ping -c 4 host
D. ping 4 host
Answer: B — Windows: ping -n 4. Linux/Mac: ping -c 4.

Q41. Which command uses TTL to discover the path to a destination?

A. ping
B. traceroute
C. arp
D. netstat
Answer: B — Traceroute manipulates TTL values to make each router along the path reveal itself.

Q42. What is the well-known port for Telnet?

A. 21
B. 22
C. 23
D. 25
Answer: C — Telnet uses port 23. It is insecure (sends data in plaintext) and should be replaced with SSH.

Q43. What is the default gateway in a subnet used for?

A. DNS resolution
B. Routing traffic outside the local subnet
C. Assigning IP addresses
D. Resolving MAC addresses
Answer: B — The default gateway is the router interface that forwards traffic destined for other subnets.

Q44. What protocol does ping use?

A. TCP
B. UDP
C. ICMP
D. ARP
Answer: C — Ping uses ICMP Echo Request/Reply messages.

Q45. What is the multicast address range in IPv4?

A. 127.0.0.0/8
B. 169.254.0.0/16
C. 224.0.0.0/4
D. 192.168.0.0/16
Answer: C — Class D multicast = 224.0.0.0 to 239.255.255.255.

Q46. What is the APIPA address range?

A. 10.0.0.0/8
B. 169.254.0.0/16
C. 172.16.0.0/12
D. 192.168.0.0/16
Answer: B — APIPA (Automatic Private IP Addressing) uses 169.254.0.0/16 when DHCP fails.

Q47. What is the size of an Ethernet frame (excluding preamble)?

A. 64-1518 bytes
B. 32-1024 bytes
C. 46-1500 bytes
D. 60-9000 bytes
Answer: A — Standard Ethernet frame is 64-1518 bytes (data field 46-1500). Jumbo frames can be larger.

Q48. What is the MTU of standard Ethernet?

A. 1500 bytes
B. 1518 bytes
C. 9000 bytes
D. 65535 bytes
Answer: A — Standard Ethernet MTU = 1500 bytes (the data field, not the whole frame).

Q49. Which address is a Layer 2 broadcast?

A. 255.255.255.255
B. FF:FF:FF:FF:FF:FF
C. 0.0.0.0
D. 00:00:00:00:00:00
Answer: B — FF:FF:FF:FF:FF:FF is the Layer 2 broadcast MAC address.

Q50. Which IPv4 address is the limited broadcast (sent to all hosts on local subnet)?

A. 0.0.0.0
B. 127.0.0.1
C. 255.255.255.255
D. 224.0.0.1
Answer: C — 255.255.255.255 is the limited broadcast — not forwarded by routers.

Q51. Which type of cable is used to connect a PC to a switch?

A. Straight-through
B. Crossover
C. Rollover
D. Console
Answer: A — PC-to-switch uses a straight-through cable. Switch-to-switch or PC-to-PC uses crossover (though modern NICs auto-MDIX).

Q52. What is the purpose of a console cable?

A. To connect two switches

B. To connect PC to router/switch for out-of-band management

C. To connect PC to internet

D. To connect router to modem

Answer: B — Console (rollover) cable is used for initial out-of-band configuration of Cisco devices.

Q53. What is the speed of standard FastEthernet?

A. 10 Mbps

B. 100 Mbps

C. 1000 Mbps

D. 10000 Mbps

Answer: B — FastEthernet = 100 Mbps. GigabitEthernet = 1000 Mbps (1 Gbps).

Q54. What does Mbps stand for?

A. Megabits per second

B. Megabytes per second

C. Million packets per second

D. Microseconds per bit

Answer: A — Mbps = Megabits per second (1 megabit = 1,000,000 bits). 8 bits = 1 byte.

Q55. Which fiber connector is most commonly used in modern networks?

A. LC

B. RJ-45

C. SC

D. BNC

Answer: A — LC (Lucent Connector) is the most common small-form fiber connector today.

Q56. What is the maximum distance for Cat6 copper cable?

A. 100 meters

B. 200 meters

C. 500 meters

D. 1000 meters

Answer: A — Cat5e/Cat6 copper cable supports up to 100 meters (328 feet) for Ethernet.

Q57. What is the difference between bandwidth and throughput?

A. Bandwidth = max capacity; Throughput = actual data transferred per second

B. Bandwidth = speed; Throughput = latency

C. They are the same thing

D. Bandwidth is physical, throughput is logical

Answer: A — Bandwidth is the theoretical maximum; throughput is what's actually achieved due to overhead, collisions, etc.

Q58. What does latency measure?

A. Speed of data transfer

B. Time delay for data to travel from source to destination

C. Number of packets per second

D. Total bandwidth used

Answer: B — Latency = delay (typically measured in ms). Lower = better.

Q59. Which Wi-Fi standard operates at 6 GHz?

A. 802.11ac

B. 802.11ax (Wi-Fi 6E)

C. 802.11n

D. 802.11g

Answer: B — Wi-Fi 6E (802.11ax extended) adds the 6 GHz band. Wi-Fi 7 (802.11be) also uses 6 GHz.

Q60. Which Wi-Fi standard introduced MIMO?

A. 802.11a

B. 802.11b

C. 802.11g

D. 802.11n

Answer: D — 802.11n (Wi-Fi 4) introduced MIMO — multiple input, multiple output antennas.

Q61. What does CSMA/CD stand for?

A. Carrier Sense Multiple Access with Collision Detection

B. Channel Sharing Multiple Access with Collision Domain

C. Carrier Sense Multi-Access Collision Domain

D. Common System Multiple Access Collision Detection

Answer: A — CSMA/CD is the protocol used by Ethernet to detect and handle collisions on shared media.

Q62. What is a collision domain?

A. A group of devices that share the same IP subnet

B. A network segment where collisions can occur if two devices transmit simultaneously

C. A group of devices connected by a router

D. A VLAN with multiple switches

Answer: B — Collision domain = a segment where collisions are possible. Each switch port = its own collision domain.

Q63. What is a broadcast domain?

A. A group of devices that receive the same broadcast frames

B. A group of routers

C. A single switch port

D. A fiber-optic segment

Answer: A — Broadcast domain = set of devices that receive a broadcast frame. Routers divide broadcast domains; switches extend them.

Q64. How many collision domains does a 24-port switch have?

A. 1

B. 12

C. 24

D. 48

Answer: C — Each switch port is its own collision domain → 24 ports = 24 collision domains.

Q65. How many broadcast domains does a single switch with no VLANs have?

A. 1

B. Depends on ports

C. Equal to port count

D. 0

Answer: A — Without VLANs, a switch is one broadcast domain.

Q66. Which device operates at Layer 4 of the OSI model?

A. Hub
B. Switch
C. Router
D. Layer 4-7 load balancer
Answer: D — Load balancers and firewalls can operate at Layer 4+ (using TCP/UDP ports).

Q67. What is the purpose of a hub?

A. To route packets
B. To repeat signals to all ports (multi-port repeater)
C. To learn MAC addresses
D. To filter broadcast traffic
Answer: B — A hub is a Layer 1 device — it just repeats incoming signals to all ports.

Q68. Which device extends a collision domain?

A. Switch
B. Router
C. Hub
D. Firewall
Answer: C — Hubs extend collision domains because all ports share one. Switches break them.

Q69. What is the difference between TCP and UDP?

A. TCP is connectionless, UDP is connection-oriented
B. TCP is reliable and ordered, UDP is best-effort and unordered
C. TCP is faster than UDP
D. UDP is more secure than TCP
Answer: B — TCP is reliable (uses ACKs, sequence numbers), UDP is fire-and-forget (faster but unreliable).

Q70. Which transport protocol is best for live video streaming?

A. TCP
B. UDP
C. ICMP
D. ARP
Answer: B — UDP is preferred for live streaming — lower latency, can tolerate some packet loss.

Q71. Which transport protocol is best for file downloads?

A. TCP
B. UDP
C. ICMP
D. ARP
Answer: A — TCP is preferred for file downloads — reliability and ordering matter.

Q72. What is multiplexing in TCP/IP?

A. Using multiple routers for redundancy
B. Using port numbers to distinguish multiple connections on the same IP
C. Combining multiple VLANs on one trunk
D. Using multiple DNS servers
Answer: B — Multiplexing uses port numbers (source + dest) to track many connections over one IP.

Q73. What is the well-known port range?

A. 0-1023
B. 1024-49151
C. 49152-65535
D. 0-65535
Answer: A — Well-known ports = 0-1023. Registered = 1024-49151. Ephemeral/dynamic = 49152-65535.

Q74. What is the ephemeral port range?

A. 0-1023
B. 1024-49151
C. 49152-65535
D. 0-65535
Answer: C — Ephemeral (dynamic/source) ports are 49152-65535 per IANA.

Q75. What does a socket consist of?

A. IP address only
B. Port number only
C. IP address + port number
D. MAC address + IP address
Answer: C — A socket = (IP address, port number). For TCP, also includes protocol.

Q76. What is a TCP flag used to begin a connection?

A. FIN
B. SYN
C. ACK
D. RST
Answer: B — SYN (synchronize) flag initiates a TCP connection.

Q77. What is a TCP flag used to gracefully close a connection?

A. SYN
B. FIN
C. RST
D. PSH
Answer: B — FIN (finish) flag is sent to gracefully close a TCP connection.

Q78. What is a TCP flag used to abruptly reset a connection?

A. SYN
B. FIN
C. RST
D. URG
Answer: C — RST (reset) flag abruptly terminates a TCP connection.

Q79. How many segments are exchanged in a TCP 3-way handshake?

A. 2
B. 3
C. 4
D. 5
Answer: B — SYN, SYN-ACK, ACK = 3 segments. Connection is then established.

Q80. Which IPv4 address is the network address of 10.1.1.0/24?

A. 10.1.1.0
B. 10.1.1.1
C. 10.1.1.254
D. 10.1.1.255
Answer: A — Network address = all host bits set to 0 = 10.1.1.0.

Q81. How many host bits are in a /28?

A. 2
B. 3
C. 4
D. 5
Answer: C — /28 = 28 network bits, so 32-28 = 4 host bits → 2^4 - 2 = 14 usable hosts.

Q82. Which IPv4 range is reserved for documentation (RFC 5737)?

A. 10.0.0.0/8
B. 172.16.0.0/12
C. 192.168.0.0/16
D. 203.0.113.0/24
Answer: D — 203.0.113.0/24 (TEST-NET-3) is reserved for documentation. Also 192.0.2.0/24 and 198.51.100.0/24.

Q83. What is the size of an ARP packet?

A. 28 bytes
B. 32 bytes
C. 64 bytes
D. Variable
Answer: A — ARP packet = 28 bytes (Ethernet payload). It does not have an IP header.

Q84. What is the destination MAC in an ARP request?

A. Source MAC
B. FF:FF:FF:FF:FF:FF (broadcast)
C. 00:00:00:00:00:00
D. Destination host MAC
Answer: B — ARP request is a broadcast — destination MAC is FF:FF:FF:FF:FF:FF.

Q85. What is the destination MAC in an ARP reply?

A. FF:FF:FF:FF:FF:FF (broadcast)
B. The original sender's MAC (unicast)
C. 00:00:00:00:00:00
D. The router's MAC
Answer: B — ARP reply is unicast — sent directly back to the original requester's MAC.

Q86. How long are ARP cache entries typically kept?

A. 4 seconds
B. 4 minutes
C. 4 hours
D. 4 days
Answer: C — Cisco devices default to 4 hours for ARP cache entries. Windows defaults to 2-10 minutes.

Q87. What does the command 'arp -a' do on Windows?

A. Adds an ARP entry

B. Displays the ARP cache

C. Deletes ARP cache

D. Sends ARP request

Answer: B — arp -a displays the current ARP cache table.

Q88. Which protocol is used for secure remote CLI access to Cisco devices?

A. Telnet

B. SSH

C. HTTP

D. FTP

Answer: B — SSH (port 22) is the secure alternative to Telnet (port 23).

Q89. Which command on Linux shows the IP configuration of all interfaces?

A. ipconfig /all

B. ifconfig -a or ip addr

C. show ip interface

D. netstat -a

Answer: B — Linux uses ifconfig or ip addr (newer). Windows uses ipconfig /all.

Q90. What is the maximum length of an IPv4 packet?

A. 1500 bytes

B. 65535 bytes

C. 9000 bytes

D. Unlimited

Answer: B — The 16-bit Total Length field in the IPv4 header allows up to 65535 bytes.

Q91. What is jumbo frame MTU?

A. 1500 bytes

B. 9000 bytes (typically)

C. 65535 bytes

D. 256 bytes

Answer: B — Jumbo frames typically use 9000 bytes MTU for better throughput in data centers.

Q92. What does the 'TTL' field in an IP packet do?

A. Limits the number of routers a packet can pass through

B. Limits the size of the packet

C. Sets the priority of the packet

D. Identifies the application

Answer: A — TTL (Time To Live) is decremented by 1 at each router; if it reaches 0, the packet is dropped (prevents loops).

Q93. What is the default TTL on Windows?

A. 32

B. 64

C. 128

D. 255

Answer: C — Windows defaults TTL to 128. Linux/macOS defaults to 64. Cisco routers default to 255.

Q94. What is the protocol field value in IPv4 header for ICMP?

A. 1
B. 6
C. 17
D. 89
Answer: A — ICMP = protocol 1. TCP = 6. UDP = 17. OSPF = 89.

Q95. What is the protocol field value in IPv4 header for TCP?

A. 1
B. 6
C. 17
D. 47
Answer: B — TCP = protocol 6 in the IPv4 header.

Q96. What is the protocol field value in IPv4 header for UDP?

A. 1
B. 6
C. 17
D. 50
Answer: C — UDP = protocol 17 in the IPv4 header.

Q97. What is the protocol field value in IPv4 header for OSPF?

A. 6
B. 17
C. 89
D. 92
Answer: C — OSPF = protocol 89. It runs directly on IP, not over TCP/UDP.

Q98. What does Wireshark do?

A. Captures and analyzes network packets
B. Generates network traffic
C. Configures routers
D. Monitors bandwidth
Answer: A — Wireshark is the world's most popular network packet analyzer.

Q99. What is the OSI layer where encryption typically occurs?

A. Layer 1
B. Layer 2
C. Layer 6 (Presentation)
D. Layer 7
Answer: C — Encryption is a Presentation layer (L6) function, though modern protocols like TLS span L4-L7.

Q100. What is the OSI layer where SSL/TLS operate?

A. L4-L7 (between TCP and application)
B. L2
C. L3
D. L1
Answer: A — TLS sits between TCP (L4) and the application (L7), often called L6 or L5/6.

Q101. What is encapsulation?

A. Adding headers/trailers as data moves down the OSI layers

B. Removing headers as data moves up

C. Encrypting data

D. Compressing data

Answer: A — Encapsulation = adding protocol headers/trailers at each layer as data flows down the stack.

Q102. What is decapsulation?

A. Adding headers as data moves down

B. Removing headers as data moves up the OSI layers

C. Encrypting data

D. Routing packets

Answer: B — Decapsulation = stripping protocol headers at each layer as data flows up the stack at the destination.

Q103. Which PDU is at the Transport layer?

A. Frame

B. Packet

C. Segment

D. Bit

Answer: C — Transport layer PDU = Segment (TCP) or Datagram (UDP).

Q104. Which PDU is at the Data Link layer?

A. Bit

B. Frame

C. Packet

D. Segment

Answer: B — Data Link layer PDU = Frame (Ethernet frame includes MAC addresses + data + FCS).

Q105. What is the FCS in an Ethernet frame?

A. Frame Check Sequence — error detection (CRC32)

B. Forwarding Control Sequence — routing decisions

C. Frame Compression Standard

D. Fast Circuit Switching

Answer: A — FCS = Frame Check Sequence, a 4-byte CRC used to detect corrupted frames.

Q106. Which field in an Ethernet frame contains the protocol of the payload?

A. Source MAC

B. Destination MAC

C. EtherType / Length

D. FCS

Answer: C — EtherType field identifies the upper-layer protocol (e.g., 0x0800 = IPv4, 0x86DD = IPv6, 0x0806 = ARP).

Q107. What is the EtherType value for IPv4?

A. 0x0800

B. 0x0806

C. 0x86DD

D. 0x8100

Answer: A — 0x0800 = IPv4. 0x0806 = ARP. 0x86DD = IPv6. 0x8100 = 802.1Q VLAN tag.

Q108. What is the EtherType value for ARP?

A. 0x0800
B. 0x0806
C. 0x86DD
D. 0x8100
Answer: B — 0x0806 = ARP.

Q109. What is the EtherType value for IPv6?

A. 0x0800
B. 0x0806
C. 0x86DD
D. 0x8100
Answer: C — 0x86DD = IPv6.

Q110. What is the EtherType value for an 802.1Q tagged frame?

A. 0x0800
B. 0x0806
C. 0x86DD
D. 0x8100
Answer: D — 0x8100 indicates an 802.1Q VLAN tag follows.

Q111. Which fiber type is used for long-distance (10+ km)?

A. Single-mode fiber (SMF)
B. Multi-mode fiber (MMF)
C. Cat6 copper
D. Coaxial
Answer: A — Single-mode fiber has a small core (~9 micron) and uses laser light — supports long distances.

Q112. Which fiber type is used for short-distance (within a data center)?

A. Single-mode fiber
B. Multi-mode fiber (MMF)
C. Coaxial
D. Twisted pair
Answer: B — Multi-mode fiber has a larger core (~50/62.5 micron) and uses LED light — cheaper, shorter distances.

Q113. What does the 'SVI' abbreviation stand for in Cisco switching?

A. Switch Virtual Interface
B. Single VLAN Interconnect
C. Standard Virtual Interconnect
D. Switch VLAN Interface
Answer: A — SVI = Switch Virtual Interface — a Layer 3 interface representing a VLAN, used for inter-VLAN routing and switch management.

Q114. What is a routed port on a switch?

A. A port that has been shutdown
B. A physical port configured as a Layer 3 interface (no switchport)
C. A port assigned to a single VLAN
D. A port on a Layer 2 switch
Answer: B — Routed port: configured with 'no switchport' — behaves like a router interface, not a switch port.

Q115. What command on Cisco disables a port?

A. disable

B. shutdown

C. no use

D. stop

Answer: B — shutdown disables the port administratively. Use 'no shutdown' to enable it.

Q116. What command on Cisco enables a port?

A. enable

B. no shutdown

C. start

D. active

Answer: B — no shutdown (often abbreviated 'no shut') brings a port up.

Q117. What does the 'duplex' setting control?

A. Speed of the link

B. Whether data can flow in one or both directions simultaneously

C. VLAN assignment

D. MAC address learning

Answer: B — Half-duplex = one direction at a time (like a walkie-talkie). Full-duplex = both directions simultaneously (like a phone call).

Q118. What happens if there is a duplex mismatch between two devices?

A. No data flows

B. Slow performance and CRC errors

C. Bandwidth doubles

D. Nothing — modern devices auto-negotiate

Answer: B — Duplex mismatch causes collisions on the full-duplex side and CRC errors on the half-duplex side — slow performance.

Q119. What is auto-MDIX?

A. Automatically detects cable type (straight/crossover) and adjusts

B. Auto-negotiates speed only

C. Auto-detects VLANs

D. Auto-assigns IP addresses

Answer: A — Auto-MDIX automatically detects and adjusts for straight-through vs crossover cables.

Q120. What is the Cisco IOS command to view the running configuration?

A. show run

B. show startup-config

C. show config

D. show ios

Answer: A — show running-config (abbreviated 'show run') displays the current active configuration in RAM.

Q121. What is the Cisco IOS command to view the startup configuration?

A. show run

B. show startup-config

C. show nvram

D. show saved

Answer: B — show startup-config (abbreviated 'show start') displays the saved configuration in NVRAM.

Q122. Where is the running-config stored?

A. NVRAM
B. RAM
C. Flash
D. TFTP server
Answer: B — Running-config lives in RAM — lost on reboot. Startup-config lives in NVRAM — persistent.

Q123. Where is the startup-config stored?

A. RAM
B. NVRAM
C. Flash
D. ROM
Answer: B — Startup-config is in NVRAM — survives reboots and power cycles.

Q124. What command saves the running configuration to startup-config?

A. copy run start
B. save config
C. write nvram
D. commit
Answer: A — copy running-config startup-config (or 'write memory' / 'wr') saves the config.

Q125. Which Cisco IOS mode is indicated by the '>' prompt?

A. User EXEC mode
B. Privileged EXEC mode
C. Global Configuration mode
D. Interface Configuration mode
Answer: A — Router> = User EXEC mode (limited commands). Router# = Privileged EXEC.

Q126. Which Cisco IOS mode is indicated by the '#' prompt?

A. User EXEC
B. Privileged EXEC
C. Global Config
D. Setup mode
Answer: B — Router# = Privileged EXEC mode (full access to view/configure).

Q127. Which Cisco IOS mode is indicated by '(config)#'?

A. User EXEC
B. Privileged EXEC
C. Global Configuration
D. Interface Config
Answer: C — Router(config)# = Global Configuration mode — changes affect the whole device.

Q128. What command moves you from User EXEC to Privileged EXEC?

A. enable
B. configure terminal
C. privilege
D. admin
Answer: A — Typing 'enable' moves you from '>' (User EXEC) to '#' (Privileged EXEC).

Q129. What command moves you from Privileged EXEC to Global Configuration?

A. enable
B. configure terminal
C. edit
D. global
Answer: B — configure terminal (abbreviated 'conf t') enters Global Configuration mode.

Q130. What is the keyboard shortcut to move back one mode in Cisco IOS?

A. Ctrl+C
B. Ctrl+Z
C. exit
D. Both exit and Ctrl+Z (different effects)
Answer: D — exit moves you back one level. Ctrl+Z jumps all the way back to Privileged EXEC.

Q131. What command displays interface status and IP address?

A. show ip interface brief
B. show interfaces
C. show ip
D. show status
Answer: A — show ip interface brief (abbreviated 'sh ip int br') gives a concise overview of all interfaces.

Q132. What is the shortcut to abort a DNS lookup in Cisco IOS?

A. Ctrl+C
B. Ctrl+Shift+6
C. Esc
D. Ctrl+Z
Answer: B — Ctrl+Shift+6 (often written as ^^) aborts DNS lookups, traceroute, ping, etc.

Q133. Which command configures an IP address on a Cisco interface?

A. ip address 192.168.1.1 255.255.255.0
B. ip 192.168.1.1/24
C. set ip 192.168.1.1 255.255.255.0
D. interface ip 192.168.1.1
Answer: A — ip address is the correct syntax (mask in dotted-decimal, not CIDR).

Q134. Which command sets a default gateway on a Cisco switch?

A. ip default-gateway 192.168.1.1
B. default-gateway 192.168.1.1
C. ip route 0.0.0.0 0.0.0.0 192.168.1.1
D. gateway 192.168.1.1
Answer: A — On a Layer 2 switch, use 'ip default-gateway'. On a Layer 3 switch/router, use 'ip route 0.0.0.0 0.0.0.0 ...'.

Q135. What is the default subnet mask for 10.0.0.0 (Class A)?

A. 255.0.0.0 (/8)
B. 255.255.0.0 (/16)
C. 255.255.255.0 (/24)
D. 255.255.255.255 (/32)
Answer: A — Class A default = /8 = 255.0.0.0.

Q136. What is the default subnet mask for 172.16.0.0 (Class B)?

A. 255.0.0.0
B. 255.255.0.0
C. 255.255.255.0
D. 255.255.255.128
Answer: B — Class B default = /16 = 255.255.0.0.

Q137. What is the IPv6 link-local address prefix?

A. 2000::/3
B. FE80::/10
C. FF00::/8
D. FC00::/7
Answer: B — FE80::/10 is link-local. Auto-configured on every IPv6 interface; not routed.

Q138. What is the IPv6 multicast address prefix?

A. FE80::/10
B. FF00::/8
C. 2000::/3
D. ::/0
Answer: B — FF00::/8 is IPv6 multicast.

Q139. What is the IPv6 unique local address (ULA) range?

A. FE80::/10
B. FF00::/8
C. FC00::/7
D. 2000::/3
Answer: C — FC00::/7 is ULA (private IPv6, similar to RFC1918 IPv4).

Q140. What is the IPv6 global unicast address range?

A. FE80::/10
B. FF00::/8
C. FC00::/7
D. 2000::/3
Answer: D — 2000::/3 is global unicast — routable on the IPv6 Internet.

Q141. How is the IPv6 address ::1 best described?

A. Default route
B. Loopback address
C. Multicast
D. Broadcast
Answer: B — ::1/128 is the IPv6 loopback address (equivalent to IPv4 127.0.0.1).

Q142. How is the IPv6 address :: best described?

A. Loopback
B. Unspecified address (like 0.0.0.0 in IPv4)
C. Default route
D. Broadcast
Answer: B — ::/128 is the unspecified address. ::/0 is the default route.

Q143. What does SLAAC stand for?

A. StateLess Address AutoConfiguration

B. Stateful Local Address Authentication Code

C. Single Link Auto Configuration

D. Standard Local Area Connection

Answer: A — SLAAC = Stateless Address AutoConfiguration. Allows IPv6 hosts to auto-configure their own addresses using Router Advertisements.

Q144. What protocol does SLAAC rely on?

A. DHCPv6

B. ICMPv6 Router Advertisement

C. ARP

D. DNS

Answer: B — SLAAC uses ICMPv6 Router Advertisement (RA) messages from routers.

Q145. What is EUI-64?

A. A method to derive the 64-bit interface ID from the 48-bit MAC address

B. A 64-bit encryption key

C. A new IPv6 header format

D. A Cisco proprietary protocol

Answer: A — EUI-64 inserts FFFE in the middle of the MAC and flips the U/L bit to create a 64-bit interface ID.

Q146. How is the IPv6 address 2001:0db8:0000:0000:0000:0000:0000:0001 shortened?

A. 2001:db8::1

B. 2001:db8:0:0:0:0:0:1 (already shortest)

C. 2001:db8:::1

D. 2001:db8:0::1

Answer: A — Leading zeros can be removed, and one run of all-zero groups can be replaced with '::'.

Q147. What does the double colon '::' represent in IPv6?

A. One or more consecutive groups of zeros (used once per address)

B. Multiple addresses

C. A wildcard

D. The end of the address

Answer: A — :: replaces one or more consecutive all-zero groups. Can only appear once per address.

Q148. What is the size of an IPv6 header (base)?

A. 20 bytes

B. 40 bytes

C. 60 bytes

D. Variable

Answer: B — IPv6 base header is 40 bytes (fixed). Extension headers can be added.

Q149. Does IPv6 use ARP?

A. Yes, same as IPv4

B. No — uses ICMPv6 Neighbor Discovery instead

C. Yes, but only for /64 subnets

D. Only on Wi-Fi

Answer: B — IPv6 does NOT use ARP. It uses ICMPv6 Neighbor Discovery Protocol (NDP) for MAC resolution.

Q150. Which ICMPv6 message is used to resolve MAC addresses in IPv6?

A. Router Advertisement

B. Neighbor Solicitation

C. Echo Request

D. Destination Unreachable

Answer: B — Neighbor Solicitation (NS) — equivalent to ARP Request. Neighbor Advertisement (NA) is the reply.

Q151. What is the destination address for an ICMPv6 Neighbor Solicitation?

A. FF:FF:FF:FF:FF:FF

B. Solicited-node multicast address

C. ::1

D. FF02::1

Answer: B — IPv6 uses solicited-node multicast (FF02::1:FFxx:xxxx) instead of broadcast.

Q152. What is the IPv6 'all-nodes' multicast address?

A. FF02::1

B. FF02::2

C. FF02::5

D. FF02::9

Answer: A — FF02::1 = all nodes on the link. FF02::2 = all routers.

Q153. What is the IPv6 'all-routers' multicast address?

A. FF02::1

B. FF02::2

C. FF02::5

D. FF02::9

Answer: B — FF02::2 = all routers on the link.

Q154. What is the OSPFv3 multicast address (all routers)?

A. FF02::1

B. FF02::2

C. FF02::5

D. FF02::9

Answer: C — FF02::5 = all OSPF routers. FF02::6 = OSPF DRs.

Q155. Which command shows the MAC address table on a Cisco switch?

A. show mac

B. show mac-address-table

C. show cam

D. show arp

Answer: B — show mac-address-table displays the Layer 2 forwarding table (CAM table).

Q156. What is the CAM table on a Cisco switch?

A. Content-Addressable Memory — the MAC address table

B. Cisco Address Map — IP routing table

C. Configuration And Management

D. Connected Active Modules

Answer: A — CAM = Content-Addressable Memory. It's the MAC address table used for Layer 2 forwarding.

Q157. How does a switch handle a frame with an unknown destination MAC?

A. Drops it

B. Floods it out all ports except the port it arrived on

C. Sends it to the router

D. Sends an ICMP error

Answer: B — Unknown unicast flooding — switch sends the frame out all active ports except the source, hoping to learn the destination.

Q158. What does a switch do when it receives a broadcast frame?

A. Drops it

B. Floods it out all ports except the source

C. Sends it to the CPU only

D. Sends it to the router

Answer: B — Broadcast frames are flooded out all ports in the same VLAN except the source port.

Q159. How does a switch learn MAC addresses?

A. By examining destination MAC of incoming frames

B. By examining source MAC of incoming frames

C. By querying the router

D. Manually configured only

Answer: B — Switch learns by examining the SOURCE MAC of each incoming frame and associating it with the ingress port.

Q160. What happens to a MAC table entry when no traffic is seen from that MAC for 5 minutes?

A. It is permanently stored

B. It is aged out and removed

C. It is moved to a backup table

D. It is broadcast to other switches

Answer: B — Default MAC age time is 5 minutes (300 seconds) on Cisco switches. Entry is removed if no traffic is seen.

Q161. Which port security violation mode drops violating frames silently without shutting the port?

A. protect

B. restrict

C. shutdown

D. disable

Answer: A — protect = silently drop. restrict = drop + log/SNMP trap. shutdown = err-disable port (default).

Q162. Which port security violation mode logs violations and increments a counter?

A. protect

B. restrict

C. shutdown

D. trap

Answer: B — restrict drops frames, logs violations, and sends SNMP traps. Port stays up.

Q163. What is the default port security violation mode?

A. protect

B. restrict

C. shutdown

D. disable

Answer: C — Default = shutdown. Port goes to err-disabled state and must be manually recovered (or via errdisable recovery).

Q164. What does STP stand for?

A. Standard Tree Protocol

B. Spanning Tree Protocol

C. Switched Trunk Protocol

D. Source Transfer Protocol

Answer: B — STP = Spanning Tree Protocol (IEEE 802.1D). Prevents Layer 2 loops.

Q165. What is the purpose of STP?

A. Encrypt Layer 2 traffic

B. Prevent Layer 2 loops in redundant networks

C. Speed up convergence

D. Allow VLAN tagging

Answer: B — STP prevents loops by blocking redundant paths, then re-activates them if the primary path fails.

Q166. What is the default STP bridge priority?

A. 0

B. 1

C. 32768

D. 65535

Answer: C — Default priority = 32768 (with VLAN ID added, e.g., 32768+10 for VLAN 10).

Q167. How is the root bridge elected in STP?

A. Lowest priority, then lowest MAC address

B. Highest priority, then highest MAC address

C. Highest port count

D. Fastest CPU

Answer: A — Lowest bridge ID = (priority, MAC). Lowest priority wins; ties broken by lowest MAC.

Q168. What is the STP port state after convergence?

A. Listening

B. Learning

C. Forwarding or Blocking

D. Disabled

Answer: C — After convergence, ports are either Forwarding (active) or Blocking (redundant backup).

Q169. What is the STP port cost for a 1 Gbps link (new 802.1D-1998 standard)?

A. 1

B. 4

C. 19

D. 100

Answer: B — 1 Gbps = cost 4 (10 Gbps = 2, 100 Mbps = 19, 10 Mbps = 100).

Q170. Which port role in STP is on the non-root switch with the lowest cost to root?

A. Designated Port
B. Root Port
C. Blocked Port
D. Alternate Port
Answer: B — Root Port = the port on a non-root switch with the lowest path cost to the root bridge.

Q171. What does RSTP stand for?

A. Rapid Spanning Tree Protocol
B. Reduced STP
C. Reliable STP
D. Routed STP
Answer: A — RSTP = Rapid STP (802.1w). Converges in <1 second vs 30-50 seconds for classic STP.

Q172. What is the hello time in STP by default?

A. 1 second
B. 2 seconds
C. 5 seconds
D. 10 seconds
Answer: B — Default STP hello time = 2 seconds. Max age = 20 seconds. Forward delay = 15 seconds.

Q173. What is the max age in STP by default?

A. 10 seconds
B. 15 seconds
C. 20 seconds
D. 30 seconds
Answer: C — Max age = 20 seconds (10× hello). After this, BPDU info is considered stale.

Q174. What is the forward delay in STP by default?

A. 5 seconds
B. 10 seconds
C. 15 seconds
D. 30 seconds
Answer: C — Forward delay = 15 seconds. Time spent in Listening and Learning states.

Q175. What is the total convergence time for classic STP when a link fails (non-direct)?

A. ~1 second
B. ~15 seconds
C. ~30 seconds
D. ~50 seconds
Answer: D — Default STP convergence: 20s max age + 15s listening + 15s learning = ~50 seconds.

Q176. Which BPDU type is sent only by the root bridge?

A. Configuration BPDU (TCN)
B. Topology Change Notification
C. TCN ACK
D. Hello BPDU
Answer: A — Only the root bridge sends Configuration BPDUs. Other switches relay them.

Q177. What is EtherChannel?

A. Bonding multiple physical links into one logical link for redundancy and bandwidth

B. Encrypting Ethernet frames

C. A protocol for wireless channels

D. A VLAN trunking method

Answer: A — EtherChannel (LACP/PAgP) bundles 2-8 physical links into one logical channel for load balancing and redundancy.

Q178. Which protocol is the open standard for EtherChannel negotiation?

A. PAgP (Cisco proprietary)

B. LACP (802.3ad)

C. DTP

D. CDP

Answer: B — LACP (Link Aggregation Control Protocol) is the IEEE 802.3ad standard.

Q179. What is the maximum number of physical links in an LACP EtherChannel?

A. 4

B. 8 active + 8 standby

C. 16

D. 32

Answer: B — LACP supports up to 16 links: 8 active + 8 standby (hot-standby).

Q180. What is the purpose of a trunk link?

A. Carry traffic for multiple VLANs between switches

B. Connect a PC to a switch

C. Provide PoE

D. Increase bandwidth

Answer: A — Trunk links carry multiple VLANs using 802.1Q (or legacy ISL) tagging.

Q181. What is the default native VLAN on Cisco switches?

A. VLAN 0

B. VLAN 1

C. VLAN 1002

D. VLAN 4094

Answer: B — Default native VLAN = 1. Best practice: change to an unused VLAN for security.

Q182. What is the size of the 802.1Q VLAN tag?

A. 2 bytes

B. 4 bytes

C. 8 bytes

D. 12 bytes

Answer: B — 802.1Q tag = 4 bytes (2 bytes TPID 0x8100 + 2 bytes TCI).

Q183. What does DTP stand for?

A. Dynamic Trunking Protocol

B. Dynamic Tunnel Protocol

C. Direct Trunk Path

D. Data Transport Protocol

Answer: A — DTP = Dynamic Trunking Protocol (Cisco proprietary). Auto-negotiates trunk status between switches.

Q184. What command disables DTP on a Cisco switch port?

A. switchport nonegotiate

B. no dtp

C. switchport mode access

D. dtp off

Answer: A — switchport nonegotiate disables DTP. Best practice on access ports for security.

Q185. What is the maximum number of VLANs supported by 802.1Q?

A. 256

B. 1001

C. 4094

D. 65535

Answer: C — 802.1Q VLAN IDs are 12 bits → 4096 values (0-4095), but 0 and 4095 are reserved → 4094 usable.

Q186. Which VLANs are reserved by default on Cisco switches?

A. VLAN 1 only

B. VLANs 1-1001

C. VLANs 1, 1002-1005

D. VLANs 0-4095

Answer: C — VLAN 1 (default), 1002-1005 (FDDI/Token Ring) are reserved by default.

Q187. What is the extended VLAN range on Cisco switches?

A. 1-1001

B. 1006-4094

C. 2000-3000

D. 4090-4094

Answer: B — Extended VLANs = 1006-4094. Require VTP transparent mode (older switches).

Q188. Which command creates VLAN 10 on a Cisco switch?

A. vlan 10 (in global config)

B. create vlan 10

C. vlan create 10

D. switchport vlan 10

Answer: A — In global config: vlan 10 — then name it (optional): name Sales.

Q189. Which command assigns a port to VLAN 10?

A. switchport vlan 10

B. switchport access vlan 10

C. vlan 10 access

D. set vlan 10

Answer: B — switchport access vlan 10 (in interface config mode).

Q190. What command sets a port as a trunk?

A. switchport trunk

B. switchport mode trunk

C. switchport trunk enable

D. trunk enable

Answer: B — switchport mode trunk (in interface config mode).

Q191. Which command sets VLAN 99 as the native VLAN on a trunk?

A. switchport native vlan 99

B. switchport trunk native vlan 99

C. native vlan 99

D. trunk native 99

Answer: B — switchport trunk native vlan 99.

Q192. What is inter-VLAN routing?

A. Routing between VLANs using a Layer 3 device (router or L3 switch)

B. Sending broadcasts between VLANs

C. Tagging frames with VLAN IDs

D. Trunking between switches

Answer: A — Inter-VLAN routing uses a router (router-on-a-stick) or L3 switch (SVI) to route between VLANs.

Q193. What is 'router-on-a-stick'?

A. A router with a single interface configured with subinterfaces for each VLAN

B. A router connected to a single switch

C. A router with one antenna

D. A routing protocol

Answer: A — Router-on-a-stick = one physical router interface with multiple subinterfaces, each in a different VLAN.

Q194. Which command creates a subinterface on a Cisco router?

A. interface gig0/0.10

B. subinterface gig0/0 10

C. interface gig0/0 vlan 10

D. create subinterface 10

Answer: A — interface GigabitEthernet0/0.10 creates subinterface .10 (typically matching the VLAN ID for clarity).

Q195. Which command configures 802.1Q encapsulation on a subinterface?

A. encapsulation dot1q 10

B. encap vlan 10

C. trunk vlan 10

D. dot1q 10

Answer: A — encapsulation dot1q 10 (where 10 is the VLAN ID). Add 'native' for the native VLAN.

Q196. What is the 'default gateway' setting on a host used for?

A. DNS resolution

B. Routing traffic to other subnets

C. ARP resolution

D. DHCP discovery

Answer: B — Default gateway = the router's IP on the local subnet. Hosts send non-local traffic to it.

Q197. What is a /32 used for?

A. A subnet with 256 hosts

B. A host-only route (single host)

C. A broadcast subnet

D. A loopback-only mask

Answer: B — /32 = 255.255.255.255 = single host. Used for host routes, loopbacks, and /32 host IPs in VPNs.

Q198. What is RFC 1918?

A. The standard defining private IPv4 address ranges

B. The Ethernet standard

C. The IPv6 specification

D. The ARP specification

Answer: A — RFC 1918 defines private IPv4 ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.

Q199. Which command verifies connectivity to a destination using ICMP?

A. traceroute

B. ping

C. arp

D. ssh

Answer: B — ping sends ICMP Echo Request and waits for Echo Reply.

Q200. Which command shows the path packets take to a destination?

A. ping

B. traceroute

C. show ip route

D. show arp

Answer: B — traceroute (or 'tracert' on Windows) shows each router hop to the destination.

Q201. What does NAT do?

A. Translates private IP to public IP for internet access

B. Encrypts traffic

C. Resolves MAC to IP

D. Assigns IP addresses

Answer: A — NAT (Network Address Translation) translates private IPv4 addresses to public ones.

Q202. What is PAT?

A. Port Address Translation — many IPs to one public IP using port numbers

B. Protocol Address Translation

C. Public Address Translation

D. Permanent Address Translation

Answer: A — PAT (also called NAT overload) maps many internal IPs to one public IP using unique source ports.

Q203. What does the Cisco command 'show version' display?

A. OS version, hardware, uptime, configuration register

B. Configuration file

C. Interface status

D. Routing table

Answer: A — show version displays IOS version, hardware info, uptime, serial numbers, config register.

Q204. What is the configuration register value that boots normally from NVRAM?

A. 0x2100

B. 0x2102

C. 0x2142

D. 0x102

Answer: B — 0x2102 = normal boot. 0x2142 = ignore NVRAM (password recovery). 0x2100 = ROMmon mode.

Q205. What is the configuration register value used for password recovery?

A. 0x2102

B. 0x2142

C. 0x2100

D. 0x2120

Answer: B — 0x2142 = bypass startup-config (used in password recovery procedure).

Q206. What is the boot sequence of a Cisco router?

A. POST → Bootloader → Locate IOS → Load IOS → Load startup-config

B. Load IOS → POST → Load config

C. POST → Load config → Boot IOS

D. Bootloader → Load config → POST → IOS

Answer: A — POST (hardware check) → Bootloader (ROMmon) → locate IOS in flash → load IOS → load startup-config into running-config.

Q207. Where is the Cisco IOS image stored?

A. RAM

B. NVRAM

C. Flash

D. ROM

Answer: C — IOS image is stored in Flash memory. Loaded into RAM at boot.

Q208. What is the Cisco Discovery Protocol (CDP)?

A. Cisco proprietary Layer 2 protocol for neighbor discovery

B. Routing protocol

C. VLAN protocol

D. Encryption protocol

Answer: A — CDP (Cisco Discovery Protocol) is L2, Cisco-proprietary, advertises device info to directly connected Cisco neighbors.

Q209. What is LLDP?

A. Link Layer Discovery Protocol — IEEE standard (802.1AB) equivalent of CDP

B. Loop Detection Protocol

C. Local Device Discovery

D. Long Distance Protocol

Answer: A — LLDP is the vendor-neutral standard equivalent of CDP.

Q210. Which command displays CDP neighbors?

A. show cdp neighbors

B. show cdp

C. show neighbors

D. show lldp neighbors

Answer: A — show cdp neighbors displays directly connected Cisco devices.

Q211. Which command displays detailed CDP info including IP addresses?

A. show cdp neighbors detail

B. show cdp ip

C. show cdp all

D. show cdp entry *

*Answer: A — show cdp neighbors detail (or 'show cdp entry *') shows IPs, capabilities, IOS version, etc.*

Q212. What is the default CDP update interval?

A. 30 seconds
B. 60 seconds
C. 90 seconds
D. 120 seconds

Answer: B — CDP sends updates every 60 seconds by default. Hold time = 180 seconds.

Q213. What is the default CDP hold time?

A. 60 seconds
B. 120 seconds
C. 180 seconds
D. 240 seconds

Answer: C — Default CDP hold time = 180 seconds (3× update interval).

Q214. Which command disables CDP globally?

A. no cdp
B. no cdp run
C. cdp off
D. disable cdp

Answer: B — no cdp run disables CDP globally. no cdp enable (in interface mode) disables it per-interface.

Q215. Why might you disable CDP on edge ports?

A. To save bandwidth
B. Security — prevent attackers from discovering device info
C. To speed up convergence
D. CDP cannot be disabled

Answer: B — CDP reveals device info (model, IOS version, IP) — useful to attackers. Disable on ports facing untrusted networks.

Q216. What is the syslog severity level for 'emergency'?

A. 0
B. 1
C. 5
D. 7

Answer: A — 0=Emergency, 1=Alert, 2=Critical, 3=Error, 4=Warning, 5=Notification, 6=Informational, 7=Debug.

Q217. What is the syslog severity level for 'debugging'?

A. 0
B. 3
C. 5
D. 7

Answer: D — Level 7 = Debugging (most verbose).

Q218. What is the syslog severity level for 'error'?

A. 1
B. 2
C. 3
D. 4

Answer: C — Level 3 = Error.

Q219. What is the syslog severity level for 'warning'?

A. 2
B. 3
C. 4
D. 5
Answer: C — Level 4 = Warning.

Q220. Which command enables syslog debugging for OSPF?

A. debug ospf
B. debug ip ospf
C. debug ospf events
D. show debug ospf
Answer: C — debug ip ospf events shows OSPF events. Use 'undebug all' or 'u all' to stop.

Q221. What does the 'terminal monitor' command do?

A. Displays terminal info
B. Sends console output to your SSH/Telnet session
C. Sets terminal type
D. Locks the terminal
Answer: B — terminal monitor enables log/debug output to be displayed on a remote (SSH/Telnet) session.

Q222. Which command synchronizes unsolicited output with your prompt?

A. logging synchronous
B. no logging console
C. terminal no monitor
D. service timestamps
Answer: A — logging synchronous (under line console 0 / line vty 0 4) prevents log messages from interrupting your typing.

Q223. What does NTP stand for?

A. Network Time Protocol
B. Network Transfer Protocol
C. Node Time Protocol
D. Network Topology Protocol
Answer: A — NTP synchronizes clocks across network devices. Uses UDP port 123.

Q224. What is the NTP default port?

A. 53
B. 123
C. 161
D. 514
Answer: B — NTP uses UDP port 123.

Q225. Which NTP stratum level indicates the most accurate source (directly attached atomic clock)?

A. 0
B. 1
C. 5
D. 16
Answer: B — Stratum 0 = atomic clock/GPS (reference). Stratum 1 = server directly connected to stratum 0. Higher = less accurate.

Q226. What does SNMP stand for?

A. Simple Network Management Protocol

B. System Network Management Protocol

C. Standard Network Monitoring Protocol

D. Simple Node Management Protocol

Answer: A — SNMP = Simple Network Management Protocol. Used for monitoring and managing network devices.

Q227. Which ports does SNMP use?

A. 161 (queries), 162 (traps)

B. 80 and 443

C. 22 and 23

D. 53 and 67

Answer: A — SNMP uses UDP 161 for polling (GET/SET) and UDP 162 for trap notifications.

Q228. What is an SNMP community string?

A. A password used to authenticate SNMP requests

B. An SNMP server address

C. An SNMP version

D. An OID

Answer: A — Community string = SNMPv1/v2c password. Common defaults: 'public' (read-only), 'private' (read-write).

Q229. Which SNMP version introduced encryption?

A. v1

B. v2c

C. v3

D. v4

Answer: C — SNMPv3 added authentication and encryption (USM: User-based Security Model).

Q230. What is an OID in SNMP?

A. Object Identifier — a unique ID for a managed object (e.g., sysUpTime)

B. Open Internet Daemon

C. Original ID

D. Operation ID

Answer: A — OID = Object Identifier. Hierarchical, dot-separated (e.g., 1.3.6.1.2.1.1.3.0 = sysUpTime).

Q231. Which command configures NTP on a Cisco device?

A. ntp server 192.168.1.5

B. ntp 192.168.1.5

C. time 192.168.1.5

D. clock ntp 192.168.1.5

Answer: A — ntp server configures the device to use that NTP server.

Q232. What does the 'show clock' command display?

A. CPU clock speed

B. Current system date and time

C. Interface clock rate

D. Memory clock

Answer: B — show clock displays the current system date/time. Add 'detail' for time source info.

Q233. What is a 'layer 3 switch'?

A. A switch that can also route packets (Layer 3 functions)
B. A 3-port switch
C. A switch with 3 VLANs
D. A switch that uses 3-tier architecture
Answer: A — L3 switch = switch + router. Can do both Layer 2 switching and Layer 3 routing in hardware.

Q234. Which command shows the IP routing table?

A. show ip route
B. show route
C. show routing table
D. show ip table
Answer: A — show ip route displays the IP routing table — the brain of routing decisions.

Q235. What does the letter 'C' mean in the routing table?

A. Connected network (directly attached)
B. Configured route
C. Classful route
D. Cached route
Answer: A — C = directly Connected. Other codes: S=Static, R=RIP, O=OSPF, D=EIGRP, B=BGP.

Q236. What does the letter 'S' mean in the routing table?

A. Static route
B. Switched route
C. Saved route
D. Subnet route
Answer: A — S = Static — manually configured route.

Q237. What does the letter 'O' mean in the routing table?

A. Open
B. OSPF route
C. Origin
D. Outgoing
Answer: B — O = OSPF. O E1/E2 = external OSPF routes.

Q238. What does the letter 'D' mean in the routing table?

A. Direct
B. Default
C. DUAL (EIGRP)
D. Dynamic
Answer: C — D = EIGRP (named after DUAL algorithm). EIGRP is Cisco proprietary (was, now RFC 7868).

Q239. What does AD stand for in routing?

A. Active Directory
B. Administrative Distance
C. Address Distribution
D. Auto-Discovery
Answer: B — Administrative Distance = trustworthiness of a route source. Lower = more trusted.

Q240. What is the default AD of a static route?

A. 0
B. 1
C. 90
D. 120
Answer: B — Static route AD = 1. Directly connected = 0 (most trusted).

Q241. What is the default AD of OSPF?

A. 90
B. 100
C. 110
D. 120
Answer: C — OSPF AD = 110.

Q242. What is the default AD of EIGRP (internal)?

A. 90
B. 100
C. 110
D. 120
Answer: A — EIGRP internal AD = 90. External = 170. Summary = 5.

Q243. What is the default AD of RIP?

A. 90
B. 100
C. 110
D. 120
Answer: D — RIP AD = 120. (High = less trusted than OSPF/EIGRP.)

Q244. What is the default AD of eBGP?

A. 20
B. 100
C. 200
D. 255
Answer: A — eBGP AD = 20 (very trusted). iBGP AD = 200.

Q245. What is the longest prefix match rule?

A. When multiple routes match, the one with the longest prefix (most specific) wins
B. The route with the highest metric wins
C. The route with the lowest AD wins
D. The most recently learned route wins
Answer: A — Longest prefix match: a /32 beats a /24 beats a /0. Most specific always wins.

Q246. What is a floating static route?

A. A static route with high AD used as a backup
B. A static route that floats between routers
C. A static route with no next-hop
D. A static route in a VLAN
Answer: A — Floating static = static route with high AD (e.g., 250) that activates only when the primary route (lower AD) disappears.

Q247. What does VRRP stand for?

A. Virtual Router Redundancy Protocol
B. Variable Routing Redundancy Protocol
C. Virtual Routing Recovery Protocol
D. VLAN Router Redundancy Protocol
Answer: A — VRRP (RFC 5798) is the open-standard FHRP. Cisco's proprietary equivalent is HSRP.

Q248. What does HSRP stand for?

A. Hot Standby Router Protocol
B. High Security Routing Protocol
C. Host Standby Routing Protocol
D. Hierarchical Source Routing Protocol
Answer: A — HSRP = Cisco proprietary FHRP. Default active virtual IP provides gateway redundancy.

Q249. What is the HSRP default priority?

A. 50
B. 100
C. 150
D. 200
Answer: B — Default HSRP priority = 100. Higher = preferred active router.

Q250. Which command configures HSRP?

A. standby 1 ip 192.168.1.1
B. hsrp 1 ip 192.168.1.1
C. vrrp 1 ip 192.168.1.1
D. glbp 1 ip 192.168.1.1
Answer: A — standby ip configures HSRP.

Q251. What command makes HSRP router primary (active)?

A. standby 1 priority 110
B. standby 1 active
C. standby 1 preempt
D. standby 1 primary
Answer: A — Higher priority wins. 'standby 1 preempt' allows the router to take over when it has higher priority.

Q252. What is the difference between HSRP active and standby routers?

A. Active forwards traffic; standby waits for active to fail
B. Active routes; standby does not
C. Both forward traffic
D. Standby is offline
Answer: A — Active = forwards traffic for the virtual IP. Standby = monitors active and takes over if it fails.

Q253. Which protocol uses UDP port 514?

A. Syslog
B. SNMP trap
C. NTP
D. DNS
Answer: A — Syslog uses UDP port 514.

Q254. What is AAA?

A. Authentication, Authorization, Accounting

B. Access, Audit, Accounting

C. Authentication, Access, Authorization

D. Auto, Audit, Authority

Answer: A — AAA = Authentication (who), Authorization (what), Accounting (when/what they did).

Q255. What does TACACS+ use for transport?

A. UDP 49

B. TCP 49

C. TCP 1812

D. UDP 1812

Answer: B — TACACS+ uses TCP 49. RADIUS uses UDP 1812/1813.

Q256. What does RADIUS use for transport?

A. TCP 49

B. UDP 49

C. UDP 1812/1813

D. TCP 1812

Answer: C — RADIUS uses UDP 1812 (auth) and 1813 (accounting).

Q257. Which AAA protocol encrypts only the password?

A. RADIUS

B. TACACS+

C. Both

D. Neither

Answer: A — RADIUS encrypts only the password field. TACACS+ encrypts the entire packet body.

Q258. Which AAA protocol is Cisco proprietary?

A. RADIUS

B. TACACS+

C. Both are open standards

D. Kerberos

Answer: B — TACACS+ is Cisco proprietary (although now widely available). RADIUS is open standard (RFC 2865).

Q259. What is the default console password if none is set?

A. cisco

B. admin

C. No password required

D. Password123

Answer: C — By default, no console password is set — anyone with physical access can connect and configure.

Q260. What does the command 'login local' do?

A. Requires login using local username/password database

B. Allows login from anywhere

C. Logs in automatically

D. Logs to local syslog

Answer: A — login local tells the line to authenticate against the local username database (created with 'username X secret Y').

Q261. Which command creates a local user with a hashed password?

A. username admin password cisco123

B. username admin secret cisco123

C. user admin password cisco123

D. create user admin

Answer: B — username secret hashes the password. Avoid 'password' (plaintext).

Q262. Which command encrypts all plaintext passwords in the config?

A. service password-encryption

B. encrypt passwords

C. secret all

D. secure passwords

Answer: A — service password-encryption encrypts plaintext passwords (console, line passwords). 'secret' passwords are already hashed.

Q263. Which hashing algorithm does Cisco 'secret' use by default?

A. MD5

B. SHA-256

C. PBKDF2

D. bcrypt

Answer: A — Older Cisco IOS uses MD5 (type 5). Newer IOS-XE defaults to SHA-256 (type 8 or 9).

Q264. What is the enable secret password used for?

A. To access Privileged EXEC mode (#)

B. To access User EXEC mode (>)

C. To access console port

D. To access SSH

Answer: A — enable secret protects access to Privileged EXEC mode.

Q265. Which command sets the enable secret?

A. enable password

B. enable secret

C. set enable

D. privilege secret

Answer: B — enable secret — uses hashing. Avoid 'enable password' (uses weak reversible encryption).

Q266. Which command configures an SSH-only VTY access?

A. transport input ssh

B. transport ssh

C. ssh only

D. no telnet

Answer: A — transport input ssh under 'line vty 0 4' restricts remote access to SSH only (disables Telnet).

Q267. What is the minimum RSA key size for SSH v2?

A. 256 bits

B. 512 bits

C. 768 bits

D. 2048 bits

Answer: C — SSH v2 requires at least 768-bit RSA keys. Cisco best practice: use 2048 or higher.

Q268. Which command generates RSA keys for SSH?

A. crypto key generate rsa

B. crypto rsa generate

C. ssh key generate

D. rsa generate

Answer: A — crypto key generate rsa modulus 2048 generates an RSA keypair (modulus 512-2048).

Q269. Which command enables SSH version 2 only?

A. ip ssh version 2

B. ssh version 2

C. ssh v2 enable

D. ip ssh 2

Answer: A — ip ssh version 2 forces SSH v2 only (more secure than v1).

Q270. What is a VLAN?

A. Virtual LAN — logical grouping of devices in the same broadcast domain

B. Virtual Logical Access Node

C. Variable Length Address Number

D. VLAN Logical Area Network

Answer: A — VLAN = Virtual LAN. Logically groups devices into broadcast domains regardless of physical location.

Q271. What does a firewall do?

A. Filters traffic based on rules

B. Routes packets

C. Assigns IP addresses

D. Resolves MAC addresses

Answer: A — Firewalls filter traffic between security zones based on policy rules (ACLs, state, app-layer).

Q272. What is a DMZ?

A. Demilitarized Zone — a network segment for public-facing services

B. Dynamic Management Zone

C. Default Management Zone

D. Direct Memory Zone

Answer: A — DMZ = a separate network segment for servers that need to be reachable from the internet (web, email).

Q273. What does IDS stand for?

A. Intrusion Detection System

B. Internal Detection System

C. Internet Defense System

D. Intrusion Defense System

Answer: A — IDS = Intrusion Detection System. Passive — detects and alerts on attacks.

Q274. What does IPS stand for?

A. Intrusion Prevention System

B. Internal Prevention System

C. Internet Protection System

D. Intrusion Protection Standard

Answer: A — IPS = Intrusion Prevention System. Active — detects and blocks attacks in real-time.

Q275. What is the difference between IDS and IPS?

A. IDS detects; IPS detects AND blocks

B. IDS is faster than IPS

C. IPS only detects; IDS blocks

D. They are identical

Answer: A — IDS = passive (alert only). IPS = active (block malicious traffic inline).

Q276. What does VPN stand for?

A. Virtual Private Network

B. Variable Private Network

C. Verified Private Network

D. Virtual Protected Network

Answer: A — VPN = Virtual Private Network. Creates encrypted tunnels over untrusted networks (Internet).

Q277. What does IPsec stand for?

A. Internet Protocol Security

B. Internal Protocol Security

C. Internet Packet Security

D. Inter-Process Security

Answer: A — IPsec = Internet Protocol Security. Suite of protocols for encrypting/authenticating IP packets.

Q278. What are the two main IPsec protocols?

A. AH (Authentication Header) and ESP (Encapsulating Security Payload)

B. TCP and UDP

C. TLS and SSL

D. AES and 3DES

Answer: A — AH = authentication only. ESP = authentication + encryption. ESP is most commonly used today.

Q279. What are the two IPsec modes?

A. Transport mode (encrypts payload only) and Tunnel mode (encrypts entire packet)

B. Fast mode and Slow mode

C. Active mode and Passive mode

D. Internal mode and External mode

Answer: A — Transport = only IP payload encrypted. Tunnel = entire original packet encrypted inside a new IP packet.

Q280. What is a site-to-site VPN?

A. Connects two LANs over the internet via encrypted tunnel between two VPN gateways

B. Single user connecting to corporate network

C. VPN between two VLANs

D. VPN using only one router

Answer: A — Site-to-site VPN = gateway-to-gateway tunnel. Both LANs appear directly connected.

Q281. What is a remote-access VPN?

A. Single user connects to corporate network via encrypted tunnel

B. VPN between two sites

C. VPN over a single VLAN

D. VPN without encryption

Answer: A — Remote-access VPN = client-to-gateway. Used by remote workers to access corporate resources.

Q282. Which port does SSL VPN (AnyConnect) use by default?

A. 443 (HTTPS)

B. 500 (IKE)

C. 4500 (NAT-T)

D. 22 (SSH)

Answer: A — SSL VPN uses HTTPS port 443. Works through firewalls that block IPsec.

Q283. What is IKE in IPsec?

A. Internet Key Exchange — protocol for negotiating security associations

B. Internal Key Encryption

C. Internet Key Encryption

D. Inline Key Exchange

Answer: A — IKE = Internet Key Exchange. Used to set up IPsec security associations (SAs).

Q284. Which ports does IKE use?

A. UDP 500 and UDP 4500

B. TCP 500 and TCP 4500

C. UDP 50 and UDP 51

D. TCP 443

Answer: A — IKE uses UDP 500. NAT-T (NAT Traversal) uses UDP 4500.

Q285. What is NAT-T in IPsec?

A. NAT Traversal — encapsulates ESP in UDP 4500 to pass through NAT devices

B. Network Address Translation Tunnel

C. Network Authentication Token

D. Native Address Translation

Answer: A — NAT-T wraps ESP packets in UDP 4500 so they can traverse NAT devices (which can't handle ESP protocol 50).

Q286. What does ACL stand for?

A. Access Control List

B. Address Control List

C. Authentication Control List

D. Application Control List

Answer: A — ACL = Access Control List. Ordered list of permit/deny rules used to filter traffic.

Q287. Which ACL type filters based on source IP only?

A. Standard ACL

B. Extended ACL

C. Reflexive ACL

D. Dynamic ACL

Answer: A — Standard ACL (1-99, 1300-1999) matches only source IP.

Q288. Which ACL type filters based on source/dest IP, protocol, and ports?

A. Standard ACL

B. Extended ACL

C. Reflexive ACL

D. Time-based ACL

Answer: B — Extended ACL (100-199, 2000-2699) matches protocol, source/dest IP, ports, etc.

Q289. Where should you place a standard ACL?

A. Close to the destination

B. Close to the source

C. In the middle of the network

D. On the router

Answer: A — Standard ACLs only know source IP — place close to destination to avoid blocking legitimate traffic from other sources.

Q290. Where should you place an extended ACL?

A. Close to the source

B. Close to the destination

C. On the core switch

D. On the firewall

Answer: A — Extended ACLs know source AND destination — place close to source to drop unwanted traffic early (saves bandwidth).

Q291. What is the implicit default action at the end of every ACL?

A. Permit any any

B. Deny any any

C. Log all traffic

D. Do nothing

Answer: B — Every ACL ends with an implicit 'deny any any'. If no rule matches, traffic is dropped.

Q292. How is an ACL applied to an interface?

A. ip access-group 10 in

B. access-group 10

C. apply acl 10

D. interface acl 10

Answer: A — ip access-group applies the ACL to an interface.

Q293. What does 'in' mean in 'ip access-group 10 in'?

A. ACL applied to incoming traffic (before routing)

B. ACL applied to outgoing traffic

C. ACL only checks inbound routing

D. ACL only allows inbound traffic

Answer: A — in = traffic arriving on the interface, before routing decision.

Q294. What is the wildcard mask 0.0.0.255 equivalent to?

A. /24

B. /16

C. /8

D. /32

Answer: A — Wildcard 0.0.0.255 = inverse of 255.255.255.0 = /24.

Q295. What is the wildcard mask for /26?

A. 0.0.0.63

B. 0.0.0.127

C. 0.0.0.31

D. 0.0.0.255

Answer: A — /26 = 255.255.255.192. Wildcard = 0.0.0.63 (255-192=63, 255-255=0).

Q296. What is the 'host' keyword in an ACL?

A. Matches a single host (equivalent to wildcard 0.0.0.0)

B. Matches any host

C. Matches all hosts

D. Matches the local router

Answer: A — 'host 192.168.1.10' = '192.168.1.10 0.0.0.0' (matches exactly that IP).

Q297. What is the 'any' keyword in an ACL?

A. Matches all IPs (equivalent to 0.0.0.0 255.255.255.255)

B. Matches any single host

C. Matches any subnet

D. Matches the local network only

Answer: A — 'any' = '0.0.0.0 255.255.255.255' (matches everything).

Q298. Which ACL type uses names instead of numbers?

A. Named ACL

B. Numbered ACL

C. Standard ACL

D. Extended ACL

Answer: A — Named ACLs use descriptive names (e.g., 'BLOCK_TELNET') instead of numbers.

Q299. Can you insert lines in the middle of a numbered ACL?

A. No — they're always appended

B. Yes — using sequence numbers

C. Yes — by deleting and recreating

D. Only with extended ACLs

Answer: A — Numbered ACLs always append at the end. Named ACLs support sequence numbers for inserting lines.

Q300. What is an ACL used for besides traffic filtering?

A. Selecting routes for redistribution, NAT, QoS marking, controlling SNMP access

B. Encrypting data

C. Assigning IP addresses

D. Routing packets

Answer: A — ACLs are also used in route filtering (distribute-list), NAT (inside source list), QoS, SNMP, etc.

Q301. Which command displays ACL matches?

A. show access-lists

B. show acl

C. show ip access-lists

D. Both A and C

Answer: D — show access-lists and show ip access-lists both display ACLs and match counters.

Q302. What is the 'established' keyword in an extended ACL?

A. Allows TCP traffic only if the TCP ACK or RST flag is set (return traffic from established connections)

B. Allows all established protocols

C. Marks traffic as established

D. Required for all ACLs

Answer: A — 'established' matches TCP segments with ACK or RST flag — used as a stateful-ish filter for return traffic.

Q303. What is a reflexive ACL?

A. An ACL that dynamically creates temporary entries for return traffic

B. An ACL that mirrors another ACL

C. An ACL that blocks all traffic

D. An ACL with no rules

Answer: A — Reflexive ACLs create temporary permit entries when an outbound session starts, then remove them when the session ends.

Q304. What does the 'log' keyword do in an ACL entry?

A. Logs matching packets to syslog/console

B. Logs to a database

C. Blocks traffic with logging

D. Has no effect

Answer: A — log keyword generates a syslog entry for each match (or first match per flow). Useful for monitoring.

Q305. What is the well-known port for SMTP?

A. 21

B. 25

C. 53

D. 110

Answer: B — SMTP (Simple Mail Transfer Protocol) uses port 25 (or 587 for submission).

Q306. What is the well-known port for FTP?

A. 20 (data), 21 (control)

B. 22

C. 23

D. 25

Answer: A — FTP uses port 21 for control and port 20 for data (active mode). Passive mode uses dynamic ports.

Q307. What is the well-known port for TFTP?

A. 21

B. 69

C. 80

D. 161

Answer: B — TFTP (Trivial FTP) uses UDP port 69. Simpler than FTP, used for config/IOS transfers.

Q308. What is the well-known port for NTP?

A. 53

B. 123

C. 514

D. 161

Answer: B — NTP uses UDP port 123.

Q309. What is the well-known port for SNMP?

A. 161 (polling), 162 (traps)

B. 53

C. 123

D. 514

Answer: A — SNMP uses UDP 161 for queries and UDP 162 for trap notifications.

Q310. What is the well-known port for Syslog?

A. 53
B. 123
C. 514
D. 161
Answer: C — Syslog uses UDP port 514.

Q311. Which protocol does DHCP use?

A. TCP 67/68
B. UDP 67/68
C. UDP 53
D. TCP 514
Answer: B — DHCP uses UDP, with server on port 67 and client on port 68.

Q312. What is the well-known port for SSH?

A. 21
B. 22
C. 23
D. 25
Answer: B — SSH uses TCP port 22.

Q313. What is the well-known port for Telnet?

A. 21
B. 22
C. 23
D. 25
Answer: C — Telnet uses TCP port 23. Insecure — prefer SSH.

Q314. What is the well-known port for POP3?

A. 25
B. 110
C. 143
D. 443
Answer: B — POP3 uses TCP port 110 (or 995 with SSL).

Q315. What is the well-known port for IMAP4?

A. 25
B. 110
C. 143
D. 443
Answer: C — IMAP4 uses TCP port 143 (or 993 with SSL).

Q316. What is the well-known port for RDP?

A. 22
B. 23
C. 3389
D. 443
Answer: C — RDP (Remote Desktop Protocol) uses TCP port 3389.

Q317. What does DNS use for typical queries?

A. TCP 53
B. UDP 53
C. TCP 80
D. UDP 80
Answer: B — DNS typically uses UDP 53. TCP 53 is used for zone transfers and large responses (>512 bytes).

Q318. What does HTTP use?

A. TCP 80
B. UDP 80
C. TCP 21
D. UDP 21
Answer: A — HTTP uses TCP port 80.

Q319. What does HTTPS use?

A. TCP 80
B. TCP 443
C. UDP 443
D. TCP 8443
Answer: B — HTTPS uses TCP port 443.

Q320. What is the difference between HTTP and HTTPS?

A. HTTPS = HTTP over TLS (encrypted)
B. HTTPS uses different headers
C. HTTPS is faster
D. There is no difference
Answer: A — HTTPS is HTTP wrapped in TLS — provides confidentiality, integrity, and authentication.

Q321. Which protocol is used to securely transfer files over SSH?

A. FTP
B. SFTP
C. TFTP
D. FTPS
Answer: B — SFTP (SSH File Transfer Protocol) uses SSH. FTPS is FTP over TLS (different protocol).

Q322. What is FTPS?

A. FTP over SSL/TLS
B. FTP over SSH
C. Trivial FTP Secure
D. Secure Trivial FTP
Answer: A — FTPS = FTP over TLS. Different from SFTP (which is a sub-protocol of SSH).

Q323. What is the well-known port for SMTP over TLS (submission)?

A. 25
B. 465
C. 587
D. 993
Answer: B — SMTPS (SMTP over implicit TLS) uses port 465. Submission with STARTTLS uses port 587.

Q324. Which port does SNMPv3 use?

A. 161/162 (same as v1/v2c)

B. 443

C. 162 only

D. 996

Answer: A — All SNMP versions use the same ports: UDP 161 (queries) and UDP 162 (traps).

Q325. What is a VLAN hop attack?

A. An attacker double-tags a frame to bypass VLAN segregation

B. An attacker floods the VLAN

C. An attacker spoofs the gateway

D. An attacker uses ARP poisoning

Answer: A — VLAN hopping: attacker sends a double-tagged frame. The first switch strips the outer tag, the second switch forwards the inner tag to a different VLAN.

Q326. How can you prevent VLAN hopping?

A. Disable auto-trunking on access ports (switchport nonegotiate), put unused ports in unused VLAN

B. Disable CDP

C. Use encryption

D. Use longer passwords

Answer: A — Disable DTP auto-negotiation, set ports to access mode, and shut down unused ports.

Q327. What is a DHCP starvation attack?

A. Attacker sends many DHCP requests with fake MACs to exhaust the DHCP pool

B. Attacker blocks DHCP traffic

C. Attacker spoofs the DHCP server

D. Attacker modifies DHCP messages

Answer: A — DHCP starvation uses tools like Yersinia to consume all DHCP leases, causing DoS for legitimate clients.

Q328. What is DHCP spoofing?

A. Attacker sets up a rogue DHCP server to hand out fake IPs/gateways

B. Attacker floods DHCP requests

C. Attacker modifies DNS

D. Attacker intercepts ARP

Answer: B — Rogue DHCP server hands attacker's IP as gateway, allowing them to intercept all traffic (man-in-the-middle).

Q329. How can you prevent DHCP attacks?

A. DHCP snooping

B. BPDU Guard

C. Port security

D. ARP inspection

Answer: A — DHCP snooping allows DHCP responses only from trusted ports, preventing rogue servers.

Q330. What is ARP poisoning/spoofing?

A. Attacker sends fake ARP replies to associate their MAC with a victim's IP

B. Attacker floods ARP requests

C. Attacker modifies the ARP table of the switch

D. Attacker blocks ARP traffic

Answer: A — ARP poisoning allows MITM attacks by sending fake ARP replies to victims.

Q331. How can you mitigate ARP spoofing?

A. Dynamic ARP Inspection (DAI)

B. BPDU Guard

C. Port security

D. DHCP snooping

Answer: A — DAI validates ARP packets against the DHCP snooping binding table.

Q332. What does BPDU Guard do?

A. Disables a port if it receives a BPDU (prevents rogue switches)

B. Blocks broadcast traffic

C. Encrypts BPDUs

D. Speeds up STP convergence

Answer: A — BPDU Guard err-disables access ports that receive BPDUs — prevents users from plugging in switches and disrupting STP.

Q333. What does BPDU Filter do?

A. Stops BPDUs from being sent or received on a port

B. Filters broadcast BPDUs

C. Encrypts BPDUs

D. Logs BPDUs

Answer: A — BPDU Filter stops BPDUs entirely. Use with caution — can cause loops if misconfigured.

Q334. What does Root Guard do?

A. Prevents a port from becoming a root port (protects root bridge placement)

B. Guards against rogue BPDUs

C. Encrypts root bridge traffic

D. Speeds up root election

Answer: A — Root Guard prevents external switches from becoming the root. If a superior BPDU is received, the port is err-disabled.

Q335. What does Loop Guard do?

A. Prevents loops caused by unidirectional link failures

B. Detects physical loops

C. Encrypts loop traffic

D. Speeds up loop resolution

Answer: A — Loop Guard detects when BPDUs stop arriving on a non-designated port (unidirectional link) and keeps the port blocking instead of going to forwarding.

Q336. What is UDLD?

A. UniDirectional Link Detection — detects one-way fiber links

B. Universal Data Link Driver

C. Unified Data Link Protocol

D. Universal Direct Link Detection

Answer: A — UDLD detects unidirectional fiber links (where one strand is broken) and disables the port to prevent loops.

Q337. What is PoE?

A. Power over Ethernet — provides DC power to devices over the same cable as data

B. Power on Ethernet

C. Packet over Ethernet

D. Port of Entry

Answer: A — PoE (IEEE 802.3af/at/bt) powers devices like IP phones, APs, and cameras over the Ethernet cable.

Q338. What is the maximum power per port for PoE+ (802.3at)?

A. 15.4 W

B. 30 W

C. 60 W

D. 100 W

Answer: B — PoE+ (802.3at) = 30W per port. Original 802.3af = 15.4W. 802.3bt Type 3 = 60W, Type 4 = 100W.

Q339. What is the maximum power per port for PoE (802.3af)?

A. 7.5 W

B. 15.4 W

C. 30 W

D. 60 W

Answer: B — Original PoE (802.3af) provides 15.4W per port (12.95W at the device).

Q340. Which command shows PoE status on a Cisco switch?

A. show power inline

B. show poe

C. show power

D. show inline-power

Answer: A — show power inline displays PoE consumption per port.

Q341. What is a 'stack' of switches?

A. Multiple switches connected to act as a single logical switch

B. Switches stacked physically on top of each other

C. A VLAN stack

D. A trunk between switches

Answer: A — Stacking (e.g., Cisco StackWise) connects switches with special cables so they act as one logical switch with one control plane.

Q342. What is the benefit of switch stacking?

A. Single management point, redundant control plane, higher port density

B. Faster forwarding

C. Better encryption

D. Reduced cost

Answer: A — Stacking allows multiple switches to be managed as one, with redundancy and easier management.

Q343. What is SD-Access?

A. Cisco's software-defined networking solution for campus networks

B. A type of VPN

C. A wireless protocol

D. A type of VLAN

Answer: A — SD-Access = Cisco's intent-based networking for campus, using DNA Center, LISP, and VXLAN.

Q344. What is SD-WAN?

A. Software-defined WAN that abstracts transport and provides centralized policy

B. A type of LAN switch

C. A wireless WAN protocol

D. A VPN type

Answer: A — SD-WAN separates the data plane (routers) from the control plane (controller), allowing policy-based routing across multiple WAN links.

Q345. What is REST API?

A. Representational State Transfer — HTTP-based API using GET/POST/PUT/DELETE

B. Remote Execution Standard API

C. Real-time Event Streaming Protocol

D. Reliable End System Transfer

Answer: A — REST = HTTP verbs (GET/POST/PUT/DELETE) on resources identified by URLs.

Q346. What is JSON?

A. JavaScript Object Notation — lightweight data interchange format

B. Java Standard Object Notation

C. Just Some Object Names

D. JavaScript Oriented Naming

Answer: A — JSON is text-based, human-readable, used widely in REST APIs.

Q347. What is YAML?

A. YAML Ain't Markup Language — human-friendly data serialization

B. Yet Another Markup Language

C. Yes And Markup Language

D. Young Application Markup Language

Answer: A — YAML is widely used for configuration (Ansible, Kubernetes).

Q348. What is the difference between JSON and XML?

A. JSON is lighter (less verbose). XML uses tags, JSON uses braces.

B. JSON supports comments, XML does not

C. JSON is older than XML

D. XML is faster than JSON

Answer: A — JSON has less overhead than XML (no closing tags). XML supports attributes and schemas more naturally.

Q349. Which HTTP method retrieves a resource without side effects?

A. GET

B. POST

C. PUT

D. DELETE

Answer: A — GET retrieves a resource. Should be idempotent (no state change). POST creates/updates.

Q350. Which HTTP method creates a new resource?

A. GET
B. POST
C. DELETE
D. HEAD
Answer: B — POST creates a new resource. PUT updates an existing one (or creates if missing).

Q351. Which HTTP status code indicates 'OK'?

A. 100
B. 200
C. 301
D. 404
Answer: B — 200 OK = successful response.

Q352. Which HTTP status code indicates 'Not Found'?

A. 200
B. 301
C. 404
D. 500
Answer: C — 404 Not Found = resource does not exist.

Q353. Which HTTP status code indicates 'Unauthorized' (needs authentication)?

A. 301
B. 401
C. 403
D. 500
Answer: B — 401 Unauthorized = authentication required. 403 Forbidden = authenticated but not allowed.

Q354. Which HTTP status code indicates 'Internal Server Error'?

A. 400
B. 404
C. 500
D. 503
Answer: C — 500 = server error. 503 = service unavailable (often overload/maintenance).

Q355. What is the default port for RESTCONF?

A. 443 (HTTPS)
B. 80 (HTTP)
C. 830 (NETCONF)
D. 22 (SSH)
Answer: A — RESTCONF uses HTTPS (port 443) by default. Data formats: JSON or XML.

Q356. What is the default port for NETCONF?

A. 22
B. 443
C. 830
D. 8080
Answer: C — NETCONF uses TCP port 830 over SSH.

Q357. What does NETCONF use for data encoding?

A. JSON only

B. XML only

C. XML by default, JSON optionally

D. YAML

Answer: B — NETCONF uses XML encoded over SSH (port 830).

Q358. What is the YANG data model?

A. Data modeling language used with NETCONF/RESTCONF to describe network configuration

B. A routing protocol

C. A type of fiber cable

D. A Cisco proprietary protocol

Answer: A — YANG (Yet Another Next Generation) models network configuration data, used by NETCONF/RESTCONF.

Q359. Which Python library is commonly used for SSH automation on Cisco devices?

A. Netmiko

B. Requests

C. BeautifulSoup

D. NumPy

Answer: A — Netmiko (built on Paramiko) provides SSH automation for network devices.

Q360. Which Python library is used for REST API calls?

A. Requests

B. Netmiko

C. Paramiko

D. Pandas

Answer: A — Requests is the de-facto Python HTTP library for REST API calls.

Q361. What is Ansible?

A. Agentless configuration management tool using SSH

B. A programming language

C. A Cisco proprietary tool

D. A network monitoring system

Answer: A — Ansible (Red Hat) uses YAML playbooks over SSH — no agent needed on devices.

Q362. What is Terraform?

A. Infrastructure-as-Code tool for declarative resource provisioning

B. A Cisco routing protocol

C. A Linux distribution

D. A monitoring tool

Answer: A — Terraform (HashiCorp) uses HCL to declaratively provision infrastructure across many providers.

Q363. What is Git?

A. Distributed version control system

B. A network protocol

C. A routing protocol

D. A type of cable

Answer: A — Git is the most widely used version control system for code and configuration.

Q364. What is CI/CD?

A. Continuous Integration / Continuous Deployment

B. Cisco Internet Configuration Daemon

C. Common Interface Configuration Data

D. Customer Initiated Channel Data

Answer: A — CI/CD = automated build, test, and deploy pipeline for software and infrastructure.

Q365. What is the difference between Git and GitHub?

A. Git is the VCS tool; GitHub is a hosting service for Git repositories

B. They are the same thing

C. Git is for code, GitHub is for documents

D. GitHub is the VCS, Git is the hosting

Answer: A — Git is the local version control tool. GitHub/GitLab/Bitbucket host Git repositories.

Q366. What is Cisco DNA Center?

A. Cisco's SDN controller for campus networks (SD-Access)

B. A type of router

C. A protocol

D. An IOS version

Answer: A — DNA Center = Cisco's intent-based networking controller. Provides automation, assurance, and SDA.

Q367. What is the difference between southbound and northbound APIs in SDN?

A. Southbound = controller to devices (OpenFlow). Northbound = apps to controller (REST).

B. Southbound = apps to controller. Northbound = controller to devices.

C. They are the same

D. Southbound is for VPNs

Answer: A — Southbound APIs (OpenFlow, NETCONF) talk to network devices. Northbound APIs (REST) talk to applications.

Q368. What is the role of the control plane?

A. Decides how to forward traffic (routing, MAC learning, STP)

B. Forwards actual packets

C. Encrypts packets

D. Monitors traffic

Answer: A — Control plane = brain (routing protocols, ARP, STP). Data plane = muscle (forwards packets).

Q369. What is the role of the data plane (forwarding plane)?

A. Forwards actual packets based on the forwarding table

B. Builds routing tables

C. Configures the device

D. Monitors the network

Answer: A — Data plane = fast path. Forwards packets at line rate based on FIB and adjacency tables.

Q370. What is the management plane?

A. Used for device management (SSH, SNMP, NETCONF, RESTCONF)

B. Same as control plane

C. Same as data plane

D. Used for routing

Answer: A — Management plane = protocols used to manage the device (SSH, SNMP, NETCONF, RESTCONF, syslog).

Q371. What is the difference between in-band and out-of-band management?

A. In-band uses the same network as data; out-of-band uses a separate network

B. In-band is faster

C. Out-of-band is slower

D. There is no difference

Answer: A — OOB management uses a dedicated network (console, separate mgmt VLAN) — accessible even when data plane is down.

Q372. What is QoS?

A. Quality of Service — prioritizes certain traffic types

B. Quality of Service — encrypts traffic

C. Quality of Service — assigns IP addresses

D. Quality of Service — routes packets

Answer: A — QoS = Quality of Service. Prioritizes traffic (voice, video) over best-effort (file transfers).

Q373. What does CoS stand for in QoS?

A. Class of Service — L2 priority (802.1p, 3 bits in 802.1Q tag)

B. Cosmic Operating System

C. Class of Service — L3 priority

D. Cycle of Service

Answer: A — CoS = 3-bit field in 802.1Q tag (PCP). Values 0-7. Layer 2 only.

Q374. What does DSCP stand for?

A. Differentiated Services Code Point — 6-bit L3 QoS marker

B. Dynamic Service Code Point

C. Distributed Service Class Protocol

D. Default Service Class Priority

Answer: A — DSCP = 6 bits in IPv4 ToS field (or IPv6 Traffic Class). 64 possible values (0-63).

Q375. Which DSCP value is typically used for voice payload?

A. EF (46) — Expedited Forwarding

B. AF31

C. CS0

D. Default (0)

Answer: A — EF (DSCP 46) is reserved for voice payload. AF41 is often used for video.

Q376. What is the recommended bandwidth allocation for voice payload in LLQ?

A. 1%

B. 5-10%

C. 33%

D. 50%

Answer: B — Voice typically gets 5-10% bandwidth reservation in Low-Latency Queuing (LLQ).

Q377. What is the difference between hard QoS and soft QoS?

A. Hard = strictly enforced bandwidth; Soft = best-effort marking

B. Hard = uses CoS, Soft = uses DSCP

C. Hard = L2, Soft = L3

D. There is no difference

Answer: A — Hard QoS reserves bandwidth (e.g., LLQ). Soft QoS just marks packets and hopes downstream devices honor it.

Q378. What is the purpose of trust boundary in QoS?

A. Where to trust markings (typically at the access switch or IP phone)

B. Where to encrypt traffic

C. Where to route packets

D. Where to apply ACLs

Answer: A — Trust boundary = the point where QoS markings are accepted. Trust the IP phone, not the user PC.

Q379. Which tool is used to view and analyze packets in real time?

A. Wireshark

B. Ping

C. Traceroute

D. NMAP

Answer: A — Wireshark is the most widely used packet analyzer.

Q380. What is NMAP used for?

A. Network discovery and security auditing (port scanner)

B. Packet capture

C. Routing protocol

D. Syslog server

Answer: A — NMAP = Network Mapper. Used for port scanning, OS fingerprinting, service detection.

Q381. What is a packet sniffer?

A. Tool that captures and analyzes network traffic

B. A routing protocol

C. A type of firewall

D. A VLAN protocol

Answer: A — Packet sniffer = tool like Wireshark/tcpdump that captures packets for analysis.

Q382. What is promiscuous mode in network capture?

A. NIC accepts all frames, not just those addressed to it

B. NIC encrypts all traffic

C. NIC routes all traffic

D. NIC blocks all traffic

Answer: A — Promiscuous mode allows a NIC to capture all traffic on the segment, not just unicast to its MAC.

Q383. What is port mirroring (SPAN)?

A. Copies traffic from one or more ports to a monitoring port

B. Duplicates ports for redundancy

C. Mirrors configurations between switches

D. Encrypts port traffic

Answer: A — SPAN (Switched Port Analyzer) copies traffic to a monitoring port for analysis. RSPAN extends across switches.

Q384. Which command configures SPAN on a Cisco switch?

A. monitor session 1 source interface Fa0/1

B. span source Fa0/1

C. mirror Fa0/1 to Fa0/2

D. capture interface Fa0/1

Answer: A — monitor session 1 source interface Fa0/1 both, then monitor session 1 destination interface Fa0/2.

Q385. What is RSPAN?

A. Remote SPAN — mirrors traffic across switches via a special VLAN

B. Reverse SPAN

C. Redundant SPAN

D. Rapid SPAN

Answer: A — RSPAN carries mirrored traffic across switches using a dedicated RSPAN VLAN.

Q386. What is a SPAN destination port?

A. Where mirrored traffic is sent (for packet capture)

B. Where original traffic comes from

C. A trunk port

D. An access port

Answer: A — SPAN destination port = monitoring port connected to a packet analyzer (Wireshark).

Q387. Which command tests layer 3 connectivity to a destination?

A. ping

B. traceroute

C. show interface

D. show arp

Answer: A — ping uses ICMP Echo Request/Reply to test reachability.

Q388. Which command tests the path to a destination?

A. traceroute

B. ping

C. show ip route

D. show interface

Answer: A — traceroute discovers each router hop on the path to the destination.

Q389. What does the 'traceroute' command use to discover hops?

A. TTL manipulation in ICMP/UDP packets

B. ARP requests

C. TCP SYN packets

D. DNS queries

Answer: A — Traceroute sends packets with increasing TTL. Each router along the path returns an ICMP Time Exceeded.

Q390. What is a VLAN database on older Cisco switches?

A. A separate file (vlan.dat) storing VLAN info

B. A routing table

C. An ARP table

D. A MAC address table

Answer: A — vlan.dat in flash stores VLAN info. Modern IOS stores VLAN config in running-config.

Q391. What is VTP?

A. VLAN Trunking Protocol — advertises VLANs between switches

B. Virtual Tunnel Protocol

C. VLAN Translation Protocol

D. Vector Trunk Protocol

Answer: A — VTP (Cisco proprietary) syncs VLAN database across switches. Risky — can wipe VLANs if misconfigured.

Q392. What are the VTP modes?

A. Server, Client, Transparent (and Off in v3)

B. Master, Slave, Backup

C. Active, Standby, Disabled

D. Root, Designated, Alternate

Answer: A — VTP modes: Server (can modify), Client (only receives), Transparent (forwards but doesn't apply), Off (v3, doesn't forward).

Q393. What is the VTP default mode?

A. Server

B. Client

C. Transparent

D. Off

Answer: A — Default = Server. New Cisco switches can wipe the VLAN database of an existing VTP domain.

Q394. What is the safest VTP mode?

A. Transparent

B. Server

C. Client

D. Off (v3) or Transparent

Answer: A — Transparent mode is safest — switch doesn't sync its VLAN DB with others, preventing accidental wipes.

Q395. What is VTP pruning?

A. Stops VLAN traffic from going to switches that don't have ports in that VLAN

B. Removes unused VLANs permanently

C. Trims MAC tables

D. Reduces STP traffic

Answer: A — VTP pruning prevents flooding unknown unicast/broadcast to switches that don't need it (no ports in that VLAN).

Q396. Which is safer: VTP version 2 or 3?

A. VTP v3 — supports VLANs 1-4094, has authentication, can be set to off

B. VTP v2 — older but more stable

C. Both are equally safe

D. VTP v1 is safest

Answer: A — VTP v3 added many safety features including per-VLAN config, better authentication, and off mode.

Q397. What does the command 'erase startup-config' do?

A. Erases the saved configuration in NVRAM

B. Erases the running config

C. Reboots the device

D. Erases the IOS image

Answer: A — erase startup-config deletes NVRAM. Reload the device after to confirm — it'll boot with no config.

Q398. What does the 'reload' command do?

A. Reboots the device

B. Reloads the config

C. Reloads the OS only

D. Reloads interfaces

Answer: A — reload reboots the Cisco device.

Q399. Which command copies an IOS image from a TFTP server to flash?

A. copy tftp flash

B. copy flash tftp

C. tftp get

D. transfer tftp

Answer: A — copy tftp flash downloads an IOS image from TFTP to flash. Then boot system flash: to use it.

Q400. Which command sets the boot image?

A. boot system flash:c2900-universalk9.bin

B. boot image c2900-universalk9.bin

C. system boot flash:c2900.bin

D. ios boot flash:c2900.bin

Answer: A — boot system flash: tells the device which IOS image to load at next boot.

Q401. Which command licenses Cisco IOS-XE?

A. license boot module c2900 technology-package securityk9

B. ios license security

C. feature security

D. enable security

Answer: A — license boot ... technology-package activates features like securityk9 (IPsec, firewall, SSH).

Q402. What is Cisco Smart Licensing?

A. Cloud-based license management (replaces traditional PAK)

B. A routing protocol

C. A new IOS version

D. A type of firewall

Answer: A — Smart Licensing reports usage to Cisco's CSSM cloud for centralized license management.

Q403. What is the difference between data, voice, and signaling VLANs?

A. Data = user traffic; Voice = IP phone voice; Signaling/control = management traffic

B. All three are the same

C. Voice VLAN uses TCP

D. Data VLAN is for SSH only

Answer: A — Best practice: separate VLANs for data, voice (voice VLAN feature), and management.

Q404. What is a voice VLAN on Cisco switches?

A. A VLAN dedicated to IP phone voice traffic, automatically configured via CDP

B. A VLAN for VoIP signaling

C. A VLAN that uses HTTPS

D. A VLAN with QoS disabled

Answer: A — Cisco voice VLAN (configured with 'switchport voice vlan N') is auto-detected by Cisco IP phones via CDP.

Q405. What is the recommended voice VLAN ID convention?

A. Different from data VLAN (e.g., data=10, voice=110)

B. Same as data VLAN

C. VLAN 1 only

D. VLAN 1002-1005

Answer: A — Common convention: data VLAN 10, voice VLAN 110 (data VLAN + 100). Helps with QoS and security.

Q406. What is the purpose of a trunk between a switch and an IP phone?

A. Carries both voice VLAN and data VLAN on a single cable

B. Carries only voice VLAN

C. Carries only data VLAN

D. Provides PoE only

Answer: A — IP phone is essentially a mini-switch: it tags voice frames and forwards untagged PC data.

Q407. What is a default route?

A. Route of last resort — used when no specific route matches

B. Route for default traffic only

C. Route with no destination

D. Route for broadcast traffic

Answer: A — Default route (0.0.0.0/0 or ::/0) catches all traffic without a more specific match.

Q408. How do you write a default route in IPv4?

A. ip route 0.0.0.0 0.0.0.0

B. ip route default

C. default-route

D. ip default

Answer: A — ip route 0.0.0.0 0.0.0.0 is the IPv4 default route.

Q409. How do you write a default route in IPv6?

A. ipv6 route ::/0

B. ipv6 route 0.0.0.0 0.0.0.0

C. ipv6 default

D. ipv6 route default

Answer: A — ipv6 route ::/0 is the IPv6 default route.

Q410. What is route recursion?

A. When a route's next-hop must be resolved by looking up another route

B. When routing loops occur

C. When a route is unreachable

D. When OSPF recalculates

Answer: A — Route recursion: if next-hop isn't directly connected, the router must look it up again in the routing table.

Q411. What is IPv6 anycast?

A. One address shared by multiple devices; routed to nearest

B. One-to-many multicast

C. Broadcast-like IPv6

D. Loopback only

Answer: A — Anycast = same address assigned to multiple nodes. Routed to topologically nearest. Used for DNS root servers.

Q412. What is IPv6 multicast scope 'ff02'?

A. Link-local scope (routers don't forward)

B. Site-local scope

C. Global scope

D. Organization scope

Answer: A — FF02 = link-local. FF05 = site-local. FF0E = global.

Q413. What is a Cisco 'banner' used for?

A. Legal notice shown before login

B. Display system info

C. Welcome message after login

D. Show interface status

Answer: A — Banner (banner motd, banner login) displays legal warning before authentication.

Q414. Which banner type displays before login prompt?

A. banner login

B. banner motd

C. banner exec

D. banner incoming

Answer: A — banner login displays before the login prompt. banner motd (Message of the Day) displays after login.

Q415. What is the proper legal banner content?

A. Warning that unauthorized access is prohibited and will be prosecuted

B. Welcome message

C. System info

D. Nothing

Answer: A — Legal banners should warn that unauthorized access is prohibited. Avoid 'welcome' wording (implied consent).

Q416. What is the enable password 'cisco' considered?

A. Weak password

B. Strong password

C. Cisco default

D. Encrypted

Answer: A — Never use 'cisco', 'admin', 'password', or any dictionary word as a password.

Q417. What is the recommended password length?

A. At least 8 characters

B. At least 12 characters

C. At least 4 characters

D. At least 16 characters

Answer: B — Modern best practice: ≥12 characters minimum, with MFA where possible.

Q418. Which command enforces password complexity on Cisco IOS?

A. security passwords min-length 12

B. password complexity on

C. secure passwords

D. strong-password enable

Answer: A — security passwords min-length enforces minimum length. Other complexity via 'security passwords' sub-commands.

Q419. What is a login block-out?

A. Locks login after N failed attempts (Cisco login block-for)

B. Blocks all logins permanently

C. Blocks external logins only

D. Blocks console access

Answer: A — login block-for attempts blocks logins after N failed attempts within a window.

Q420. Which command shows failed login attempts?

A. show login failures

B. show login

C. show auth failures

D. show security

Answer: A — show login failures displays recent failed login attempts.

Q421. What is MFA?

A. Multi-Factor Authentication — requires 2+ factors (password + token, etc.)

B. Multiple Frequency Authentication

C. Master File Authentication

D. Multi-Function Adapter

Answer: A — MFA combines: something you know (password), have (token), are (biometric).

Q422. Which Cisco feature dynamically assigns VLANs based on user identity (802.1X)?

A. VLAN Assignment (VSA in RADIUS)

B. Dynamic Trunking

C. VLAN Membership Policy Server (VMPS)

D. Both A and C

Answer: D — 802.1X with RADIUS VSAs and legacy VMPS both allow dynamic VLAN assignment based on user identity.

Q423. What is 802.1X?

A. Port-based network access control (PNAC)

B. Wireless encryption standard

C. VLAN tagging protocol

D. Spanning tree variant

Answer: A — 802.1X (EAPOL) authenticates users/devices before allowing traffic on a switch port.

Q424. What are the three 802.1X components?

A. Supplicant (client), Authenticator (switch), Authentication Server (RADIUS)

B. Client, Switch, Server

C. User, Router, Firewall

D. Host, AP, WLC

Answer: A — Supplicant (client software) ↔ Authenticator (switch/AP) ↔ Authentication Server (RADIUS).

Q425. What is EAP?

A. Extensible Authentication Protocol — framework for authentication methods

B. Encryption Authentication Protocol

C. Extended Access Protocol

D. Ethernet Authentication Protocol

Answer: A — EAP is a framework supporting many methods: EAP-TLS, PEAP, EAP-TTLS, EAP-FAST, etc.

Q426. Which EAP method uses TLS certificates on both client and server?

A. EAP-TLS

B. PEAP

C. EAP-TTLS

D. EAP-FAST

Answer: A — EAP-TLS requires certificates on both sides — strongest security, hardest to deploy.

Q427. Which EAP method wraps EAP in TLS and only requires a server certificate?

A. PEAP

B. EAP-TLS

C. LEAP

D. EAP-MD5

Answer: A — PEAP (Protected EAP) only requires server cert; client uses traditional credentials inside TLS tunnel.

Q428. What does CoA stand for in RADIUS?

A. Change of Authorization — allows server to push policy changes mid-session

B. Class of Authorization

C. Cache of Authentication

D. Centralized Office Authentication

Answer: A — CoA lets RADIUS server dynamically change session attributes (e.g., VLAN change after posture check).

Q429. What is Zero Trust?

A. Security model assuming no implicit trust — every access is verified

B. Trust nothing, ever

C. Disable all access

D. Encrypt all traffic

Answer: A — Zero Trust: 'never trust, always verify'. Each access request is authenticated/authorized regardless of network location.

Domain 2: Network Access

200 questions

Q430. Which command configures a port as an access port in VLAN 10?

A. switchport mode access; switchport access vlan 10

B. switchport access vlan 10; switchport mode access

C. vlan 10 access

D. switchport vlan 10 access

Answer: A — Set the port mode first, then assign VLAN. Both commands are needed in interface config mode.

Q431. What is the purpose of the 'switchport mode trunk' command?

A. Configures the port to carry multiple VLANs

B. Disables the port

C. Assigns VLAN 1 to the port

D. Enables port security

Answer: A — Trunk mode carries multiple VLANs using 802.1Q tagging between switches.

Q432. Which command enables port security on a Cisco switch port?

A. switchport port-security

B. port-security enable

C. switchport security

D. security port

Answer: A — switchport port-security enables port security. Other sub-commands set max MACs, violation mode, etc.

Q433. What is the default maximum number of MAC addresses on a port with port security enabled?

A. 1

B. 5

C. 10

D. Unlimited

Answer: A — Default max MAC count = 1. Change with 'switchport port-security maximum '.

Q434. Which port security mode sends an SNMP trap when violated but doesn't shut the port?

A. restrict

B. protect

C. shutdown

D. trap

Answer: A — restrict drops frames, logs violations, and sends SNMP trap. Port stays up.

Q435. Which command sets a static secure MAC on a port?

A. switchport port-security mac-address AAAA.AAAA.AAAA

B. mac-address secure AAAA.AAAA.AAAA

C. secure-mac AAAA.AAAA.AAAA

D. switchport mac AAAA.AAAA.AAAA

Answer: A — switchport port-security mac-address statically configures a secure MAC.

Q436. What is sticky MAC learning?

A. Dynamically learned MACs are saved to running-config

B. MACs are permanently stored in NVRAM

C. MACs cannot be deleted

D. MACs are shared between ports

Answer: A — 'switchport port-security mac-address sticky' converts learned MACs to static, saved to running-config.

Q437. How do you recover a port from err-disabled state due to port security?

A. shutdown then no shutdown

B. reload the switch

C. no port-security

D. clear errdisable

Answer: A — Manually: shutdown then no shutdown. Or automate with 'errdisable recovery cause psecure-violation'.

Q438. Which command enables automatic errdisable recovery for port security violations?

A. errdisable recovery cause psecure-violation

B. errdisable recovery psecure

C. no errdisable

D. auto recovery psecure

Answer: A — errdisable recovery cause psecure-violation + interval auto-recovers the port.

Q439. What is the default errdisable recovery interval?

A. 30 seconds

B. 60 seconds

C. 300 seconds

D. Never (disabled by default)

Answer: C — Default interval = 300 seconds (5 min). Recovery is disabled by default — must enable per cause.

Q440. Which command shows err-disabled ports?

A. show interfaces status

B. show errdisable

C. show port-security

D. show interfaces errdisable

Answer: A — show interfaces status shows err-disabled ports with reason in the status column.

Q441. What is DHCP snooping?

A. L2 security feature that prevents rogue DHCP servers

B. DHCP server feature

C. DHCP client feature

D. DHCP relay feature

Answer: A — DHCP snooping filters DHCP messages — only trusted ports can send DHCP offers.

Q442. Which ports should be configured as DHCP snooping trusted?

A. Ports connected to DHCP servers or uplinks to other switches

B. All access ports

C. Only the management port

D. No ports

Answer: A — Trust ports facing the DHCP server or uplinks. Access ports (with clients) stay untrusted.

Q443. What does DHCP snooping build?

A. DHCP binding table (MAC, IP, VLAN, interface, lease)

B. MAC address table

C. ARP cache

D. Routing table

Answer: A — DHCP snooping binding table is used by DAI and IP Source Guard for validation.

Q444. Which command enables DHCP snooping for VLAN 10?

A. ip dhcp snooping vlan 10

B. dhcp snooping vlan 10

C. snooping dhcp vlan 10

D. ip snooping 10

Answer: A — ip dhcp snooping (global) + ip dhcp snooping vlan enables snooping for that VLAN.

Q445. What is Dynamic ARP Inspection (DAI)?

A. L2 security feature that validates ARP packets against the DHCP snooping table

B. Encrypts ARP packets

C. Speeds up ARP

D. Prevents broadcast ARP

Answer: A — DAI intercepts ARP packets and validates against the DHCP snooping binding table.

Q446. Which command enables DAI on VLAN 10?

A. ip arp inspection vlan 10

B. arp inspection vlan 10

C. ip dhcp snooping vlan 10

D. arp secure vlan 10

Answer: A — ip arp inspection vlan enables DAI. Requires DHCP snooping to be enabled for binding table.

Q447. What is IP Source Guard?

A. L2 feature that filters traffic based on source IP+MAC matching the DHCP binding table

B. Encrypts source IP

C. Filters destination IP

D. Speeds up source routing

Answer: A — IP Source Guard drops traffic where source IP/MAC doesn't match the DHCP snooping binding.

Q448. Which command enables IP Source Guard on an interface?

A. ip verify source

B. ip source guard

C. verify source

D. source guard

Answer: A — ip verify source [tracking] [port-security] enables IP Source Guard. Often combined with port-security.

Q449. What does 802.1X provide?

A. Port-based authentication

B. VLAN tagging

C. QoS marking

D. Spanning tree

Answer: A — 802.1X = port-based network access control using EAPoL.

Q450. Which STP bridge ID has lower priority?

A. Lower priority value
B. Higher priority value
C. Lower MAC address only
D. Higher bridge ID
Answer: A — Lower priority value wins. Ties broken by lower MAC address.

Q451. What is the bridge ID composed of?

A. Priority + MAC address
B. IP + MAC
C. Priority + IP
D. MAC + port count
Answer: A — Bridge ID = 2-byte priority + 6-byte MAC = 8 bytes total.

Q452. Which STP port state forwards data and learns MACs?

A. Forwarding
B. Learning
C. Listening
D. Blocking
Answer: A — Forwarding = both learns MACs and forwards frames. Final stable state for active ports.

Q453. Which STP port state learns MACs but does NOT forward?

A. Learning
B. Listening
C. Forwarding
D. Blocking
Answer: A — Learning state: switch learns MACs but doesn't forward user data yet (15s default).

Q454. Which STP port state listens to BPDUs but does NOT learn MACs or forward?

A. Listening
B. Learning
C. Forwarding
D. Disabled
Answer: A — Listening state: only processes BPDUs. No MAC learning or forwarding. 15s default.

Q455. Which STP port state blocks user data but still receives BPDUs?

A. Blocking
B. Listening
C. Learning
D. Forwarding
Answer: A — Blocking: port stays quiet (no data, no learning), but receives BPDUs to maintain loop-free topology.

Q456. What does the 'spanning-tree vlan 1 root primary' command do?

A. Sets the switch as root bridge for VLAN 1
B. Disables STP
C. Forgets root
D. Speeds up STP
Answer: A — Macro command that lowers priority enough to become root. 'secondary' sets backup.

Q457. What is the priority after 'spanning-tree vlan 1 root primary'?

A. 24576 (or lower)

B. 32768

C. 8192

D. 4096

Answer: A — Root primary lowers priority to 24576 (or 4096 less than current root's priority).

Q458. Which STP version converges in <1 second?

A. RSTP (802.1w)

B. STP (802.1D)

C. MST

D. PVST+

Answer: A — RSTP converges in under 1 second via proposal/agreement handshake.

Q459. What new port roles did RSTP introduce?

A. Alternate and Backup

B. Designated and Root

C. Forwarding and Blocking

D. Master and Slave

Answer: A — RSTP Alternate = secondary path to root (replaces blocked role). Backup = secondary path on same segment.

Q460. What does 'spanning-tree mode rapid-pvst' do?

A. Enables Cisco's per-VLAN RSTP

B. Enables MST

C. Disables STP

D. Speeds up CDP

Answer: A — rapid-pvst = Per-VLAN Rapid Spanning Tree. Cisco's per-VLAN RSTP implementation.

Q461. What is MST?

A. Multiple Spanning Tree (802.1s) — maps multiple VLANs to fewer STP instances

B. Multi-Source Tree

C. Master Spanning Tree

D. Modified Spanning Tree

Answer: A — MST reduces STP instances by grouping VLANs into instances. Useful for large networks.

Q462. Which command enables PortFast?

A. spanning-tree portfast

B. portfast enable

C. spanning-tree fast

D. no spanning-tree

Answer: A — spanning-tree portfast (interface) skips Listening/Learning, goes straight to Forwarding. Only on access ports!

Q463. Why should PortFast only be enabled on access ports?

A. To prevent loops — port skips listening/learning, dangerous on trunks

B. It only works on access ports

C. It's required by 802.1D

D. It's a Cisco proprietary limitation

Answer: A — PortFast bypasses STP listening/learning. On a trunk or switch port, this could create loops.

Q464. What does BPDU Guard do when enabled with PortFast?

A. Err-disables the port if a BPDU is received

B. Forwards BPDUs

C. Logs BPDUs only

D. Speeds up BPDU transmission

Answer: A — BPDU Guard protects against rogue switches on access ports. Port is err-disabled if BPDU received.

Q465. How do you globally enable BPDU Guard on all PortFast ports?

A. spanning-tree portfast bpduguard default

B. bpduguard enable

C. spanning-tree bpduguard

D. no bpduguard

Answer: A — Global: spanning-tree portfast bpduguard default. Per-port: spanning-tree bpduguard enable.

Q466. What is the recommended EtherChannel load balancing method?

A. Source and destination IP (default on most platforms)

B. Source MAC only

C. Destination MAC only

D. Round-robin

Answer: A — Default is src-dst IP. Choice depends on traffic patterns; goal is even distribution.

Q467. Which LACP mode actively initiates EtherChannel negotiation?

A. Active

B. Passive

C. Auto

D. Desirable

Answer: A — LACP active initiates. Passive waits. Both ends must agree (active+passive or active+active work).

Q468. Which PAgP mode actively initiates?

A. Desirable

B. Auto

C. Active

D. Passive

Answer: A — PAgP desirable initiates. Auto waits. PAgP is Cisco proprietary; prefer LACP.

Q469. Which EtherChannel mode forces the channel on without negotiation?

A. On

B. Active

C. Auto

D. Desirable

Answer: A — 'on' forces EtherChannel without negotiation protocol. Both ends must be 'on'. Used with static LAG.

Q470. What is the maximum active links in an LACP bundle?

A. 8 (with up to 8 standby)

B. 4

C. 16

D. Unlimited

Answer: A — LACP: 8 active + 8 standby. Active links are those with lowest priority numbers.

Q471. Which command verifies EtherChannel status?

A. show etherchannel summary

B. show lacp

C. show port-channel

D. show etherchannel

Answer: A — show etherchannel summary gives a concise view of all bundles and their member ports.

Q472. Which EtherChannel load-balancing method is best for traffic to many servers from one client?

A. Source IP

B. Destination IP

C. Source MAC

D. Round-robin

Answer: B — If single client to many servers, use destination IP to distribute across links.

Q473. Which command configures a Layer 3 EtherChannel?

A. interface port-channel 1; no switchport; ip address ...

B. interface port-channel 1; switchport mode trunk

C. channel-group 1 mode active

D. no switchport

Answer: A — L3 EtherChannel: 'no switchport' on the port-channel interface, then assign IP.

Q474. What happens if you don't match the configuration on all member ports of an EtherChannel?

A. EtherChannel won't form (suspended)

B. Only first port is used

C. Bandwidth doubles

D. Nothing

Answer: A — All member ports must have matching speed/duplex/VLAN/trunk config or EtherChannel won't form.

Q475. Which Wi-Fi standard introduced OFDM and 54 Mbps?

A. 802.11a/g

B. 802.11b

C. 802.11n

D. 802.11ax

Answer: A — 802.11a (5 GHz) and 802.11g (2.4 GHz) introduced OFDM and 54 Mbps.

Q476. Which Wi-Fi standard is also known as Wi-Fi 6?

A. 802.11ax

B. 802.11ac

C. 802.11n

D. 802.11be

Answer: A — Wi-Fi 6 = 802.11ax. Wi-Fi 7 = 802.11be. Wi-Fi 5 = 802.11ac. Wi-Fi 4 = 802.11n.

Q477. Which Wi-Fi standard operates in the 6 GHz band?

A. Wi-Fi 6E (802.11ax extended)

B. Wi-Fi 5

C. Wi-Fi 4

D. Wi-Fi 3

Answer: A — Wi-Fi 6E adds the 6 GHz band to Wi-Fi 6. Wi-Fi 7 also uses 6 GHz.

Q478. What is the maximum theoretical speed of Wi-Fi 6 (802.11ax)?

A. 9.6 Gbps
B. 3.5 Gbps
C. 1.3 Gbps
D. 600 Mbps
Answer: A — Wi-Fi 6 max = 9.6 Gbps (theoretical, multi-user aggregate).

Q479. What frequency bands does Wi-Fi use?

A. 2.4 GHz, 5 GHz, and 6 GHz
B. 900 MHz only
C. 60 GHz only
D. 5 GHz only
Answer: A — Wi-Fi uses 2.4 GHz, 5 GHz, and (since Wi-Fi 6E) 6 GHz bands.

Q480. Which Wi-Fi standard introduced MU-MIMO (Multi-User MIMO)?

A. 802.11ac (Wi-Fi 5)
B. 802.11g
C. 802.11n
D. 802.11b
Answer: A — 802.11ac introduced downlink MU-MIMO. 802.11ax added uplink MU-MIMO (OFDMA also).

Q481. Which Wi-Fi authentication uses a RADIUS server?

A. WPA2-Enterprise (802.1X)
B. WPA2-Personal (PSK)
C. WEP
D. Open
Answer: A — Enterprise mode uses 802.1X + RADIUS. Personal mode uses Pre-Shared Key (PSK).

Q482. Which Wi-Fi encryption is considered broken and should not be used?

A. WEP
B. WPA2-Personal
C. WPA3-Personal
D. WPA3-Enterprise
Answer: A — WEP can be cracked in minutes. WPA is also weak. Use WPA2 or WPA3.

Q483. Which Wi-Fi security standard introduced SAE (Simultaneous Authentication of Equals)?

A. WPA3
B. WPA2
C. WEP
D. 802.1X
Answer: A — WPA3 replaces PSK with SAE, preventing offline dictionary attacks.

Q484. What is the OWE (Opportunistic Wireless Encryption)?

A. Provides encryption on open networks (RFC 8110)
B. A WEP replacement
C. A WPA3 mode
D. An enterprise feature
Answer: A — OWE encrypts individual sessions on open hotspots without password — used in Hotspot 2.0.

Q485. Which wireless topology uses a controller to manage multiple APs?

A. Centralized (lightweight APs + WLC)

B. Autonomous APs only

C. Mesh without controller

D. Ad-hoc

Answer: A — Centralized: lightweight APs (LAPs) managed by WLC. Common in enterprise.

Q486. What protocol does Cisco use for AP-WLC communication?

A. CAPWAP (Control and Provisioning of Wireless Access Points)

B. LWAPP (legacy)

C. Both A and B (CAPWAP is the standard, evolved from LWAPP)

D. SNMP

Answer: C — CAPWAP (RFC 5415/5416) is the standard. LWAPP is Cisco's older protocol, now replaced by CAPWAP.

Q487. Which ports does CAPWAP use?

A. UDP 5246 (control) and UDP 5247 (data)

B. TCP 80

C. UDP 500

D. TCP 443

Answer: A — CAPWAP control = UDP 5246, data = UDP 5247. Both can be encrypted with DTLS.

Q488. What is wireless roaming?

A. Client moves from one AP to another without losing connection

B. Client changes Wi-Fi channels

C. Client moves from 2.4 to 5 GHz

D. WLC failover

Answer: A — Roaming allows clients to switch APs seamlessly as they move. Fast roaming = 802.11r.

Q489. Which roaming standard reduces handoff time for voice clients?

A. 802.11r (Fast BSS Transition)

B. 802.11k

C. 802.11v

D. 802.11w

Answer: A — 802.11r = Fast BSS Transition. Reduces roaming time from 100s of ms to <50ms.

Q490. What is PMF (Protected Management Frames)?

A. 802.11w — encrypts management frames to prevent deauth attacks

B. Encryption for data frames

C. A new wireless standard

D. A roaming protocol

Answer: A — PMF (802.11w) protects deauth/disassoc frames from forgery, preventing deauth DoS attacks.

Q491. Which wireless band has the longest range but lowest speed?

A. 2.4 GHz

B. 5 GHz

C. 6 GHz

D. 60 GHz

Answer: A — Lower frequency = better range/wall penetration, but more interference and lower speeds.

Q492. Which wireless band has the highest speed but shortest range?

A. 6 GHz
B. 5 GHz
C. 2.4 GHz
D. 900 MHz
Answer: A — Higher frequency = more bandwidth but shorter range and worse wall penetration.

Q493. Which Wi-Fi channels don't overlap in 2.4 GHz?

A. 1, 6, 11
B. 1, 5, 9
C. 2, 7, 12
D. 3, 8, 13
Answer: A — Only channels 1, 6, 11 (or 1, 5, 9, 13 in some regions) are non-overlapping in 2.4 GHz.

Q494. How many non-overlapping channels does 5 GHz have?

A. ~23 (depending on region and channel width)
B. 3
C. 11
D. 1
Answer: A — 5 GHz has many more non-overlapping channels than 2.4 GHz (23+ at 20 MHz channel width).

Q495. What is the default channel width in 802.11ac?

A. 20 MHz
B. 40 MHz
C. 80 MHz
D. 160 MHz
Answer: C — 802.11ac default = 80 MHz. Can use 20/40/80/160 MHz channels.

Q496. Which wireless AP mode connects wireless clients to a wired network?

A. Local mode (also called Normal mode)
B. FlexConnect mode
C. Monitor mode
D. Sniffer mode
Answer: A — Local mode = traditional AP that tunnels traffic to WLC via CAPWAP.

Q497. Which AP mode is used in branch offices where WAN may fail?

A. FlexConnect (formerly H-REAP)
B. Local mode
C. Monitor mode
D. Rogue detector
Answer: A — FlexConnect: AP can switch client traffic locally if WLC is unreachable. Good for branches.

Q498. Which AP mode only monitors for rogue APs?

A. Monitor mode
B. Local mode
C. FlexConnect
D. Sniffer
Answer: A — Monitor mode: AP doesn't serve clients, just listens for rogue APs and interferers.

Q499. What is BSSID in Wi-Fi?

A. Basic Service Set Identifier — usually the AP's MAC for each SSID

B. Network name (SSID)

C. Encryption type

D. Channel number

Answer: A — BSSID = MAC address of the AP's radio for each SSID. SSID is the human-readable name.

Q500. What is ESSID?

A. Extended SSID — same SSID across multiple APs (roaming network)

B. Encryption SSID

C. Enterprise SSID

D. Extended BSSID

Answer: A — ESSID is the SSID shared across multiple APs in an extended service set (roaming network).

Q501. Which command shows wireless clients connected to a WLC?

A. show client summary

B. show wireless client

C. show wlan

D. show ap summary

Answer: A — show client summary lists all clients connected to the WLC.

Q502. What is RSSI?

A. Received Signal Strength Indicator

B. Radio Signal Strength Index

C. Relative Signal Strength Indicator

D. Receiver Signal Source Index

Answer: A — RSSI measures signal strength in dBm. Closer to 0 = stronger (-30 dBm is excellent, -80 dBm is weak).

Q503. What is the SNR in wireless?

A. Signal-to-Noise Ratio — difference between signal and noise floor

B. Signal Network Range

C. Signal Network Routing

D. Source Network Reach

Answer: A — SNR (dB) = signal (dBm) - noise (dBm). Higher = better. ≥25 dB is good for voice.

Q504. What is the noise floor?

A. Background RF energy level in the area

B. Maximum signal level

C. Average client signal

D. Channel bandwidth

Answer: A — Noise floor = ambient RF energy. Typical -90 to -100 dBm. Higher noise = poorer performance.

Q505. What is a wireless site survey?

A. Process of measuring RF coverage and interference to plan AP placement

B. Survey of user opinions

C. Survey of cabling

D. Survey of switch capacity

Answer: A — Site survey determines AP locations, channels, and power for optimal coverage.

Q506. What is co-channel interference (CCI)?

A. Two APs on the same channel can hear each other, reducing performance

B. Interference from microwaves

C. Interference from Bluetooth

D. Interference between 2.4 and 5 GHz

Answer: A — CCI = same-channel APs/clients contend for airtime. Reduce by reusing channels carefully.

Q507. What is adjacent channel interference?

A. Overlap between channels with insufficient spacing (e.g., 20 MHz channels at 1 and 3)

B. Same-channel interference

C. Interference from non-Wi-Fi sources

D. Interference between 5 GHz and 6 GHz

Answer: A — Adjacent channel interference from overlapping frequencies — worse than CCI. Avoid overlapping channels.

Q508. What is the benefit of band steering?

A. Pushes dual-band clients to less-crowded 5 GHz

B. Speeds up 2.4 GHz

C. Increases range

D. Reduces roaming time

Answer: A — Band steering encourages 5 GHz clients to use 5 GHz (less congested, more channels, faster).

Q509. What is a hidden node problem in Wi-Fi?

A. Two clients can't hear each other but both can reach the AP, causing collisions

B. An AP that doesn't broadcast SSID

C. A client that hides its MAC

D. A weak signal

Answer: A — Hidden node: clients can't sense each other's transmissions, causing collisions. Solved with RTS/CTS.

Q510. What is the maximum transmit power for 2.4 GHz Wi-Fi in the US (FCC)?

A. 1 W (30 dBm)

B. 100 mW (20 dBm)

C. 10 mW (10 dBm)

D. Unlimited

Answer: A — FCC limit for 2.4 GHz = 1 W (30 dBm) EIRP. Most enterprise APs use 50-100 mW.

Q511. What is EIRP?

A. Effective Isotropic Radiated Power — TX power + antenna gain - cable loss

B. Effective Internet Routing Protocol

C. Encryption Isotropic Radio Power

D. End-to-End Radio Power

Answer: A — EIRP = TX power (dBm) + antenna gain (dBi) - cable loss (dB). Regulated by FCC/etc.

Q512. Which antenna type focuses energy in one direction for point-to-point links?

A. Directional (e.g., parabolic, Yagi)

B. Omnidirectional

C. Dipole

D. Patch

Answer: A — Directional antennas concentrate energy in one direction, increasing range in that direction.

Q513. Which antenna type provides 360° coverage for general office use?

A. Omnidirectional

B. Directional

C. Yagi

D. Parabolic

Answer: A — Omnidirectional antennas (most APs) radiate in all horizontal directions.

Q514. What is antenna gain measured in?

A. dBi (decibels relative to isotropic radiator)

B. Watts

C. dBm

D. MHz

Answer: A — Antenna gain is measured in dBi (relative to a theoretical isotropic antenna).

Q515. What does +3 dB represent?

A. Doubling of power

B. Halving of power

C. 10x power

D. 100x power

Answer: A — +3 dB = 2x power. +10 dB = 10x. -3 dB = 1/2. -10 dB = 1/10.

Q516. What is the relationship between mW and dBm at 1 mW?

A. 1 mW = 0 dBm

B. 1 mW = 1 dBm

C. 1 mW = -1 dBm

D. 1 mW = 10 dBm

Answer: A — dBm = 10 × log10(mW). 1 mW = 0 dBm. 10 mW = 10 dBm. 100 mW = 20 dBm. 1000 mW = 30 dBm.

Q517. What is the transmit power of 100 mW in dBm?

A. 20 dBm

B. 10 dBm

C. 30 dBm

D. 0 dBm

Answer: A — 100 mW = 10 × log10(100) = 20 dBm.

Q518. What is 30 dBm in mW?

A. 1000 mW (1 W)

B. 100 mW

C. 10 mW

D. 1 mW

Answer: A — 30 dBm = 1000 mW = 1 W.

Q519. Which Cisco WLC feature provides guest access isolation?

A. Guest LAN with isolation

B. RF profile

C. FlexConnect group

D. AP group

Answer: A — Guest WLAN isolates guest traffic from corporate using separate VLAN + ACLs.

Q520. What is a wireless mesh?

A. APs relay traffic to each other back to a wired AP (RAP)

B. Wi-Fi over Wi-Fi

C. A type of antenna

D. A roaming protocol

Answer: A — Mesh: MAPs (Mesh APs) connect wirelessly to RAPs (Root APs) to extend coverage without cabling.

Q521. Which interface on a WLC is used for management?

A. Management interface

B. Service-port

C. Dynamic interface

D. Virtual interface

Answer: A — Management interface: used for HTTPS/SSH management, CAPWAP, RADIUS, etc.

Q522. What is the WLC virtual interface used for?

A. DHCP relay, client web auth, guest portal

B. Management

C. Console

D. CAPWAP

Answer: A — Virtual interface is a dummy IP (e.g., 1.1.1.1) used as source for DHCP/relay/web auth to clients.

Q523. What is the WLC service-port used for?

A. Out-of-band management and disaster recovery

B. Data traffic

C. CAPWAP

D. DHCP relay

Answer: A — Service-port is an OOB management interface (doesn't process data traffic).

Q524. How many WLANs can a Cisco WLC support by default?

A. 16 (older models), 512+ (newer)

B. 1

C. 5

D. 100

Answer: A — Modern WLCs support hundreds of WLANs. Limit varies by model and license.

Q525. What is a WLAN ID?

A. Numeric identifier (1-512) for each WLAN on WLC

B. SSID

C. VLAN ID

D. AP ID

Answer: A — WLAN ID is internal to WLC. SSID is what clients see. VLAN is mapped to WLAN.

Q526. Which command on Cisco WLC creates a WLAN?

A. Via GUI: WLANs → New, or 'config wlan create ' via CLI

B. wlan create

C. ssid create

D. create wlan

Answer: A — WLANs are usually created via GUI, but CLI also works: config wlan create .

Q527. What is the difference between 802.11ac Wave 1 and Wave 2?

A. Wave 2 adds MU-MIMO and 160 MHz channels

B. Wave 2 is faster but only on 2.4 GHz

C. Wave 1 is faster

D. No difference

Answer: A — Wave 2 (2016) added MU-MIMO (downlink), 4 spatial streams, 160 MHz channels.

Q528. What is the maximum number of spatial streams in 802.11ax?

A. 8

B. 4

C. 16

D. 1

Answer: A — 802.11ax supports up to 8 spatial streams (theoretical max ~9.6 Gbps aggregate).

Q529. Which feature reduces Wi-Fi contention by dividing channels into sub-channels?

A. OFDMA (802.11ax)

B. MU-MIMO

C. Beamforming

D. Channel bonding

Answer: A — OFDMA splits a 20 MHz channel into smaller Resource Units, allowing simultaneous multi-user transmissions.

Q530. What is beamforming?

A. AP focuses RF energy toward specific clients to improve signal

B. Shaping the antenna

C. Blocking interference

D. Increasing power

Answer: A — Beamforming (802.11ac/ax) uses multiple antennas to focus signal toward clients, improving range/speed.

Q531. What is BSS coloring in Wi-Fi 6?

A. Each BSS gets a 'color' tag to reduce co-channel interference

B. Color-codes Wi-Fi signals

C. Adds visual color to AP LEDs

D. Assigns colors to clients

Answer: A — BSS Color (802.11ax) lets a device identify which BSS a transmission belongs to, allowing spatial reuse.

Q532. What is target wake time (TWT) in Wi-Fi 6?

A. AP schedules when clients can wake/sleep — saves battery for IoT devices

B. Schedules AP reboots

C. Schedules channel scanning

D. Schedules roaming

Answer: A — TWT allows AP and client to negotiate wake times — major battery saving for IoT.

Q533. What is WPA3-Personal's key improvement over WPA2-Personal?

A. SAE replaces PSK — prevents offline dictionary attacks

B. Longer passwords

C. Faster encryption

D. Better range

Answer: A — SAE (Simultaneous Authentication of Equals) forces interactive attack, blocking offline cracking.

Q534. What is the minimum password length for WPA3-Personal?

A. 8 characters (no upper limit)

B. 12 characters

C. 16 characters

D. 4 characters

Answer: A — WPA3 mandates minimum 8 chars. SAE prevents brute force even with weak passwords (but longer is better).

Q535. What is the difference between 192-bit and 128-bit WPA3-Enterprise?

A. 192-bit uses CNSA suite (NSA-approved) for high-security environments

B. 192-bit is faster

C. 128-bit is for government use

D. No difference

Answer: A — WPA3-Enterprise 192-bit mode uses CNSA suite (AES-256, SHA-384, etc.) for high-assurance environments.

Q536. What is Opportunistic Wireless Encryption (OWE)?

A. Unauthenticated encryption for open networks (RFC 8110)

B. A WPA3 variant

C. A WEP successor

D. Same as WPS

Answer: A — OWE uses Diffie-Hellman to encrypt open networks without password. Used in Hotspot 2.0.

Q537. What is WPS and why is it insecure?

A. Wi-Fi Protected Setup — has WPS PIN vulnerability that can be brute-forced

B. Wi-Fi Power Saver

C. Wireless Profile Standard

D. Wi-Fi Protected Security

Answer: A — WPS PIN is 8 digits but reduces to ~11,000 attempts. Should be disabled.

Q538. What is the maximum distance for typical Wi-Fi coverage (indoor)?

A. ~30-100 meters (depending on environment and standard)

B. 1 km

C. 10 km

D. 100 km

Answer: A — Indoor Wi-Fi range: 30-100m (2.4GHz better than 5/6GHz for range).

Q539. What is the difference between autonomous and lightweight APs?

A. Autonomous: standalone, fully configured. Lightweight: managed by WLC.

B. Autonomous are smaller

C. Lightweight are faster

D. No difference

Answer: A — Autonomous APs do everything themselves. Lightweight APs (LAPs) rely on WLC for config/control.

Q540. Which mode do most enterprise deployments use today?

A. Lightweight APs with WLC (and SD-Access)

B. Autonomous APs

C. Ad-hoc mode

D. Mesh with no controller

Answer: A — Enterprise: lightweight APs + WLC, increasingly managed by DNA Center for SD-Access.

Q541. What is Cisco FlexConnect?

A. AP mode that switches client traffic locally when WAN is down

B. A wireless encryption

C. A roaming protocol

D. An antenna type

Answer: A — FlexConnect (formerly H-REAP): AP can locally switch traffic if WLC unreachable. Good for branches.

Q542. Which AP mode is used for RF spectrum analysis?

A. Sniffer mode

B. Local mode

C. Monitor mode

D. FlexConnect

Answer: A — Sniffer mode: AP captures 802.11 frames like a wireless Wireshark for troubleshooting.

Q543. Which command shows the AP configuration summary?

A. show ap config summary

B. show ap summary

C. show wireless ap

D. show ap

Answer: A — show ap config summary shows detailed AP configuration. show ap summary gives brief overview.

Q544. What is the Cisco WLC GUI default URL?

A. https://

B. http://

C. https://:8443

D. ftp://

Answer: A — WLC GUI is HTTPS on the management IP. Default credentials depend on model.

Q545. What is the default Cisco WLC admin password (newer models)?

A. Admin123! (varies by model and version)

B. cisco

C. admin

D. Cisco123

Answer: A — Modern Cisco WLCs require complex passwords on first login. Older default was admin/admin.

Q546. Which feature allows you to give different VLANs to different clients on the same SSID?

A. Identity-based networking (RADIUS returns VLAN ID)

B. Multiple SSIDs

C. Trunking

D. VLAN hopping

Answer: A — RADIUS can return VLAN ID as a VSA — different users on same SSID get different VLANs (dynamic VLAN assignment).

Q547. What is a guest portal?

A. Web page that guests must accept before internet access

B. Admin login page

C. AP config page

D. WLC dashboard

Answer: A — Guest portal: captive web page where guests accept terms or enter credentials before access.

Q548. What is 802.11k?

A. Radio Resource Measurement — helps clients find better APs for roaming

B. Fast roaming

C. Wireless management frame protection

D. WPA3

Answer: A — 802.11k provides neighbor reports to clients, helping them find target APs faster.

Q549. What is 802.11v?

A. BSS Transition Management — AP suggests better APs to clients

B. Fast roaming

C. Encryption

D. QoS

Answer: A — 802.11v: AP can send BSS Transition requests to clients to optimize roaming.

Q550. What is the Cisco CleanAir feature?

A. Identifies and classifies non-Wi-Fi RF interference (microwaves, Bluetooth, etc.)

B. Cleans the air filter

C. A Wi-Fi encryption

D. A roaming protocol

Answer: A — CleanAir: Cisco WLC/AP feature that identifies sources of RF interference (microwave, Bluetooth, etc.).

Q551. What is the RRM (Radio Resource Management) in Cisco WLC?

A. Auto-configures AP transmit power and channels

B. Manages RADIUS

C. Routes wireless traffic

D. Encrypts traffic

Answer: A — RRM (Auto-RF) dynamically adjusts AP power and channels based on RF environment.

Q552. What does 'Tx Power Level 1' mean on a Cisco WLC?

A. Maximum power

B. Minimum power

C. Average power

D. Disabled

Answer: A — Level 1 = max power. Higher numbers = lower power. Levels 1-8 typically.

Q553. What is the default channel width for 802.11ax?

A. 20 MHz (auto-bonding up to 80)
B. 40 MHz
C. 80 MHz
D. 160 MHz
Answer: A — Default = 20 MHz. Bonding can go to 40/80/160 MHz if there's clean spectrum.

Q554. Which command shows CDP neighbors on a Cisco switch?

A. show cdp neighbors
B. show neighbors
C. show cdp
D. show cdp all
Answer: A — show cdp neighbors lists directly connected Cisco devices.

Q555. Which command disables CDP on a specific interface?

A. no cdp enable
B. cdp off
C. no cdp
D. disable cdp
Answer: A — no cdp enable in interface config disables CDP on that interface.

Q556. Which command disables CDP globally?

A. no cdp run
B. no cdp
C. cdp disable
D. no cdp global
Answer: A — no cdp run in global config disables CDP globally.

Q557. Which command enables LLDP globally?

A. lldp run
B. lldp enable
C. enable lldp
D. lldp global
Answer: A — lldp run enables LLDP globally. Per-interface: lldp transmit / lldp receive.

Q558. What is the default LLDP transmit interval?

A. 30 seconds
B. 60 seconds
C. 5 seconds
D. 120 seconds
Answer: A — LLDP sends advertisements every 30 seconds by default. Hold time = 120 seconds.

Q559. What is the default LLDP hold time?

A. 120 seconds
B. 30 seconds
C. 60 seconds
D. 180 seconds
Answer: A — Default LLDP hold time = 120 seconds (4× transmit interval).

Q560. Which command shows LLDP neighbors?

A. show lldp neighbors
B. show ldp neighbors
C. show neighbors lldp
D. show lldp
Answer: A — show lldp neighbors displays LLDP-discovered neighbors.

Q561. What does UDLD aggressive mode do?

A. Err-disables port after 8 consecutive missed UDLD messages (faster detection)
B. Sends more BPDUs
C. Increases transmit power
D. Disables port permanently
Answer: A — UDLD aggressive detects one-way links even when link stays up but traffic flows one direction.

Q562. What is the default UDLD message interval?

A. 15 seconds
B. 60 seconds
C. 1 second
D. 5 seconds
Answer: A — UDLD default interval = 15 seconds. Detection time = 3 intervals = 45 seconds.

Q563. Which command enables UDLD on a fiber interface?

A. udld enable
B. udld port
C. udld aggressive
D. no udld
Answer: A — udld port enables normal mode; udld port aggressive enables aggressive mode.

Q564. Which command enables UDLD globally on all fiber ports?

A. udld enable
B. udld aggressive
C. udld global
D. no udld
Answer: A — udld enable in global config enables UDLD on all fiber ports by default.

Q565. What is the difference between T568A and T568B?

A. Color order of pairs 2 and 3 is swapped (orange/green)
B. T568A is for fiber
C. T568B is for coax
D. No difference
Answer: A — T568A and T568B differ in pair 2/3 color order. Both work; pick one and be consistent.

Q566. Which pins are used by PoE mode 802.3af (Mode A)?

A. Pins 1-2 and 3-6 (same as data)
B. Pins 4-5 and 7-8 only
C. All 8 pins
D. Pins 1-2 only
Answer: A — 802.3af Mode A: power on data pairs (1-2, 3-6). Mode B: power on spare pairs (4-5, 7-8).

Q567. What is the maximum distance for a Cat6 cable?

A. 100 meters (328 feet)
B. 50 meters
C. 200 meters
D. 1000 meters
Answer: A — Standard Cat6 supports up to 100m for Ethernet (90m permanent + 10m patch).

Q568. Which cable type is best for high-interference environments?

A. STP (Shielded Twisted Pair)
B. UTP (Unshielded)
C. Coax
D. Fiber
Answer: A — STP has shielding to reduce EMI. Fiber is best but more expensive.

Q569. What is the speed of 10GBASE-T over Cat6a?

A. 10 Gbps up to 100m
B. 10 Gbps up to 55m
C. 1 Gbps only
D. 100 Mbps
Answer: A — 10GBASE-T: Cat6a = 100m, Cat6 = 55m, Cat7 = 100m.

Q570. Which cable type is used for console connections to Cisco devices?

A. Rollover cable (8-pin, reversed pinout)
B. Straight-through Ethernet
C. Crossover Ethernet
D. Fiber
Answer: A — Rollover cable connects PC serial (DB9 or USB) to Cisco console port (RJ-45).

Q571. What is the default baud rate for Cisco console?

A. 9600
B. 19200
C. 115200
D. 4800
Answer: A — Default console: 9600 baud, 8 data bits, no parity, 1 stop bit (8N1).

Q572. Which console settings are needed for Cisco device console access?

A. 9600 baud, 8 data bits, no parity, 1 stop bit, no flow control
B. 115200 baud, 7E1
C. 19200 baud, 8N1
D. 4800 baud, 8N1
Answer: A — Standard Cisco console: 9600 8N1, no flow control.

Q573. What does 'speed 100' on a Cisco interface do?

A. Forces 100 Mbps
B. Sets speed to 100 Gbps
C. Increases speed by 100%
D. Sets MTU to 100
Answer: A — speed 100 forces the port to 100 Mbps. 'speed auto' (default) auto-negotiates.

Q574. What does 'duplex full' on a Cisco interface do?

A. Forces full-duplex

B. Forces half-duplex

C. Sets duplex to auto

D. Disables duplex

Answer: A — duplex full forces full-duplex. Best practice: leave on auto for GigE+ (auto-MDIX requires auto).

Q575. What happens if one side is set to 100/full and the other to auto?

A. Duplex mismatch — auto side picks half-duplex (collisions and CRC errors)

B. Both sides pick full-duplex

C. Link goes down

D. Nothing — auto negotiates correctly

Answer: A — When one side is forced, the auto side can't detect duplex and defaults to half-duplex. Mismatch!

Q576. Which command shows interface error counters?

A. show interfaces

B. show errors

C. show interface counters errors

D. Both A and C

Answer: D — show interfaces shows errors like CRC, runts, giants, collisions. 'show int counters errors' is more concise.

Q577. What do CRC errors typically indicate?

A. Cable issues, EMI, duplex mismatch (half side)

B. Routing problems

C. VLAN misconfig

D. DNS issues

Answer: A — CRC errors = corrupted frames. Often a Layer 1 issue: bad cable, EMI, or duplex mismatch.

Q578. What do runts indicate?

A. Undersized frames (<64 bytes), often from collision or duplex mismatch

B. Oversized frames

C. Routing errors

D. DNS errors

Answer: A — Runts = frames smaller than 64 bytes. Often caused by collisions on a half-duplex side of a mismatch.

Q579. What do giants indicate?

A. Oversized frames (>1518 bytes), often from jumbo frames or NIC bug

B. Small frames

C. Collision issues

D. Routing issues

Answer: A — Giants = frames larger than 1518 bytes. Can be legitimate (jumbo) or NIC driver bug.

Q580. What do late collisions indicate?

A. Duplex mismatch or cable too long — collision after 64 bytes of frame

B. Normal operation

C. Cable too short

D. Speed mismatch

Answer: A — Late collisions happen after 64 bytes. Should NEVER happen in properly working switched network. Often duplex mismatch.

Q581. Which command clears interface counters?

A. clear counters

B. reset counters

C. clear interface counters

D. no counters

Answer: A — clear counters resets all interface statistics for a fresh measurement.

Q582. What is the default Cisco switch port mode (access/trunk)?

A. Dynamic auto (will become access or trunk based on neighbor)

B. Access

C. Trunk

D. Disabled

Answer: A — Default = dynamic auto. Waits for the other side to negotiate. Best practice: explicitly set mode.

Q583. Which DTP mode actively tries to become a trunk?

A. dynamic desirable

B. dynamic auto

C. access

D. trunk

Answer: A — dynamic desirable actively initiates trunk negotiation. Auto waits. Best practice: hard-code mode + nonegotiate.

Q584. What is the safest configuration for a trunk port?

A. switchport mode trunk + switchport nonegotiate

B. switchport mode dynamic desirable

C. switchport mode dynamic auto

D. switchport mode access

Answer: A — Hard-code trunk mode and disable DTP (nonegotiate) for security and predictability.

Q585. What is the safest configuration for an access port?

A. switchport mode access + switchport nonegotiate + shutdown unused

B. switchport mode trunk

C. switchport mode dynamic auto

D. switchport mode dynamic desirable

Answer: A — Hard-code access mode, disable DTP, and shut down unused ports for security.

Q586. Which command lists all VLANs on a switch?

A. show vlan brief

B. show vlans

C. show vlan all

D. show vlan list

Answer: A — show vlan brief lists all VLANs, names, status, and assigned ports in concise format.

Q587. Which command displays VLAN trunk info?

A. show interfaces trunk

B. show trunk

C. show vlan trunk

D. show trunk status

Answer: A — show interfaces trunk shows trunk ports, native VLAN, allowed VLANs, etc.

Q588. How do you allow only VLANs 10 and 20 on a trunk?

A. switchport trunk allowed vlan 10,20

B. switchport trunk allow 10 20

C. trunk allow 10,20

D. switchport allowed 10,20

Answer: A — switchport trunk allowed vlan 10,20 restricts the trunk to only those VLANs.

Q589. How do you add VLAN 30 to an existing allowed VLAN list?

A. switchport trunk allowed vlan add 30

B. switchport trunk allowed vlan 10,20,30

C. trunk add 30

D. vlan add 30

Answer: A — Use 'add' keyword. Without 'add', you'd replace the list and possibly break other VLANs.

Q590. What is the default native VLAN on Cisco switches?

A. VLAN 1

B. VLAN 99

C. VLAN 1002

D. VLAN 4094

Answer: A — Default native VLAN = 1. Best practice: change to unused VLAN for security (prevent VLAN hopping).

Q591. Why change the native VLAN from 1?

A. Security — VLAN 1 carries CDP, STP, and other management traffic

B. Faster performance

C. Required by 802.1Q

D. Better QoS

Answer: A — VLAN 1 is the default for management/control traffic. Moving native to unused VLAN reduces attack surface.

Q592. How do you set VLAN 99 as native on a trunk?

A. switchport trunk native vlan 99

B. switchport native vlan 99

C. native vlan 99

D. trunk native 99

Answer: A — switchport trunk native vlan 99. Must match on both ends of the trunk!

Q593. What happens if native VLANs don't match on both ends of a trunk?

A. Traffic from one VLAN leaks into the other (native mismatch)

B. Trunk fails completely

C. All traffic is dropped

D. No effect

Answer: A — Native VLAN mismatch: traffic from VLAN X on one side ends up in VLAN Y on the other. CDP warns about this.

Q594. Which command shuts down an unused VLAN?

A. shutdown (in vlan config mode)

B. no vlan 10

C. vlan shutdown

D. disable vlan

Answer: A — In VLAN config mode: shutdown takes the VLAN administratively down.

Q595. Which command deletes VLAN 10 from the running-config?

A. no vlan 10

B. delete vlan 10

C. remove vlan 10

D. no vlan 10 config

Answer: A — no vlan 10 (in global config) removes VLAN 10 from running-config.

Q596. Where is the vlan.dat file stored?

A. Flash memory

B. NVRAM

C. RAM

D. ROM

Answer: A — On older Cisco switches, VLAN info is stored in flash as vlan.dat. Modern IOS uses running-config.

Q597. How do you delete all VLANs and reset to default?

A. delete flash:vlan.dat + reload

B. erase vlan.dat

C. no vlan all

D. clear vlan

Answer: A — delete flash:vlan.dat then reload the switch to clear VLAN database.

Q598. What does VTP pruning do?

A. Stops VLAN flooding to switches that don't need it

B. Removes unused VLANs permanently

C. Trims MAC tables

D. Reduces STP traffic

Answer: A — VTP pruning prevents unknown unicast/broadcast flooding to switches without ports in that VLAN.

Q599. How do you enable VTP pruning?

A. vtp pruning

B. pruning enable

C. vtp prune

D. no vlan prune

Answer: A — vtp pruning (in global config, on VTP server). Pruning applies to the entire VTP domain.

Q600. What VTP version added support for extended VLANs (1006-4094)?

A. VTP v3

B. VTP v2

C. VTP v1

D. All versions

Answer: A — VTP v3 supports extended VLANs, private VLANs, MST, and adds better authentication.

Q601. What is a private VLAN (PVLAN)?

A. Subdivides a VLAN into isolated/promiscuous sub-VLANs for extra isolation

B. A private VLAN for one user

C. An encrypted VLAN

D. A VLAN on a private switch

Answer: A — PVLANs isolate hosts within the same VLAN. Promiscuous (talk to all), community (talk to group), isolated (only promiscuous).

Q602. What is the difference between PVLAN isolated and community?

A. Isolated can't talk to anyone (only promiscuous); community can talk within the group

B. Isolated is faster

C. Community is encrypted

D. No difference

Answer: A — Isolated: only communicates with promiscuous port. Community: members talk to each other + promiscuous.

Q603. What is a promiscuous PVLAN port?

A. Port that can talk to all PVLAN hosts (router/firewall/uplink)

B. Most secure port

C. Isolated port

D. Community port

Answer: A — Promiscuous ports (typically routers/uplinks) can talk to all isolated/community ports in the PVLAN.

Q604. What is the use case for PVLANs?

A. ISP hosting — isolate customers in same subnet

B. Home networks

C. Small offices

D. Wireless networks

Answer: A — PVLANs let service providers isolate customer hosts in the same IP subnet without consuming extra VLANs.

Q605. Which command configures a PVLAN host port?

A. switchport mode private-vlan host

B. switchport pvlan host

C. private-vlan host

D. switchport mode pvlan-host

Answer: A — switchport mode private-vlan host + association with primary/secondary VLANs.

Q606. What is SPAN?

A. Switched Port Analyzer — port mirroring for traffic capture

B. Spanning Tree Analyzer

C. Switched Packet Analyzer

D. Standard Port Access Network

Answer: A — SPAN copies traffic from source ports/VLANs to a destination port connected to a packet analyzer.

Q607. What is RSPAN?

A. Remote SPAN — mirrors across switches via a dedicated RSPAN VLAN

B. Reverse SPAN

C. Rapid SPAN

D. Redundant SPAN

Answer: A — RSPAN extends SPAN across switches using a special RSPAN VLAN to carry mirrored traffic.

Q608. What is VSPAN?

A. SPAN source is a VLAN instead of a port

B. Voice SPAN

C. Virtual SPAN

D. VLAN-aware SPAN

Answer: A — VSPAN = SPAN with VLAN as source. All traffic in that VLAN is mirrored.

Q609. Which command configures SPAN source Fa0/1 to destination Fa0/2?

A. monitor session 1 source interface Fa0/1 both; monitor session 1 destination interface Fa0/2

B. span Fa0/1 to Fa0/2

C. mirror Fa0/1 Fa0/2

D. monitor Fa0/1 to Fa0/2

Answer: A — Configure session number, source, and destination. 'both' = RX+TX direction.

Q610. Which command displays SPAN sessions?

A. show monitor

B. show span

C. show monitor session

D. Both A and C

Answer: D — show monitor or show monitor session displays SPAN configuration.

Q611. What is the difference between SPAN and a network tap?

A. SPAN is software (switch feature); tap is hardware (passive inline device)

B. SPAN is more accurate

C. Taps can modify traffic

D. SPAN works on fiber only

Answer: A — SPAN is software-based on the switch. Taps are physical inline devices — no impact on switch CPU, fully passive.

Q612. What is a router-on-a-stick?

A. Router with subinterfaces for inter-VLAN routing

B. A router with one cable

C. A router with a single antenna

D. A standalone router

Answer: A — Router-on-a-stick: one physical interface with multiple subinterfaces, each handling one VLAN's traffic.

Q613. How do you configure a router-on-a-stick for VLAN 10?

A. interface Fa0/0.10; encapsulation dot1q 10; ip address 192.168.10.1 255.255.255.0

B. interface Fa0/0 vlan 10

C. subinterface Fa0/0.10

D. vlan 10 interface Fa0/0

Answer: A — Subinterface number (often = VLAN ID) + 802.1Q encapsulation + IP for that VLAN's gateway.

Q614. What is an SVI?

A. Switch Virtual Interface — Layer 3 interface for a VLAN

B. Standard Virtual Interface

C. Switched VLAN Interface

D. System Virtual Interface

Answer: A — SVI = virtual interface representing a VLAN. Used for inter-VLAN routing on L3 switches and switch management.

Q615. Which command creates an SVI for VLAN 10?

A. interface vlan 10

B. svi 10

C. interface sv10

D. vlan interface 10

Answer: A — interface vlan 10 creates the SVI. Then assign IP: ip address 192.168.10.1 255.255.255.0.

Q616. What is required for an SVI to be up/up?

A. VLAN exists, has at least one active port, and 'no shutdown' on SVI

B. VLAN must be the native VLAN

C. VLAN must be in VTP server mode

D. SVI is always up

Answer: A — SVI is up/up only if VLAN exists, has at least one port in spanning-tree forwarding, and is not shut down.

Q617. What is a routed port on a switch?

A. Physical port configured as Layer 3 (no switchport)

B. Port assigned to a VLAN

C. Trunk port

D. Console port

Answer: A — Routed port: 'no switchport' turns a switch port into a router port (no Layer 2 switching).

Q618. Which command makes Fa0/24 a routed port?

A. no switchport

B. switchport mode routed

C. ip routed

D. no switchport mode access

Answer: A — no switchport (in interface config) converts the port to Layer 3.

Q619. When would you use a routed port vs an SVI?

A. Routed port for point-to-point L3 links (between switches/routers). SVI for default gateways of VLANs.

B. Always use SVI

C. Always use routed port

D. They are the same

Answer: A — Routed ports for L3 point-to-point (like router-to-router). SVIs for default gateways of access VLANs.

Q620. What is the difference between 'ip routing' and 'ip default-gateway'?

A. ip routing enables L3 routing. ip default-gateway is for L2 switches to reach remote networks.

B. They are the same

C. ip default-gateway is for routers

D. ip routing is only for switches

Answer: A — L2 switches without ip routing use ip default-gateway for management traffic. L3 devices use ip routing + ip route.

Q621. Which command enables IP routing on a Layer 3 switch?

A. ip routing

B. routing enable

C. ip route enable

D. no ip routing off

Answer: A — ip routing (in global config) enables Layer 3 routing on a switch.

Q622. What is the default state of 'ip routing' on Cisco L3 switches?

A. Disabled (must be enabled manually)

B. Enabled

C. Auto-detected

D. Depends on license

Answer: A — Default = disabled. Most L3 switches need 'ip routing' enabled explicitly to route between VLANs.

Q623. What is the default state of 'ip routing' on Cisco routers?

A. Enabled

B. Disabled

C. Auto

D. Varies by model

Answer: A — Routers always have ip routing enabled by default (it's their main job).

Q624. Which command enables IPv6 routing?

A. ipv6 unicast-routing

B. ipv6 routing

C. ip routing ipv6

D. ipv6 enable

Answer: A — ipv6 unicast-routing (global config) enables IPv6 routing. 'ipv6 enable' only enables IPv6 on an interface.

Q625. What does the 'ipv6 enable' interface command do?

A. Enables IPv6 on the interface and generates a link-local address

B. Enables IPv6 routing globally

C. Assigns a global IPv6 address

D. Disables IPv6

Answer: A — ipv6 enable activates IPv6 processing on an interface and creates a link-local (FE80::) address.

Q626. Which command assigns an IPv6 address to an interface?

A. ipv6 address 2001:db8::1/64

B. ip address 2001:db8::1/64

C. ipv6 address 2001:db8::1 255.255.255.0

D. address ipv6 2001:db8::1

Answer: A — ipv6 address /. CIDR notation, not dotted-decimal mask.

Q627. How do you enable SLAAC on a Cisco interface?

A. Just configure ipv6 address autoconfig (or rely on RA)

B. ipv6 slaac

C. ipv6 router-advertisement

D. ipv6 address slaac

Answer: A — SLAAC happens automatically if ipv6 enable is set and router advertisements are on (default).

Q628. Which command shows IPv6 interface info?

A. show ipv6 interface brief

B. show ipv6 interface

C. show interface ipv6

D. Both A and B

Answer: D — show ipv6 interface brief for concise, show ipv6 interface for detailed.

Q629. Which command shows the IPv6 routing table?

A. show ipv6 route

B. show ipv6 routing

C. show ip route ipv6

D. show ipv6 table

Answer: A — show ipv6 route displays IPv6 routing table.

Domain 3: IP Connectivity

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Q630. What does a router use to make forwarding decisions?

A. Routing table (longest prefix match)

B. MAC address table

C. ARP cache

D. DNS cache

Answer: A — Router looks up destination IP in routing table. Longest prefix match wins.

Q631. What is the longest prefix match rule?

A. Most specific route (longest mask) wins

B. Lowest metric wins

C. Lowest AD wins

D. First learned route wins

Answer: A — If multiple routes match destination, the one with longest prefix (most specific) is used.

Q632. What is Administrative Distance (AD)?

A. Trustworthiness of a route source (lower = more trusted)

B. Hop count

C. Bandwidth metric

D. Latency

Answer: A — AD is a Cisco-proprietary ranking of route source trust. Directly connected = 0, static = 1, etc.

Q633. What is the default AD of a directly connected route?

A. 0

B. 1

C. 90

D. 120

Answer: A — Directly connected routes have AD = 0 (most trusted).

Q634. What is the default AD of a static route?

A. 0

B. 1

C. 5

D. 90

Answer: B — Static route AD = 1 (more trusted than any dynamic protocol).

Q635. What is the default AD of OSPF (internal)?

A. 90

B. 100

C. 110

D. 120

Answer: C — OSPF internal = 110.

Q636. What is the default AD of EIGRP (internal)?

A. 90
B. 100
C. 110
D. 120
Answer: A — EIGRP internal = 90.

Q637. What is the default AD of EIGRP (external)?

A. 90
B. 100
C. 170
D. 255
Answer: C — EIGRP external = 170 (less trusted than OSPF internal).

Q638. What is the default AD of RIP?

A. 90
B. 100
C. 110
D. 120
Answer: D — RIP AD = 120. High = less trusted than OSPF/EIGRP.

Q639. What is the default AD of eBGP?

A. 20
B. 100
C. 200
D. 255
Answer: A — eBGP AD = 20 (very trusted). iBGP AD = 200.

Q640. What is the default AD of iBGP?

A. 20
B. 100
C. 200
D. 255
Answer: C — iBGP AD = 200 (less trusted than most IGP routes).

Q641. What is the default AD of a floating static route?

A. 1 (default)
B. Custom (typically 200-254)
C. 0
D. 255
Answer: B — Floating static = static with high AD (e.g., 250) used as backup when dynamic route fails.

Q642. What is route metric?

A. Cost/distance to reach destination (used to choose between routes from same protocol)
B. Same as AD
C. Hop count only
D. Bandwidth
Answer: A — Metric is protocol-specific (e.g., OSPF cost, EIGRP composite, RIP hop count) and chooses best route within protocol.

Q643. What is the metric for RIP?

A. Hop count

B. Cost

C. Bandwidth + delay

D. Load

Answer: A — RIP uses hop count (max 15, 16 = unreachable).

Q644. What is the metric for OSPF?

A. Cost (based on bandwidth: ref_bw / interface_bw)

B. Hop count

C. Delay

D. Bandwidth

Answer: A — OSPF cost = reference bandwidth (default 100 Mbps) / interface bandwidth.

Q645. What is the default OSPF reference bandwidth?

A. 100 Mbps

B. 1 Gbps

C. 10 Gbps

D. 10 Mbps

Answer: A — Default ref-bw = 100 Mbps. Modern networks should use 'auto-cost reference-bandwidth 10000' for 10 Gbps.

Q646. What is the metric for EIGRP?

A. Composite (bandwidth + delay, optionally load + reliability + MTU)

B. Hop count

C. Cost

D. Latency

Answer: A — EIGRP uses a composite metric. Default = bandwidth + delay. Others can be added but usually aren't.

Q647. What is the metric for BGP?

A. Multi-attribute (AS-path, local-pref, MED, weight, etc.)

B. Hop count

C. Bandwidth

D. Delay

Answer: A — BGP uses path-vector logic with multiple attributes (not a simple metric).

Q648. What does a static route look like?

A. ip route 192.168.1.0 255.255.255.0 10.0.0.1

B. ip route 192.168.1.0/24 10.0.0.1

C. static route 192.168.1.0 10.0.0.1

D. route 192.168.1.0 10.0.0.1

Answer: A — ip route . Mask is dotted-decimal, not CIDR.

Q649. What is a recursive static route?

A. Static route whose next-hop must be resolved via another route lookup

B. Static route that loops

C. Static route with no next-hop

D. Static route in a VLAN

Answer: A — If next-hop IP isn't directly connected, the router does another lookup to find how to reach it (recursion).

Q650. Which is preferred: static route to next-hop IP or to exit interface?

A. Next-hop IP (better for multi-access networks)

B. Exit interface (better for point-to-point)

C. Both are equal

D. Neither is preferred

Answer: A — Next-hop IP is more flexible. Exit interface on multi-access causes ARP for every destination (bad).

Q651. What problem occurs with static routes pointing to a multi-access exit interface?

A. Excessive ARP entries (one per destination)

B. Routing loops

C. Duplicate routes

D. High CPU

Answer: A — On Ethernet, 'ip route 0.0.0.0 0.0.0.0 Fa0/0' makes router ARP for every destination IP — wastes memory.

Q652. What is the IPv4 default route command?

A. ip route 0.0.0.0 0.0.0.0

B. ip route 0/0

C. ip default

D. default route

Answer: A — 0.0.0.0 0.0.0.0 = match any destination. Used as route of last resort.

Q653. What is the IPv6 default route command?

A. ipv6 route ::/0

B. ipv6 route 0.0.0.0 0.0.0.0

C. ipv6 default

D. default ipv6

Answer: A — ::/0 = match any IPv6 destination.

Q654. How do you configure a floating static route?

A. Add AD at end: ip route 192.168.1.0 255.255.255.0 10.0.0.1 250

B. Use 'floating' keyword

C. Configure two routes with same metric

D. Configure route on standby router only

Answer: A — Add AD parameter at the end (higher than dynamic protocol's AD) so it's only used when primary fails.

Q655. Which command verifies a static route is in the routing table?

A. show ip route static

B. show static

C. show ip static

D. show route static

Answer: A — show ip route static shows only static routes. 'show ip route' shows everything.

Q656. What does 'S*' mean in the routing table?

A. Candidate default static route

B. Saved static

C. Source static

D. Stale static

Answer: A — S = candidate default route (static route to 0.0.0.0/0).*

Q657. What does the routing code 'C' mean?

A. Connected
B. Static
C. OSPF
D. Cached
Answer: A — C = directly connected network.

Q658. What does the routing code 'L' mean?

A. Local — IP assigned to an interface (/32 host route)
B. Local LAN
C. Linked route
D. Loopback
Answer: A — L = local /32 route for the IP on an interface. Added automatically in newer IOS.

Q659. What does the routing code 'S' mean?

A. Static route
B. Saved route
C. Source route
D. Subnet route
Answer: A — S = static route (manually configured).

Q660. What does the routing code 'O' mean?

A. OSPF route
B. Origin route
C. Outgoing
D. Open
Answer: A — O = OSPF internal route.

Q661. What does the routing code 'O E1' mean?

A. OSPF external type 1 route
B. OSPF external type 2 route
C. OSPF edge route
D. OSPF express 1
Answer: A — O E1 = OSPF external type 1. Cost = external cost + internal cost. O E2 (default) = only external cost.

Q662. What does the routing code 'O E2' mean?

A. OSPF external type 2 (default for external routes)
B. OSPF external type 1
C. OSPF edge route type 2
D. OSPF express 2
Answer: A — O E2 = external cost only (no internal cost added). Default for OSPF external routes.

Q663. What does the routing code 'O IA' mean?

A. OSPF inter-area route
B. OSPF internal area
C. OSPF interface area
D. OSPF isolated area
Answer: A — O IA = OSPF inter-area route (route from another OSPF area).

Q664. What does the routing code 'D' mean?

A. EIGRP (DUAL)
B. Direct
C. Default
D. Dynamic
Answer: A — D = EIGRP (named after DUAL algorithm).

Q665. What does the routing code 'D EX' mean?

A. EIGRP external route (redistributed)
B. EIGRP express
C. EIGRP extended
D. EIGRP exception
Answer: A — D EX = EIGRP external (redistributed from another protocol).

Q666. What does the routing code 'B' mean?

A. BGP route
B. Backup route
C. Broadcast route
D. Best route
Answer: A — B = BGP route.

Q667. What does the routing code 'R' mean?

A. RIP route
B. Redistributed
C. Recursive
D. Reserved
Answer: A — R = RIP route.

Q668. Which command displays the routing table?

A. show ip route
B. show route
C. show routing-table
D. show ip table
Answer: A — show ip route is the primary command. Add protocol name to filter (show ip route ospf).

Q669. Which command shows only OSPF-learned routes?

A. show ip route ospf
B. show ospf routes
C. show ip ospf
D. show ip route o
Answer: A — show ip route ospf filters routing table to only OSPF entries.

Q670. Which command displays a specific route?

A. show ip route 192.168.1.0
B. show ip route specific 192.168.1.0
C. show route 192.168.1.0
D. find route 192.168.1.0
Answer: A — show ip route shows the matching route and detailed info.

Q671. Which command displays the default gateway/route?

A. show ip route 0.0.0.0

B. show ip route 0.0.0.0 0.0.0.0

C. show default-gateway

D. Both A and B work; also 'show ip route | include 0.0.0.0*'

Answer: D — Multiple ways work. Easiest: 'show ip route 0.0.0.0 0.0.0.0' or 'show ip route | incl 0.0.0.0'

Q672. What is a gateway of last resort?

A. The router where default route points

B. Last router in network

C. Border router

D. DHCP server

Answer: A — Gateway of last resort = next-hop of the default route. Shown in 'show ip route' header.

Q673. Which command shows CDP neighbors' IP addresses?

A. show cdp neighbors detail

B. show cdp entry *

C. show cdp ip

D. Both A and B

*Answer: D — Both 'show cdp neighbors detail' and 'show cdp entry *' show IPs of neighbors.*

Q674. What is OSPF?

A. Open Shortest Path First — link-state IGP (RFC 2328)

B. Open Static Path Finder

C. Optimal Shortest Path First

D. Open Source Path Finder

Answer: A — OSPF = link-state IGP. Uses Dijkstra's SPF algorithm. OSPFv2 = IPv4, OSPFv3 = IPv6 (and IPv4 with AF).

Q675. What algorithm does OSPF use?

A. Dijkstra's Shortest Path First (SPF)

B. Bellman-Ford

C. Diffusing Update Algorithm (DUAL)

D. Hop count

Answer: A — OSPF uses Dijkstra's SPF algorithm to compute shortest path tree from each router's perspective.

Q676. What does OSPF use as its metric?

A. Cost (based on interface bandwidth)

B. Hop count

C. Delay

D. Bandwidth + delay

Answer: A — Cost = ref-bw (default 100 Mbps) / interface bandwidth. Higher speed = lower cost = preferred.

Q677. What is the OSPF cost of a 100 Mbps link (default ref-bw 100 Mbps)?

A. 1

B. 10

C. 100

D. 1000

Answer: A — 100/100 = 1. Cost of 10 Mbps = 10. Cost of 1 Gbps = 1 (problem! use auto-cost ref-bw 10000).

Q678. What is the OSPF cost of a 1 Gbps link with default ref-bw (100 Mbps)?

A. 1 (rounded down — should be 0.1)

B. 10

C. 100

D. 1000

Answer: A — 100/1000 = 0.1, rounded to 1. Problem: 100M and 1G both have cost 1. Fix: 'auto-cost reference-bandwidth 10000'.

Q679. Which command sets the OSPF reference bandwidth?

A. auto-cost reference-bandwidth 10000

B. ospf reference-bw 10000

C. auto-cost ref-bw 10000

D. reference-bandwidth 10000

Answer: A — auto-cost reference-bandwidth in OSPF router config. Should match on all OSPF routers.

Q680. What is the OSPF router ID?

A. Unique 32-bit ID identifying each OSPF router (looks like IP)

B. MAC address

C. Hostname

D. ASN

Answer: A — Router ID = highest IP on loopback, else highest IP on physical interface, else manually set.

Q681. How is the OSPF router ID selected by default?

A. 1. Manually configured (if set) → 2. Highest loopback IP → 3. Highest physical IP

B. 1. Highest physical IP → 2. Manually configured

C. Random

D. Lowest IP

Answer: A — Order: manual → highest loopback → highest physical. Best practice: set manually or use loopback.

Q682. Which command manually sets the OSPF router ID?

A. router-id 1.1.1.1

B. ospf router-id 1.1.1.1

C. router ospf id 1.1.1.1

D. set router-id 1.1.1.1

Answer: A — router-id under router ospf . Must clear OSPF process for it to take effect.

Q683. Which command clears the OSPF process (to apply new router ID)?

A. clear ip ospf process

B. clear ospf

C. reset ospf

D. reload

Answer: A — clear ip ospf process restarts OSPF — applies new RID, re-elects DR/BDR.

Q684. What is the OSPF process ID?

A. Local identifier for OSPF instance on a router (1-65535)

B. Same as router ID

C. Same as area ID

D. Same as ASN

Answer: A — Process ID is locally significant only — doesn't need to match between routers. Used to allow multiple OSPF instances.

Q685. Which command enables OSPF process 1?

A. router ospf 1

B. ospf 1

C. router ospf process 1

D. ospf enable 1

Answer: A — router ospf enters OSPF router config mode.

Q686. Which command enables OSPF on network 192.168.1.0/24 in area 0?

A. network 192.168.1.0 0.0.0.255 area 0

B. network 192.168.1.0 255.255.255.0 area 0

C. network 192.168.1.0/24 area 0

D. ospf network 192.168.1.0 area 0

Answer: A — network area . Use WILDCARD mask, not subnet mask!

Q687. Which command is the modern way to enable OSPF on an interface?

A. interface Fa0/0; ip ospf 1 area 0

B. interface Fa0/0; ospf 1 area 0

C. router ospf 1; interface Fa0/0 area 0

D. interface Fa0/0; network 1 area 0

Answer: A — Modern interface-level config: ip ospf area under the interface.

Q688. What is OSPF area 0?

A. Backbone area — all other areas must connect to it

B. First area

C. Default area

D. Largest area

Answer: A — Area 0 (0.0.0.0) is the OSPF backbone. All non-backbone areas must connect to it (directly or via virtual link).

Q689. What is an ABR?

A. Area Border Router — connects OSPF areas (has interfaces in 2+ areas)

B. Area Border Region

C. Automatic Backup Router

D. Aggregated Backbone Router

Answer: A — ABR has interfaces in 2+ areas, at least one of which is area 0. Maintains LSDBs for each connected area.

Q690. What is an ASBR?

A. Autonomous System Boundary Router — injects external routes into OSPF

B. Area Source Boundary Router

C. Auto Static Boundary Router

D. Active Source Boundary Router

Answer: A — ASBR redistributes routes from other protocols (or static/connected) into OSPF. Creates external LSAs.

Q691. What is the OSPF Hello interval default on broadcast networks?

A. 10 seconds

B. 30 seconds

C. 60 seconds

D. 5 seconds

Answer: A — Default hello = 10s on broadcast/point-to-point. 30s on NBMA. Dead = 4×hello.

Q692. What is the OSPF dead interval default on broadcast networks?

A. 40 seconds

B. 30 seconds

C. 120 seconds

D. 20 seconds

Answer: A — Default dead = 4×hello = 40s on broadcast. If no hello in 40s, neighbor considered dead.

Q693. What happens if OSPF hello/dead intervals don't match between neighbors?

A. Neighbors won't form adjacency

B. Neighbors form but with errors

C. Slower convergence

D. Nothing

Answer: A — Hello/dead intervals must match. Also area ID, auth, stub flag, and subnet mask (on broadcast).

Q694. Which OSPF network type is the default on Ethernet?

A. Broadcast (elects DR/BDR)

B. Point-to-point

C. NBMA

D. Point-to-multipoint

Answer: A — Ethernet default = broadcast. DR/BDR elected to reduce LSA flooding.

Q695. Which OSPF network type is the default on Point-to-Point links (PPP/HDLC)?

A. Point-to-point (no DR/BDR)

B. Broadcast

C. NBMA

D. Point-to-multipoint

Answer: A — P2P links have only 2 routers — no need for DR/BDR. Adjacency forms directly.

Q696. What is the OSPF DR?

A. Designated Router — represents the multi-access network, generates Network LSA

B. Default Router

C. Direct Router

D. Dynamic Router

Answer: A — DR reduces LSA flooding on multi-access networks. All routers sync with DR.

Q697. What is the OSPF BDR?

A. Backup Designated Router — takes over if DR fails

B. Border Designated Router

C. Backup Default Router

D. Backup Direct Router

Answer: A — BDR is hot standby for DR. All routers also sync with BDR for fast failover.

Q698. How is the OSPF DR elected?

A. Highest priority (default 1), then highest router ID

B. Lowest priority, then lowest RID

C. First router to boot

D. Highest IP address only

Answer: A — Priority 0 = never DR. Default priority = 1. Higher priority wins. Ties → highest RID.

Q699. What is the OSPF priority of a router that can never become DR?

A. 0
B. 1
C. 100
D. 255
Answer: A — Priority 0 = ineligible to be DR/BDR. Use on edge routers you don't want as DR.

Q700. Which command sets the OSPF priority to 0?

A. ip ospf priority 0
B. ospf priority 0
C. priority 0
D. ip ospf 0 priority
Answer: A — ip ospf priority <0-255> in interface config. Default = 1.

Q701. When can a new OSPF router preempt as DR?

A. Never — DR election is non-preemptive. Only triggers if DR/BDR fail.
B. Always
C. Only if priority is highest
D. Only if router ID is highest
Answer: A — OSPF DR election is non-preemptive. Even a router with higher priority won't preempt an existing DR.

Q702. What is an OSPF LSA?

A. Link State Advertisement — OSPF's data unit describing part of the topology
B. Link State Acknowledgment
C. Link State Algorithm
D. Loop State Advertisement
Answer: A — LSAs are flooded through the area, building the Link State Database (LSDB).

Q703. What is a Type 1 LSA (Router LSA)?

A. Describes router's links within an area (from every OSPF router)
B. Describes network (from DR)
C. Describes summary (from ABR)
D. Describes external (from ASBR)
Answer: A — Type 1 = Router LSA. Flooded within area only. Lists all links, interfaces, and neighbors.

Q704. What is a Type 2 LSA (Network LSA)?

A. Describes a multi-access network (from DR)
B. Describes router links
C. Describes summary routes
D. Describes external routes
Answer: A — Type 2 = Network LSA. Generated by DR. Lists all routers on the multi-access segment.

Q705. What is a Type 3 LSA (Summary LSA)?

A. Describes networks in other areas (from ABR)
B. Describes router links
C. Describes multi-access network
D. Describes external ASBR
Answer: A — Type 3 = Summary LSA. ABR advertises inter-area networks. Flooded within area.

Q706. What is a Type 4 LSA (ASBR Summary LSA)?

A. Advertises location of ASBR (from ABR)

B. Describes router links

C. Describes multi-access network

D. Describes external routes

Answer: A — Type 4 = ASBR Summary. Tells other areas how to reach the ASBR.

Q707. What is a Type 5 LSA (External LSA)?

A. Describes external networks (from ASBR)

B. Describes router links

C. Describes multi-access network

D. Describes summary routes

Answer: A — Type 5 = External LSA. Flooded throughout AS (except stub areas).

Q708. What is a Type 7 LSA?

A. Used in NSSA areas to carry external routes

B. Same as Type 5

C. Used in stub areas

D. Used in backbone only

Answer: A — Type 7 = NSSA External LSA. Created by ASBR in NSSA. Converted to Type 5 by ABR.

Q709. What is an OSPF stub area?

A. Area that doesn't accept Type 5 (external) LSAs; default route injected instead

B. Area with no routers

C. Area without ABR

D. Backup area

Answer: A — Stub area blocks external LSAs. ABR injects default route. Reduces LSDB size.

Q710. What is an OSPF totally stubby area?

A. Blocks Type 3, 4, and 5 LSAs — only intra-area routes + default

B. Same as stub

C. Only blocks Type 5

D. Blocks Type 1 and 2

Answer: A — Totally stubby = Cisco proprietary. Blocks all inter-area + external LSAs. Just intra-area + default.

Q711. What is an OSPF NSSA?

A. Not-So-Stubby Area — like stub but allows ASBR (Type 7 LSAs)

B. Network Secure Service Area

C. Network Segment Service Area

D. Non-Stub Service Area

Answer: A — NSSA: external routes allowed (Type 7) but no Type 5. Used when stub area needs to import some externals.

Q712. What is an OSPF totally NSSA?

A. Like NSSA but also blocks Type 3 LSAs (only intra-area + default + Type 7)

B. Same as NSSA

C. Same as totally stubby

D. Same as backbone

Answer: A — Totally NSSA = NSSA + blocks inter-area summaries. Cisco proprietary. Most restrictive external-allowing area.

Q713. Which command configures an area as stub?

A. area 1 stub

B. stub area 1

C. area 1 stubby

D. area stub 1

Answer: A — area stub (under router ospf). Must be on all routers in the area.

Q714. Which command configures a totally stubby area?

A. area 1 stub no-summary

B. area 1 totally-stub

C. area 1 stub no-lsa

D. area 1 totally-stubby

Answer: A — area 1 stub no-summary on ABR. Just 'area 1 stub' on internal routers.

Q715. Which command configures an NSSA?

A. area 1 nssa

B. nssa area 1

C. area 1 not-so-stubby

D. area 1 nssa-only

Answer: A — area nssa on all routers in the area.

Q716. What is OSPF virtual link?

A. Connects an area to backbone through a non-backbone area

B. Connects two areas directly

C. Encrypts OSPF

D. Speeds up convergence

Answer: A — Virtual link tunnels OSPF through a transit area to connect a disconnected area to backbone. Temporary fix only.

Q717. Which command creates an OSPF virtual link?

A. area 1 virtual-link

B. virtual-link 1

C. area 1 virtual 2

D. ospf virtual 1

Answer: A — area virtual-link . Configure on both ABRs.

Q718. What is OSPF SPF throttle?

A. Limits how often SPF is calculated to protect CPU

B. Speeds up SPF

C. Disables SPF

D. Encrypts SPF

Answer: A — SPF throttle prevents router from running SPF too often during topology changes.

Q719. What is OSPF graceful restart (NSF)?

A. Allows OSPF to keep forwarding during RP failover (with state preservation)

B. Restarts OSPF immediately

C. Disables OSPF

D. Reloads router

Answer: A — NSF (Non-Stop Forwarding) keeps forwarding using stale LSDB while OSPF restarts. Requires helper routers.

Q720. What is OSPF authentication type 0?

A. Null (no authentication)

B. Simple password

C. MD5

D. SHA-256

Answer: A — Type 0 = null. Type 1 = simple (clear text). Type 2 = MD5.

Q721. What is OSPF authentication type 1?

A. Simple password (clear text)

B. Null

C. MD5

D. SHA-256

Answer: A — Type 1 = clear text password. Not secure — prefer MD5.

Q722. What is OSPF authentication type 2?

A. MD5

B. Null

C. Simple password

D. SHA-256

Answer: A — Type 2 = MD5 cryptographic authentication. Most common secure option.

Q723. Which command enables OSPF MD5 authentication on an interface?

A. ip ospf message-digest-key 1 md5 + ip ospf authentication message-digest

B. ip ospf auth md5

C. ospf authentication md5

D. ip ospf md5

Answer: A — Two commands: define the key, then enable message-digest mode. Both must be on the interface.

Q724. Which command displays OSPF neighbors?

A. show ip ospf neighbor

B. show ospf neighbors

C. show ip ospf

D. show neighbor ospf

Answer: A — show ip ospf neighbor shows all OSPF neighbors and their states.

Q725. What OSPF neighbor state means full adjacency formed?

A. FULL

B. UP

C. DONE

D. ESTAB

Answer: A — FULL = LSDBs are fully synchronized. This is the desired state for DR/BDR relationships.

Q726. What OSPF neighbor state means 2-way communication but no full adjacency?

A. 2WAY

B. EXSTART

C. INIT

D. FULL

Answer: A — 2WAY = neighbors see each other but didn't form full adjacency (normal between DROthers on broadcast).

Q727. What OSPF neighbor state means just received a Hello?

A. INIT
B. 2WAY
C. EXSTART
D. FULL
Answer: A — INIT = local router received hello from neighbor but neighbor doesn't list local RID yet.

Q728. What OSPF neighbor state means starting LDB exchange?

A. EXSTART
B. INIT
C. 2WAY
D. FULL
Answer: A — EXSTART = negotiating master/slave and initial DBD sequence numbers.

Q729. What OSPF neighbor state means exchanging LSA headers?

A. EXCHANGE
B. EXSTART
C. LOADING
D. FULL
Answer: A — EXCHANGE = routers are exchanging Database Description (DBD) packets with LSA headers.

Q730. What OSPF neighbor state means requesting specific LSAs?

A. LOADING
B. EXCHANGE
C. FULL
D. INIT
Answer: A — LOADING = sending Link State Requests (LSR) for specific LSAs seen in DBDs.

Q731. Which command shows OSPF interface details?

A. show ip ospf interface
B. show ospf interface
C. show ip ospf
D. show interface ospf
Answer: A — show ip ospf interface [brief] shows OSPF config on each interface (cost, area, network type, DR/BDR).

Q732. Which command shows OSPF LSDB?

A. show ip ospf database
B. show ospf database
C. show ip ospf
D. show ip ospf lsdb
Answer: A — show ip ospf database displays the Link State Database.

Q733. Which command shows OSPF route table (just OSPF routes)?

A. show ip route ospf
B. show ospf routes
C. show ip ospf route
D. show ip route | section ospf
Answer: A — show ip route ospf filters routing table to OSPF entries only.

Q734. Which command shows OSPF process info (router ID, areas, SPF)?

A. show ip ospf

B. show ospf

C. show ip ospf process

D. show ip ospf summary

Answer: A — show ip ospf shows OSPF process info: router ID, areas, SPF runs, ABR/ASBR status.

Q735. What is the default OSPF cost of an interface?

A. 100 / bandwidth (Mbps), rounded to integer

B. 1

C. 10

D. 0

Answer: A — Default cost = reference-bandwidth (100) / interface-bandwidth (Mbps). 1Gbps = 0.1 → rounds to 1 (problem).

Q736. Which command manually sets the OSPF cost on an interface?

A. ip ospf cost 50

B. ospf cost 50

C. ip cost 50 ospf

D. cost 50

Answer: A — ip ospf cost <1-65535> in interface config overrides the auto-calculated cost.

Q737. What is OSPF passive interface?

A. Interface that's part of OSPF but doesn't send hellos (no neighbors)

B. Disabled interface

C. Slower interface

D. Backup interface

Answer: A — Passive interface: included in OSPF (advertises network) but doesn't send/receive hellos. Use on LANs without OSPF neighbors.

Q738. Which command makes an interface passive in OSPF?

A. passive-interface Fa0/0 (under router ospf)

B. ospf passive Fa0/0

C. interface passive Fa0/0

D. no ospf hello Fa0/0

Answer: A — passive-interface in router ospf config. Or 'passive-interface default' + 'no passive-interface X' for whitelist.

Q739. What is the best practice for OSPF passive interfaces?

A. Make all LAN interfaces passive, only WAN/inter-router interfaces active

B. All interfaces passive

C. No passive interfaces

D. Only loopback passive

Answer: A — Best practice: 'passive-interface default' then 'no passive-interface ' to whitelist active interfaces.

Q740. What is OSPF summarization?

A. Advertising a single summary route instead of multiple specific routes

B. Reducing LSA count

C. Encrypting LSAs

D. Speeding up SPF

Answer: A — Summarization reduces routing table size and LSA flooding. Done at ABR (inter-area) or ASBR (external).

Q741. Where do you configure OSPF inter-area summarization?

A. On the ABR

B. On any router

C. On the ASBR

D. On the DR

Answer: A — ABR summarizes between areas. ASBR summarizes external routes. Both configured under router ospf.

Q742. Which command summarizes area 1 routes to 192.168.0.0/22 at the ABR?

A. area 1 range 192.168.0.0 255.255.252.0

B. summary 192.168.0.0/22 area 1

C. area 1 summary 192.168.0.0 255.255.252.0

D. summarize area 1 192.168.0.0/22

Answer: A — area range under router ospf. Advertises summary, suppresses specifics.

Q743. Which command summarizes external routes on the ASBR?

A. summary-address 192.168.0.0 255.255.252.0

B. area 0 range 192.168.0.0 255.255.252.0

C. summarize external 192.168.0.0/22

D. summary 192.168.0.0/22

Answer: A — summary-address under router ospf. Summarizes external (Type 5) routes.

Q744. What is OSPF route filtering?

A. Block specific routes from being advertised or installed in routing table

B. Encrypt routes

C. Speed up routing

D. Reduce MTU

Answer: A — Filter routes using distribute-list, route-map, or prefix-list.

Q745. What is OSPF default route origination?

A. ASBR injects default route into OSPF (default-information originate)

B. OSPF creates default automatically

C. ABR creates default

D. DR creates default

Answer: A — default-information originate [always] under router ospf. 'always' advertises default even if no default in RIB.

Q746. Which command injects a default route into OSPF?

A. default-information originate

B. default-route advertise

C. ip default-network

D. advertise default

Answer: A — default-information originate [always] under router ospf. Requires existing default route unless 'always' used.

Q747. What is RIP?

A. Routing Information Protocol — distance-vector IGP, max 15 hops

B. Rapid IP Protocol

C. Router IP Protocol

D. Relay IP Protocol

Answer: A — RIP = distance-vector, hop count metric, max 15 hops. v1 = classful, v2 = classless (supports VLSM).

Q748. What is the maximum hop count in RIP?

A. 15 (16 = unreachable)

B. 16

C. 100

D. 255

Answer: A — RIP max hop count = 15. 16+ is considered infinite (unreachable).

Q749. What is the RIP update interval?

A. 30 seconds

B. 60 seconds

C. 90 seconds

D. 10 seconds

Answer: A — RIP sends full routing table every 30 seconds by default (periodic updates).

Q750. What is RIP split horizon?

A. Don't advertise a route back out the interface you learned it on

B. Split updates by half

C. Send only half the routes

D. Advertise only some routes

Answer: A — Split horizon prevents loops by not re-advertising a route back out the interface it was learned from.

Q751. What is RIP route poisoning?

A. Advertise a failed route with metric 16 (infinity) to propagate failure

B. Encrypt routes

C. Speed up convergence

D. Block bad routes

Answer: A — Route poisoning: failed routes are advertised with infinite metric (16) so neighbors remove them quickly.

Q752. What is RIP poison reverse?

A. Advertise failed route back out the interface it was learned on (with metric 16)

B. Block all reverse routes

C. Speed up reverse traffic

D. Encrypt reverse traffic

Answer: A — Poison reverse = exception to split horizon: explicitly advertise failed route back with infinite metric to ensure everyone knows.

Q753. What is RIP hold-down timer?

A. Time to wait before believing a worse route for an existing destination

B. Time to keep route after deletion

C. Update interval

D. Hello interval

Answer: A — Hold-down = 180s. Suppresses route updates for a destination after a route goes bad, preventing flapping.

Q754. What is the RIP garbage collection (flush) timer?

A. 240 seconds — when route is finally removed from RIP table

B. 30 seconds

C. 60 seconds

D. 120 seconds

Answer: A — After hold-down, RIP waits until 240s total before finally removing the route.

Q755. Which RIP version supports VLSM and CIDR?

A. RIPv2

B. RIPv1

C. Both

D. Neither

Answer: A — RIPv2 supports VLSM (sends subnet mask in updates). RIPv1 is classful.

Q756. Which RIP version supports authentication?

A. RIPv2

B. RIPv1

C. Both

D. Neither

Answer: A — RIPv2 supports MD5 authentication. RIPv1 has no authentication.

Q757. Does RIP support IPv6?

A. Yes, RIPng

B. No, only IPv4

C. Yes, with RIPv3

D. Only with extensions

Answer: A — RIPng (next generation) supports IPv6. Uses UDP 521 (vs 520 for IPv4).

Q758. Which command enables RIPv2?

A. router rip; version 2

B. router rip 2

C. rip v2

D. router rip; no auto-summary; version 2

Answer: D — router rip + version 2 + no auto-summary (to support VLSM properly).

Q759. What does 'no auto-summary' do in RIPv2?

A. Disables automatic classful summarization at major network boundaries

B. Speeds up convergence

C. Encrypts routes

D. Disables RIP

Answer: A — RIPv2 defaults to auto-summary (classful). 'no auto-summary' allows VLSM and discontiguous subnets.

Q760. What is EIGRP?

A. Enhanced Interior Gateway Routing Protocol — Cisco advanced distance-vector

B. External IGMP Routing Protocol

C. Encrypted Interior Gateway Routing Protocol

D. Extended IP Gateway Routing Protocol

Answer: A — EIGRP = Cisco proprietary (now RFC 7868). Advanced distance-vector. Uses DUAL algorithm. Reliable transport.

Q761. What algorithm does EIGRP use?

A. DUAL (Diffusing Update Algorithm)

B. Dijkstra SPF

C. Bellman-Ford

D. Hop count

Answer: A — DUAL guarantees loop-free paths and provides fast convergence via feasible successors.

Q762. What is the EIGRP metric?

A. Composite: bandwidth + delay (default), can include load, reliability, MTU

B. Hop count

C. Cost

D. Latency

Answer: A — EIGRP composite metric. Default K-values: K1 (bandwidth) and K3 (delay) = 1, others = 0.

Q763. What is an EIGRP feasible successor?

A. Backup route that meets feasibility condition (loop-free)

B. Backup router

C. Failed route

D. Alternative path

Answer: A — Feasible successor = backup route kept in topology table. Installed in routing table if successor fails.

Q764. What is an EIGRP successor?

A. Primary route (lowest metric) installed in routing table

B. Backup route

C. Failed route

D. Neighbor router

Answer: A — Successor = best path to destination. Installed in routing table.

Q765. What is the EIGRP feasibility condition?

A. Reported distance < feasible distance (RD < FD)

B. RD > FD

C. RD = FD

D. Hop count = 0

Answer: A — FC: neighbor's reported distance must be less than local feasible distance. Guarantees loop-free path.

Q766. What is reported distance (RD) in EIGRP?

A. Distance from neighbor to destination

B. Distance from local router to destination

C. Total metric

D. Hop count

Answer: A — RD = neighbor's metric to reach destination. Sent to us in updates.

Q767. What is feasible distance (FD) in EIGRP?

A. Local router's metric to reach destination

B. Neighbor's metric

C. Total path metric

D. Hop count

Answer: A — FD = local router's best (lowest) metric to reach destination.

Q768. What is the EIGRP hello interval default?

A. 5 seconds (LAN), 60 seconds (T1 and slower WAN)

B. 10 seconds

C. 30 seconds

D. 60 seconds

Answer: A — EIGRP hello = 5s on fast links, 60s on slow (NBMA ≤T1). Hold = 3×hello.

Q769. What is the EIGRP hold timer default?

A. 15 seconds (3× hello on LAN), 180 seconds on slow WAN

B. 30 seconds

C. 60 seconds

D. 120 seconds

Answer: A — Hold = 3× hello. If no hello in hold time, neighbor is declared dead.

Q770. What is the EIGRP AS number?

A. Autonomous System number — must match between EIGRP neighbors

B. Area ID

C. Process ID

D. Router ID

Answer: A — AS number must match between routers for EIGRP neighborship. Range 1-65535.

Q771. Which command enables EIGRP for AS 100?

A. router eigrp 100

B. eigrp 100

C. router eigrp as 100

D. eigrp enable 100

Answer: A — router eigrp enters EIGRP config mode. Then 'no auto-summary' + network statements.

Q772. How do you enable EIGRP named mode (modern)?

A. router eigrp

B. router eigrp named

C. eigrp name

D. router eigrp named

Answer: A — router eigrp (no AS) uses named mode. Multi-AF (IPv4 + IPv6) under one process.

Q773. What does EIGRP use as its transport?

A. RTP (Reliable Transport Protocol) — its own protocol over IP protocol 88

B. TCP

C. UDP

D. ICMP

Answer: A — EIGRP uses RTP (Cisco proprietary) over IP protocol 88. Reliable for updates, unreliable for hellos.

Q774. What is the EIGRP protocol number?

A. 88

B. 89

C. 6

D. 17

Answer: A — EIGRP = IP protocol 88. OSPF = 89.

Q775. What is EIGRP stub?

A. Router that only advertises limited routes (e.g., connected + summary)

B. Backup router

C. Edge router only

D. Internal router only

Answer: A — Stub routers don't act as transit. Reduces query scope, faster convergence. Common at branch offices.

Q776. Which command configures an EIGRP stub?

A. eigrp stub connected

B. stub eigrp

C. eigrp stub

D. no transit eigrp

Answer: A — eigrp stub [connected | summary | static | redistributed | receive-only | leak-map].

Q777. What is BGP?

A. Border Gateway Protocol — path-vector EGP for inter-AS routing

B. Border Gateway Protocol — link-state IGP

C. Best Gateway Protocol

D. Backup Gateway Protocol

Answer: A — BGP = path-vector protocol used between autonomous systems (the protocol of the Internet).

Q778. What is an autonomous system (AS) in BGP?

A. Group of networks under a single administration with a unified routing policy

B. Area ID

C. Network segment

D. VLAN

Answer: A — AS = set of networks under one administration. Each AS has a unique ASN (2-byte: 1-65535, 4-byte: 1-4294967295).

Q779. What is the difference between eBGP and iBGP?

A. eBGP = between different ASes; iBGP = within same AS

B. eBGP is faster

C. iBGP is for external

D. No difference

Answer: A — eBGP: neighbors in different ASes. iBGP: neighbors in same AS. Different TTL/loop prevention rules.

Q780. What ports does BGP use?

A. TCP 179

B. UDP 179

C. TCP 89

D. UDP 89

Answer: A — BGP uses TCP port 179 for its transport.

Q781. What is the BGP AS path attribute?

A. List of ASes a route has traversed (loop prevention + shortest path selection)

B. Hop count

C. Metric

D. Delay

Answer: A — AS-path lists ASNs the route passed through. Router rejects routes containing its own AS (loop prevention).

Q782. What is the BGP local preference attribute?

A. Local preference for outbound path selection (higher = preferred)

B. Inbound path selection

C. AS-path length

D. MED

Answer: A — LOCAL_PREF = local to AS, advertised in iBGP. Higher = preferred for outbound traffic.

Q783. What is BGP MED?

A. Multi-Exit Discriminator — suggests inbound path to neighboring AS (lower = preferred)

B. Outbound path selection

C. AS-path

D. Local preference

Answer: A — MED = optional non-transitive. Lower MED = preferred for inbound. Like a hint to neighbor AS.

Q784. What is BGP weight?

A. Cisco-proprietary, local-only (not advertised). Higher = preferred.

B. Standard attribute

C. Advertised to neighbors

D. Same as local preference

Answer: A — Weight = Cisco-only, local to router, not advertised. Highest weight wins in BGP path selection.

Q785. What is the BGP next-hop-self attribute?

A. Sets next-hop to local router when advertising to iBGP neighbor

B. Sets next-hop to self for eBGP

C. Encrypts next-hop

D. Speeds up BGP

Answer: A — next-hop-self is used in iBGP because by default, eBGP next-hop is passed unchanged into iBGP (often unreachable).

Q786. What is the BGP rule of synchronization?

A. Old rule: iBGP route must be in IGP before BGP uses it (now disabled by default)

B. BGP must sync with NTP

C. BGP must sync between peers

D. BGP must use sync routing tables

Answer: A — Synchronization = old rule requiring IGP to know BGP route before using it. Disabled by default in modern IOS.

Q787. What is BGP route reflector?

A. iBGP router that reflects routes to other iBGP peers (avoids full mesh)

B. BGP backup

C. BGP encryption

D. BGP load balancer

Answer: A — Route reflector allows iBGP to scale by avoiding the full-mesh requirement. RR forwards iBGP routes to clients.

Q788. What is a BGP confederation?

A. Sub-divides a large AS into smaller sub-ASes to reduce iBGP mesh

B. BGP encryption

C. BGP load balancer

D. BGP failover

Answer: A — Confederations split a big AS into sub-ASes. eBGP between sub-ASes, iBGP within.

Q789. Which command enables BGP with AS 65001?

A. router bgp 65001

B. bgp 65001

C. router bgp as 65001

D. enable bgp 65001

Answer: A — router bgp enters BGP config mode.

Q790. Which command configures a BGP neighbor?

A. neighbor 192.168.1.1 remote-as 65002

B. bgp neighbor 192.168.1.1 as 65002

C. neighbor 192.168.1.1

D. peer 192.168.1.1 as 65002

Answer: A — neighbor remote-as establishes BGP peering.

Q791. What is FHRP?

A. First Hop Redundancy Protocol — provides gateway redundancy

B. Fast Hop Routing Protocol

C. Forwarding Hop Redundancy Protocol

D. First Hop Routing Protocol

Answer: A — FHRP = gateway redundancy (HSRP, VRRP, GLBP). Lets hosts use a virtual IP that floats between routers.

Q792. What is HSRP?

A. Hot Standby Router Protocol — Cisco proprietary FHRP

B. Open standard FHRP

C. Hashed Source Routing Protocol

D. High Speed Routing Protocol

Answer: A — HSRP = Cisco proprietary. Active/standby model. Default hello 3s, hold 10s.

Q793. What is VRRP?

A. Virtual Router Redundancy Protocol — open standard (RFC 5798) FHRP

B. Cisco proprietary FHRP

C. Voice Redundancy Routing Protocol

D. Vector Routing Redundancy Protocol

Answer: A — VRRP = open standard equivalent of HSRP. Master/backup model. Default hello 1s.

Q794. What is GLBP?

A. Gateway Load Balancing Protocol — Cisco proprietary FHRP with load balancing

B. Open standard

C. Generic Load Balancing Protocol

D. Global Load Balancing Protocol

Answer: A — GLBP = Cisco proprietary. Single virtual IP, multiple AVGs (Active Virtual Gateways) for load balancing.

Q795. What is the default HSRP priority?

A. 100

B. 50

C. 150

D. 200

Answer: A — Default HSRP priority = 100. Higher priority = preferred active router.

Q796. What is the HSRP virtual MAC address format?

A. 0000.0C07.AC (e.g., 0000.0C07.AC0A for group 10)

B. 0000.5E00.01

C. 0011.2233.4455

D. FFFF.FFFF.FFFF

Answer: A — HSRP v1 MAC: 0000.0C07.ACxx where xx = group number in hex. HSRP v2: 0000.0C9F.F.

Q797. What is the VRRP virtual MAC address format?

A. 0000.5E00.01

B. 0000.0C07.AC

C. 0011.2233.4455

D. FFFF.FFFF.FFFF

Answer: A — VRRP MAC: 0000.5E00.01xx (for IPv4) or 0000.5E00.02xx (for IPv6).

Q798. What is HSRP preempt?

A. Allows a higher-priority router to take over as active when it comes online

B. Prevents failover

C. Speeds up hello

D. Disables HSRP

Answer: A — preempt allows the preferred router (higher priority) to become active when it returns. Without preempt, current active stays.

Q799. Which command configures HSRP group 1 with virtual IP 192.168.1.1?

A. standby 1 ip 192.168.1.1

B. hsrp 1 ip 192.168.1.1

C. vrrp 1 ip 192.168.1.1

D. standby ip 192.168.1.1

Answer: A — standby ip in interface config.

Q800. Which command sets HSRP priority to 110?

A. standby 1 priority 110

B. hsrp priority 110

C. priority 110

D. standby priority 110

Answer: A — standby priority <0-255>. Higher = preferred active.

Q801. Which command enables HSRP preempt?

A. standby 1 preempt

B. preempt

C. hsrp preempt

D. standby preempt

Answer: A — standby preempt. Add delay for stable failover: 'standby 1 preempt delay minimum 30'.

Q802. What is HSRP interface tracking?

A. Lowers priority if a tracked interface goes down (forces failover)

B. Tracks interface counters

C. Logs interface state

D. Encrypts interface traffic

Answer: A — Tracks interface state. If tracked interface goes down, HSRP priority is reduced (default 10), triggering failover.

Q803. Which command tracks interface Fa0/0 with HSRP decrement of 50?

A. standby 1 track Fa0/0 50

B. track Fa0/0 decrement 50

C. standby track Fa0/0 50

D. hsrp track Fa0/0 50

Answer: A — standby track . Newer IOS: track interface ; standby track .

Q804. Which command displays HSRP status?

A. show standby

B. show hsrp

C. show standby brief

D. Both A and C

Answer: D — show standby (detailed) or show standby brief (concise). Both show HSRP group state.

Q805. What is HSRP version 2?

A. Supports more groups (0-4095), millisecond hello, IPv6

B. Faster encryption

C. Same as v1 but IPv6 only

D. Cisco proprietary

Answer: A — HSRPv2: groups 0-4095 (matches VLAN IDs), millisecond timers, IPv6 support. Use 'standby version 2'.

Q806. What is VRRP master?

A. Equivalent of HSRP active router (forwards traffic for virtual IP)

B. Backup router

C. All routers in VRRP

D. VRRP controller

Answer: A — VRRP master = active router. Highest priority (default 100). All others are backup.

Q807. What is GLBP AVG?

A. Active Virtual Gateway — answers ARP for virtual IP, assigns AVFs to clients

B. Always Valid Gateway

C. Available Virtual Gateway

D. Auto Voice Gateway

Answer: A — AVG = the GLBP router that responds to ARP for virtual IP and assigns different MACs to different clients.

Q808. What is GLBP AVF?

A. Active Virtual Forwarder — actually forwards traffic for one of up to 4 MACs

B. Always Valid Forwarder

C. Active Voice Forwarder

D. Auto Virtual Forwarder

Answer: A — AVF = forwarder. Up to 4 AVFs per group, each with its own virtual MAC. Provides load balancing.

Q809. Which command displays VRRP status?

A. show vrrp

B. show vrrp brief

C. show standby

D. Both A and B

Answer: D — show vrrp or show vrrp brief displays VRRP group info.

Q810. Which command displays GLBP status?

A. show glbp

B. show glbp brief

C. show standby

D. Both A and B

Answer: D — show glbp or show glbp brief displays GLBP group state.

Q811. What is ICMP redirect?

A. Router tells host to use a different next-hop for that destination

B. Redirects ICMP

C. Blocks ICMP

D. Encrypts ICMP

Answer: A — ICMP redirect: when router receives a packet and forwards it back out the same interface, it tells the host to use a better next-hop.

Q812. What is the IPv6 equivalent of ARP?

A. ICMPv6 Neighbor Discovery (NDP)

B. ARPv6

C. DHCPv6

D. RARPv6

Answer: A — IPv6 doesn't use ARP. Uses ICMPv6 NDP (Neighbor Discovery Protocol) for MAC resolution.

Q813. What is IPv6 neighbor cache?

A. Equivalent of ARP cache — maps IPv6 to MAC

B. MAC table

C. Routing table

D. DNS cache

Answer: A — IPv6 neighbor cache = ARP equivalent. View with 'show ipv6 neighbors'.

Q814. Which command displays the IPv6 neighbor table?

A. show ipv6 neighbors

B. show ipv6 arp

C. show arp v6

D. show ipv6 neighbor-cache

Answer: A — show ipv6 neighbors displays the IPv6 neighbor cache (IPv6 to MAC mappings).

Q815. Which command displays IPv6 routing table?

A. show ipv6 route

B. show ipv6 routing

C. show ip route ipv6

D. show ipv6 table

Answer: A — show ipv6 route displays the IPv6 routing table.

Q816. What is IPv6 link-local address used for?

A. Neighbor discovery, default gateway for hosts on the local link

B. Internet routing

C. Multicast

D. Loopback only

Answer: A — Link-local (FE80::/10) only valid on the local link. Used for NDP, OSPFv3 adjacencies, host gateway.

Q817. What is IPv6 anycast?

A. Same address shared by multiple nodes; routed to nearest

B. Multicast

C. Broadcast

D. Unicast

Answer: A — Anycast = one address, multiple devices. Used for DNS root servers, load balancing.

Q818. Which routing protocol is recommended for modern IPv6 networks?

A. OSPFv3

B. OSPFv2

C. RIP

D. Static only

Answer: A — OSPFv3 supports IPv6 natively. OSPFv3 with Address Families also supports IPv4 in modern IOS.

Q819. What is OSPFv3?

A. OSPF version 3 — supports IPv6 (and IPv4 with Address Family)

B. OSPF for IPv4 only

C. OSPF v3 = same as v2 with extensions

D. OSPF for large enterprises

Answer: A — OSPFv3 (RFC 5340) supports IPv6. With AF extension (RFC 5838), also supports IPv4 in same process.

Q820. Which command enables OSPFv3 on an interface?

A. ipv6 ospf 1 area 0

B. ospf 1 area 0

C. ipv6 ospf area 0

D. ospfv3 1 area 0

Answer: A — ipv6 ospf area on the interface. OSPFv3 is interface-based, not network-based.

Q821. Which command enables OSPFv3 process 1?

A. ipv6 router ospf 1
B. router ospf 1
C. router ospfv3 1
D. ipv6 ospf 1
Answer: A — ipv6 router ospf enters OSPFv3 router config.

Q822. Which protocol uses multicast FF02::5 (all OSPF routers)?

A. OSPF
B. EIGRP
C. RIP
D. BGP
Answer: A — OSPF uses FF02::5 (all OSPF routers) and FF02::6 (DR/BDR).

Q823. Which protocol uses multicast FF02::A (all EIGRP routers)?

A. EIGRP
B. OSPF
C. RIPng
D. BGP
Answer: A — EIGRP uses FF02::A for IPv6.

Q824. Which protocol uses multicast FF02::9 (RIPng)?

A. RIPng
B. EIGRP
C. OSPF
D. BGP
Answer: A — RIPng uses FF02::9.

Q825. What is the IPv6 default route?

A. ::/0
B. 0.0.0.0/0
C. FE80::/10
D. FF00::/8
Answer: A — ::/0 = IPv6 default route (matches any IPv6 destination).

Q826. Which command creates an IPv6 static route?

A. ipv6 route 2001:db8:1::/64 2001:db8::1
B. ipv6 route 2001:db8:1:: 2001:db8::1
C. ip route 2001:db8:1::/64 2001:db8::1
D. route ipv6 2001:db8:1::/64 2001:db8::1
Answer: A — ipv6 route / .

Q827. What is route redistribution?

A. Sharing routes between different routing protocols
B. Encrypting routes
C. Speeding up convergence
D. Filtering routes
Answer: A — Redistribution: import routes from one protocol into another. E.g., OSPF into BGP, static into OSPF.

Q828. Which command redistributes static routes into OSPF?

A. redistribute static subnets

B. redistribute static

C. import static

D. static to ospf

Answer: A — redistribute static [subnets] under router ospf. 'subnets' keyword required for classful OSPF to advertise subnets.

Q829. Which command redistributes OSPF into EIGRP?

A. redistribute ospf 1 metric 10000 100 255 1 1500

B. redistribute ospf

C. import ospf to eigrp

D. ospf to eigrp

Answer: A — redistribute ospf metric . Metrics must be set for EIGRP to use route.

Q830. What is a route map used for in redistribution?

A. Filter/modify routes during redistribution

B. Encrypt routes

C. Speed up routes

D. Block all routes

Answer: A — Route-maps allow conditional redistribution (permit/deny routes, set metrics, set tags).

Q831. What is the seed metric?

A. Starting metric assigned to redistributed routes

B. Initial route cost

C. Encrypted metric

D. Backup metric

Answer: A — Seed metric = metric assigned to external routes when redistributed. Some protocols (RIP, EIGRP) need it set; OSPF defaults to 20.

Q832. What is the default seed metric when redistributing into OSPF?

A. 20 (Type 2)

B. 1

C. 100

D. 0

Answer: A — OSPF default external metric = 20, Type 2 (E2). Type 2 only counts external cost, not internal.

Q833. What is the default seed metric when redistributing into EIGRP?

A. None (must be set manually)

B. 0

C. 100

D. 1000

Answer: A — EIGRP has no default — must specify metric (BW, delay, reliability, load, MTU) or routes won't be installed.

Q834. What is the default seed metric when redistributing into RIP?

A. 0 (treated as infinite — must be set)

B. 1

C. 10

D. 100

Answer: A — RIP default seed = 0 = infinite. Must set: 'redistribute ospf 1 metric 5'.

Q835. What is route tagging?

A. Marks routes with a tag to prevent redistribution loops

B. Encrypts routes

C. Speeds up routing

D. Tags for QoS only

Answer: A — Tags (e.g., 110 for OSPF-originated, 90 for EIGRP) prevent route feedback loops in mutual redistribution.

Q836. What is a routing loop?

A. Packet circles between routers forever (or until TTL expires)

B. Fast routing

C. Encrypted routing

D. Backup routing

Answer: A — Loop = packet forwarded back to a router it already visited. Mitigated by TTL, split horizon, poison reverse, etc.

Q837. What is TTL used for in IP?

A. Prevents packets from circulating forever (decremented at each router)

B. Encrypts packets

C. Speeds up packets

D. Routes packets

Answer: A — TTL (Time To Live) = max hops. Each router decrements by 1. If 0, packet is dropped (ICMP Time Exceeded).

Q838. What is the maximum TTL value?

A. 255

B. 128

C. 64

D. 256

Answer: A — TTL is an 8-bit field → max value = 255. Common defaults: Windows=128, Linux=64, Cisco=255.

Q839. Which ICMP message is sent when TTL reaches 0?

A. Time Exceeded

B. Echo Reply

C. Destination Unreachable

D. Source Quench

Answer: A — ICMP Type 11 = Time Exceeded. Used by traceroute to discover hops.

Q840. What does 'show ip protocols' display?

A. Routing protocol parameters (timers, networks, redistribution, filters)

B. Interface status

C. ARP cache

D. MAC table

Answer: A — show ip protocols shows running routing protocols and their configuration.

Q841. What is a debug command used for?

A. Real-time troubleshooting of protocol events

B. Permanent logging

C. Config backup

D. Performance monitoring

Answer: A — debug shows real-time events. Use sparingly — can flood console and impact CPU. Always 'undebug all' after.

Q842. Which command disables all debugs?

A. undebug all (or 'u all')

B. no debug

C. debug off

D. stop debug

Answer: A — undebug all (abbreviated 'u all') disables all active debugs. Always run this after debugging.

Q843. What is OSPF SPF delay?

A. Initial wait before running SPF after topology change (default 5s)

B. Hello timer

C. Dead timer

D. LSA refresh

Answer: A — SPF delay = 5s default. Wait before running SPF to batch multiple changes. Can be tuned.

Q844. What is OSPF LSA refresh interval?

A. 30 minutes — each LSA refreshed every 30 min

B. 30 seconds

C. 30 hours

D. 1 hour

Answer: A — Each LSA is refreshed every 30 minutes (LSRefreshTime). Even if no change, LSAs are re-flooded.

Q845. What is the OSPF MaxAge?

A. 60 minutes — LSAs removed from LSDB after 1 hour without refresh

B. 60 seconds

C. 60 hours

D. 1 day

Answer: A — MaxAge = 3600s (60 min). If LSA not refreshed in 60 min, it's flushed from LSDB.

Q846. What is the OSPF multicast address for all OSPF routers (IPv4)?

A. 224.0.0.5

B. 224.0.0.6

C. 224.0.0.9

D. 224.0.0.10

Answer: A — 224.0.0.5 = all OSPF routers. 224.0.0.6 = DR/BDR only.

Q847. What is the OSPF multicast address for DR/BDR (IPv4)?

A. 224.0.0.6

B. 224.0.0.5

C. 224.0.0.9

D. 224.0.0.10

Answer: A — 224.0.0.6 = all OSPF DRs (DR and BDR listen).

Q848. What is the EIGRP multicast address (IPv4)?

A. 224.0.0.10
B. 224.0.0.5
C. 224.0.0.6
D. 224.0.0.9
Answer: A — EIGRP uses 224.0.0.10.

Q849. What is the RIP multicast address (RIPv2)?

A. 224.0.0.9
B. 224.0.0.5
C. 224.0.0.10
D. 224.0.0.6
Answer: A — RIPv2 uses 224.0.0.9 (vs RIPv1 uses 255.255.255.255 broadcast).

Q850. What is the OSPF multicast for all routers in IPv6?

A. FF02::5
B. FF02::6
C. FF02::A
D. FF02::9
Answer: A — FF02::5 = all OSPFv3 routers. FF02::6 = DR/BDR.

Q851. What is the EIGRP multicast for IPv6?

A. FF02::A
B. FF02::5
C. FF02::9
D. FF02::1
Answer: A — FF02::A = all EIGRP routers (IPv6).

Q852. What is the RIPng multicast?

A. FF02::9
B. FF02::5
C. FF02::A
D. FF02::1
Answer: A — RIPng uses FF02::9.

Q853. Which command shows OSPF routes that failed to install in routing table?

A. show ip ospf rib
B. show ip ospf route
C. show ospf failed
D. show ip route ospf rejected
Answer: A — show ip ospf rib shows OSPF's local RIB including routes not installed in global RIB (with reason).

Q854. What is OSPF forwarding address?

A. Used in Type 5 LSAs to allow routers to forward directly to ASBR (or next hop)
B. OSPF router ID
C. OSPF area ID
D. OSPF cost
Answer: A — FA in Type 5 LSA can be non-zero, letting routers send packets directly to external next-hop instead of via ASBR.

Q855. What is OSPF sham link?

A. Used in MPLS L3 VPN to make PE-to-CE route appear as intra-area

B. OSPF virtual link

C. OSPF encryption

D. OSPF backup link

Answer: A — Sham link: used in MPLS VPN backdoor scenarios to make OSPF prefer MPLS path over backdoor.

Q856. What is OSPF Graceful Shutdown?

A. Router sends LSAs with max age before going down, speeding convergence

B. OSPF shutdown

C. OSPF reboot

D. OSPF encryption

Answer: A — Graceful shutdown = flush own LSAs before going down so neighbors know immediately.

Q857. What is the OSPF TTL security check?

A. Drops OSPF packets with TTL < 255 - hops (prevents remote attacks)

B. Encrypts OSPF

C. Speeds up OSPF

D. Disables OSPF

Answer: A — TTL security: TTL must be ≥ 255 - . For directly connected neighbors, TTL must be 255.

Q858. What is the BGP BGP-4 protocol?

A. Current version of BGP (RFC 4271), supports CIDR

B. Old version

C. Cisco proprietary

D. Same as OSPF

Answer: A — BGP-4 = current standard. Supports CIDR, route aggregation, path attributes. The Internet routing protocol.

Q859. What is the BGP AS number range (2-byte)?

A. 1-65535

B. 0-65535

C. 1-1024

D. 1-32768

Answer: A — 2-byte AS: 1-65535. Reserved: 23456, 64512-65534 (private), 65535.

Q860. What are the private AS numbers (2-byte)?

A. 64512-65534

B. 1-1024

C. 100-199

D. 65000-65099

Answer: A — Private AS: 64512-65534 (like RFC 1918 for IPs). Not advertised to global Internet.

Q861. What is the BGP AS number range (4-byte)?

A. 1-4294967295

B. 1-65535

C. 1-100000

D. 1-1048576

Answer: A — 4-byte AS: 1-4294967295. Private: 4200000000-4294967294.

Q862. What is BGP path selection order?

A. Weight → Local Pref → Locally originated → AS_PATH length → Origin → MED → eBGP over iBGP → Lowest IGP metric

B. AS_PATH → MED → Local Pref

C. Lowest router ID first

D. First received route

Answer: A — BGP uses 13+ attributes in specific order. Weight, LP, locally originated, AS_PATH, origin, MED, eBGP>iBGP, IGP metric, etc.

Q863. What is BGP route flap damping?

A. Suppresses unstable routes that flap frequently

B. Speeds up BGP

C. Encrypts BGP routes

D. Reduces BGP load

Answer: A — Damping: routes that flap get penalty. Above suppress limit, route is suppressed for time. Currently discouraged by RFC.

Q864. What is BGP graceful restart?

A. Allows BGP to keep forwarding during restart if neighbor supports

B. BGP reboot

C. BGP shutdown

D. BGP encryption

Answer: A — Graceful restart: router signals capability to keep forwarding during restart. Neighbor maintains routes.

Q865. What is BGP BMP?

A. BGP Monitoring Protocol — monitors BGP sessions without affecting them

B. BGP Management Protocol

C. BGP Message Protocol

D. Border Monitoring Protocol

Answer: A — BMP = passive monitoring of BGP. Sends BGP updates to monitoring station. No impact on routing.

Q866. Which is the most preferred BGP attribute?

A. Weight (Cisco-proprietary, local only)

B. Local Preference

C. AS_PATH length

D. MED

Answer: A — BGP path selection order: Weight > LOCAL_PREF > locally originated > AS_PATH > Origin > MED > eBGP>iBGP > IGP metric.

Q867. What does BGP 'network' command do?

A. Advertises a network if it exists in the routing table

B. Creates a network

C. Encrypts a network

D. Blocks a network

Answer: A — network mask advertises a prefix if it exactly matches an entry in the IP routing table.

Q868. Which command advertises 192.168.1.0/24 in BGP?

A. network 192.168.1.0 mask 255.255.255.0

B. network 192.168.1.0/24

C. advertise 192.168.1.0 255.255.255.0

D. network 192.168.1.0

Answer: A — BGP uses explicit mask (not CIDR). Must exactly match the routing table entry.

Q869. What is BGP aggregation?

A. Combine multiple specific prefixes into one summary

B. Encrypt routes

C. Speed up BGP

D. Filter routes

Answer: A — aggregate-address [summary-only] creates a summary. summary-only suppresses more specific routes.

Q870. What is the BGP 'summary-only' keyword?

A. Suppresses more specific routes, only advertises summary

B. Encrypts the summary

C. Speeds up summary

D. Required for all summaries

Answer: A — summary-only: only the aggregate is advertised, specifics are suppressed (still in local BGP RIB).

Q871. Which command displays BGP summary?

A. show ip bgp summary

B. show bgp summary

C. show ip bgp

D. show bgp

Answer: A — show ip bgp summary gives concise view: neighbors, AS, state, prefixes received.

Q872. Which command displays the BGP table?

A. show ip bgp

B. show bgp table

C. show ip bgp rib

D. show bgp routes

Answer: A — show ip bgp displays the BGP RIB — all BGP-learned routes.

Q873. Which command displays BGP neighbors?

A. show ip bgp neighbors

B. show bgp neighbors

C. show ip bgp summary

D. Both A and C

Answer: D — show ip bgp neighbors (detailed) or show ip bgp summary (concise list).

Q874. What is BGP community?

A. Tag attached to routes for grouping/policy

B. AS path

C. Encryption

D. Local pref

Answer: A — Communities (32-bit, often format AS:value) tag routes for bulk policy application. E.g., no-export = don't send to eBGP.

Q875. What is the BGP 'no-export' community?

A. Don't advertise this route to eBGP peers

B. Don't advertise to iBGP

C. Don't advertise at all

D. Encrypt the route

Answer: A — no-export: route can be shared within AS but not advertised to eBGP peers.

Q876. What is the BGP 'local-AS' community?

A. Don't advertise outside local AS (used in confederations)

B. Same as no-export

C. Encrypt route

D. Advertise only locally

Answer: A — local-AS: keep within sub-AS in confederations. Don't advertise to eBGP (including confederation eBGP).

Q877. What is the BGP 'no-advertise' community?

A. Don't advertise to any BGP peer

B. Don't advertise to eBGP

C. Don't advertise to iBGP

D. Encrypt route

Answer: A — no-advertise: most restrictive — don't share with any BGP peer.

Q878. What is IPv6 over IPv4 tunneling?

A. Encapsulates IPv6 packets in IPv4 to cross IPv4-only networks

B. Encrypts IPv6

C. Speeds up IPv6

D. Routes IPv4 over IPv6

Answer: A — 6to4, 6rd, ISATAP, GRE: encapsulate IPv6 in IPv4 to traverse IPv4 networks during migration.

Q879. What is NAT64?

A. Translates IPv6 to IPv4 (and vice versa) for v6-only to v4-only communication

B. Encrypts IPv6

C. Speeds up IPv6

D. Routes IPv6 only

Answer: A — NAT64 + DNS64 allows IPv6-only clients to reach IPv4-only servers.

Q880. What is dual-stack?

A. Running IPv4 and IPv6 simultaneously on the same network

B. Two routers

C. Two switches

D. Two VLANs

Answer: A — Dual-stack: devices run both IPv4 and IPv6. Most common migration strategy. Apps choose protocol.

Q881. What is the OSPF mtu-ignore command?

A. Ignores MTU mismatch in OSPF DBD exchange (allows adjacency even with MTU mismatch)

B. Disables OSPF

C. Speeds up OSPF

D. Encrypts OSPF

Answer: A — OSPF requires MTU match for full adjacency. mtu-ignore bypasses check (use cautiously — may cause packet drops).

Q882. What is OSPF SPF throttling?

A. Limits how often SPF is calculated to protect CPU

B. Speeds up SPF

C. Disables SPF

D. Encrypts SPF

Answer: A — SPF throttle: 5s initial, 10s second, 10s subsequent (default). Prevents SPF from running too often.

Q883. What is OSPF LSA throttling?

A. Limits how fast same LSA can be re-originated

B. Limits LSA size

C. Disables LSAs

D. Encrypts LSAs

Answer: A — LSA throttle: prevents LSA flapping. Initial delay, then min interval between re-originations.

Q884. What is OSPF flood pacing?

A. Paces flooding of LSAs to avoid bursts

B. Speeds up flooding

C. Encrypts flooding

D. Disables flooding

Answer: A — Flood pacing (default 33ms) groups LSAs into batches to reduce overhead.

Q885. What is the OSPF neighbor state 'ATTEMPT'?

A. Used on NBMA networks — neighbor not yet reachable but trying

B. Same as INIT

C. Failed state

D. Success state

Answer: A — ATTEMPT = NBMA only, neighbor is being attempted (no hello received yet).

Q886. What is a transit area in OSPF?

A. Area through which a virtual link passes (not stub/NSSA)

B. Backbone area

C. Stub area

D. NSSA

Answer: A — Transit area: used by virtual link to connect a disconnected area to backbone. Cannot be a stub/NSSA.

Q887. What is OSPF SPF algorithm?

A. Dijkstra's shortest path algorithm

B. Bellman-Ford

C. DUAL

D. Path-vector

Answer: A — Dijkstra's SPF computes shortest path from a source to all nodes in a graph (link-state DB).

Q888. What is the OSPF cost formula?

A. reference-bandwidth / interface-bandwidth (Mbps)

B. interface-bandwidth / reference-bandwidth

C. 100 / hop count

D. hop count / 100

Answer: A — Cost = ref-bw / interface-bw. Default ref-bw = 100 Mbps. Higher speed = lower cost = preferred.

Q889. What is the default OSPF cost of a 10 Mbps interface?

A. 10
B. 1
C. 100
D. 1000
Answer: A — 100/10 = 10.

Q890. What is the default OSPF cost of a 64 Kbps interface?

A. 1562
B. 100
C. 10
D. 1
Answer: A — 100/0.064 = 1562.

Q891. What is the OSPF 'show ip ospf border-routers' command?

A. Displays ABRs and ASBRs in the OSPF domain
B. Displays all routers
C. Displays DR/BDR
D. Displays neighbors
Answer: A — show ip ospf border-routers shows ABRs and ASBRs reachable in the OSPF topology.

Q892. What is the 'show ip ospf max-metric' command used for?

A. Displays max-metric configuration (router in stub-router mode)
B. Displays max metric ever seen
C. Displays max LSA
D. Displays max cost
Answer: A — max-metric advertises all LSAs with max metric, making router less preferred (used in maintenance).

Q893. What is OSPF stub-router (max-metric)?

A. Router advertises its LSAs with max metric, making it less preferred for transit
B. Stub area router
C. Backup router
D. Edge router
Answer: A — Stub-router mode (max-metric LSA): used during maintenance/reboot to gracefully drain traffic.

Q894. What is OSPF fast hello (sub-second)?

A. Hellos sent every <1 second for fast failure detection
B. Faster SPF
C. Faster LSA
D. Faster routing
Answer: A — Sub-second hello: configure hello = e.g., 250ms, dead = 1s. Faster failure detection.

Q895. What is BFD?

A. Bidirectional Forwarding Detection — sub-second failure detection
B. Backup Forwarding Daemon
C. Binary Frame Detection
D. Best Forwarding Decision
Answer: A — BFD provides fast (sub-50ms) failure detection independent of routing protocol hellos. Can be used by OSPF, BGP, etc.

Q896. Which command enables BFD on an OSPF interface?

A. ip ospf bfd

B. bfd enable

C. ospf bfd

D. bfd interface

Answer: A — ip ospf bfd (interface config). Then 'bfd' globally configured. Many protocols can share BFD.

Q897. What is the difference between ip sla and ip sla schedule?

A. ip sla defines the test; ip sla schedule activates it

B. Same command

C. ip sla is for routing

D. ip sla schedule is for monitoring

Answer: A — ip sla defines the probe. ip sla schedule sets when to run. Used for SLA monitoring, failover triggers.

Q898. What is IP SLA used for?

A. Generate test traffic to measure latency, jitter, reachability

B. Encrypt traffic

C. Speed up routing

D. Filter traffic

Answer: A — IP SLA probes network performance. Can trigger route changes via object tracking.

Q899. What is object tracking?

A. Tracks state of an IP SLA, interface, or route; can influence routing decisions

B. Tracks MAC addresses

C. Tracks VLANs

D. Tracks QoS

Answer: A — Object tracking tracks an IP SLA, interface, route, etc. Can be linked to static route or HSRP for failover.

Q900. Which command tracks IP SLA operation 1?

A. track 1 ip sla 1 reachability

B. track ip sla 1

C. ip sla track 1

D. track 1 sla

Answer: A — track ip sla reachability. Then use the track ID in static route, HSRP, etc.

Q901. Which command links a static route to tracked object 1?

A. ip route 0.0.0.0 0.0.0.0 10.0.0.1 track 1

B. ip route 0.0.0.0 0.0.0.0 10.0.0.1 sla 1

C. ip route track 1 0.0.0.0 0.0.0.0 10.0.0.1

D. ip sla route 1

Answer: A — ip route ... track . Route is installed only when tracked object is up.

Domain 4: IP Services

136 questions

Q902. What is DHCP?

A. Dynamic Host Configuration Protocol — assigns IP, mask, gateway, DNS to clients
B. DNS Host Configuration Protocol
C. Dynamic Host Control Protocol
D. Distributed Host Configuration Protocol
Answer: A — DHCP automates IP configuration. Uses UDP 67 (server) and 68 (client).

Q903. What are the four DHCP message types (DORA)?

A. Discover, Offer, Request, Acknowledge
B. Demand, Offer, Reply, Ack
C. Discover, Open, Receive, Accept
D. Detect, Offer, Reply, Allocate
Answer: A — DORA: client DISCOVERs, server OFFERs, client REQUESTs, server ACKs.

Q904. What is DHCP Discover?

A. Client broadcast to find available DHCP servers
B. Server message
C. Client confirmation
D. Server confirmation
Answer: A — DISCOVER = client broadcasts to find DHCP servers. Sent to 255.255.255.255 UDP/68.

Q905. What is DHCP Offer?

A. Server response offering an IP address and config
B. Client request
C. Server acknowledgment
D. Client broadcast
Answer: A — OFFER: server proposes IP, mask, gateway, DNS, lease time. May be unicast or broadcast.

Q906. What is DHCP Request?

A. Client requests the offered IP (broadcast, in case multiple servers offered)
B. Server confirms
C. Client declines
D. Server offer
Answer: A — REQUEST: client accepts one offer, broadcasts so other servers know to retract their offers.

Q907. What is DHCP Acknowledge?

A. Server confirms IP assignment and parameters
B. Client acknowledgment
C. Server offer
D. Client request
Answer: A — ACK: server confirms. Client now has IP and config. Lease timer starts.

Q908. What is DHCP Relay?

A. Router forwards DHCP broadcast from client to server in another subnet

B. DHCP server backup

C. DHCP encryption

D. DHCP load balancer

Answer: A — ip helper-address on router interface forwards DHCP broadcasts to remote DHCP server as unicast.

Q909. Which command configures DHCP relay?

A. ip helper-address 192.168.1.5

B. dhcp relay 192.168.1.5

C. ip dhcp-relay 192.168.1.5

D. helper-address 192.168.1.5

Answer: A — ip helper-address on the interface receiving client broadcasts.

Q910. What UDP port does DHCP server use?

A. 67

B. 68

C. 514

D. 53

Answer: A — DHCP server listens on UDP 67. DHCP client uses UDP 68.

Q911. What UDP port does DHCP client use?

A. 68

B. 67

C. 514

D. 53

Answer: A — DHCP client uses UDP 68 (source) and sends to UDP 67 (server).

Q912. What is DHCP lease time?

A. Duration the client can use the assigned IP

B. Time to first DHCP packet

C. Time to renew

D. Time to decline

Answer: A — Lease time = how long client can use IP. Typical: 1 day (router), 8 days (Windows server), 1 hour (guest WLAN).

Q913. What is the default Cisco router DHCP lease time?

A. 1 day

B. 8 days

C. 1 hour

D. 30 minutes

Answer: A — Default = 1 day (24 hours). Configurable: lease [hours] [minutes] or lease infinite.

Q914. What is DHCP renewal (T1)?

A. At 50% lease time, client tries to renew with original server

B. At 100% lease

C. At 25% lease

D. At 75% lease

Answer: A — T1 timer (50% lease): client unicasts REQUEST to original server to renew.

Q915. What is DHCP rebinding (T2)?

A. At 87.5% lease time, client broadcasts to any server to extend

B. At 50% lease

C. At 75% lease

D. At 100% lease

Answer: A — T2 timer (87.5% lease): client broadcasts to any DHCP server to extend or get new IP.

Q916. What happens when DHCP lease expires?

A. Client must release IP and request a new one

B. Client keeps IP forever

C. Server renews automatically

D. Nothing

Answer: A — If lease expires (100%), client must release IP and start DHCP discovery again.

Q917. Which command configures a Cisco router as DHCP server?

A. ip dhcp pool

B. dhcp server enable

C. ip dhcp server

D. dhcp pool

Answer: A — ip dhcp pool creates DHCP pool. Then configure network, default-router, dns-server, lease.

Q918. Which command defines the DHCP pool's network?

A. network 192.168.1.0 255.255.255.0

B. subnet 192.168.1.0/24

C. network 192.168.1.0/24

D. dhcp-network 192.168.1.0

Answer: A — network defines the subnet to assign addresses from.

Q919. Which command sets the DHCP default gateway?

A. default-router 192.168.1.1

B. gateway 192.168.1.1

C. default-gateway 192.168.1.1

D. router 192.168.1.1

Answer: A — default-router sets the default gateway option (DHCP option 3).

Q920. Which command excludes IPs from DHCP pool?

A. ip dhcp excluded-address 192.168.1.1 192.168.1.10

B. excluded 192.168.1.1 192.168.1.10

C. no dhcp 192.168.1.1 192.168.1.10

D. deny 192.168.1.1 192.168.1.10

Answer: A — ip dhcp excluded-address [] (in global config) reserves IPs not to assign.

Q921. Which command displays DHCP bindings on a Cisco router?

A. show ip dhcp binding

B. show dhcp binding

C. show ip dhcp

D. show dhcp

Answer: A — show ip dhcp binding shows assigned IPs, client MACs, lease expiration.

Q922. Which command displays DHCP conflicts?

A. show ip dhcp conflict

B. show dhcp conflict

C. show ip dhcp

D. show dhcp errors

Answer: A — show ip dhcp conflict shows IPs the router found were already in use (ping check).

Q923. What is DHCP option 82?

A. Relay Agent Information — adds circuit ID and remote ID for client identification

B. DNS server option

C. Default gateway option

D. Lease time option

Answer: A — Option 82 (Relay Agent Info): relay adds info about which switch/port received the DHCP. Used for security/policy.

Q924. What is DNS?

A. Domain Name System — translates domain names to IP addresses

B. Dynamic Name Service

C. Domain Network Service

D. Distributed Name System

Answer: A — DNS = translates human-readable names (example.com) to IPs (93.184.216.34). Uses UDP/TCP 53.

Q925. Which port does DNS use?

A. 53

B. 67

C. 80

D. 443

Answer: A — DNS uses port 53 (TCP for zone transfers/large responses, UDP for typical queries).

Q926. What is a DNS A record?

A. Maps hostname to IPv4 address

B. Maps hostname to IPv6

C. Mail exchange record

D. Alias record

Answer: A — A record = Address record (IPv4). AAAA = IPv6. MX = mail. CNAME = alias. TXT = text.

Q927. What is a DNS AAAA record?

A. Maps hostname to IPv6 address

B. Maps hostname to IPv4

C. Mail exchange

D. Alias

Answer: A — AAAA (quad-A) = IPv6 address record.

Q928. What is a DNS MX record?

A. Mail Exchange — identifies mail servers for the domain

B. Mail eXtra

C. Multiple X

D. Mail eXchange IPv6

Answer: A — MX records point to mail servers. Has priority value (lower = preferred).

Q929. What is a DNS CNAME record?

A. Canonical Name — alias pointing to another hostname

B. Common Name

C. Custom Name

D. Canonical Network

Answer: A — CNAME = alias. E.g., www.example.com → example.com. Used for aliasing.

Q930. What is a DNS PTR record?

A. Pointer — reverse DNS lookup (IP to name)

B. Public Translation Record

C. Path Record

D. Private Record

Answer: A — PTR = reverse DNS. Maps IP to hostname. Used in anti-spam checks.

Q931. What is a DNS TXT record?

A. Text — arbitrary text (SPF, DKIM, domain verification)

B. Transfer

C. Translation

D. Trace

Answer: A — TXT records hold text. Common uses: SPF (sender policy framework), DKIM, domain ownership verification.

Q932. What is a DNS NS record?

A. Name Server — identifies authoritative DNS servers for the zone

B. Network Service

C. Next Server

D. Name Space

Answer: A — NS records list authoritative nameservers for a domain.

Q933. What is DNS SOA record?

A. Start of Authority — defines the zone and its properties

B. Source of Authority

C. Standard of Authority

D. Single Origin Address

Answer: A — SOA = master record for a zone. Contains primary NS, admin email, serial, refresh, retry, expire, TTL.

Q934. What is a DNS recursive query?

A. Resolver does all the work and returns final answer to client

B. Client does the recursion

C. Server does one lookup

D. No recursion

Answer: A — Recursive: client asks resolver, resolver asks root → TLD → authoritative, returns final IP.

Q935. What is a DNS iterative query?

A. Server returns best answer it has (may be referral to another server)

B. Server does all the work

C. Same as recursive

D. No lookup

Answer: A — Iterative: server returns referral if it doesn't have answer. Client (or recursive resolver) follows referrals.

Q936. What is the DNS root server count?

A. 13 logical (using anycast, hundreds of physical instances)

B. 1

C. 100

D. 1000

Answer: A — 13 logical root servers (a.root-servers.net through m.root-servers.net). Anycast provides geographic distribution.

Q937. What is a DNS forwarder?

A. DNS server that forwards queries it can't resolve to another DNS server

B. DNS backup

C. DNS encryption

D. DNS load balancer

Answer: A — Forwarder: if DNS server can't resolve locally, forwards to upstream (e.g., ISP DNS, Google 8.8.8.8).

Q938. What is DNS caching?

A. DNS resolvers cache results to reduce query time and bandwidth

B. DNS backup

C. DNS encryption

D. DNS load balancer

Answer: A — Caching: results cached based on TTL. Reduces latency and root server load. Typical cache hit >95%.

Q939. What is DNS TTL?

A. Time To Live — how long a record can be cached (in seconds)

B. Total Time Limit

C. Transmission Time Limit

D. Transfer Time Limit

Answer: A — TTL = how long resolvers can cache the answer. Lower TTL = faster DNS changes propagate. Common: 300-86400s.

Q940. Which command configures a Cisco router to use DNS server 8.8.8.8?

A. ip name-server 8.8.8.8

B. dns 8.8.8.8

C. ip dns 8.8.8.8

D. name-server 8.8.8.8

Answer: A — ip name-server [IP2 IP3 ...]. Router uses these for DNS resolution (e.g., for pings by name).

Q941. Which command enables DNS on a Cisco router?

A. ip domain-lookup

B. dns enable

C. ip dns

D. domain-lookup

Answer: A — ip domain-lookup enables DNS resolution. Default is on.

Q942. Which command configures the router's domain name?

A. ip domain-name example.com

B. domain-name example.com

C. ip domain example.com

D. hostname-domain example.com

Answer: A — ip domain-name sets the default domain (appended to short hostnames).

Q943. What is NTP?

A. Network Time Protocol — synchronizes clocks across network devices

B. Network Transfer Protocol

C. Network Time Pass-through

D. Network Topology Protocol

Answer: A — NTP synchronizes device clocks. Uses UDP 123. Critical for syslog, certs, Kerberos.

Q944. Which port does NTP use?

A. 123

B. 53

C. 514

D. 161

Answer: A — NTP uses UDP port 123.

Q945. What is an NTP stratum?

A. Distance from reference clock (0 = atomic clock, 1 = directly attached, etc.)

B. Network segment

C. Time zone

D. Server type

Answer: A — Stratum 0 = reference (atomic, GPS). Stratum 1 = directly attached. Higher = further from source.

Q946. What is the maximum NTP stratum?

A. 16 (means unsynchronized)

B. 15

C. 32

D. 255

Answer: A — Stratum 16 = unsynchronized. Valid range for servers: 1-15.

Q947. Which command configures NTP server on a Cisco device?

A. ntp server 192.168.1.5

B. ntp 192.168.1.5

C. time-server 192.168.1.5

D. clock ntp 192.168.1.5

Answer: A — ntp server [version] [key] [source].

Q948. Which command displays NTP status?

A. show ntp status

B. show ntp

C. show ntp associations

D. Both A and C

Answer: D — show ntp status (sync state, stratum, offset) and show ntp associations (peer list).

Q949. Which command displays NTP associations?

A. show ntp associations

B. show ntp peers

C. show ntp status

D. show ntp servers

*Answer: A — show ntp associations shows NTP peers and their state (st = stratum, * = current sync source).*

Q950. What is NTP authentication?

A. MD5-based auth to prevent rogue NTP servers

B. Encryption of NTP traffic

C. Speed up NTP

D. Authenticate NTP source port

Answer: A — NTP auth uses MD5 (optionally SHA1) to ensure time comes from trusted source.

Q951. What is the default NTP version on Cisco?

A. 4

B. 3

C. 2

D. 1

Answer: A — Default = v4 (RFC 5905). Supports IPv6, auto-authentication.

Q952. What is SNTP?

A. Simple NTP — simplified client-only version of NTP

B. Secure NTP

C. Standard NTP

D. Single NTP

Answer: A — SNTP = simplified NTP for clients that don't need full NTP complexity. Less accurate, no server mode.

Q953. What is syslog?

A. Standard for sending log messages over IP (UDP 514)

B. System Log

C. Server Log

D. Standard Log

Answer: A — Syslog = standard logging protocol. Uses UDP 514 (or TCP for reliable). Severity 0-7.

Q954. Which port does syslog use?

A. 514

B. 53

C. 123

D. 161

Answer: A — Syslog uses UDP 514 by default. Some implementations use TCP for reliability.

Q955. What are the syslog severity levels?

A. 0-Emergency, 1-Alert, 2-Critical, 3-Error, 4-Warning, 5-Notification, 6-Informational, 7-Debugging

B. 0-7 in opposite order

C. 1-10

D. High to Low

Answer: A — 0 = most severe (emerg), 7 = least (debug). Lower number = more severe.

Q956. Which syslog level is 'warning'?

A. 4
B. 3
C. 5
D. 2
Answer: A — Level 4 = Warning. Common default logging level.

Q957. Which syslog level is 'error' (also called 'err' or 'error')?

A. 3
B. 4
C. 5
D. 2
Answer: A — Level 3 = Error.

Q958. Which syslog level is 'debugging'?

A. 7
B. 0
C. 5
D. 6
Answer: A — Level 7 = Debugging (most verbose). Use only when troubleshooting — generates lots of output.

Q959. Which syslog level is 'emergency'?

A. 0
B. 1
C. 7
D. 5
Answer: A — Level 0 = Emergency (system unusable). Highest severity.

Q960. Which command enables logging to a syslog server?

A. logging 192.168.1.10
B. syslog 192.168.1.10
C. log server 192.168.1.10
D. logging host 192.168.1.10
Answer: A — logging sends syslog to that server. Multiple servers can be configured.

Q961. Which command sets the logging level for syslog?

A. logging trap informational
B. logging level informational
C. logging 6
D. trap informational
Answer: A — logging trap sets the severity threshold for messages sent to syslog server.

Q962. Which command enables timestamp in syslog messages?

A. service timestamps log datetime msec
B. timestamps enable
C. log timestamps
D. service log timestamps
Answer: A — service timestamps log datetime msec [localtime] [show-timezone]. Adds precise timestamps to log messages.

Q963. Which command sets the source interface for syslog?

A. logging source-interface Fa0/0

B. logging source Fa0/0

C. source-interface syslog Fa0/0

D. logging interface Fa0/0

Answer: A — logging source-interface sets which interface IP is used as syslog source.

Q964. What is FTP?

A. File Transfer Protocol — TCP ports 20 (data) and 21 (control)

B. Fast Transfer Protocol

C. File Transfer Port

D. Full Transfer Protocol

Answer: A — FTP = TCP 21 control, 20 data (active mode). Uses cleartext passwords — prefer SFTP/FTPS.

Q965. What is TFTP?

A. Trivial FTP — UDP port 69, simple file transfer (no auth, no directory)

B. Trusted FTP

C. TCP FTP

D. Tunneled FTP

Answer: A — TFTP = UDP 69. Simple, no auth, no directory listing. Used for IOS image transfer, config backup.

Q966. Which port does TFTP use?

A. 69

B. 21

C. 20

D. 80

Answer: A — TFTP uses UDP port 69.

Q967. What is the difference between FTP active and passive mode?

A. Active: server connects to client for data. Passive: client connects to server (works through NAT).

B. Active is faster

C. Passive is faster

D. No difference

Answer: A — Active: server initiates data connection (fails through NAT). Passive: client initiates both (NAT-friendly).

Q968. What is HTTP?

A. HyperText Transfer Protocol — TCP 80

B. HyperText Transmission Protocol

C. High Transfer Protocol

D. Host Transfer Protocol

Answer: A — HTTP = TCP 80. Cleartext protocol for web. HTTPS (TCP 443) = HTTP over TLS.

Q969. What is HTTPS?

A. HTTP Secure — HTTP over TLS, TCP 443

B. HyperText Transfer Protocol Secure

C. HTTP Standard Protocol Enhanced

D. HyperText Transfer Protocol Standard

Answer: A — HTTPS = HTTP encrypted with TLS. Uses TCP 443.

Q970. What is SSH?

A. Secure Shell — encrypted remote login, TCP 22

B. Secure Socket Handler

C. System Shell

D. Standard Shell

Answer: A — SSH = TCP 22. Replaces Telnet (cleartext). SSH v2 is the modern secure standard.

Q971. Which port does SSH use?

A. 22

B. 23

C. 21

D. 25

Answer: A — SSH uses TCP port 22.

Q972. What is Telnet?

A. Cleartext remote login protocol — TCP 23 (insecure)

B. Telnet Encryption Login

C. Trusted Ethernet Login

D. TCP Enabled Login

Answer: A — Telnet = TCP 23. Sends data (including passwords) in cleartext. Avoid — use SSH.

Q973. Which port does Telnet use?

A. 23

B. 22

C. 21

D. 80

Answer: A — Telnet uses TCP port 23.

Q974. What is SNMP?

A. Simple Network Management Protocol — UDP 161/162, monitors network devices

B. Secure Network Management Protocol

C. Standard Node Management Protocol

D. Simple Node Monitoring Protocol

Answer: A — SNMP = UDP 161 (queries) and 162 (traps). Used by NMS to poll devices and receive alerts.

Q975. Which ports does SNMP use?

A. 161 (queries), 162 (traps)

B. 161 only

C. 162 only

D. 53

Answer: A — SNMP uses UDP 161 for GET/SET queries and UDP 162 for trap notifications.

Q976. What is SNMP v1?

A. First version, basic, uses community strings (cleartext)

B. Encrypted

C. Most secure

D. Modern standard

Answer: A — SNMPv1: simple, community strings, no encryption. Largely deprecated.

Q977. What is SNMP v2c?

A. Adds bulk transfers, still uses community strings (cleartext)

B. Encrypted

C. Uses SHA auth

D. Uses AES encryption

Answer: A — SNMPv2c: added GetBulk for efficiency, but still uses community strings (no real security).

Q978. What is SNMP v3?

A. Adds user-based security (authentication + encryption)

B. Same as v2c

C. Removes security

D. Only for IPv6

Answer: A — SNMPv3 = USM (User-based Security Model). Supports MD5/SHA auth, DES/3DES/AES encryption. Use this!

Q979. What is an SNMP community string?

A. Password for SNMPv1/v2c authentication

B. Server address

C. OID identifier

D. Trap destination

Answer: A — Community string = SNMPv1/2c password. Common: 'public' (read), 'private' (write). Insecure.

Q980. What is the default SNMP community?

A. public (read-only)

B. private (read-write)

C. admin

D. cisco

Answer: A — Default = 'public' (read-only). Should be changed for security.

Q981. What is an SNMP trap?

A. Unsolicited notification sent by device to NMS (UDP 162)

B. SNMP query

C. SNMP response

D. SNMP table

Answer: A — Trap: device sends alert to NMS when an event occurs (link down, BGP peer lost, etc.).

Q982. What is an SNMP inform?

A. Trap with acknowledgment (reliable, requires response from NMS)

B. Same as trap

C. SNMP query

D. Encryption

Answer: A — Inform = trap that NMS must ACK. If no ACK, device retransmits. More reliable than traps.

Q983. What is an OID?

A. Object Identifier — hierarchical unique ID for a managed object

B. Object Identification Data

C. Operational ID

D. Originated Identifier

Answer: A — OID = dotted hierarchical identifier (e.g., 1.3.6.1.2.1.1.5.0 = sysName). Standardized by IETF.

Q984. What is SNMP MIB?

A. Management Information Base — collection of OIDs the device supports

B. Memory Information Block

C. Management Interface Block

D. Module Information Base

Answer: A — MIB = hierarchical database of OIDs. Defines what data the device exposes via SNMP.

Q985. Which command enables SNMP v2c read-only community 'public'?

A. snmp-server community public RO

B. snmp community public ro

C. snmp-server public ro

D. snmp public RO

Answer: A — snmp-server community RO [acl] configures read-only community (optionally with ACL).

Q986. Which command sets the SNMP contact?

A. snmp-server contact

B. snmp contact

C. snmp-server location contact

D. contact

Answer: A — snmp-server contact sets the sysContact field.

Q987. Which command sets the SNMP location?

A. snmp-server location

B. snmp location

C. snmp-server site

D. location

Answer: A — snmp-server location sets the sysLocation field.

Q988. Which command enables SNMP trap for link state changes?

A. snmp-server enable traps snmp linkdown linkup

B. snmp traps link

C. snmp-server trap link

D. trap link

Answer: A — snmp-server enable traps enables traps of that type. Then 'snmp-server host version 2c '.

Q989. Which command sets the SNMP trap destination?

A. snmp-server host 192.168.1.10 version 2c public

B. snmp-server trap 192.168.1.10

C. snmp host 192.168.1.10

D. trap-host 192.168.1.10

Answer: A — snmp-server host version [traps|informs] defines trap destination.

Q990. What is NAT?

A. Network Address Translation — translates IPs (typically private to public)

B. Network Authentication Translation

C. Node Address Translation

D. Network Address Tracker

Answer: A — NAT translates source/destination IPs at packet time. Solves IPv4 shortage and provides some security.

Q991. What are the main NAT types?

A. Static, Dynamic, PAT (overload)

B. Public, Private

C. TCP, UDP

D. Manual, Auto

Answer: A — Static = 1:1 mapping. Dynamic = many-to-many pool. PAT = many-to-one (overload, uses ports).

Q992. What is PAT?

A. Port Address Translation — many internal IPs to one public IP, using unique source ports

B. Public Address Translation

C. Permanent Address Translation

D. Protocol Address Translation

Answer: A — PAT (NAT overload) is what most home routers use. Many internal IPs share one public IP via port multiplexing.

Q993. Which command enables PAT (NAT overload)?

A. ip nat inside source list 1 interface Fa0/0 overload

B. nat overload list 1

C. ip nat pat list 1

D. ip nat overload

Answer: A — ip nat inside source list interface overload. The 'overload' keyword enables PAT.

Q994. Which command marks an interface as NAT inside?

A. ip nat inside

B. nat inside

C. interface nat inside

D. ip inside nat

Answer: A — ip nat inside (interface config). The interface facing internal network.

Q995. Which command marks an interface as NAT outside?

A. ip nat outside

B. nat outside

C. interface nat outside

D. ip outside nat

Answer: A — ip nat outside (interface config). The interface facing the public Internet.

Q996. Which command configures a static NAT entry?

A. ip nat inside source static 192.168.1.10 203.0.113.10

B. static nat 192.168.1.10 203.0.113.10

C. ip nat static 192.168.1.10 203.0.113.10

D. nat static 192.168.1.10

Answer: A — ip nat inside source static creates a permanent 1:1 mapping.

Q997. Which command configures a static PAT entry (port forward)?

A. ip nat inside source static tcp 192.168.1.10 80 203.0.113.10 80

B. static pat tcp 192.168.1.10 80 203.0.113.10 80

C. ip nat pat tcp 192.168.1.10 80 203.0.113.10 80

D. port-forward 80 192.168.1.10

Answer: A — ip nat inside source static tcp .

Q998. Which command displays active NAT translations?

A. show ip nat translations

B. show nat

C. show ip nat

D. show nat translations

Answer: A — show ip nat translations displays current NAT table (active mappings).

Q999. Which command displays NAT statistics?

A. show ip nat statistics

B. show nat stats

C. show ip nat stats

D. show ip nat

Answer: A — show ip nat statistics shows hits, misses, translations, pool usage.

Q1000. Which command clears all dynamic NAT translations?

A. clear ip nat translation *

B. clear nat

C. clear ip nat *

D. no ip nat translation

*Answer: A — clear ip nat translation * removes all dynamic translations. Static entries remain.*

Q1001. What is NAT pool?

A. Range of public IPs used for dynamic NAT

B. Backup NAT

C. NAT for VPN

D. Static NAT

Answer: A — NAT pool: range of public IPs (e.g., 203.0.113.10-203.0.113.20). Used for dynamic 1:1 mapping (not PAT).

Q1002. Which command configures a NAT pool?

A. ip nat pool MYPOOL 203.0.113.10 203.0.113.20 netmask 255.255.255.0

B. nat pool MYPOOL 203.0.113.10/28

C. ip nat MYPOOL 203.0.113.10-20

D. pool nat MYPOOL 203.0.113.10

Answer: A — ip nat pool netmask [type rotary].

Q1003. What is NAT64?

A. Translates IPv6 to IPv4 (and vice versa) for v6-only clients to reach v4-only servers

B. NAT for IPv4 only

C. NAT for 64-bit IPs

D. NAT for TCP/IPv6

Answer: A — NAT64 + DNS64 allows IPv6-only networks to communicate with IPv4-only services.

Q1004. What is QoS?

A. Quality of Service — prioritizes certain traffic (voice, video) over others

B. Quality of Service — encrypts traffic

C. Quality of Service — routes traffic

D. Quality of Service — speeds up traffic

Answer: A — QoS prioritizes traffic types. Voice/video need low latency; file transfers tolerate delay.

Q1005. What is the difference between IntServ and DiffServ?

A. IntServ = per-flow reservation (RSVP). DiffServ = per-hop marking (DSCP).

B. IntServ is faster

C. DiffServ is per-flow

D. Same thing

Answer: A — IntServ: RSVP reserves bandwidth per flow. DiffServ: marks packets (DSCP), each router applies QoS. DiffServ is dominant.

Q1006. What is CoS?

A. Class of Service — L2 priority (802.1p, 3 bits in 802.1Q tag)

B. Code of Service

C. Class of Service — L3

D. Cosmic OS

Answer: A — CoS = 3-bit field in 802.1Q tag (PCP). Values 0-7. Layer 2 only.

Q1007. What is DSCP?

A. Differentiated Services Code Point — 6-bit L3 QoS marker

B. Differentiated Service Code Priority

C. Dynamic Service Code Point

D. Distributed Service Class Protocol

Answer: A — DSCP = 6 bits in IPv4 ToS / IPv6 Traffic Class. 64 values. Common: EF (46), AF41 (34), CS0 (0).

Q1008. What is EF DSCP value?

A. 46

B. 0

C. 8

D. 10

Answer: A — EF (Expedited Forwarding) = 46. Used for voice payload.

Q1009. What is AF41?

A. Assured Forwarding class 4, drop precedence 1 — recommended for video

B. Audio Forwarding

C. Application Forwarding

D. Available Forwarding

Answer: A — AF41 (DSCP 34) is recommended for interactive video. AF classes: AF1x, AF2x, AF3x, AF4x.

Q1010. What is the trust boundary?

A. Where QoS markings are accepted (typically access switch or IP phone)

B. Where QoS is encrypted

C. Where QoS is blocked

D. Where QoS is calculated

Answer: A — Trust boundary = where markings are trusted. Trust IP phone, re-mark user PC traffic.

Q1011. What is LLQ?

A. Low Latency Queuing — strict priority queue for voice/video + CBWFQ for others

B. Low Loss Queuing

C. Logical Layer Queuing

D. Long Latency Queuing

Answer: A — LLQ = CBWFQ + strict priority queue. Priority queue is served first (up to bandwidth limit), rest get CBWFQ.

Q1012. What is CBWFQ?

A. Class-Based Weighted Fair Queuing — assigns bandwidth per class

B. Class-Based Weighted Fair Queuing

C. Cisco-Based Weighted Fair Queuing

D. Custom Bandwidth Weighted Fair Queuing

Answer: A — CBWFQ: each traffic class gets guaranteed bandwidth. No strict priority (LLQ adds that).

Q1013. What is shaping vs policing?

A. Shaping buffers excess traffic; policing drops or remarks it

B. Shaping drops, policing buffers

C. Same thing

D. Shaping is for inbound, policing for outbound

Answer: A — Shaping = buffer/smooth (delays excess). Policing = drop or re-mark (immediate). Shaping is outbound only.

Q1014. What is Cisco Discovery Protocol (CDP)?

A. Cisco proprietary L2 protocol for neighbor discovery

B. Common Discovery Protocol

C. Cisco Device Protocol

D. Cisco Detection Protocol

Answer: A — CDP (L2, Cisco proprietary) advertises device info to directly connected Cisco neighbors.

Q1015. What is LLDP?

A. Link Layer Discovery Protocol — IEEE 802.1AB (vendor-neutral CDP equivalent)

B. Local Loop Detection Protocol

C. Low Latency Discovery Protocol

D. Logical Layer Detection Protocol

Answer: A — LLDP = IEEE standard (802.1AB) equivalent of CDP. Supported by all vendors.

Q1016. What is the default CDP update interval?

A. 60 seconds

B. 30 seconds

C. 120 seconds

D. 5 seconds

Answer: A — CDP sends updates every 60 seconds by default. Hold time = 180s (3× interval).

Q1017. What is the default CDP hold time?

A. 180 seconds

B. 60 seconds

C. 120 seconds

D. 240 seconds

Answer: A — CDP hold = 180s (3× update interval). If no update in 180s, neighbor entry removed.

Q1018. What is the default LLDP update interval?

A. 30 seconds

B. 60 seconds

C. 5 seconds

D. 120 seconds

Answer: A — LLDP sends every 30s. Hold time = 120s (4× interval).

Q1019. What is the default LLDP hold time?

A. 120 seconds
B. 30 seconds
C. 60 seconds
D. 180 seconds
Answer: A — LLDP hold = 120s (4× interval).

Q1020. Which command disables CDP globally?

A. no cdp run
B. cdp disable
C. no cdp
D. cdp off
Answer: A — no cdp run (global config) disables CDP on all interfaces.

Q1021. Which command disables CDP on a specific interface?

A. no cdp enable
B. cdp off
C. no cdp
D. cdp disable interface
Answer: A — no cdp enable (interface config) disables CDP on that interface only.

Q1022. Which command enables LLDP globally?

A. lldp run
B. lldp enable
C. enable lldp
D. lldp global
Answer: A — lldp run enables LLDP globally. Then per-interface: lldp transmit, lldp receive.

Q1023. What is the Cisco IOS 'service' command used for?

A. Global services like timestamps, password-encryption, tcp-keepalives
B. Service routing
C. Service ACL
D. Service NAT
Answer: A — service configures global services (timestamps, password-encryption, etc.).

Q1024. What does 'service password-encryption' do?

A. Encrypts plaintext passwords in the config
B. Encrypts all data
C. Encrypts routing updates
D. Encrypts SSH sessions
Answer: A — Encrypts passwords that would otherwise be cleartext (console, vty, username password). Use 'secret' for stronger hashing.

Q1025. What does 'no service password-encryption' do?

A. Disables password encryption (existing encrypted passwords stay encrypted)
B. Decrypts all passwords
C. Removes all passwords
D. Has no effect
Answer: A — Disables encryption for future passwords. Already-encrypted passwords remain encrypted.

Q1026. What does 'service timestamps log datetime' do?

A. Adds timestamps to log messages using system clock

B. Encrypts timestamps

C. Speeds up logs

D. Disables timestamps

Answer: A — Adds datetime to syslog messages. Add 'msec localtime show-timezone' for precision.

Q1027. What does 'no ip domain-lookup' do?

A. Disables DNS resolution on the router

B. Disables domain

C. Disables DNS server

D. Disables NTP

Answer: A — Disables DNS lookups (prevents the annoying 'Translating...domain server' timeout when you mistype a command).

Q1028. What is 'ip http server'?

A. Enables HTTP server on Cisco device (web UI)

B. Enables HTTP client

C. Disables HTTP

D. HTTP ACL

Answer: A — Enables HTTP server for web-based management. Insecure — use 'ip http secure-server' (HTTPS) instead.

Q1029. What is 'ip http secure-server'?

A. Enables HTTPS server on Cisco device (secure web UI)

B. Disables HTTPS

C. HTTPS client

D. HTTPS encryption key

Answer: A — Enables HTTPS web management. Requires crypto keys (RSA/ECDSA) to be generated first.

Q1030. Which command generates RSA keys for HTTPS/SSH?

A. crypto key generate rsa modulus 2048

B. crypto rsa generate 2048

C. generate rsa 2048

D. ip rsa key generate

Answer: A — crypto key generate rsa modulus . Generates RSA keys for SSH/HTTPS. Use 2048+ bits.

Q1031. What is SCP?

A. Secure Copy Protocol — file transfer over SSH

B. Simple Copy Protocol

C. Secure Copy Port

D. Source Copy Protocol

Answer: A — SCP = file transfer using SSH (TCP 22). Secure alternative to TFTP/FTP.

Q1032. Which command enables SCP server on Cisco?

A. ip scp server enable

B. scp enable

C. scp server

D. ip scp enable

Answer: A — ip scp server enable allows SCP file transfers to/from the device.

Q1033. What is 'archive' used for on Cisco?

A. Configuration archive and versioning

B. IOS image archive

C. Log archive

D. Backup routing archive

Answer: A — archive feature saves running-config on changes. Useful for change tracking and rollback.

Q1034. Which command auto-saves config on changes?

A. archive; write-memory; time-period 60

B. auto-save 60

C. archive auto 60

D. config-archive 60

Answer: A — Under 'archive' config: 'write-memory' (save on write mem), 'time-period ' (auto-save interval).

Q1035. What is the 'configure replace' command?

A. Replaces running-config with a saved file (rollback)

B. Replaces IOS image

C. Replaces device

D. Replaces interface

Answer: A — configure replace [force] replaces running-config with the specified file. Used for config rollback.

Q1036. What is the 'archive config' command?

A. Saves current running-config to archive

B. Archives interface

C. Archives IOS

D. Archives log

Answer: A — archive config (privileged EXEC) saves current running-config to the archive path.

Q1037. What is Cisco Catalyst Center (DNA Center)?

A. Cisco's SDN controller for campus (SD-Access)

B. A type of switch

C. A routing protocol

D. A monitoring tool

Answer: A — DNA Center = SDN controller for campus networks. Provides automation, assurance, fabric (SD-Access).

Domain 5: Security Fundamentals

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Q1038. What does the 'C' in CIA triad stand for?

A. Confidentiality

B. Capability

C. Connection

D. Compliance

Answer: A — C = Confidentiality. Only authorized users can read data. Encryption provides this.

Q1039. What does the 'I' in CIA triad stand for?

A. Integrity

B. Internet

C. Internal

D. Identity

Answer: A — I = Integrity. Data has not been altered. Hashing (SHA, MD5) provides this.

Q1040. What does the 'A' in CIA triad stand for?

A. Availability

B. Authentication

C. Authorization

D. Accounting

Answer: A — A = Availability. Data/services are accessible when needed. Redundancy, backups provide this.

Q1041. What is authentication?

A. Proving who you are (identity)

B. Deciding what you can do

C. Logging what you did

D. Encrypting your data

Answer: A — Authentication = proving identity (username + password, cert, biometric).

Q1042. What is authorization?

A. Deciding what you can do (permissions)

B. Proving who you are

C. Logging actions

D. Encrypting data

Answer: A — Authorization = what resources/actions you're allowed. Happens after authentication.

Q1043. What is accounting?

A. Logging what you did (audit trail)

B. Proving identity

C. Setting permissions

D. Encrypting data

Answer: A — Accounting = recording actions for audit/billing. AAA = Authentication, Authorization, Accounting.

Q1044. What does AAA stand for?

A. Authentication, Authorization, Accounting

B. Access, Audit, Accounting

C. Auto, Auth, Audit

D. Authentication, Access, Authorization

Answer: A — AAA = Authentication (who), Authorization (what), Accounting (when/what they did).

Q1045. What is the difference between authentication and authorization?

A. Auth = who you are; Authz = what you can do

B. Same thing

C. Auth = what you can do; Authz = who you are

D. Auth is for users, Authz is for admins

Answer: A — Authentication verifies identity. Authorization determines permissions. AuthN happens first.

Q1046. What is RADIUS?

A. Open standard AAA protocol (RFC 2865), UDP 1812/1813

B. Cisco proprietary

C. TCP-based

D. Encryption protocol

Answer: A — RADIUS = open standard AAA. Uses UDP 1812 (auth) and 1813 (accounting). Only encrypts password.

Q1047. What is TACACS+?

A. Cisco proprietary AAA protocol, TCP 49

B. Open standard

C. UDP-based

D. Routing protocol

Answer: A — TACACS+ = Cisco proprietary AAA. Uses TCP 49. Encrypts entire packet body.

Q1048. Which AAA protocol encrypts only the password?

A. RADIUS

B. TACACS+

C. Both

D. Neither

Answer: A — RADIUS encrypts only the password field. TACACS+ encrypts the entire body.

Q1049. Which AAA protocol encrypts the entire packet body?

A. TACACS+

B. RADIUS

C. Both

D. Neither

Answer: A — TACACS+ encrypts the entire packet body. RADIUS only encrypts the password.

Q1050. Which AAA protocol is better for device administration (per-command authorization)?

A. TACACS+

B. RADIUS

C. Both equally

D. Kerberos

Answer: A — TACACS+ supports per-command authorization (admin can allow 'show run' but not 'reload'). RADIUS doesn't.

Q1051. Which AAA protocol is better for network access (VPN, Wi-Fi)?

A. RADIUS
B. TACACS+
C. Both equally
D. LDAP
Answer: A — RADIUS is widely supported for network access (WPA2-Enterprise, VPN, 802.1X).

Q1052. Which AAA protocol uses TCP?

A. TACACS+ (TCP 49)
B. RADIUS (UDP 1812)
C. Both use TCP
D. Neither
Answer: A — TACACS+ uses TCP 49 (reliable). RADIUS uses UDP (best-effort).

Q1053. Which AAA protocol uses UDP?

A. RADIUS (UDP 1812/1813)
B. TACACS+ (TCP 49)
C. Both use UDP
D. Neither
Answer: A — RADIUS uses UDP 1812 (auth) and 1813 (accounting).

Q1054. What is Kerberos?

A. SSO protocol using tickets (symmetric key crypto)
B. AAA protocol
C. Routing protocol
D. Encryption standard
Answer: A — Kerberos = SSO using tickets (TGT, service tickets). Used in Active Directory. Strong security.

Q1055. What is LDAP?

A. Lightweight Directory Access Protocol — directory service (TCP 389)
B. Local Directory Authentication Protocol
C. Layered Data Access Protocol
D. Logical Domain Access Protocol
Answer: A — LDAP = TCP 389 (or 636 for LDAPS). Used to query Active Directory and other directories.

Q1056. Which port does LDAPS use?

A. 636
B. 389
C. 443
D. 646
Answer: A — LDAPS (LDAP over SSL/TLS) uses TCP 636.

Q1057. What is Active Directory?

A. Microsoft directory service for user/computer management
B. Cisco routing protocol
C. Linux authentication
D. Network monitoring tool
Answer: A — AD = Microsoft's directory service. Uses Kerberos + LDAP. Common in enterprise environments.

Q1058. What is SSO?

A. Single Sign-On — one set of credentials for multiple apps

B. Secure Sign-On

C. Single Sign-Off

D. Standard Sign-On

Answer: A — SSO = one login for multiple services. Reduces password fatigue and improves security.

Q1059. What is MFA?

A. Multi-Factor Authentication — combines 2+ factors

B. Multiple Frequency Authentication

C. Master File Authentication

D. Multi-Function Adapter

Answer: A — MFA: something you know (password) + have (token) + are (biometric).

Q1060. What are the three MFA factors?

A. Knowledge (password), Possession (token), Inherence (biometric)

B. Username, Password, PIN

C. Card, Token, Code

D. Email, SMS, Phone

Answer: A — Three factors: knowledge, possession, inherence. Two-factor uses two of these.

Q1061. What is a firewall?

A. Device that filters traffic based on rules

B. Routing device

C. DHCP server

D. DNS server

Answer: A — Firewall enforces access policy between security zones (internal, DMZ, Internet).

Q1062. What is a stateful firewall?

A. Tracks connection state (allows return traffic for established connections)

B. Stateless filter

C. Encrypts traffic

D. Routes packets

Answer: A — Stateful firewall maintains state table. Allows return traffic for connections initiated from inside.

Q1063. What is a stateless firewall (packet filter)?

A. Filters each packet independently (no state tracking)

B. Tracks connections

C. Encrypts traffic

D. Application-aware

Answer: A — Stateless = ACL-based, no state. Faster but less secure — can't differentiate new vs return traffic.

Q1064. What is a next-gen firewall (NGFW)?

A. Application-aware firewall with IPS, malware analysis, user identity

B. Faster firewall

C. Hardware-only firewall

D. Cloud-only firewall

Answer: A — NGFW = application-aware + IPS + URL filtering + user identity. Examples: Cisco Firepower, Palo Alto.

Q1065. What is a DMZ?

A. Demilitarized Zone — network segment for public-facing services (web, email)

B. Dynamic Management Zone

C. Default Management Zone

D. Direct Memory Zone

Answer: A — DMZ = isolated segment for services that need internet access. Reduces exposure of internal LAN.

Q1066. What is IDS?

A. Intrusion Detection System — passive, alerts on suspicious traffic

B. Internal Detection System

C. Internet Defense System

D. Intrusion Defense System

Answer: A — IDS = passive sensor. Analyzes traffic, alerts on matches. Doesn't block.

Q1067. What is IPS?

A. Intrusion Prevention System — active, blocks suspicious traffic inline

B. Internal Prevention System

C. Internet Protection System

D. Intrusion Protection Standard

Answer: A — IPS = inline sensor. Can block (drop) malicious traffic in real-time.

Q1068. What is the difference between IDS and IPS?

A. IDS detects and alerts; IPS detects AND blocks (inline)

B. IDS is faster

C. IPS only detects

D. They are identical

Answer: A — IDS = passive (alert only). IPS = active (block/drop). IPS is inline.

Q1069. What is a VPN?

A. Virtual Private Network — encrypted tunnel over untrusted network

B. Variable Private Network

C. Verified Private Network

D. Virtual Protected Network

Answer: A — VPN creates encrypted tunnel over Internet. Provides confidentiality and integrity.

Q1070. What is IPsec?

A. Internet Protocol Security — protocol suite for encrypting/authenticating IP packets

B. Internal Protocol Security

C. Internet Packet Security

D. Inter-Process Security

Answer: A — IPsec = suite of protocols (AH, ESP, IKE). Encrypts/authenticates IP traffic.

Q1071. What are the two IPsec main protocols?

A. AH (Authentication Header) and ESP (Encapsulating Security Payload)

B. TCP and UDP

C. TLS and SSL

D. AES and 3DES

Answer: A — AH = authentication only (rarely used). ESP = auth + encryption (most common).

Q1072. What does ESP provide?

A. Authentication + encryption (confidentiality + integrity)

B. Authentication only

C. Encryption only

D. Routing

Answer: A — ESP (protocol 50) provides confidentiality (encryption) and integrity (auth).

Q1073. What does AH provide?

A. Authentication + integrity only (no encryption)

B. Encryption + auth

C. Encryption only

D. Routing only

Answer: A — AH (protocol 51) provides authentication and integrity but no encryption. Rarely used today.

Q1074. What are IPsec transport and tunnel modes?

A. Transport = encrypt payload only. Tunnel = encrypt entire original packet.

B. Transport = encrypt everything. Tunnel = payload only.

C. Both encrypt everything

D. Same thing

Answer: A — Transport mode: encrypts only IP payload (used host-to-host). Tunnel mode: encrypts whole packet, used gateway-to-gateway.

Q1075. What is IKE?

A. Internet Key Exchange — protocol for negotiating IPsec SAs

B. Internal Key Encryption

C. Internet Key Encryption

D. Inline Key Exchange

Answer: A — IKE (UDP 500/4500) negotiates security parameters (algorithms, keys) before IPsec establishes.

Q1076. Which ports does IKE use?

A. UDP 500 (and 4500 for NAT-T)

B. TCP 500

C. UDP 100

D. TCP 443

Answer: A — IKE uses UDP 500. NAT-T wraps ESP in UDP 4500 to pass through NAT.

Q1077. What is NAT-T in IPsec?

A. NAT Traversal — wraps ESP in UDP 4500 so it can pass through NAT

B. Network Address Translation Tunnel

C. Network Authentication Token

D. Native Address Translation

Answer: A — NAT-T detects NAT and encapsulates ESP in UDP 4500 so NAT devices can handle it.

Q1078. What is a site-to-site VPN?

A. Connects two sites via encrypted tunnel between VPN gateways

B. Single user to corporate

C. VPN between two VLANs

D. VPN using only one router

Answer: A — Site-to-site = gateway-to-gateway. Both sites appear directly connected.

Q1079. What is a remote-access VPN?

A. Single user connects to corporate network via encrypted tunnel

B. Connects two sites

C. VPN between VLANs

D. VPN without encryption

Answer: A — Remote-access = client-to-gateway. Used by remote workers. Examples: Cisco AnyConnect, OpenVPN.

Q1080. What is SSL VPN?

A. VPN using TLS/SSL (HTTPS) — works through browsers and firewalls

B. Same as IPsec

C. Slower VPN

D. Hardware-only VPN

Answer: A — SSL VPN uses HTTPS (TCP 443). Clientless mode works in browser. Cisco AnyConnect is SSL VPN.

Q1081. What is Cisco AnyConnect?

A. Cisco's SSL/IPsec VPN client for remote access

B. Routing protocol

C. Encryption algorithm

D. Switch feature

Answer: A — AnyConnect = Cisco's remote access VPN client. Supports SSL/TLS and IPsec IKEv2.

Q1082. What is a VPN split tunnel?

A. Only corporate traffic goes through VPN; internet goes direct

B. Splits VPN in half

C. Two VPNs at once

D. Backup VPN

Answer: A — Split tunnel: corporate traffic encrypted, internet traffic direct. Saves bandwidth but less secure.

Q1083. What is a VPN full tunnel?

A. All traffic (including internet) goes through VPN

B. Half traffic through VPN

C. Backup VPN

D. Faster VPN

Answer: A — Full tunnel: all client traffic goes through VPN. More secure (corporate controls apply) but uses more bandwidth.

Q1084. What is a pre-shared key (PSK)?

A. Shared secret string used for authentication

B. Public key

C. Cert-based auth

D. Token-based auth

Answer: A — PSK = both sides have same secret string. Simple but less secure than certificates.

Q1085. What is PKI?

A. Public Key Infrastructure — system for managing digital certificates

B. Private Key Infrastructure

C. Public Key Integration

D. Private Key Integration

Answer: A — PKI = CAs (Certificate Authorities), certificates, key pairs. Used for SSL/TLS, IPsec certs, code signing.

Q1086. What is a CA?

A. Certificate Authority — issues and signs digital certificates

B. Crypto Algorithm

C. Cisco Authentication

D. Common Authentication

Answer: A — CA = trusted third party that issues certificates. Examples: DigiCert, Let's Encrypt.

Q1087. What is a digital certificate?

A. Binds public key to identity (signed by CA)

B. Password

C. Encryption key

D. Token ID

Answer: A — Certificate = X.509 document signed by CA, binding public key to identity (server name).

Q1088. What is TLS?

A. Transport Layer Security — encryption protocol (successor to SSL)

B. Transfer Layer Security

C. Transmission Layer Standard

D. Transport Layer Standard

Answer: A — TLS = current standard (TLS 1.3 is latest). SSL is deprecated. Used for HTTPS, SMTPS, etc.

Q1089. Which TLS version is currently recommended?

A. TLS 1.3 (or 1.2 minimum)

B. TLS 1.0

C. TLS 1.1

D. SSL 3.0

Answer: A — TLS 1.3 is latest and most secure. TLS 1.2 minimum acceptable. TLS 1.0/1.1 deprecated (2020).

Q1090. What hashing algorithm is considered weak today?

A. MD5 (and SHA-1)

B. SHA-256

C. SHA-3

D. BLAKE2

Answer: A — MD5 and SHA-1 are cryptographically weak. Use SHA-256 or SHA-3.

Q1091. What is a salt in password hashing?

A. Random value added to password before hashing (prevents rainbow tables)

B. Encryption key

C. Password policy

D. Token

Answer: A — Salt = random per-user value added to password before hashing. Prevents rainbow table attacks.

Q1092. What is bcrypt?

A. Slow password hashing function (designed to resist brute force)

B. Encryption algorithm

C. Hash function

D. Routing protocol

Answer: A — bcrypt = adaptive password hashing. Slow on purpose to resist brute force. Uses salt.

Q1093. What is a VLAN hopping attack?

A. Attacker sends double-tagged frame to bypass VLAN isolation

B. VLAN flood

C. VLAN encryption attack

D. VLAN MAC spoofing

Answer: A — Double-tagged frame: outer tag is native VLAN (stripped by switch), inner tag reaches different VLAN.

Q1094. How do you prevent VLAN hopping?

A. Disable auto-trunking, set native VLAN to unused, shut unused ports

B. Encrypt VLANs

C. Use longer passwords

D. Disable CDP

Answer: A — Set ports to access mode + nonegotiate, change native VLAN to unused, shut unused ports.

Q1095. What is a DHCP starvation attack?

A. Attacker floods DHCP requests to exhaust the pool

B. Rogue DHCP server

C. DHCP encryption attack

D. DHCP snooping

Answer: A — DHCP starvation: tools like Yersinia send many requests with fake MACs. DoS for DHCP clients.

Q1096. What is DHCP spoofing?

A. Attacker runs rogue DHCP server to assign fake IPs/gateway

B. DHCP flood

C. DHCP encryption attack

D. DHCP starvation

Answer: A — Rogue DHCP gives attacker IP as gateway, enabling MITM.

Q1097. How do you prevent DHCP attacks?

A. DHCP snooping (only trusted ports can send offers)

B. Stronger passwords

C. Encryption

D. Disable DHCP

Answer: A — DHCP snooping prevents rogue servers. Configure trusted ports only on switch uplinks/DHCP server ports.

Q1098. What is ARP spoofing/poisoning?

A. Attacker sends fake ARP replies to redirect traffic (MITM)

B. ARP flood

C. ARP encryption

D. ARP filter

Answer: A — ARP poisoning: attacker claims to be the gateway MAC. Traffic goes through attacker.

Q1099. How do you mitigate ARP spoofing?

A. Dynamic ARP Inspection (DAI) using DHCP snooping table

B. Disable ARP

C. Strong passwords

D. BPDU Guard

Answer: A — DAI validates ARP packets against DHCP snooping binding table. Drops invalid ARP.

Q1100. What is MAC flooding?

A. Attacker floods switch with fake MACs to overflow CAM table (causes switch to act as hub)

B. Encrypt MAC

C. Spoof ARP

D. DHCP flood

Answer: A — MAC flooding fills CAM table. Once full, switch floods all ports → attacker captures all traffic.

Q1101. How do you mitigate MAC flooding?

A. Port security (limit MACs per port)

B. BPDU Guard

C. DAI

D. DHCP snooping

Answer: A — Port security limits MAC count per port. Violations trigger protect/restrict/shutdown.

Q1102. What is MAC spoofing?

A. Attacker changes their MAC to impersonate another device

B. MAC flood

C. MAC encryption

D. MAC learning

Answer: A — MAC spoofing: change MAC to bypass port security or impersonate legitimate device.

Q1103. What is a replay attack?

A. Attacker captures and retransmits a legitimate message

B. Captures and modifies

C. Encrypts message

D. Blocks traffic

Answer: A — Replay: attacker re-sends captured valid packet (e.g., bank transfer). Prevented by nonces/timestamps.

Q1104. What is a man-in-the-middle (MITM) attack?

A. Attacker intercepts and possibly modifies traffic between two parties

B. DDoS attack

C. Recon attack

D. Insider threat

Answer: A — MITM: attacker sits between client and server. Can read/modify encrypted traffic if cert not validated.

Q1105. What is a DDoS attack?

A. Distributed Denial of Service — many sources flood target with traffic

B. Single-source DoS

C. Insider attack

D. Recon attack

Answer: A — DDoS = many compromised hosts (botnet) flood target. Hard to block since traffic comes from many IPs.

Q1106. What is a SYN flood attack?

A. Attacker sends many SYN packets without completing handshake (exhausts server resources)

B. UDP flood

C. ICMP flood

D. HTTP flood

Answer: A — SYN flood: server allocates resources for each half-open connection. Eventually can't accept new connections.

Q1107. How do you mitigate SYN flood?

A. TCP SYN cookies, SYN proxy, rate limiting

B. Strong passwords

C. Encryption

D. Disable TCP

Answer: A — SYN cookies: server encodes state in SYN-ACK sequence number. No per-connection state until ACK received.

Q1108. What is a phishing attack?

A. Fake emails/sites trick users into revealing credentials

B. Network attack

C. DDoS

D. Recon

Answer: A — Phishing = social engineering via email. Variants: spear phishing (targeted), whaling (executives).

Q1109. What is ransomware?

A. Malware that encrypts files and demands payment

B. Network attack

C. DDoS

D. Spyware

Answer: A — Ransomware = encrypts files, demands cryptocurrency ransom. Defenses: backups, patching, email filtering.

Q1110. What is malware?

A. Malicious software (viruses, worms, Trojans, spyware)

B. Network attack

C. Routing protocol

D. Encryption weakness

Answer: A — Malware = umbrella term for viruses, worms, Trojans, spyware, ransomware, etc.

Q1111. What is social engineering?

A. Manipulating people into revealing info or taking actions

B. Network attack

C. DDoS

D. Hardware attack

Answer: A — Social engineering = psychological manipulation. Phishing, pretexting, baiting, tailgating.

Q1112. What is tailgating?

A. Attacker follows authorized person through secure door

B. Network attack

C. DDoS

D. Software attack

Answer: A — Tailgating = unauthorized person follows authorized through door. Defenses: badges, mantraps, education.

Q1113. What is zero trust?

A. Security model that trusts nothing by default; verifies every access

B. Disable all access

C. Encrypt all traffic

D. Block all external

Answer: A — Zero Trust: 'never trust, always verify'. Each request authenticated/authorized regardless of network location.

Q1114. What are the core principles of Zero Trust?

A. Verify explicitly, use least privilege, assume breach

B. Trust but verify

C. Block all external

D. Encrypt all internal

Answer: A — Zero Trust: verify explicitly (identity + context), least privilege access, assume breach (segment, monitor).

Q1115. What is least privilege?

A. Give users only the access they need (no more)

B. Remove all privileges

C. Give everyone admin

D. Use long passwords

Answer: A — Least privilege: minimum access needed for job. Reduces attack surface and impact of compromised accounts.

Q1116. What is segmentation?

A. Divide network into smaller zones to limit lateral movement

B. Encrypt traffic

C. Speed up network

D. Reduce broadcast

Answer: A — Segmentation limits attacker movement. VLANs, firewalls, micro-segmentation (Cisco TrustSec).

Q1117. What is micro-segmentation?

A. Fine-grained per-workload segmentation (down to individual VMs/containers)

B. VLAN segmentation

C. Subnet segmentation

D. Hardware segmentation

Answer: A — Micro-segmentation: per-workload policies. Cisco TrustSec (SGT tags), VMware NSX, Cisco ACI.

Q1118. What is Cisco TrustSec?

A. Cisco's security group tagging (SGT) for scalable micro-segmentation

B. Routing protocol

C. Switching protocol

D. Encryption protocol

Answer: A — TrustSec: tags packets with Security Group Tags. Policies based on tags, not IPs. Scales to large networks.

Q1119. What is an ACL?

A. Access Control List — ordered permit/deny rules for traffic filtering

B. Address Control List

C. Authentication Control List

D. Application Control List

Answer: A — ACL filters traffic. Standard (source only), Extended (src/dst/proto/ports), Named (descriptive name).

Q1120. Which ACL type matches source IP only?

A. Standard (1-99, 1300-1999)

B. Extended

C. Reflexive

D. Dynamic

Answer: A — Standard ACL = source IP only. Limited use; place close to destination.

Q1121. Which ACL type matches protocol, src/dst IP, ports?

A. Extended (100-199, 2000-2699)

B. Standard

C. Reflexive

D. Time-based

Answer: A — Extended ACL = more granular. Place close to source to drop early.

Q1122. Where should you place a standard ACL?

A. Close to the destination

B. Close to the source

C. On the core

D. On the firewall

Answer: A — Standard ACL only knows source — place close to destination to avoid blocking legitimate traffic from other sources.

Q1123. Where should you place an extended ACL?

A. Close to the source

B. Close to the destination

C. On the core

D. On the firewall

Answer: A — Extended ACL knows src + dst — place close to source to drop unwanted traffic early.

Q1124. What is the implicit rule at the end of every ACL?

A. Deny any any (deny all)

B. Permit any any

C. Log all traffic

D. Do nothing

Answer: A — Every ACL ends with implicit 'deny any any'. If no rule matches, traffic is dropped.

Q1125. Which command applies an ACL to interface Fa0/0 for inbound traffic?

A. ip access-group 10 in

B. access-group 10 in

C. apply acl 10 in

D. ip acl 10 in

Answer: A — ip access-group applies ACL to interface.

Q1126. Which command creates a named extended ACL?

A. ip access-list extended BLOCK_TELNET

B. access-list extended BLOCK_TELNET

C. ip acl extended BLOCK_TELNET

D. extended acl BLOCK_TELNET

Answer: A — ip access-list extended enters named ACL config mode.

Q1127. Which command denies Telnet from 192.168.1.0/24 to anywhere?

A. deny tcp 192.168.1.0 0.0.0.255 any eq 23

B. deny tcp 192.168.1.0 255.255.255.0 any eq 23

C. deny udp 192.168.1.0 0.0.0.255 any eq 23

D. deny tcp 192.168.1.0/24 any eq 23

Answer: A — Extended ACL: deny tcp eq . Telnet = port 23.

Q1128. What does the 'established' keyword do in an extended ACL?

A. Matches TCP segments with ACK or RST flag set (return traffic)

B. Blocks all established

C. Required for all ACLs

D. Speeds up ACL

Answer: A — 'established' = match return traffic for already-established TCP connections. Primitive stateful filtering.

Q1129. What is a reflexive ACL?

A. Dynamically creates temporary entries for return traffic

B. Mirrors another ACL

C. Blocks all traffic

D. Has no rules

Answer: A — Reflexive ACL: when outbound session starts, creates temp permit for return traffic. Removed when session ends.

Q1130. What is port security?

A. Cisco switch feature that limits MACs per port

B. Router ACL feature

C. Wireless security

D. VPN feature

Answer: A — Port security limits MAC count per port. Prevents MAC flooding and unauthorized device connection.

Q1131. What are the port security violation modes?

A. protect, restrict, shutdown

B. drop, log, alert

C. block, warn, kill

D. allow, deny, log

Answer: A — protect = silent drop. restrict = drop + log + SNMP trap. shutdown = err-disable port (default).

Q1132. What does port security 'shutdown' do?

A. Err-disables the port (all traffic stops, requires manual recovery)

B. Drops violating frames only

C. Logs but allows

D. Sends SNMP trap only

Answer: A — shutdown = port goes to err-disabled state. All traffic stops. Requires 'shutdown' then 'no shutdown' to recover.

Q1133. What does port security 'restrict' do?

A. Drops violating frames, logs violation, sends SNMP trap (port stays up)

B. Err-disables port

C. Silently drops

D. Allows all traffic

Answer: A — restrict = drop + log + trap. Port stays up for legitimate MACs.

Q1134. What does port security 'protect' do?

A. Silently drops violating frames (no log, no trap)

B. Logs and drops

C. Err-disables port

D. Allows all traffic

Answer: A — protect = silent drop. No logging. Useful when you don't want alerts.

Q1135. What is the default port security violation mode?

A. shutdown

B. restrict

C. protect

D. log

Answer: A — Default = shutdown. Port is err-disabled. Most secure but requires manual recovery.

Q1136. Which command enables port security on an interface?

A. switchport port-security

B. port-security enable

C. switchport security

D. security port

Answer: A — switchport port-security enables port security. Must be on an access port.

Q1137. Which command sets max MAC count to 2?

A. switchport port-security maximum 2

B. port-security max 2

C. switchport max 2

D. maximum 2

Answer: A — switchport port-security maximum <1-8192> sets max MAC count (default 1).

Q1138. Which command enables sticky MAC learning?

A. switchport port-security mac-address sticky

B. mac sticky

C. switchport sticky

D. sticky mac enable

Answer: A — Sticky learning: dynamically learned MACs are saved to running-config as static.

Q1139. Which command sets a static secure MAC?

A. switchport port-security mac-address AAAA.AAAA.AAAA

B. mac secure AAAA.AAAA.AAAA

C. secure-mac AAAA.AAAA.AAAA

D. switchport mac AAAA.AAAA.AAAA

Answer: A — Statically configures a secure MAC on the port.

Q1140. What is BPDU Guard?

A. Err-disables access port if BPDU received (prevents rogue switches)

B. Encrypts BPDUs

C. Speeds up STP

D. Filters BPDUs

Answer: A — BPDU Guard: protects access ports from rogue switches. Port is err-disabled on BPDU receipt.

Q1141. What is BPDU Filter?

A. Stops BPDUs from being sent/received on a port

B. Logs BPDUs

C. Encrypts BPDUs

D. Speeds up BPDUs

Answer: A — BPDU Filter: stops BPDUs entirely. Use with caution — can cause loops if misconfigured.

Q1142. What is Root Guard?

A. Prevents external switches from becoming root (protects root placement)

B. Protects against rogue BPDUs

C. Encrypts root traffic

D. Speeds up root election

Answer: A — Root Guard: if superior BPDU received on port, port is err-disabled (root-bridge inconsistent).

Q1143. What is Loop Guard?

A. Prevents loops from unidirectional link failures

B. Detects physical loops

C. Encrypts loop traffic

D. Speeds up loop resolution

Answer: A — Loop Guard: if BPDUs stop arriving on non-designated port, keeps port blocking instead of transitioning.

Q1144. What is UDLD?

A. UniDirectional Link Detection — detects one-way fiber links

B. Universal Data Link Driver

C. Unified Data Link Protocol

D. Universal Direct Link Detection

Answer: A — UDLD detects broken fiber strands causing one-way communication. Prevents loops.

Q1145. What is DHCP snooping?

A. L2 security feature: blocks rogue DHCP servers, builds binding table

B. DHCP encryption

C. DHCP load balancer

D. DHCP relay

Answer: A — DHCP snooping: only trusted ports can send DHCP offers. Builds binding table used by DAI.

Q1146. What is Dynamic ARP Inspection (DAI)?

A. Validates ARP packets against DHCP snooping binding table

B. Encrypts ARP

C. Speeds up ARP

D. Prevents broadcast ARP

Answer: A — DAI prevents ARP spoofing. Drops ARP packets whose IP/MAC doesn't match DHCP binding table.

Q1147. What is IP Source Guard?

A. Filters traffic based on source IP+MAC matching DHCP binding

B. Encrypts source IP

C. Filters destination IP

D. Speeds up source routing

Answer: A — IP Source Guard drops traffic where source IP/MAC doesn't match DHCP snooping binding table.

Q1148. What is 802.1X?

A. Port-based authentication using EAPoL

B. Encryption standard

C. VLAN tagging

D. STP variant

Answer: A — 802.1X (PNAC) authenticates client before allowing traffic on switch port.

Q1149. What are the 802.1X components?

A. Supplicant, Authenticator, Authentication Server

B. Client, Switch, Router

C. User, AP, WLC

D. PC, Switch, Server

Answer: A — Supplicant (client) ↔ Authenticator (switch/AP) ↔ Authentication Server (RADIUS).

Q1150. What is EAP?

A. Extensible Authentication Protocol — framework for various auth methods

B. Encryption Auth Protocol

C. Extended Access Protocol

D. Ethernet Auth Protocol

Answer: A — EAP framework supports: EAP-TLS, PEAP, EAP-TTLS, EAP-FAST, EAP-MD5, etc.

Q1151. Which EAP method requires certificates on both client and server?

A. EAP-TLS

B. PEAP

C. EAP-TTLS

D. EAP-FAST

Answer: A — EAP-TLS = mutual cert auth. Most secure. Requires PKI for client certs.

Q1152. Which EAP method wraps EAP in TLS (only server cert required)?

A. PEAP

B. EAP-TLS

C. LEAP

D. EAP-MD5

Answer: A — PEAP: server cert establishes TLS tunnel, then traditional auth (MS-CHAPv2) inside.

Q1153. What is CoA in RADIUS?

A. Change of Authorization — server pushes policy changes mid-session

B. Class of Authorization

C. Cache of Authentication

D. Centralized Office Authentication

Answer: A — CoA: RADIUS server can change VLAN, ACL, session timeout on a live session. Used by NAC.

Q1154. What is NAC?

A. Network Access Control — posture check + policy enforcement

B. Network Authentication Cache

C. Node Access Controller

D. Network Address Converter

Answer: A — NAC: checks device posture (AV, patch level) before granting access. Cisco ISE is a NAC product.

Q1155. What is Cisco ISE?

A. Cisco Identity Services Engine — NAC, AAA, policy enforcement

B. Internal Service Engine

C. Internet Service Equipment

D. Integrated Switch Engine

Answer: A — ISE = Cisco's NAC/AAA product. Provides 802.1X, profiling, posture, guest management, BYOD.

Q1156. What is MAC Authentication Bypass (MAB)?

A. Authenticates devices by MAC when they don't support 802.1X

B. Encrypts MAC addresses

C. Bypasses authentication entirely

D. Blocks all MACs

Answer: A — MAB: fallback for devices (printers, IoT) that don't have 802.1X supplicant. Authenticates by MAC.

Q1157. What is WebAuth (local web authentication)?

A. User authenticates via web portal (captive portal)

B. 802.1X variant

C. MAC auth

D. Certificate auth

Answer: A — WebAuth: user redirected to web page to enter credentials. Used for guest access.

Q1158. What is a security zone?

A. Network segment with similar security requirements

B. VLAN only

C. Subnet only

D. Firewall

Answer: A — Security zones: inside, outside, DMZ. Firewalls enforce policies between zones.

Q1159. What is a bastion host?

A. Hardened server exposed to internet (typically in DMZ)

B. Backup server

C. Internal server

D. Database server

Answer: A — Bastion: hardened, exposed to internet. Used for jump hosts, public services.

Q1160. What is a jump host (jump server)?

A. Hardened server used to access internal servers (no direct access)

B. Public web server

C. Database server

D. Backup server

Answer: A — Jump host: admin SSHs to jump host, then to internal servers. Centralizes logging and access control.

Q1161. What is privileged access management (PAM)?

A. Manages and audits admin credentials

B. Personal Access Manager

C. Password Alteration Module

D. Public Access Mechanism

Answer: A — PAM = vaults admin passwords, rotates them, logs sessions. Examples: CyberArk, BeyondTrust.

Q1162. What is SIEM?

A. Security Information and Event Management — central log collection and analysis

B. Secure Internal Email Manager

C. Standard Interface Engine Module

D. System Integrity Event Monitor

Answer: A — SIEM: collects logs from many sources, correlates, alerts on threats. Examples: Splunk, QRadar, Sentinel.

Q1163. What is SOAR?

A. Security Orchestration, Automation, and Response

B. Secure Online Access Router

C. System Operations and Recovery

D. Standard Operating Audit Report

Answer: A — SOAR: automates incident response playbooks. Connects SIEM + tools + automation. Examples: Phantom, XSOAR.

Q1164. What is EDR?

A. Endpoint Detection and Response — advanced endpoint security

B. Encryption Decryption Routine

C. Extended Detection Report

D. Edge Data Router

Answer: A — EDR: replaces traditional AV. Behavioral analysis, threat hunting, response. Examples: CrowdStrike, Carbon Black.

Q1165. What is XDR?

A. Extended Detection and Response — correlates endpoint + network + cloud + email

B. Extra Detection Report

C. Extended Data Routing

D. External Detection Routine

Answer: A — XDR = EDR + network + cloud + email telemetry in one platform. Better correlation than siloed tools.

Q1166. What is threat intelligence?

A. Info about known threats (IOCs, TTPs) used to detect/prevent attacks

B. Threat guessing

C. Threat encryption

D. Threat blocking

Answer: A — Threat intel: IP/URL/domain/file hash IOCs. Sources: commercial (FireEye), open (MISP, VirusTotal).

Q1167. What is an IOC?

A. Indicator of Compromise — artifact indicating attack (IP, hash, URL)

B. Internal Operations Center

C. Input/Output Controller

D. Independent Operations Check

Answer: A — IOC: malicious IP, file hash, URL, domain. Used to detect compromises.

Q1168. What is malware analysis?

A. Examining malware to understand behavior and indicators

B. Encrypting malware

C. Removing malware

D. Speeding up malware

Answer: A — Malware analysis: static (disassemble) + dynamic (sandbox execution). Reveals IOCs, attack techniques.

Q1169. What is a honeypot?

A. Decoy system designed to attract attackers and study their behavior

B. Real production system

C. Backup system

D. Encryption device

Answer: A — Honeypot = decoy. Attackers waste time on fake systems while real ones are protected. Alerts on access.

Q1170. What is penetration testing?

A. Authorized simulated attack to find vulnerabilities

B. Actual attack

C. Vulnerability scanning only

D. Security policy

Answer: A — Pen test: ethical hacking. Black box (no info), gray box (some), white box (full info).

Q1171. What is vulnerability scanning?

A. Automated scan for known vulnerabilities (Nessus, OpenVAS)

B. Manual exploitation

C. Encryption check

D. Network mapping

Answer: A — Vuln scanning: automated tool checks for CVEs. Different from pen test (which actively exploits).

Q1172. What is CVE?

A. Common Vulnerabilities and Exposures — standardized vulnerability IDs

B. Common Vulnerability Encryption

C. Centralized Vulnerability Engine

D. Certified Vulnerability Evaluation

Answer: A — CVE = unique ID for publicly disclosed vulnerabilities. Format: CVE-YEAR-NUMBER.

Q1173. What is CVSS?

A. Common Vulnerability Scoring System — 0-10 severity score

B. Common Vulnerability Scoring Standard

C. Centralized Vulnerability Score System

D. Certified Vulnerability Severity

Answer: A — CVSS = 0-10 score. 9-10 critical, 7-8.9 high, 4-6.9 medium, 0-3.9 low. Helps prioritize.

Q1174. What is patch management?

A. Process of identifying, testing, deploying patches

B. Encrypting patches

C. Removing patches

D. Vulnerability scanning

Answer: A — Patch management: scan for missing patches, test, deploy, verify. Critical for security.

Q1175. What is hardening?

A. Reducing attack surface by disabling unnecessary services/features

B. Encrypting everything

C. Speeding up the system

D. Backing up the system

Answer: A — Hardening: disable unused services, remove default accounts, enable firewall, restrict access.

Q1176. What is CIS Benchmark?

A. Configuration hardening guidelines from Center for Internet Security

B. Cisco Internal Standard

C. Common Internet Standard

D. Certified Industry Standard

Answer: A — CIS Benchmarks: best-practice configurations for OS, apps, network devices. Free.

Q1177. What is the principle of defense in depth?

A. Multiple layers of security (no single point of failure)

B. Encryption only

C. Single strong control

D. Block all external

Answer: A — Defense in depth: firewall + IPS + endpoint + user training + backups. If one fails, others protect.

Q1178. What is encryption?

A. Process of converting plaintext to ciphertext using a key

B. Compressing data

C. Hashing data

D. Signing data

Answer: A — Encryption = confidentiality. Reversible with key. AES, RSA, ECC are common algorithms.

Q1179. What is hashing?

A. One-way function producing fixed-size output from input

B. Two-way encryption

C. Data compression

D. Authentication

Answer: A — Hashing = integrity check. Same input always produces same output. Can't reverse to get input.

Q1180. What is symmetric encryption?

A. Same key for encryption and decryption

B. Different keys for enc/dec

C. No keys

D. Public key only

Answer: A — Symmetric: AES, DES, 3DES. Fast, but key distribution is challenge. Used for bulk data.

Q1181. What is asymmetric encryption?

A. Two keys: public (encrypt) and private (decrypt)

B. Same key for enc/dec

C. No keys

D. Hash only

Answer: A — Asymmetric: RSA, ECC, DH. Slow but solves key distribution. Used for key exchange, digital signatures.

Q1182. Which is faster: symmetric or asymmetric encryption?

A. Symmetric (100-1000x faster)

B. Asymmetric

C. Same speed

D. Depends on key size

Answer: A — Symmetric is much faster. Used for bulk data. Asymmetric used for key exchange + signing.

Q1183. What is AES?

A. Advanced Encryption Standard — symmetric, block cipher (128/192/256-bit)

B. Asymmetric Encryption Standard

C. Advanced Encryption System

D. Asynchronous Encryption Service

Answer: A — AES = current standard (replaced DES). Symmetric. Key sizes: 128, 192, 256 bits. AES-256 most secure.

Q1184. What is RSA?

A. Asymmetric encryption algorithm (Rivest-Shamir-Adleman)

B. Routing Security Algorithm

C. Random Symmetric Algorithm

D. Robust Security Algorithm

Answer: A — RSA = asymmetric. Used for key exchange, digital signatures, encryption. Key size 2048+ recommended.

Q1185. What is Diffie-Hellman (DH)?

A. Key exchange algorithm — two parties derive shared secret over insecure channel

B. Encryption algorithm

C. Hash function

D. Authentication

Answer: A — DH: lets two parties derive shared secret without sharing it. Used in TLS, IPsec, SSH.

Q1186. What is SHA-256?

A. Secure Hash Algorithm producing 256-bit output

B. Symmetric Hash Algorithm

C. Standard Hash Application

D. Secure Hash Authority

Answer: A — SHA-256 = current recommended hash. 256-bit output. Used in TLS, certs, Bitcoin.

Q1187. What is HMAC?

A. Hash-based Message Authentication Code — hash with secret key

B. Hash MAC address

C. Hardware MAC

D. High-speed MAC

Answer: A — HMAC = hash + secret key. Provides integrity + authenticity. Used in IPsec, TLS.

Q1188. What is a digital signature?

A. Hash encrypted with sender's private key

B. Handwritten signature scanned

C. Encryption key

D. Hash value

Answer: A — Digital signature: proves authenticity, integrity, non-repudiation. Verified with sender's public key.

Q1189. What is non-repudiation?

A. Sender can't deny having sent the message (proven by digital signature)

B. Encryption strength

C. Hash strength

D. Key length

Answer: A — Non-repudiation: digital signature proves sender sent it. Sender can't later deny.

Q1190. What is steganography?

A. Hiding data inside other data (e.g., text in image)

B. Encrypting data

C. Hashing data

D. Compressing data

Answer: A — Steganography = hidden communication. Different from encryption (which hides content, not existence).

Q1191. What is a certificate revocation list (CRL)?

A. List of revoked certificates published by CA

B. Cache Removal List

C. Cipher Recovery List

D. Common Routing List

Answer: A — CRL: list of certs revoked before expiry. Clients check CRL to verify cert validity.

Q1192. What is OCSP?

A. Online Certificate Status Protocol — real-time cert revocation check

B. Online Certificate Service Provider

C. Open Crypto Standard Protocol

D. Online Cipher Service Provider

Answer: A — OCSP: real-time cert status check (better than CRL). 'Good', 'Revoked', 'Unknown'.

Q1193. What is a security baseline?

A. Minimum security configuration standard for systems

B. Security baseline measurement

C. Backup security config

D. Encryption standard

Answer: A — Baseline = minimum security config. Used as starting point, then customized per system.

Q1194. What is the principle of least privilege (PoLP)?

A. Give users only the access they need (and no more)

B. Remove all privileges

C. Give everyone admin

D. Use long passwords

Answer: A — PoLP: minimum access for job function. Reduces attack surface and blast radius of compromised accounts.

Q1195. What is separation of duties (SoD)?

A. Critical tasks split across multiple people (prevents single point of trust)

B. Job rotation

C. Department isolation

D. Role separation

Answer: A — SoD: e.g., developer can't deploy to production; different person approves. Prevents fraud/errors.

Q1196. What is an incident response plan?

A. Documented process for handling security incidents

B. Backup plan

C. Encryption plan

D. Network plan

Answer: A — IR plan: preparation, identification, containment, eradication, recovery, lessons learned (NIST).

Q1197. What are the NIST IR phases?

A. Preparation, Identification, Containment, Eradication, Recovery, Lessons Learned

B. Detect, Respond, Recover

C. Plan, Do, Check, Act

D. Initiate, Investigate, Resolve

Answer: A — NIST SP 61: 6 phases — Preparation → Identification → Containment → Eradication → Recovery → Lessons Learned.

Q1198. What is a security operations center (SOC)?

A. Team that monitors and responds to security events 24/7

B. Backup operations center

C. Network ops center

D. Disaster recovery site

Answer: A — SOC: 24/7 monitoring, detection, response. Uses SIEM, threat intel, playbooks. Tier 1/2/3 analysts.

Q1199. What is a CSIRT?

A. Computer Security Incident Response Team

B. Cisco Security Incident Response Tool

C. Critical System Incident Response Team

D. Central Security Incident Recovery Team

Answer: A — CSIRT = team that handles security incidents. May be internal or external (e.g., CERT/CC).

Q1200. What is GDPR?

A. General Data Protection Regulation — EU privacy law

B. General Data Privacy Directive

C. Global Data Protection Rule

D. General Database Protection Rule

Answer: A — GDPR: EU privacy law (2018). Gives users rights over their data. Heavy fines for violations.

Q1201. What is HIPAA?

A. Health Insurance Portability and Accountability Act (US healthcare)

B. Health Information Privacy Act

C. Healthcare Information Protection Act

D. Health Insurance Privacy and Authentication Act

Answer: A — HIPAA: US healthcare privacy/security law. Requires safeguards for PHI (Protected Health Information).

Q1202. What is PCI DSS?

A. Payment Card Industry Data Security Standard

B. Personal Credit Information Data Standard

C. Payment Card Industry Digital Security Standard

D. Private Card Information Data Security Standard

Answer: A — PCI DSS: required for any org handling credit cards. 12 requirements, quarterly scans, annual audits.

Q1203. What is SOC 2?

A. Service Organization Control 2 — audits service providers' security

B. Security Operations Center 2

C. Standard Operational Control 2

D. System Operations Center 2

Answer: A — SOC 2: audit report on security, availability, processing integrity, confidentiality, privacy. Common for SaaS.

Q1204. What is ISO 27001?

A. International standard for information security management systems (ISMS)

B. International Security Operations standard

C. Information Security Objective 27001

D. Internet Security Operations 27001

Answer: A — ISO 27001 = certifiable standard for ISMS. Organizations can be certified. Comprehensive security framework.

Q1205. What is the MITRE ATT&CK; framework?

A. Knowledge base of attacker tactics and techniques

B. MITRE Attack Detection Standard

C. Multi-instance Threat Response Engine

D. Microsoft Threat Response Engine

Answer: A — ATT&CK;: taxonomy of adversary tactics (columns) and techniques (rows). Used for detection, IR, threat intel.

Q1206. What is the Cyber Kill Chain?

A. Lockheed Martin's 7-stage model of cyber attacks

B. Network attack graph

C. Encryption protocol

D. Cisco IR framework

Answer: A — Kill Chain: Recon → Weaponize → Deliver → Exploit → Install → C2 → Actions on Objectives.

Q1207. What is multifactor authentication (MFA)?

A. Using 2+ factors: something you know/have/are

B. Two passwords

C. Longer password

D. Password rotation

Answer: A — MFA: combines factors. If one is compromised, attacker still can't access. Reduces 99.9% of account attacks per MS.

Domain 6: Automation & Programmability

110 questions

Q1208. What is SDN?

A. Software-Defined Networking — separates control plane from data plane

B. Software Distribution Network

C. Secure Data Network

D. Static Dynamic Network

Answer: A — SDN centralizes control plane in a controller. Devices become simple forwarders. OpenFlow, NETCONF, RESTCONF.

Q1209. What are the three SDN layers?

A. Application, Control, Infrastructure

B. Physical, Data Link, Network

C. Access, Distribution, Core

D. LAN, MAN, WAN

Answer: A — SDN: Application (apps) ↔ Control (controller) ↔ Infrastructure (devices).

Q1210. What is the SDN northbound API?

A. Interface between Application and Control layers (often REST)

B. Between Control and Infrastructure

C. Between data and control planes

D. External API

Answer: A — Northbound: apps talk to controller via REST/RESTCONF. Southbound: controller talks to devices via OpenFlow/NETCONF.

Q1211. What is the SDN southbound API?

A. Interface between Control and Infrastructure layers (OpenFlow, NETCONF)

B. Between Application and Control

C. External API

D. Customer-facing API

Answer: A — Southbound: controller to devices. OpenFlow (CE), NETCONF, RESTCONF, YANG.

Q1212. What is OpenFlow?

A. Open standard southbound SDN protocol (ONF)

B. Cisco proprietary

C. Northbound API

D. REST API

Answer: A — OpenFlow = first SDN southbound protocol. Modifies flow tables in switches. Open Networking Foundation.

Q1213. What is NETCONF?

A. Network Configuration Protocol — XML over SSH (TCP 830)

B. Network Encryption Configuration

C. Network Communication Protocol

D. Network Connection Protocol

Answer: A — NETCONF = IETF standard for config. Uses SSH (port 830). Data encoded in XML. YANG models the data.

Q1214. What port does NETCONF use?

A. 830 (TCP over SSH)

B. 443

C. 22

D. 8080

Answer: A — NETCONF = TCP 830 over SSH.

Q1215. What is RESTCONF?

A. REST-based API for NETCONF-like operations over HTTPS (443)

B. Replacement for REST

C. Routing protocol

D. Cisco proprietary

Answer: A — RESTCONF = HTTP(S) interface to NETCONF data stores. Uses REST principles. JSON or XML encoded.

Q1216. What port does RESTCONF use?

A. 443 (HTTPS)

B. 830

C. 22

D. 80

Answer: A — RESTCONF uses HTTPS (port 443 by default).

Q1217. What is YANG?

A. Yet Another Next Generation — data modeling language for NETCONF/RESTCONF

B. Yet Another Network Gateway

C. Yield-Aware Network Generation

D. Young Application Network Group

Answer: A — YANG = data modeling language. Models configuration and state data. Used by NETCONF and RESTCONF.

Q1218. What is JSON?

A. JavaScript Object Notation — lightweight data interchange format

B. Java Standard Object Notation

C. Just Some Object Names

D. JavaScript Oriented Naming

Answer: A — JSON = human-readable, lightweight. Used widely in REST APIs. Native to JavaScript.

Q1219. What is XML?

A. eXtensible Markup Language — tag-based data format

B. X-Markup Language

C. Extended Modern Language

D. External Module Language

Answer: A — XML = tag-based, more verbose than JSON. Used in SOAP, NETCONF, SVG, Office docs.

Q1220. What is YAML?

A. YAML Ain't Markup Language — human-friendly data serialization

B. Yet Another Markup Language

C. Yes And Markup Language

D. Young Application Markup Language

Answer: A — YAML = indentation-based, human-readable. Used in Ansible, Kubernetes, CI/CD configs.

Q1221. Which data format is most human-readable?

A. YAML
B. JSON
C. XML
D. Binary

Answer: A — YAML uses indentation (no braces/tags). Most readable for humans; harder for parsers (whitespace-sensitive).

Q1222. Which data format has the smallest file size?

A. JSON (less verbose than XML)
B. XML
C. YAML
D. All similar

Answer: A — JSON has less overhead than XML (no closing tags). YAML similar to JSON. Binary smallest overall.

Q1223. Which data format uses tags?

A. XML
B. JSON
C. YAML
D. All

Answer: A — XML uses opening + closing tags. JSON uses braces. YAML uses indentation.

Q1224. Which data format uses braces and brackets?

A. JSON
B. XML
C. YAML
D. All

Answer: A — JSON: {} for objects, [] for arrays. XML uses tags. YAML uses indentation.

Q1225. Which data format uses indentation (whitespace)?

A. YAML
B. JSON
C. XML
D. All

Answer: A — YAML uses indentation to define structure. No braces or tags. Whitespace-sensitive.

Q1226. What is REST?

A. Representational State Transfer — HTTP-based API style
B. Remote Execution Standard
C. Real-time Event Streaming Transfer
D. Reliable End System Transfer

Answer: A — REST = architectural style using HTTP verbs (GET/POST/PUT/DELETE) on resource URIs.

Q1227. Which HTTP method retrieves a resource?

A. GET
B. POST
C. PUT
D. DELETE

Answer: A — GET = retrieve. Should be safe and idempotent (no state change).

Q1228. Which HTTP method creates a new resource?

A. POST
B. GET
C. DELETE
D. HEAD
Answer: A — POST = create new resource. Not idempotent (multiple POSTs create multiple resources).

Q1229. Which HTTP method updates an existing resource?

A. PUT
B. GET
C. DELETE
D. HEAD
Answer: A — PUT = update (or create if missing). Idempotent (multiple PUTs = same result).

Q1230. Which HTTP method deletes a resource?

A. DELETE
B. GET
C. POST
D. PUT
Answer: A — DELETE = remove resource. Idempotent.

Q1231. Which HTTP method retrieves only headers (no body)?

A. HEAD
B. GET
C. OPTIONS
D. TRACE
Answer: A — HEAD = like GET but only headers. Useful for checking existence without downloading body.

Q1232. Which HTTP method asks which methods are allowed?

A. OPTIONS
B. GET
C. HEAD
D. TRACE
Answer: A — OPTIONS = list supported methods for a resource. Used in CORS preflight.

Q1233. What is HTTP status code 200?

A. OK (successful)
B. Created
C. Not Found
D. Server Error
Answer: A — 200 OK = successful response.

Q1234. What is HTTP status code 201?

A. Created (POST/PUT success, new resource created)
B. OK
C. Bad Request
D. Not Found
Answer: A — 201 Created = resource successfully created.

Q1235. What is HTTP status code 204?

A. No Content (success, no body to return)

B. Created

C. Not Found

D. Server Error

Answer: A — 204 No Content = success but no body. Common for DELETE operations.

Q1236. What is HTTP status code 301?

A. Moved Permanently (redirect)

B. OK

C. Not Found

D. Server Error

Answer: A — 301 = permanent redirect. Client should use new URL.

Q1237. What is HTTP status code 400?

A. Bad Request (client error, malformed request)

B. OK

C. Unauthorized

D. Not Found

Answer: A — 400 = client sent invalid request (syntax error, missing required field).

Q1238. What is HTTP status code 401?

A. Unauthorized (authentication required)

B. Forbidden

C. Not Found

D. Bad Request

Answer: A — 401 = must authenticate. Sends WWW-Authenticate header.

Q1239. What is HTTP status code 403?

A. Forbidden (authenticated but not allowed)

B. Unauthorized

C. Not Found

D. Bad Request

Answer: A — 403 = authenticated but lacks permission. Different from 401 (not authenticated).

Q1240. What is HTTP status code 404?

A. Not Found (resource doesn't exist)

B. OK

C. Bad Request

D. Server Error

Answer: A — 404 = resource not found. Most common client error.

Q1241. What is HTTP status code 500?

A. Internal Server Error (server-side problem)

B. Bad Request

C. Not Found

D. OK

Answer: A — 500 = generic server error. Indicates bug in server code.

Q1242. What is HTTP status code 503?

A. Service Unavailable (server overloaded or down)

B. Server Error

C. Not Found

D. OK

Answer: A — 503 = temporary unavailability. Often used during maintenance or overload.

Q1243. What is HTTP status code 429?

A. Too Many Requests (rate limited)

B. Not Found

C. Bad Request

D. Server Error

Answer: A — 429 = client sent too many requests. Includes Retry-After header.

Q1244. What is an API?

A. Application Programming Interface — defined way for software to communicate

B. Application Protocol Interface

C. Advanced Programming Interface

D. Auto Programming Interface

Answer: A — API = set of rules for how software components interact. REST APIs are common in networking.

Q1245. What is an API key?

A. Unique string identifying the caller (often used for auth and rate limiting)

B. Encryption key

C. Server key

D. Hash key

Answer: A — API key = secret string sent in request header. Identifies caller. Used for auth and quota.

Q1246. What is OAuth 2.0?

A. Authorization framework — delegated access (e.g., 'log in with Google')

B. Encryption protocol

C. Authentication protocol

D. Routing protocol

Answer: A — OAuth 2.0 = delegated authorization. Lets user grant third-party app access without sharing password.

Q1247. What is JWT?

A. JSON Web Token — compact, signed token for auth

B. Java Web Token

C. JavaScript Web Token

D. JSON Web Transfer

Answer: A — JWT = JSON-based token. Header, payload, signature. Used for stateless auth in REST APIs.

Q1248. What is a webhook?

A. HTTP callback — system sends HTTP POST to your URL when event happens

B. Web hook

C. Web cache

D. Web request

Answer: A — Webhook = reverse API. Instead of polling, server pushes events to your endpoint. Examples: GitHub, Stripe.

Q1249. What is rate limiting?

A. Restricting number of API calls per time window

B. Encrypting API calls

C. Caching API calls

D. Compressing API calls

Answer: A — Rate limiting protects APIs from abuse/DoS. Returns 429 if exceeded. May include Retry-After.

Q1250. What is API pagination?

A. Returning large result sets in chunks (pages)

B. Encrypting pages

C. Caching pages

D. Routing pages

Answer: A — Pagination: large responses split into pages (?page=2&limit;=50). Avoids huge single response.

Q1251. What is idempotency?

A. Same operation repeated gives same result (no extra effect)

B. Same input always fails

C. Operation can't be repeated

D. Operation is fast

Answer: A — Idempotent: GET, PUT, DELETE are idempotent. POST is not (each call creates new resource).

Q1252. What is the Cisco DevNet?

A. Cisco's developer program with sandboxes, learning, APIs

B. Cisco dev tools

C. Cisco network devices

D. Cisco certification

Answer: A — DevNet = Cisco's developer program. Free sandboxes, API docs, learning labs, certifications.

Q1253. What is Cisco DNA Center?

A. Cisco's SDN controller for campus (SD-Access)

B. Cisco router

C. Cisco switch

D. Cisco firewall

Answer: A — DNA Center = intent-based networking controller. Provides automation, assurance, fabric.

Q1254. What is Cisco SD-Access?

A. Cisco's SDN solution for campus networks (built on DNA Center, LISP, VXLAN)

B. Cisco switching protocol

C. Cisco wireless

D. Cisco routing protocol

Answer: A — SD-Access = Cisco's campus SDN. Uses LISP for control plane, VXLAN for data plane, ISE for policy.

Q1255. What is Cisco ACI?

A. Cisco's SDN solution for data centers (APIC controller, VXLAN)

B. Cisco's campus SDN

C. Cisco routing protocol

D. Cisco wireless

Answer: A — ACI = Application Centric Infrastructure. Data center SDN. APIC controller + VXLAN + spine-leaf.

Q1256. What is Cisco SD-WAN?

A. Cisco's SDN solution for WAN (vManage, vSmart, vBond)

B. Cisco campus SDN

C. Cisco wireless

D. Cisco data center SDN

Answer: A — SD-WAN: separates data plane (routers) from control plane (vSmart). Policy-based routing across WAN links.

Q1257. What is LISP?

A. Locator/ID Separation Protocol — separates identity from location

B. List Processing

C. Lightweight Internet Service Protocol

D. Local IP Service Protocol

Answer: A — LISP = used in Cisco SD-Access. Maps endpoint ID (EID) to routing locator (RLOC).

Q1258. What is VXLAN?

A. Virtual eXtensible LAN — encapsulates L2 frames in UDP for overlay networks

B. VPN Extension LAN

C. Virtual X LAN

D. Variable Length LAN

Answer: A — VXLAN = encapsulates L2 in UDP 4789. Allows L2 extension across L3. 16M VNIs (vs 4094 VLANs).

Q1259. What is Python?

A. General-purpose programming language widely used for automation

B. Type of snake

C. Cisco proprietary language

D. Database language

Answer: A — Python = most popular automation language. Simple syntax, huge ecosystem (Netmiko, NAPALM, Nornir).

Q1260. Which Python library is commonly used for SSH network automation?

A. Netmiko (built on Paramiko)

B. Requests

C. BeautifulSoup

D. NumPy

Answer: A — Netmiko = wrapper around Paramiko for network devices. Supports Cisco, Juniper, Arista, etc.

Q1261. Which Python library is used for REST API calls?

A. Requests

B. Netmiko

C. Paramiko

D. Pandas

Answer: A — Requests = de-facto Python HTTP library. Simple API: requests.get(url), requests.post(url, json=data).

Q1262. Which Python library is used for RESTCONF?

A. Requests (with auth, headers, JSON)

B. Netmiko

C. Paramiko

D. PyEAPI

Answer: A — Requests works for RESTCONF — just set headers (Accept: application/yang-data+json, Content-Type).

Q1263. Which Python library parses XML?

A. xml.etree.ElementTree (built-in)

B. Requests

C. NumPy

D. Pandas

Answer: A — Standard library: xml.etree.ElementTree. Also: lxml (faster, more features).

Q1264. Which Python library parses JSON?

A. json (built-in)

B. Requests

C. BeautifulSoup

D. lxml

Answer: A — Standard library: json. json.loads(string) → dict. json.dumps(obj) → string.

Q1265. Which Python library parses YAML?

A. PyYAML (pyyaml)

B. json

C. Requests

D. Netmiko

Answer: A — PyYAML: import yaml; yaml.safe_load(stream). Use safe_load (not load) for security.

Q1266. What is Ansible?

A. Agentless config management tool using SSH (Red Hat)

B. Cisco proprietary

C. Compiled language

D. Database

Answer: A — Ansible = agentless (uses SSH/WinRM). YAML playbooks. Idempotent. Owned by Red Hat.

Q1267. What language do Ansible playbooks use?

A. YAML

B. JSON

C. XML

D. Python

Answer: A — Ansible playbooks are YAML. Templates use Jinja2. Modules can be written in Python.

Q1268. Is Ansible agentless?

A. Yes — uses SSH/WinRM, no agent on target

B. No — requires agent

C. Both agent and agentless modes

D. Only on Windows

Answer: A — Ansible = agentless. SSH to Linux, WinRM to Windows. No daemon to install on target.

Q1269. What is an Ansible playbook?

A. YAML file defining tasks to execute on hosts

B. Configuration file

C. Python script

D. Encryption key

Answer: A — Playbook = YAML file with one or more plays. Each play has tasks (modules) executed on hosts.

Q1270. What is an Ansible module?

A. Reusable code unit that performs a specific task

B. Configuration file

C. Network device

D. Encryption key

Answer: A — Module = reusable code (e.g., ios_command, copy, apt). Idempotent. Written in Python.

Q1271. What is an Ansible inventory?

A. File listing hosts/groups to manage

B. List of playbooks

C. List of modules

D. List of tasks

Answer: A — Inventory = INI or YAML file listing hosts and groups. Examples: [routers], [switches], [webservers].

Q1272. What is idempotency in Ansible?

A. Running playbook multiple times produces same result (no unwanted changes)

B. Each run changes something

C. Each run fails

D. Each run is faster

Answer: A — Idempotent: only changes system if needed. Re-running playbook doesn't redo already-applied changes.

Q1273. What is Jinja2?

A. Python templating engine used by Ansible

B. JavaScript library

C. Cisco protocol

D. Database

Answer: A — Jinja2 = templating language. Used by Ansible for variable substitution, loops, conditionals.

Q1274. What is Terraform?

A. Infrastructure-as-Code tool (HashiCorp) using HCL

B. Cisco proprietary

C. Database

D. Compiled language

Answer: A — Terraform = declarative IaC. Defines desired state in HCL. Creates/updates/destroys resources.

Q1275. What language does Terraform use?

A. HCL (HashiCorp Configuration Language)

B. YAML

C. JSON

D. Python

Answer: A — Terraform uses HCL — declarative, block-based. JSON also supported.

Q1276. What is Git?

A. Distributed version control system

B. Programming language

C. Network protocol

D. Routing protocol

Answer: A — Git = most popular VCS. Created by Linus Torvalds (2005). Tracks changes, branches, merges.

Q1277. What is GitHub?

A. Cloud hosting service for Git repositories

B. Programming language

C. Database

D. Network device

Answer: A — GitHub (Microsoft) = most popular Git hosting. Also: GitLab, Bitbucket. Add collaboration tools.

Q1278. What is GitLab?

A. Git hosting + CI/CD + DevOps platform

B. Programming language

C. Database

D. Compiler

Answer: A — GitLab = Git hosting with built-in CI/CD, issue tracking, container registry. Self-hosted or SaaS.

Q1279. What is a Git commit?

A. Snapshot of changes saved to local repository

B. Push to remote

C. Branch

D. Tag

Answer: A — Commit = save changes to local repo with message. Includes changes since last commit.

Q1280. What is a Git push?

A. Upload local commits to remote repository

B. Download from remote

C. Branch

D. Tag

Answer: A — Push = upload commits to remote (GitHub/GitLab). Counterpart: pull = download.

Q1281. What is a Git pull?

A. Download and merge changes from remote

B. Upload changes

C. Create commit

D. Create branch

Answer: A — Pull = fetch (download) + merge. Gets latest changes from remote into local branch.

Q1282. What is a Git branch?

A. Independent line of development

B. Commit

C. Tag

D. Remote

Answer: A — Branch = parallel line of work. Main/master = default. Feature branches for new work.

Q1283. What is a Git merge?

A. Combine changes from one branch into another

B. Split branches

C. Delete branch

D. Push changes

Answer: A — Merge: combine branches. Fast-forward (if linear) or 3-way (if divergent, creates merge commit).

Q1284. What is a Git clone?

A. Download full repository (history + files) to local

B. Create new repo

C. Commit changes

D. Branch

Answer: A — Clone = copy entire repo (all history) from remote to local. Creates new local working copy.

Q1285. What is a Git fork?

A. Personal copy of someone else's repository (on hosting service)

B. Branch

C. Commit

D. Tag

Answer: A — Fork = server-side copy of repo (typically on GitHub). Lets you modify without affecting original.

Q1286. What is a pull request (PR)?

A. Request to merge your branch/fork into another repo

B. Download changes

C. Branch

D. Commit

Answer: A — PR = proposed change. Reviewed before merge. Common in open-source and team workflows.

Q1287. What is a merge conflict?

A. When two changes can't be auto-merged (require manual resolution)

B. Failed push

C. Failed pull

D. Failed commit

Answer: A — Conflict: same lines changed differently in two branches. Git pauses merge, asks user to resolve.

Q1288. What is CI/CD?

A. Continuous Integration / Continuous Deployment (or Delivery)

B. Cisco Internet Configuration Daemon

C. Common Interface Configuration Data

D. Customer Initiated Channel Data

Answer: A — CI = auto-build/test on commit. CD = auto-deploy after CI passes. Automates software delivery.

Q1289. What is the difference between Continuous Delivery and Continuous Deployment?

A. Delivery: auto-prepares release, manual deploy. Deployment: auto-deploys.

B. Same thing

C. Delivery is faster

D. Deployment is faster

Answer: A — Delivery: every change is deployable (manual go). Deployment: every change IS deployed (after tests).

Q1290. What is Jenkins?

A. Open-source CI/CD automation server

B. Programming language

C. Database

D. Network device

Answer: A — Jenkins = popular CI/CD server. Plugins for everything. Pipelines defined in Jenkinsfile (Groovy).

Q1291. What is a Docker container?

A. Lightweight, portable package of app + dependencies

B. Virtual machine

C. Hypervisor

D. Compiler

Answer: A — Container = app + deps packaged. Shares host OS kernel. Lighter than VM. Docker is the platform.

Q1292. What is Kubernetes?

A. Container orchestration platform

B. Container runtime

C. Container image

D. Container registry

Answer: A — K8s = orchestrates containers at scale. Auto-scaling, load balancing, rolling updates, self-healing.

Q1293. What is the difference between a container and a VM?

A. Containers share host OS kernel; VMs have full guest OS

B. VMs are faster

C. Same thing

D. Containers need hypervisor

Answer: A — Container: lightweight, shares OS, fast startup. VM: heavier, full OS, slower startup, better isolation.

Q1294. What is a Dockerfile?

A. Text file with instructions to build a Docker image

B. Container image

C. Container runtime

D. Container registry

Answer: A — Dockerfile = recipe for building image. Each instruction adds a layer.

Q1295. What is a Docker image?

A. Read-only template used to create containers

B. Running container

C. Container runtime

D. Container network

Answer: A — Image = template (read-only). Container = running instance of image. Images stored in registry.

Q1296. What is a Docker registry?

A. Storage and distribution system for Docker images

B. Container runtime

C. Container

D. Hypervisor

Answer: A — Registry = stores images. Docker Hub (public), Harbor, ECR, GCR, ACR (private).

Q1297. What is Docker Hub?

A. Public Docker registry (default)

B. Container runtime

C. Hypervisor

D. Compiler

Answer: A — Docker Hub = public registry at hub.docker.com. Default registry for Docker commands.

Q1298. What is a YAML manifest in Kubernetes?

A. Declarative config file defining desired state of resources

B. Container image

C. Container runtime

D. Container network

Answer: A — K8s manifest = YAML describing Deployment, Service, Pod, etc. Apply with 'kubectl apply -f file.yaml'.

Q1299. What is kubectl?

A. Command-line tool to interact with Kubernetes cluster

B. Container runtime

C. Container image

D. Container registry

Answer: A — kubectl = K8s CLI. Examples: kubectl get pods, kubectl apply -f, kubectl logs.

Q1300. What is a Kubernetes Pod?

A. Smallest deployable unit; contains one or more containers

B. Container

C. Image

D. Service

Answer: A — Pod = group of 1+ containers sharing network/storage. Smallest K8s unit. Usually 1 container per pod.

Q1301. What is a Kubernetes Deployment?

A. Manages a set of identical pods (replicas)

B. Container

C. Pod

D. Image

Answer: A — Deployment = declarative pod management. Ensures N replicas running. Handles rolling updates.

Q1302. What is a Kubernetes Service?

A. Stable network endpoint for accessing pods (which have changing IPs)

B. Container

C. Pod

D. Image

Answer: A — Service = stable IP/DNS for pod set. Types: ClusterIP, NodePort, LoadBalancer.

Q1303. What is a CI/CD pipeline?

A. Automated sequence of build, test, deploy steps

B. Network pipeline

C. Encryption pipeline

D. Database pipeline

Answer: A — Pipeline = stages (build → test → deploy). Each stage runs only if previous succeeds.

Q1304. What is infrastructure as code (IaC)?

A. Managing infrastructure via declarative code files

B. Manual infrastructure config

C. Hardware spec

D. Network diagram

Answer: A — IaC = define infra in code (Terraform, Ansible, CloudFormation). Reproducible, version-controlled.

Q1305. What is ChatOps?

A. Integrating ops tools into chat (Slack, Teams)

B. Chat protocol

C. Chat encryption

D. Chat filter

Answer: A — ChatOps: run commands, view alerts in chat. Bot translates chat commands to API calls.

Q1306. What is observability?

A. Knowing system state from external outputs (logs, metrics, traces)

B. Monitoring

C. Encryption

D. Routing

Answer: A — Observability = logs + metrics + traces. Lets you understand system behavior. Tools: Prometheus, Grafana, Jaeger.

Q1307. What is Prometheus?

A. Open-source time-series metrics database + scraping

B. Monitoring protocol

C. Routing protocol

D. Database

Answer: A — Prometheus = CNCF project. Pulls metrics via HTTP. PromQL query language. Common with Grafana.

Q1308. What is Grafana?

A. Open-source dashboarding and visualization tool

B. Database

C. Routing protocol

D. Monitoring protocol

Answer: A — Grafana = visualization for time-series data. Connects to Prometheus, InfluxDB, Elasticsearch, etc.

Q1309. What is the ELK stack?

A. Elasticsearch, Logstash, Kibana — log analysis stack

B. Encryption Logging Key

C. Extended Layer Kernel

D. Enterprise Linux Kernel

Answer: A — ELK: Elasticsearch (search/store), Logstash (collect/parse), Kibana (visualize). Now called Elastic Stack.

Q1310. What is Splunk?

A. Commercial SIEM/log analysis platform

B. Open-source database

C. Routing protocol

D. Compiler

Answer: A — Splunk = commercial log analytics + SIEM. Widely used in enterprise. SPL query language.

Q1311. What is model-driven programmability?

A. Managing devices via data models (YANG) over NETCONF/RESTCONF

B. Manual CLI config

C. Scripting only

D. Hardware programming

Answer: A — MDP = use structured data models (YANG) + APIs (NETCONF/RESTCONF) instead of CLI screen-scraping.

Q1312. What is screen scraping?

A. Parsing CLI output as text (fragile, breaks on format changes)

B. Network protocol

C. Encryption

D. Database query

Answer: A — Screen scraping = parse CLI text. Brittle — minor format change breaks script. Avoid in favor of APIs.

Q1313. What is Cisco pyATS?

A. Cisco's Python automation and testing framework

B. Cisco router

C. Cisco switch

D. Cisco protocol

Answer: A — pyATS = Cisco's Python framework for network testing/automation. Genie parser library included.

Q1314. What is NAPALM?

A. Cross-vendor network automation library (Python)

B. Cisco proprietary

C. Database

D. Routing protocol

Answer: A — NAPALM = Network Automation and Programmability Abstraction Layer. Uniform API across vendors.

Q1315. What is Nornir?

A. Python multi-threaded automation framework

B. Cisco proprietary

C. Database

D. Compiler

Answer: A — Nornir = parallel SSH automation framework. Uses Netmiko/NAPALM under the hood. Faster than serial.

Q1316. What is the difference between imperative and declarative?

A. Imperative: how to do it (steps). Declarative: what to do (desired state).

B. Same thing

C. Imperative is faster

D. Declarative is faster

Answer: A — Imperative: Ansible (mostly). Declarative: Terraform, Kubernetes, Cisco DNA Center intent-based.

Q1317. What is intent-based networking (IBN)?

A. Network configures itself based on business intent (policy)

B. Manual configuration

C. CLI scripting

D. Hardware programming

Answer: A — IBN = translate business intent to network policy. Cisco DNA Center, SD-Access. Captures 'what' not 'how'.

PART 20

Career Roadmap

From help desk to enterprise network engineer

Chapter 20: Your Networking Career Roadmap

Earning your CCNA is just the beginning. This chapter maps out the career paths available to network engineers, from entry-level help desk roles through senior architect positions. We'll cover typical job titles, salary ranges, skill requirements, certifications, and what to learn next. Plus: resume templates, interview questions, and a home lab guide.

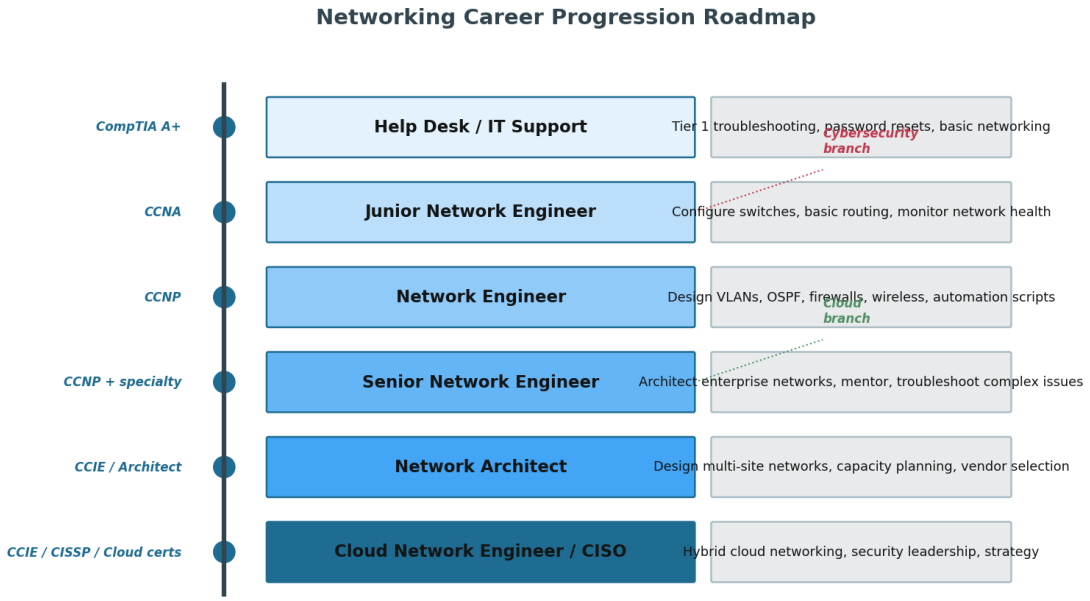


Figure: Networking career progression: Help Desk → Junior NetEng → NetEng → Senior → Architect. Side branches to cybersecurity and cloud.

Career Paths

Stage	Typical Title	Years	Salary (US)	Key Skills
1	Help Desk / IT Support	0-2	$40-55K	Troubleshooting, Windows, basic networking
2	Junior Network Engineer	1-3	$55-75K	CCNA, switching, routing, basic firewalls
3	Network Engineer	3-7	$75-110K	CCNP, OSPF/BGP, automation, security
4	Senior Network Engineer	7-12	$110-145K	Architecture, mentoring, complex troubleshooting
5	Network Architect	12+	$140-200K+	Enterprise design, vendor selection, strategy
6	Principal/Cloud NetEng	15+	$180-250K+	Hybrid cloud, multi-cloud networking, leadership

Help Desk → Network Engineer Path

Most network engineers start at help desk. This is actually a great foundation — you learn how users interact with technology, how to troubleshoot systematically, and how networks fail in the real world. Use your help desk time to study for CCNA, volunteer for network-related tickets, and shadow the network team. After 1-2 years and a CCNA, you're ready to apply for junior network engineer roles.

Junior Network Engineer Path

As a junior network engineer, you'll handle routine tasks: switch port configurations, VLAN changes, firewall rule additions, troubleshooting connectivity issues. You'll work under a senior engineer who reviews your changes. Focus on: mastering the CLI, understanding the existing network design, learning the change management process, and starting to learn automation (Python, Ansible). Start studying for CCNP after 1-2 years.

NOC Engineer Path

NOC (Network Operations Center) engineers monitor networks 24/7, respond to alerts, and escalate complex issues. Often a stepping stone to network engineer roles. Excellent for learning monitoring tools (SolarWinds, PRTG, Splunk), incident response, and how networks behave under stress. Many NOCs require shift work (nights/weekends) but pay well and offer rapid learning.

Cybersecurity Path

Network engineering is a natural stepping stone to cybersecurity. After CCNA, consider: CCNA CyberOps, CompTIA Security+, then CCNP Security or CISSP. Roles: SOC analyst, security engineer, penetration tester, security architect. Network knowledge is fundamental to security — you can't secure what you don't understand.

Cloud Networking Path

As workloads move to AWS, Azure, and GCP, cloud networking skills are in high demand. Path: CCNA → AWS Certified Advanced Networking Specialty → Azure Network Engineer Associate. Roles: Cloud Network Engineer, Cloud Architect, DevOps Engineer (with networking focus). Hybrid cloud (connecting on-prem to cloud) is especially hot.

Resume Template for Junior Network Engineer

```
JANE NETWORKENGINEER
San Francisco, CA | jane@example.com | (555) 123-4567
LinkedIn: linkedin.com/in/janenetworkengineer
GitHub: github.com/janenetworkengineer

========================================
SUMMARY
========================================
CCNA-certified IT professional with 2+ years of help desk experience
and a passion for networking. Built and documented 50+ Packet Tracer
lab topologies covering VLANs, OSPF, ACLs, and IPsec VPNs. Seeking
junior network engineer role to apply hands-on skills in enterprise
environment.

========================================
CERTIFICATIONS
========================================
• Cisco Certified Network Associate (CCNA 200-301) — 2025
• CompTIA Network+ — 2024
• CompTIA A+ — 2023

========================================
TECHNICAL SKILLS
========================================
Networking: TCP/IP, OSPF, EIGRP, BGP, VLANs, STP, EtherChannel,
            HSRP/VRRP, ACLs, NAT, VPN (IPsec, SSL), QoS
Protocols: DNS, DHCP, HTTP/S, SSH, SNMP, Syslog, NTP, RADIUS
Hardware:  Cisco Catalyst switches (2960, 9200, 9300),
           Cisco ISR routers, Cisco ASA/Firepower
Tools:     Wireshark, Cisco Packet Tracer, GNS3, Putty, Nmap
Automation: Python (Netmiko, NAPALM), Ansible basics, Git
OS:        Windows Server, Linux (Ubuntu, CentOS)

========================================
EXPERIENCE
========================================
IT Support Specialist | Acme Corp, San Francisco | 2023 – Present
• Resolve 30+ daily help desk tickets (Tier 1-2) with 95% first-call
  resolution rate.
• Configure and troubleshoot 200+ user workstations, including VPN
  access and network printer setup.
• Assist network team with switch port configurations, VLAN changes,
  and basic troubleshooting — completed 50+ network tickets.
• Maintain inventory of 500+ networked devices using SolarWinds.
• Documented common troubleshooting procedures — reduced Tier 1
  escalation rate by 30%.

Help Desk Technician | TechStart Inc | 2022 – 2023
• Handled 50+ daily support calls for password resets, software
  installs, and connectivity issues.
• Achieved 4.8/5 customer satisfaction score.
• Promoted to Tier 2 after 6 months.

========================================
EDUCATION
========================================
Bachelor of Science in Information Technology
```

```
State University | 2018 – 2022 | GPA 3.6

========================================
PROJECTS
========================================
Enterprise Network Simulation (Packet Tracer)
github.com/janenetworkengineer/enterprise-lab
• Built 10-device enterprise network with 3-tier design, OSPF, BGP,
  redundant links, VPN, and QoS.
• Documented full IP plan, VLAN plan, and configuration files.

Python Network Automation
github.com/janenetworkengineer/net-automation
• Scripts using Netmiko to back up configs, audit VLANs, and push
  changes to 10+ devices.
• Reduced config backup time from 2 hours to 5 minutes.
```

Common Interview Questions

Be prepared to answer these in a network engineer interview:

Q1. Walk me through the OSI model and what happens at each layer.

Q2. Explain the TCP three-way handshake.

Q3. What's the difference between TCP and UDP? When would you use each?

Q4. How does ARP work? Walk me through the ARP process.

Q5. How does a switch learn MAC addresses?

Q6. Explain the difference between a hub, switch, and router.

Q7. What is a VLAN? Why use them?

Q8. Explain 802.1Q trunking.

Q9. How does STP prevent loops? Walk me through the root bridge election.

Q10. What's the difference between RSTP and classic STP?

Q11. Explain OSPF areas. Why use multi-area?

Q12. How does OSPF elect a DR/BDR?

Q13. What's the OSPF cost formula? What's the default reference bandwidth?

Q14. What is administrative distance? Why does it matter?

Q15. Explain longest-prefix-match.

Q16. What is NAT? Difference between static, dynamic, and PAT?

Q17. How does DHCP work? Walk through DORA.

Q18. What is a default gateway? Why do hosts need one?

Q19. Explain the difference between standard and extended ACLs.

Q20. Where should you place a standard ACL? An extended ACL? Why?

Q21. What is HSRP? How does it work?

Q22. How does a site-to-site VPN differ from a remote-access VPN?

Q23. What is IPsec? AH vs ESP? Transport vs Tunnel mode?

Q24. Explain the CIA triad.

Q25. What's the difference between IDS and IPS?

Q26. What is port security? What are the violation modes?

Q27. How would you troubleshoot a PC that can't reach the Internet?

Q28. Walk through troubleshooting a slow network.

Q29. What is SDN? How does it differ from traditional networking?

Q30. What is REST? What HTTP methods does it use?

Q31. Explain JSON vs XML vs YAML.

Q32. How would you secure a Cisco switch? (Hardening best practices.)

Q33. Tell me about a network problem you solved. Walk me through your process.

Q34. How do you stay current with networking technology?

Q35. Describe a network you designed or built. What did you learn?

Home Lab Guide

Building a home lab is the single best thing you can do for your networking career. Theory only goes so far — you need hands-on practice. Here are three options at different price points:

Option	Cost	What You Get	Best For
Packet Tracer (Free)	$0	Cisco simulator, limited but covers CCNA	Beginners, CCNA study
GNS3 + Cisco IOS images	$0-200	Real IOS images in VMs, full CLI	CCNP-level routing
EVE-NG	$0-500	Multi-vendor simulator, powerful	CCNP/CCIE, multi-vendor
Used Cisco gear (eBay)	$200-1000	Real switches/routers, physical	Touching real hardware
Cloud lab (Cisco DevNet sandbox)	$0	Free Cisco sandboxes, real gear	Automation practice

Recommended Home Lab Build (Budget $300)

- 2× Cisco Catalyst 2960 switches (used, eBay): ~$50 each
- 1× Cisco ISR 1841 or 1921 router: ~$50
- 2× Cisco 3560 L3 switches (for routing labs): ~$60 each
- Console cable + USB adapter: ~$15
- Cat5e/Cat6 cables: ~$20
- Old laptop with GNS3/EVE-NG: $0 (repurpose)

Total: ~$305 — enough for CCNA + CCNP Routing & Switching.

Portfolio Projects to Build

Document these projects on GitHub to show employers what you've built:

- **Multi-VLAN Enterprise:** 3+ switches, 5+ VLANs, inter-VLAN routing, ACLs
- **OSPF Multi-Area:** 4+ routers, multi-area design, summarization

- **HSRP/Redundancy:** 2 routers with HSRP, track upstream, failover test
- **Site-to-Site VPN:** 2 routers, IPsec tunnel, encrypt between sites
- **Python Automation:** Netmiko scripts for backup, audit, config push
- **Ansible Playbook:** Configure 5+ devices with one playbook
- **Monitoring Stack:** LibreNMS or Zabbix monitoring your lab
- **Documentation:** Full topology, IP plan, configs in Git repo

Next Certifications After CCNA

Certification	Focus	Difficulty	Time	Value
CCNP Enterprise	Advanced routing, switching, wireless	Hard	6-12 months	High
CCNA Security (legacy) / CCNP Security	Network security, firewalls, IPS	Medium-Hard	6-12 months	High
CompTIA Security+	Broad security fundamentals	Medium	2-3 months	Good baseline
CCNA CyberOps	SOC analyst skills	Medium	3-4 months	Specialized
AWS Advanced Networking Specialty	Cloud networking on AWS	Hard	4-6 months	Very high (cloud)
Azure Network Engineer Associate	Cloud networking on Azure	Medium-Hard	3-4 months	High (cloud)
CISSP	Security management	Very hard	12+ months	High (management)
CCIE Enterprise Infrastructure	Expert level, lab exam	Extremely hard	12-24 months	Top tier

Final Words

Networking is a fantastic career — intellectually stimulating, well-paid, and constantly evolving. The CCNA is your ticket in, but curiosity and continuous learning are what will take you to the top. Build labs, document your work, network with other engineers (yes, humans networking about networks), and never stop learning. Welcome to the field.